29. $\vec{\mathbf{F}}(x,y) = xy\,\hat{\mathbf{i}} + (x+y)\,\hat{\mathbf{j}}, \quad C : x^2 + y^2 = 4$

30. $\vec{\mathbf{F}}(x,y) = (x^{3/2} - 3y)\,\hat{\mathbf{i}} + (6x + 5\sqrt{y})\,\hat{\mathbf{j}}, \quad C$: boundary of a triangle with vertices $(0,0)$, $(5,0)$, and $(0,5)$

31. Evaluate $\int_C (2x^3 - y^3)\,dx + (x^3 + y^3)\,dy$, where C is a unit circle oriented in the counterclockwise direction.

32. A particle starts at point $(-2,0)$, moves along the x-axis to $(2,0)$, and then travels along semicircle $y = \sqrt{4 - x^2}$ to the starting point. Use Green's theorem to find the work done on this particle by force field $\vec{\mathbf{F}}(x,y) = x\,\hat{\mathbf{i}} + (x^3 + 3xy^2)\,\hat{\mathbf{j}}$.

33. David and Sandra are skating on a frictionless pond in the wind. David skates on the inside, going along a circle of radius 2 in a counterclockwise direction. Sandra skates once around a circle of radius 3, also in the counterclockwise direction. Suppose the force of the wind at point (x,y) is $\vec{\mathbf{F}}(x,y) = (x^2y + 10y)\,\hat{\mathbf{i}} + (x^3 + 2xy^2)\,\hat{\mathbf{j}}$. Use Green's theorem to determine who does more work.

34. Use Green's theorem to find the work done by force field $\vec{\mathbf{F}}(x,y) = (3y - 4x)\,\hat{\mathbf{i}} + (4x - y)\,\hat{\mathbf{j}}$ when an object moves once counterclockwise around ellipse $4x^2 + y^2 = 4$.

35. Use Green's theorem to evaluate line integral $\oint_C e^{2x}\sin 2y\,dx + e^{2x}\cos 2y\,dy$, where C is ellipse $9(x-1)^2 + 4(y-3)^2 = 36$ oriented counterclockwise.

36. Evaluate line integral $\oint_C y^2\,dx + x^2\,dy$, where C is the boundary of a triangle with vertices $(0,0)$, $(1,1)$, and $(1,0)$, with the counterclockwise orientation.

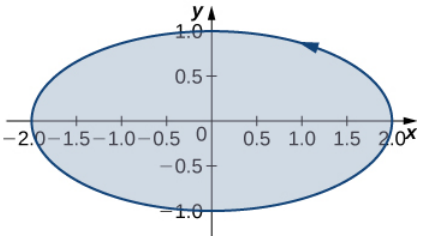
37. Use Green's theorem to evaluate line integral $\int_C \vec{\mathbf{h}} \cdot d\vec{\mathbf{r}}$ if $\vec{\mathbf{h}}(x,y) = e^y\,\hat{\mathbf{i}} - \sin \pi x\,\hat{\mathbf{j}}$, where C is a triangle with vertices $(1,0)$, $(0,1)$, and $(-1,0)$, traversed counterclockwise.

38. Use Green's theorem to evaluate line integral $\int_C \sqrt{1 + x^3}\,dx + 2xy\,dy$ where C is a triangle with vertices $(0,0)$, $(1,0)$, and $(1,3)$ oriented clockwise.

39. Use Green's theorem to evaluate line integral $\int_C x^2y\,dx - xy^2\,dy$ where C is a circle $x^2 + y^2 = 4$ oriented counterclockwise.

40. Use Green's theorem to evaluate line integral $\int_C \left(3y - e^{\sin x}\right)\,dx + \left(7x + \sqrt{y^4 + 1}\right)\,dy$ where C is circle $x^2 + y^2 = 9$ oriented in the counterclockwise direction.

41. Use Green's theorem to evaluate line integral $\int_C (3x - 5y)\,dx + (x - 6y)\,dy$, where C is ellipse $\frac{x^2}{4} + y^2 = 1$ and is oriented in the counterclockwise direction.



42. Let C be a triangular closed curve from $(0,0)$ to $(1,0)$ to $(1,1)$ and finally back to $(0,0)$. Let $\vec{\mathbf{F}}(x,y) = 4y\,\hat{\mathbf{i}} + 6x^2\,\hat{\mathbf{j}}$. Use Green's theorem to evaluate $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$.

43. Use Green's theorem to evaluate line integral $\oint_C y\,dx - x\,dy$, where C is circle $x^2 + y^2 = a^2$ oriented in the clockwise direction.

44. Use Green's theorem to evaluate line integral $\oint_C (y + x)\,dx + (x + \sin y)\,dy$, where C is any smooth simple closed curve joining the origin to itself oriented in the counterclockwise direction.

45. Use Green's theorem to evaluate line integral $\oint_C (y - \ln(x^2 + y^2))\ dx + \left(2 \arctan \frac{y}{x}\right)\ dy$, where C is the positively oriented circle $(x - 2)^2 + (y - 3)^2 = 1$.

46. Use Green's theorem to evaluate $\oint_C xy\ dx + x^3y^3\ dy$, where C is a triangle with vertices $(0, 0)$, $(1, 0)$, and $(1, 2)$ with positive orientation.

47. Use Green's theorem to evaluate line integral $\int_C \sin y\ dx + x \cos y\ dy$, where C is ellipse $x^2 + xy + y^2 = 1$ oriented in the counterclockwise direction.

48. Let $\vec{\mathbf{F}}(x, y) = (\cos(x^5) - 13y^3)\ \hat{\mathbf{i}} + 13x^3\ \hat{\mathbf{j}}$. Find the counterclockwise circulation $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where C is a curve consisting of the line segment joining $(-2, 0)$ and $(-1, 0)$, half circle $y = \sqrt{1 - x^2}$, the line segment joining $(1, 0)$ and $(2, 0)$, and half circle $y = \sqrt{4 - x^2}$.

49. Use Green's theorem to evaluate line integral $\int_C \sin(x^3)\ dx + 2ye^{x^2}\ dy$, where C is a triangular closed curve that connects the points $(0, 0)$, $(2, 2)$, and $(0, 2)$ counterclockwise.

50. Let C be the boundary of square $0 \le x \le \pi,\ 0 \le y \le \pi$, traversed counterclockwise. Use Green's theorem to find $\int_C \sin(x + y)\ dx + \cos(x + y)\ dy$.

51. Use Green's theorem to evaluate line integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y) = (y^2 - x^2)\ \hat{\mathbf{i}} + (x^2 + y^2)\ \hat{\mathbf{j}}$, and C is a triangle bounded by $y = 0,\ x = 3$, and $y = x$, oriented counterclockwise.

52. Use Green's Theorem to evaluate integral $\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$, where $\vec{\mathbf{F}}(x, y) = (xy^2)\ \hat{\mathbf{i}} + x\ \hat{\mathbf{j}}$, and C is a unit circle oriented in the counterclockwise direction.

53. Use Green's theorem in a plane to evaluate line integral $\oint_C (xy + y^2)\ dx + x^2\ dy$, where C is a closed curve of a region bounded by $y = x$ and $y = x^2$ oriented in the counterclockwise direction.

54. Calculate the outward flux of $\vec{\mathbf{F}}(x, y) = -x\ \hat{\mathbf{i}} + 2y\ \hat{\mathbf{j}}$ over a square with corners $(\pm 1, \pm 1)$, where the unit normal is outward pointing and oriented in the counterclockwise direction.

55. [T] Let C be circle $x^2 + y^2 = 4$ oriented in the counterclockwise direction. Evaluate $\oint_C \left[(3y - e^{\arctan x})\ dx + (7x + \sqrt{y^4 + 1})\ dy\right]$ using a computer algebra system.

56. Find the flux of field $\vec{\mathbf{F}}(x, y) = -x\ \hat{\mathbf{i}} + y\ \hat{\mathbf{j}}$ across $x^2 + y^2 = 16$ oriented in the counterclockwise direction.

57. Let $\vec{\mathbf{F}} = (y^2 - x^2)\ \hat{\mathbf{i}} + (x^2 + y^2)\ \hat{\mathbf{j}}$, and let C be a triangle bounded by $y = 0,\ x = 3$, and $y = x$ oriented in the counterclockwise direction. Find the outward flux of $\vec{\mathbf{F}}$ through C.

58. [T] Let C be unit circle $x^2 + y^2 = 1$ traversed once counterclockwise. Evaluate $\int_C \left[-y^3 + \sin(xy) + xy \cos(xy)\right]\ dx + \left[x^3 + x^2 \cos(xy)\right]\ dy$ by using a computer algebra system.

59. [T] Find the outward flux of vector field $\vec{\mathbf{F}}(x, y) = xy^2\ \hat{\mathbf{i}} + x^2y\ \hat{\mathbf{j}}$ across the boundary of annulus $R = \{(x, y) : 1 \le x^2 + y^2 \le 4\} = \{(r, \theta) : 1 \le r \le 2,\ 0 \le \theta \le 2\pi\}$ using a computer algebra system.

60. Consider region R bounded by parabolas $y = x^2$ and $x = y^2$. Let C be the boundary of R oriented counterclockwise. Use Green's theorem to evaluate $\oint_C \left(y + e^{\sqrt{x}}\right)\ dx + (2x + \cos(y^2))\ dy$.

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16.5: Divergence and Curl

Learning Objectives

- Determine divergence from the formula for a given vector field.
- Determine curl from the formula for a given vector field.
- Use the properties of curl and divergence to determine whether a vector field is conservative.

In this section, we examine two important operations on a vector field: divergence and curl. They are important to the field of calculus for several reasons, including the use of curl and divergence to develop some higher-dimensional versions of the Fundamental Theorem of Calculus. In addition, curl and divergence appear in mathematical descriptions of fluid mechanics, electromagnetism, and elasticity theory, which are important concepts in physics and engineering. We can also apply curl and divergence to other concepts we already explored. For example, under certain conditions, a vector field is conservative if and only if its curl is zero.

In addition to defining curl and divergence, we look at some physical interpretations of them, and show their relationship to conservative and source-free vector fields.

Divergence

Divergence is an operation on a vector field that tells us how the field behaves toward or away from a point. Locally, the divergence of a vector field $\vec{\mathbf{F}}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ at a particular point P is a measure of the "outflowing-ness" of the vector field at P. If $\vec{\mathbf{F}}$ represents the velocity of a fluid, then the divergence of $\vec{\mathbf{F}}$ at P measures the net rate of change with respect to time of the amount of fluid flowing away from P (the tendency of the fluid to flow "out of" P). In particular, if the amount of fluid flowing into P is the same as the amount flowing out, then the divergence at P is zero.

Definition: divergence in $\mathbb{R}^3$

If $\vec{\mathbf{F}} = \langle P, Q, R \rangle$ is a vector field in $\mathbb{R}^3$ and P_x, Q_y, and R_z all exist, then the divergence of $\vec{\mathbf{F}}$ is defined by

$$\operatorname{div} F = P_x + Q_y + R_z \tag{16.5.1}$$

$$= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}. \tag{16.5.2}$$

Note the divergence of a vector field is not a vector field, but a scalar function. In terms of the gradient operator

$$\vec{\nabla} = \langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \rangle$$

divergence can be written symbolically as the dot product

$$\operatorname{div} \vec{\mathbf{F}} = \vec{\nabla} \cdot \vec{\mathbf{F}}.$$

Note this is merely helpful notation, because the dot product of a vector of operators and a vector of functions is not meaningfully defined given our current definition of dot product.

If $\vec{\mathbf{F}} = \langle P, Q \rangle$ is a vector field in $\mathbb{R}^2$, and P_x and Q_y both exist, then the divergence of $\vec{\mathbf{F}}$ is defined similarly as

$$\begin{aligned} \operatorname{div} \vec{\mathbf{F}} &= P_x + Q_y \\ &= \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} \\ &= \vec{\nabla} \cdot \vec{\mathbf{F}}. \end{aligned}$$

To illustrate this point, consider the two vector fields in Figure 16.5.1. At any particular point, the amount flowing in is the same as the amount flowing out, so at every point the "outflowing-ness" of the field is zero. Therefore, we expect the divergence of both fields to be zero, and this is indeed the case, as

$$\text{div}(\langle 1,2\rangle) = \frac{\partial}{\partial x}(1) + \frac{\partial}{\partial y}(2) = 0$$

and

$$\text{div}(\langle -y,x\rangle) = \frac{\partial}{\partial x}(-y) + \frac{\partial}{\partial y}(x) = 0.$$

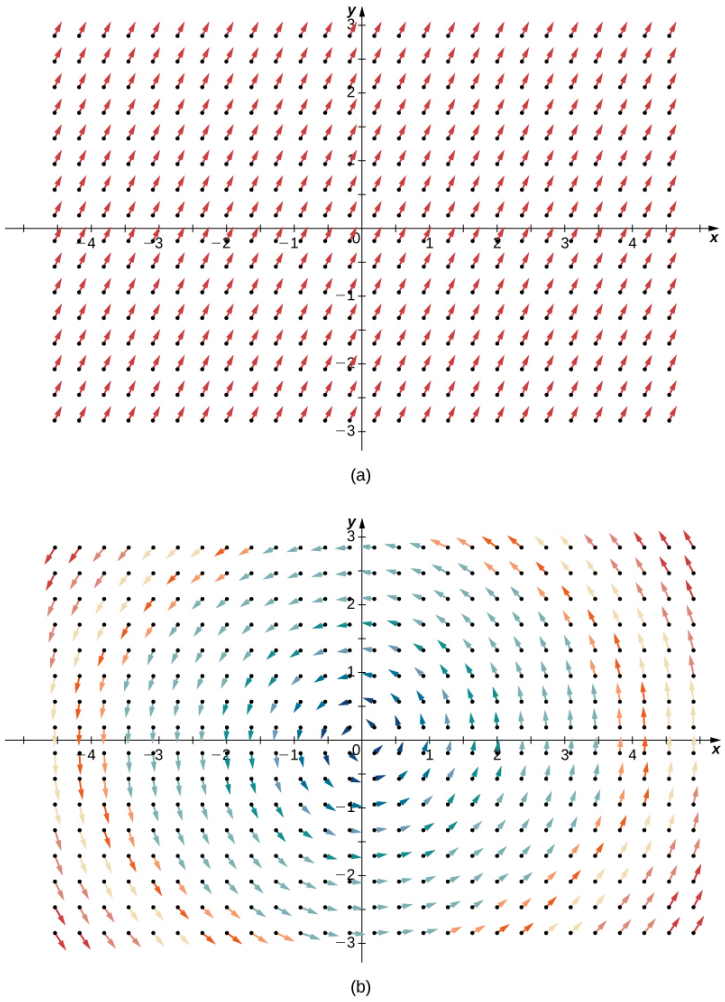


Figure 16.5.1: (a) Vector field $\langle 1,2\rangle$ has zero divergence. (b) Vector field $\langle -y,x\rangle$ also has zero divergence.

By contrast, consider radial vector field $\vec{\mathbf{R}}(x,y) = \langle -x,-y\rangle$ in Figure 16.5.2. At any given point, more fluid is flowing in than is flowing out, and therefore the "outgoingness" of the field is negative. We expect the divergence of this field to be negative, and this is indeed the case, as

$$\text{div}(\vec{\mathbf{R}}) = \frac{\partial}{\partial x}(-x) + \frac{\partial}{\partial y}(-y) = -2.$$

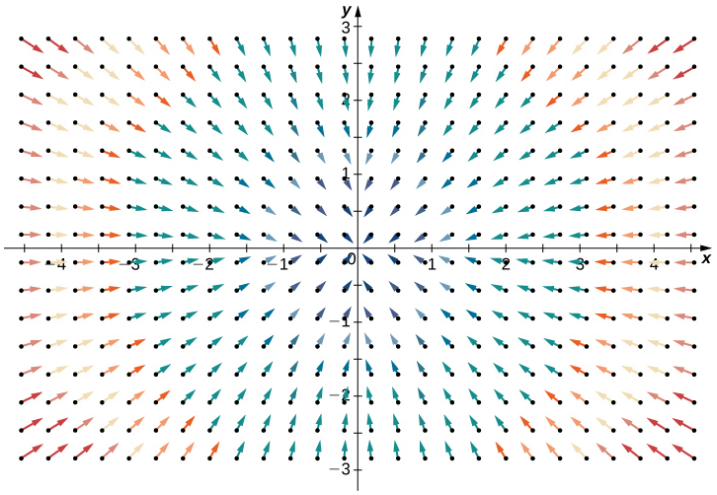


Figure 16.5.2: This vector field has negative divergence.

To get a global sense of what divergence is telling us, suppose that a vector field in $\mathbb{R}^2$ represents the velocity of a fluid. Imagine taking an elastic circle (a circle with a shape that can be changed by the vector field) and dropping it into a fluid. If the circle maintains its exact area as it flows through the fluid, then the divergence is zero. This would occur for both vector fields in Figure 16.5.1. On the other hand, if the circle's shape is distorted so that its area shrinks or expands, then the divergence is not zero. Imagine dropping such an elastic circle into the radial vector field in Figure 16.5.2 so that the center of the circle lands at point $(3, 3)$. The circle would flow toward the origin, and as it did so the front of the circle would travel more slowly than the back, causing the circle to "scrunch" and lose area. This is how you can see a negative divergence.

Example 16.5.1: Calculating Divergence at a Point

If $\vec{\mathbf{F}}(x, y, z) = e^x\hat{i} + yz\hat{j} - yz^2\hat{k}$, then find the divergence of $\vec{\mathbf{F}}$ at $(0, 2, -1)$.

Solution

The divergence of $\vec{\mathbf{F}}$ is

$$\frac{\partial}{\partial x}(e^x) + \frac{\partial}{\partial y}(yz) - \frac{\partial}{\partial z}(yz^2) = e^x + z - 2yz.$$

Therefore, the divergence at $(0, 2, -1)$ is $e^0 - 1 + 4 = 4$. If $\vec{\mathbf{F}}$ represents the velocity of a fluid, then more fluid is flowing out than flowing in at point $(0, 2, -1)$.

Exercise 16.5.1

Find $\operatorname{div} \vec{\mathbf{F}}$ for

$$\vec{\mathbf{F}}(x, y, z) = \langle xy,\ 5 - z^2,\ x^2 + y^2 \rangle.$$

Hint

Follow Example 16.5.1.

Answer

$\operatorname{div} \vec{\mathbf{F}} = y$

One application for divergence occurs in physics, when working with magnetic fields. A magnetic field is a vector field that models the influence of electric currents and magnetic materials. Physicists use divergence in Gauss's law for magnetism, which states that if $\vec{\mathbf{B}}$ is a magnetic field, then $\vec{\nabla} \cdot \vec{\mathbf{B}} = 0$; in other words, the divergence of a magnetic field is zero.

Example 16.5.2: Determining Whether a Field Is Magnetic

Is it possible for $\vec{\mathbf{F}}(x,y) = \langle x^2 y,\ y - xy^2 \rangle$ to be a magnetic field?

Solution

If $\vec{\mathbf{F}}$ were magnetic, then its divergence would be zero. The divergence of $\vec{\mathbf{F}}$ is

$$\frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(y - xy^2) = 2xy + 1 - 2xy = 1$$

and therefore $\vec{\mathbf{F}}$ cannot model a magnetic field (Figure 16.5.3).

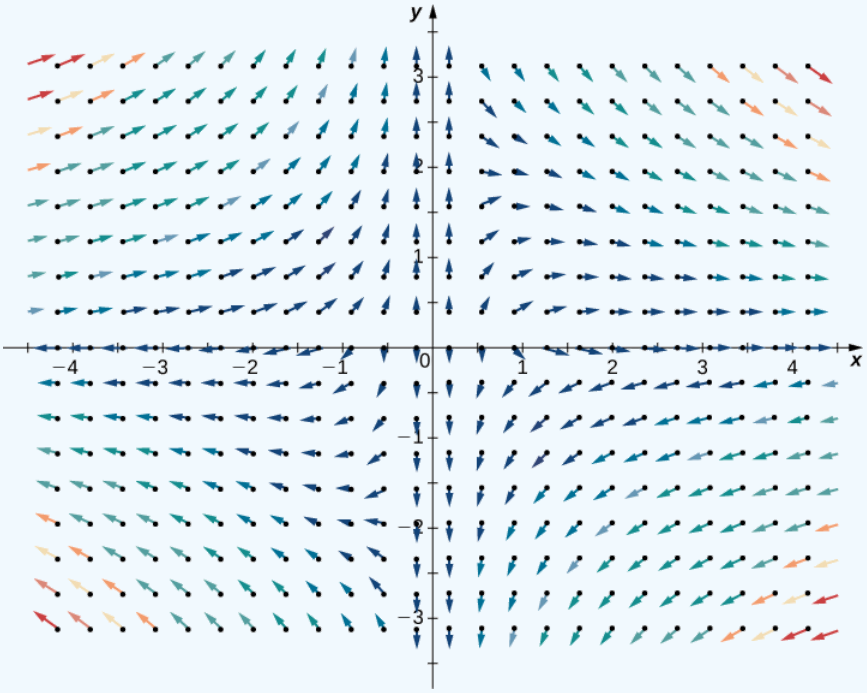


Figure 16.5.3: The divergence of vector field $\vec{\mathbf{F}}(x,y) = \langle x^2 y,\ y - xy^2 \rangle$ is one, so it cannot model a magnetic field.

Another application for divergence is detecting whether a field is source free. Recall that a source-free field is a vector field that has a stream function; equivalently, a source-free field is a field with a flux that is zero along any closed curve. The next two theorems say that, under certain conditions, source-free vector fields are precisely the vector fields with zero divergence.

Theorem: Divergence of a Source-Free Vector Field

If $\vec{\mathbf{F}} = \langle P, Q \rangle$ is a source-free continuous vector field with differentiable component functions, then $\operatorname{div} \vec{\mathbf{F}} = 0$.

Proof

Since $\vec{\mathbf{F}}$ is source free, there is a function $g(x,y)$ with $g_y = P$ and $-g_x = Q$. Therefore, $\vec{\mathbf{F}} = \langle g_y, -g_x \rangle$ and $\operatorname{div} \vec{\mathbf{F}} = g_{yx} - g_{xy} = 0$ by Clairaut's theorem.

□

The converse of Divergence of a Source-Free Vector Field is true on simply connected regions, but the proof is too technical to include here. Thus, we have the following theorem, which can test whether a vector field in $\mathbb{R}^2$ is source free.

Theorem: Divergence Test for Source-Free Vector Fields

Let $\vec{\mathbf{F}} = \langle P, Q \rangle$ be a continuous vector field with differentiable component functions with a domain that is simply connected. Then, $\operatorname{div} \vec{\mathbf{F}} = 0$ if and only if $\vec{\mathbf{F}}$ is source free.

Example 16.5.3: Determining Whether a Field Is Source Free

Is field $\vec{\mathbf{F}}(x, y) = \langle x^2 y,\ 5 - xy^2 \rangle$ source free?

Solution

Note the domain of $\vec{\mathbf{F}}$ is $\mathbb{R}^2$ which is simply connected. Furthermore, $\vec{\mathbf{F}}$ is continuous with differentiable component functions. Therefore, we can use the Divergence Test for Source-Free Vector Fields to analyze $\vec{\mathbf{F}}$. The divergence of $\vec{\mathbf{F}}$ is

$$\frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(5 - xy^2) = 2xy - 2xy = 0.$$

Therefore, $\vec{\mathbf{F}}$ is source free by the Divergence Test for Source-Free Vector Fields.

Exercise 16.5.2

Let $\vec{\mathbf{F}}(x, y) = \langle -ay, bx \rangle$ be a rotational field where a and b are positive constants. Is $\vec{\mathbf{F}}$ source free?

Hint

Calculate the divergence.

Answer

Yes

Recall that the flux form of Green's theorem says that

$$\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \iint_D P_x + Q_y\, dA,$$

where C is a simple closed curve and D is the region enclosed by C. Since $P_x + Q_y = \operatorname{div} \vec{\mathbf{F}}$, Green's theorem is sometimes written as

$$\oint_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, ds = \iint_D \operatorname{div} \vec{\mathbf{F}}\, dA.$$

Therefore, Green's theorem can be written in terms of divergence. If we think of divergence as a derivative of sorts, then Green's theorem says the "derivative" of $\vec{\mathbf{F}}$ on a region can be translated into a line integral of $\vec{\mathbf{F}}$ along the boundary of the region. This is analogous to the Fundamental Theorem of Calculus, in which the derivative of a function f on a line segment $[a, b]$ can be translated into a statement about f on the boundary of $[a, b]$. Using divergence, we can see that Green's theorem is a higher-dimensional analog of the Fundamental Theorem of Calculus.

We can use all of what we have learned in the application of divergence. Let $\vec{\mathbf{v}}$ be a vector field modeling the velocity of a fluid. Since the divergence of $\vec{\mathbf{v}}$ at point P measures the "outflowing-ness" of the fluid at P, $\operatorname{div} v(P) > 0$ implies that more fluid is flowing out of P than flowing in. Similarly, $\operatorname{div} v(P) < 0$ implies the more fluid is flowing in to P than is flowing out, and $\operatorname{div} \vec{\mathbf{v}}(P) = 0$ implies the same amount of fluid is flowing in as flowing out.

Example 16.5.4: Determining Flow of a Fluid

Suppose $\vec{\mathbf{v}}(x, y) = \langle -xy, y \rangle,\ y > 0$ models the flow of a fluid. Is more fluid flowing into point $(1, 4)$ than flowing out?

Solution

To determine whether more fluid is flowing into $(1, 4)$ than is flowing out, we calculate the divergence of $\vec{\mathbf{v}}$ at $(1, 4)$:

$$div(\vec{\mathbf{v}}) = \frac{\partial}{\partial x}(-xy) + \frac{\partial}{\partial y}(y) = -y + 1.$$

To find the divergence at $(1, 4)$ substitute the point into the divergence: $-4 + 1 = -3$. Since the divergence of $\vec{\mathbf{v}}$ at $(1, 4)$ is negative, more fluid is flowing in than flowing out (Figure 16.5.4).

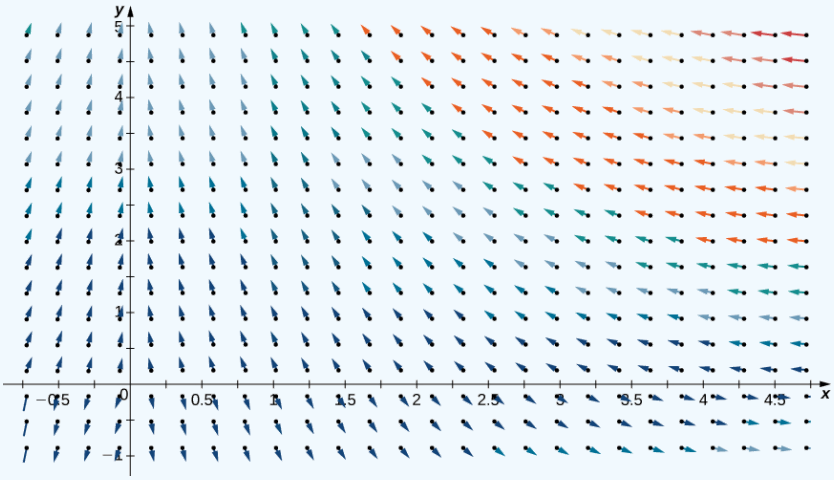


Figure 16.5.4: Vector field $\vec{\mathbf{v}}(x, y) = \langle -xy, y \rangle$ has negative divergence at $(1, 4)$

? Exercise 16.5.3

For vector field $\vec{\mathbf{v}}(x, y) = \langle -xy, y \rangle$, $y > 0$, find all points P such that the amount of fluid flowing in to P equals the amount of fluid flowing out of P.

Hint

Find where the divergence is zero.

Answer

All points on line $y = 1$.

Curl

The second operation on a vector field that we examine is the curl, which measures the extent of rotation of the field about a point. Suppose that $\vec{\mathbf{F}}$ represents the velocity field of a fluid. Then, the curl of $\vec{\mathbf{F}}$ at point P is a vector that measures the tendency of particles near P to rotate about the axis that points in the direction of this vector. The magnitude of the curl vector at P measures how quickly the particles rotate around this axis. In other words, the curl at a point is a measure of the vector field's "spin" at that point. Visually, imagine placing a paddlewheel into a fluid at P, with the axis of the paddlewheel aligned with the curl vector (Figure 16.5.5). The curl measures the tendency of the paddlewheel to rotate.

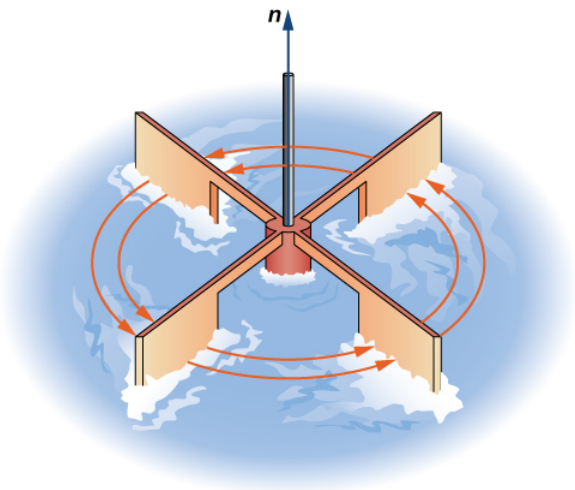


Figure 16.5.5: To visualize curl at a point, imagine placing a small paddlewheel into the vector field at a point.

Consider the vector fields in Figure 16.5.1. In part (a), the vector field is constant and there is no spin at any point. Therefore, we expect the curl of the field to be zero, and this is indeed the case. Part (b) shows a rotational field, so the field has spin. In particular, if you place a paddlewheel into a field at any point so that the axis of the wheel is perpendicular to a plane, the wheel rotates counterclockwise. Therefore, we expect the curl of the field to be nonzero, and this is indeed the case (the curl is $2\hat{\mathbf{k}}$).

To see what curl is measuring globally, imagine dropping a leaf into the fluid. As the leaf moves along with the fluid flow, the curl measures the tendency of the leaf to rotate. If the curl is zero, then the leaf doesn't rotate as it moves through the fluid.

Definition: Curl

If $\vec{\mathbf{F}} = \langle P, Q, R \rangle$ is a vector field in $\mathbb{R}^3$, and P_x, Q_y, and R_z all exist, then the curl of $\vec{\mathbf{F}}$ is defined by

$$\operatorname{curl} \vec{\mathbf{F}} = (R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}} \tag{16.5.3}$$

$$= \left(\frac{\partial R}{\partial y} - \frac{\partial Q}{\partial z}\right)\hat{\mathbf{i}} + \left(\frac{\partial P}{\partial z} - \frac{\partial R}{\partial x}\right)\hat{\mathbf{j}} + \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{\mathbf{k}}. \tag{16.5.4}$$

Note that the curl of a vector field is a vector field, in contrast to divergence.

The definition of curl can be difficult to remember. To help with remembering, we use the notation $\vec{\nabla} \times \vec{\mathbf{F}}$ to stand for a "determinant" that gives the curl formula:

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

The determinant of this matrix is

$$(R_y - Q_z)\,\hat{\mathbf{i}} - (R_x - P_z)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}} = (R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}} = \operatorname{curl} \vec{\mathbf{F}}.$$

Thus, this matrix is a way to help remember the formula for curl. Keep in mind, though, that the word *determinant* is used very loosely. A determinant is not really defined on a matrix with entries that are three vectors, three operators, and three functions.

If $\vec{\mathbf{F}} = \langle P, Q \rangle$ is a vector field in $\mathbb{R}^2$, then the curl of $\vec{\mathbf{F}}$, by definition, is

$$\operatorname{curl} \vec{\mathbf{F}} = (Q_x - P_y)\,\hat{\mathbf{k}} = \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right)\hat{\mathbf{k}}.$$

Example 16.5.5: Finding the Curl of a Three-Dimensional Vector Field

Find the curl of $\vec{\mathbf{F}}(x,y,z) = \langle P, Q, R\rangle = \langle x^2 z, e^y + xz, xyz\rangle$.

Solution

The curl is

$$\begin{aligned}\operatorname{curl}\vec{\mathbf{F}} &= \vec{\nabla}\times\vec{\mathbf{F}} \\ &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix} \\ &= (R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}} \\ &= (xz - x)\,\hat{\mathbf{i}} + (x^2 - yz)\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}.\end{aligned}$$

Exercise 16.5.4

Find the curl of $\vec{\mathbf{F}}(x,y,z) = \langle \sin x\,\cos z,\ \sin y\,\sin z,\ \cos x\,\cos y\rangle$ at point $\left(0, \frac{\pi}{2}, \frac{\pi}{2}\right)$.

Hint

Find the determinant of matrix $\vec{\nabla}\times\vec{\mathbf{F}}$.

Answer

$-\hat{i}$

Example 16.5.6: Finding the Curl of a Two-Dimensional Vector Field

Find the curl of $\vec{\mathbf{F}}(x,y) = \langle P, Q\rangle = \langle y, 0\rangle$.

Solution

Notice that this vector field consists of vectors that are all parallel. In fact, each vector in the field is parallel to the *x*-axis. This fact might lead us to the conclusion that the field has no spin and that the curl is zero. To test this theory, note that

$$\operatorname{curl}\vec{\mathbf{F}} = (Q_x - P_y)\,\hat{\mathbf{k}} = -\hat{\mathbf{k}} \neq \vec{\mathbf{0}}.$$

Therefore, this vector field does have spin. To see why, imagine placing a paddlewheel at any point in the first quadrant (Figure 16.5.6). The larger magnitudes of the vectors at the top of the wheel cause the wheel to rotate. The wheel rotates in the clockwise (negative) direction, causing the coefficient of the curl to be negative.

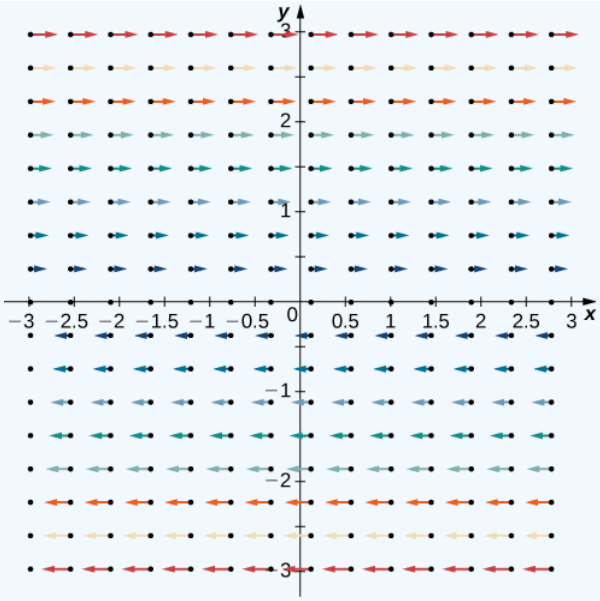


Figure 16.5.6: Vector field $\vec{\mathbf{F}}(x,y) = \langle y, 0\rangle$ consists of vectors that are all parallel.

Note that if $\vec{\mathbf{F}} = \langle P, Q\rangle$ is a vector field in a plane, then $\operatorname{curl} \vec{\mathbf{F}} \cdot \hat{\mathbf{k}} = (Q_x - P_y)\,\hat{\mathbf{k}} \cdot \hat{\mathbf{k}} = Q_x - P_y$. Therefore, the circulation form of Green's theorem is sometimes written as

$$\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_D \operatorname{curl} \vec{\mathbf{F}} \cdot \hat{\mathbf{k}}\, dA,$$

where C is a simple closed curve and D is the region enclosed by C. Therefore, the circulation form of Green's theorem can be written in terms of the curl. If we think of curl as a derivative of sorts, then Green's theorem says that the "derivative" of $\vec{\mathbf{F}}$ on a region can be translated into a line integral of $\vec{\mathbf{F}}$ along the boundary of the region. This is analogous to the Fundamental Theorem of Calculus, in which the derivative of a function f on line segment $[a,b]$ can be translated into a statement about f on the boundary of $[a,b]$. Using curl, we can see the circulation form of Green's theorem is a higher-dimensional analog of the Fundamental Theorem of Calculus.

We can now use what we have learned about curl to show that gravitational fields have no "spin." Suppose there is an object at the origin with mass m_1 at the origin and an object with mass m_2. Recall that the gravitational force that object 1 exerts on object 2 is given by field

$$\vec{\mathbf{F}}(x,y,z) = -Gm_1m_2 \left\langle \frac{x}{(x^2+y^2+z^2)^{3/2}}, \frac{y}{(x^2+y^2+z^2)^{3/2}}, \frac{z}{(x^2+y^2+z^2)^{3/2}} \right\rangle.$$

Example 16.5.7: Determining the Spin of a Gravitational Field

Show that a gravitational field has no spin.

Solution

To show that $\vec{\mathbf{F}}$ has no spin, we calculate its curl. Let

- $P(x,y,z) = \dfrac{x}{(x^2+y^2+z^2)^{3/2}}$,
- $Q(x,y,z) = \dfrac{y}{(x^2+y^2+z^2)^{3/2}}$, and
- $R(x,y,z) = \dfrac{z}{(x^2+y^2+z^2)^{3/2}}$.

Then,

$$\begin{aligned}\operatorname{curl}\vec{\mathbf{F}} &= -Gm_1m_2[(R_y-Q_z)\hat{\mathbf{i}}+(P_z-R_x)\hat{\mathbf{j}}+(Q_x-P_y)\hat{\mathbf{k}}] \\ &= -Gm_1m_2\begin{pmatrix} \left(\frac{-3yz}{(x^2+y^2+z^2)^{5/2}}-\left(\frac{-3yz}{(x^2+y^2+z^2)^{5/2}}\right)\right)\hat{\mathbf{i}} \\ +\left(\frac{-3xz}{(x^2+y^2+z^2)^{5/2}}-\left(\frac{-3xz}{(x^2+y^2+z^2)^{5/2}}\right)\right)\hat{\mathbf{j}} \\ +\left(\frac{-3xy}{(x^2+y^2+z^2)^{5/2}}-\left(\frac{-3xy}{(x^2+y^2+z^2)^{5/2}}\right)\right)\hat{\mathbf{k}}\end{pmatrix} \\ &= \vec{\mathbf{0}}.\end{aligned}$$

Since the curl of the gravitational field is zero, the field has no spin.

? Exercise 16.5.7

Field $\vec{\mathbf{v}}(x,y)=\langle -\frac{y}{x^2+y^2},\frac{x}{x^2+y^2}\rangle$ models the flow of a fluid. Show that if you drop a leaf into this fluid, as the leaf moves over time, the leaf does not rotate.

Hint

Calculate the curl.

Answer

$\operatorname{curl}\vec{\mathbf{v}}=\vec{\mathbf{0}}$

Using Divergence and Curl

Now that we understand the basic concepts of divergence and curl, we can discuss their properties and establish relationships between them and conservative vector fields.

If $\vec{\mathbf{F}}$ is a vector field in $\mathbb{R}^3$ then the curl of $\vec{\mathbf{F}}$ is also a vector field in $\mathbb{R}^3$. Therefore, we can take the divergence of a curl. The next theorem says that the result is always zero. This result is useful because it gives us a way to show that some vector fields are not the curl of any other field. To give this result a physical interpretation, recall that divergence of a velocity field $\vec{\mathbf{v}}$ at point P measures the tendency of the corresponding fluid to flow out of P. Since $\operatorname{div}(\operatorname{curl}\vec{\mathbf{v}})=0$, the net rate of flow in vector field $\operatorname{curl}\vec{\mathbf{v}}$ at any point is zero. Taking the curl of vector field $\vec{\mathbf{F}}$ eliminates whatever divergence was present in $\vec{\mathbf{F}}$.

Theorem: Divergence of the Curl

Let $\vec{\mathbf{F}}=\langle P,Q,R\rangle$ be a vector field in $\mathbb{R}^3$ such that the component functions all have continuous second-order partial derivatives. Then,

$$\operatorname{div}(\operatorname{curl}\vec{\mathbf{F}})=\vec{\nabla}\cdot(\vec{\nabla}\times\vec{\mathbf{F}})=0.$$

Proof

By the definitions of divergence and curl, and by Clairaut's theorem,

$$\begin{aligned}\operatorname{div}(\operatorname{curl}\vec{\mathbf{F}}) &= \operatorname{div}[(R_y-Q_z)\hat{\mathbf{i}}+(P_z-R_x)\hat{\mathbf{j}}+(Q_x-P_y)\hat{\mathbf{k}}] \\ &= R_{yx}-Q_{xz}+P_{yz}-R_{yx}+Q_{zx}-P_{zy} \\ &= 0.\end{aligned}$$

□

Example 16.5.8: Showing That a Vector Field Is Not the Curl of Another

Show that $\vec{\mathbf{F}}(x,y,z) = e^x\,\hat{\mathbf{i}} + yz\,\hat{\mathbf{j}} + xz^2\,\hat{\mathbf{k}}$ is not the curl of another vector field. That is, show that there is no other vector $\vec{\mathbf{G}}$ with $\text{curl}\,\vec{\mathbf{G}} = \vec{\mathbf{F}}$.

Solution

Notice that the domain of $\vec{\mathbf{F}}$ is all of $\mathbb{R}^3$ and the second-order partials of $\vec{\mathbf{F}}$ are all continuous. Therefore, we can apply the previous theorem to $\vec{\mathbf{F}}$.

The divergence of $\vec{\mathbf{F}}$ is $e^x + z + 2xz$. If $\vec{\mathbf{F}}$ were the curl of vector field $\vec{\mathbf{G}}$, then $\text{div}\,\vec{\mathbf{F}} = \text{div}(\text{curl}\,\vec{\mathbf{G}}) = 0$. But, the divergence of $\vec{\mathbf{F}}$ is not zero, and therefore $\vec{\mathbf{F}}$ is not the curl of any other vector field.

Exercise 16.5.8

Is it possible for $\vec{\mathbf{G}}(x,y,z) = \langle \sin x,\ \cos y,\ \sin(xyz)\rangle$ to be the curl of a vector field?

Hint

Find the divergence of $\vec{\mathbf{G}}$.

Answer

No.

With the next two theorems, we show that if $\vec{\mathbf{F}}$ is a conservative vector field then its curl is zero, and if the domain of $\vec{\mathbf{F}}$ is simply connected then the converse is also true. This gives us another way to test whether a vector field is conservative.

Theorem: Curl of a Conservative Vector Field

If $\vec{\mathbf{F}} = \langle P, Q, R\rangle$ is conservative, then $\text{curl}\,\vec{\mathbf{F}} = \vec{\mathbf{0}}$.

Proof

Since conservative vector fields satisfy the cross-partials property, all the cross-partials of $\vec{\mathbf{F}}$ are equal. Therefore,

$$\begin{aligned}\text{curl}\,\vec{\mathbf{F}} &= (R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}}\\ &= \vec{\mathbf{0}}.\end{aligned}$$

□

The same theorem is true for vector fields in a plane.

Since a conservative vector field is the gradient of a scalar function, the previous theorem says that $\text{curl}\,(\vec{\nabla} f) = \vec{\mathbf{0}}$ for any scalar function f. In terms of our curl notation, $\vec{\nabla} \times \vec{\nabla}(f) = \vec{\mathbf{0}}$. This equation makes sense because the cross product of a vector with itself is always the zero vector. Sometimes equation $\vec{\nabla} \times \vec{\nabla}(f) = \vec{\mathbf{0}}$ is simplified as $\vec{\nabla} \times \vec{\nabla} = \vec{\mathbf{0}}$.

Theorem: Curl Test for a Conservative Field

Let $\vec{\mathbf{F}} = \langle P, Q, R\rangle$ be a vector field in space on a simply connected domain. If $\text{curl}\,\vec{\mathbf{F}} = \vec{\mathbf{0}}$, then $\vec{\mathbf{F}}$ is conservative.

Proof

Since $\text{curl}\,\vec{\mathbf{F}} = \vec{\mathbf{0}}$, we have that $R_y = Q_z$, $P_z = R_x$, and $Q_x = P_y$. Therefore, $\vec{\mathbf{F}}$ satisfies the cross-partials property on a simply connected domain, and the Cross-Partial Property of Conservative Fields implies that $\vec{\mathbf{F}}$ is conservative.

□

The same theorem is also true in a plane. Therefore, if $\vec{\mathbf{F}}$ is a vector field in a plane or in space and the domain is simply connected, then $\vec{\mathbf{F}}$ is conservative if and only if $\text{curl}\,\vec{\mathbf{F}} = \vec{\mathbf{0}}$.

Example 16.5.9: Testing Whether a Vector Field Is Conservative

Use the curl to determine whether $\vec{\mathbf{F}}(x, y, z) = \langle yz, xz, xy\rangle$ is conservative.

Solution

Note that the domain of $\vec{\mathbf{F}}$ is all of $\mathbb{R}^3$ which is simply connected (Figure 16.5.7). Therefore, we can test whether $\vec{\mathbf{F}}$ is conservative by calculating its curl.

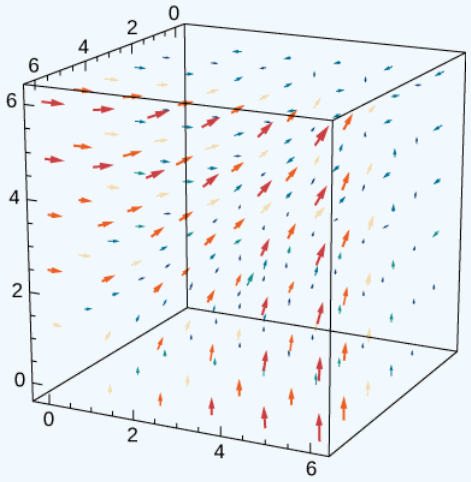


Figure 16.5.7: The curl of vector field $\vec{\mathbf{F}}(x, y, z) = \langle yz, xz, xy\rangle$ is zero.

The curl of $\vec{\mathbf{F}}$ is

$$\left(\frac{\partial}{\partial y}xy - \frac{\partial}{\partial z}xz\right)\hat{\mathbf{i}} + \left(\frac{\partial}{\partial y}yz - \frac{\partial}{\partial z}xy\right)\hat{\mathbf{j}} + \left(\frac{\partial}{\partial y}xz - \frac{\partial}{\partial z}yz\right)\hat{\mathbf{k}}$$
$$= (x - x)\hat{\mathbf{i}} + (y - y)\hat{\mathbf{j}} + (z - z)\hat{\mathbf{k}} = \vec{\mathbf{0}}.$$

Thus, $\vec{\mathbf{F}}$ is conservative.

We have seen that the curl of a gradient is zero. What is the divergence of a gradient? If f is a function of two variables, then $\text{div}(\vec{\nabla} f) = \vec{\nabla} \cdot (\vec{\nabla} f) = f_{xx} + f_{yy}$. We abbreviate this "double dot product" as $\vec{\nabla}^2$. This operator is called the *Laplace operator*, and in this notation Laplace's equation becomes $\vec{\nabla}^2 f = 0$. Therefore, a harmonic function is a function that becomes zero after taking the divergence of a gradient.

Similarly, if f is a function of three variables then

$$\text{div}(\vec{\nabla} f) = \vec{\nabla} \cdot (\vec{\nabla} f) = f_{xx} + f_{yy} + f_{zz}.$$

Using this notation we get Laplace's equation for harmonic functions of three variables:

$$\vec{\nabla}^2 f = 0.$$

Harmonic functions arise in many applications. For example, the potential function of an electrostatic field in a region of space that has no static charge is harmonic.

Example 16.5.10: Finding a Potential Function

Is it possible for $f(x, y) = x^2 + x - y$ to be the potential function of an electrostatic field that is located in a region of $\mathbb{R}^2$ free of static charge?

Solution

If f were such a potential function, then f would be harmonic. Note that $f_{xx} = 2$ and $f_{yy} = 0$, and so $f_{xx} + f_{yy} \neq 0$. Therefore, f is not harmonic and f cannot represent an electrostatic potential.

? Exercise 16.5.10

Is it possible for function $f(x, y) = x^2 - y^2 + x$ to be the potential function of an electrostatic field located in a region of $\mathbb{R}^2$ free of static charge?

Hint

Determine whether the function is harmonic.

Answer

Yes.

Key Concepts

- The divergence of a vector field is a scalar function. Divergence measures the "outflowing-ness" of a vector field. If $\vec{\mathbf{v}}$ is the velocity field of a fluid, then the divergence of $\vec{\mathbf{v}}$ at a point is the outflow of the fluid less the inflow at the point.
- The curl of a vector field is a vector field. The curl of a vector field at point P measures the tendency of particles at P to rotate about the axis that points in the direction of the curl at P.
- A vector field with a simply connected domain is conservative if and only if its curl is zero.

Key Equations

- **Curl**

$$\vec{\nabla} \times \vec{\mathbf{F}} = (R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}}$$

- **Divergence**

$$\vec{\nabla} \cdot \vec{\mathbf{F}} = P_x + Q_y + R_z$$

- **Divergence of curl is zero**

$$\vec{\nabla} \cdot (\vec{\nabla} \times \vec{\mathbf{F}}) = 0$$

- **Curl of a gradient is the zero vector**

$$\vec{\nabla} \times (\vec{\nabla} f) = 0$$

Glossary

curl

the curl of vector field $\vec{\mathbf{F}} = \langle P, Q, R \rangle$, denoted $\vec{\nabla} \times \vec{\mathbf{F}}$ is the "determinant" of the matrix

$$\begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}.$$

and is given by the expression $(R_y - Q_z)\,\hat{\mathbf{i}} + (P_z - R_x)\,\hat{\mathbf{j}} + (Q_x - P_y)\,\hat{\mathbf{k}}$; it measures the tendency of particles at a point to rotate about the axis that points in the direction of the curl at the point

divergence

the divergence of a vector field $\vec{\mathbf{F}} = \langle P, Q, R \rangle$, denoted $\vec{\nabla} \times \vec{\mathbf{F}}$, is $P_x + Q_y + R_z$; it measures the "outflowing-ness" of a vector field

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16.5E: Exercises for Section 16.5

For the following exercises, determine whether the statement is *True* or *False*.

1. If the coordinate functions of $\vec{\mathbf{F}} : \mathbb{R}^3 \to \mathbb{R}^3$ have continuous second partial derivatives, then $\text{curl}\,(\text{div}\,\vec{\mathbf{F}})$ equals zero.

2. $\vec{\nabla} \cdot (x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}) = 1$.

Answer

False

3. All vector fields of the form $\vec{\mathbf{F}}(x, y, z) = f(x)\,\hat{\mathbf{i}} + g(y)\,\hat{\mathbf{j}} + h(z)\,\hat{\mathbf{k}}$ are conservative.

4. If $\text{curl}\,\vec{\mathbf{F}} = \vec{\mathbf{0}}$, then $\vec{\mathbf{F}}$ is conservative.

Answer

True

5. If $\vec{\mathbf{F}}$ is a constant vector field then $\text{div}\,\vec{\mathbf{F}} = 0$.

6. If $\vec{\mathbf{F}}$ is a constant vector field then $\text{curl}\,\vec{\mathbf{F}} = \vec{\mathbf{0}}$.

Answer

True

For the following exercises, find the curl of $\vec{\mathbf{F}}$.

7. $\vec{\mathbf{F}}(x, y, z) = xy^2z^4\,\hat{\mathbf{i}} + (2x^2y + z)\,\hat{\mathbf{j}} + y^3z^2\,\hat{\mathbf{k}}$

8. $\vec{\mathbf{F}}(x, y, z) = x^2z\,\hat{\mathbf{i}} + y^2x\,\hat{\mathbf{j}} + (y + 2z)\,\hat{\mathbf{k}}$

Answer

$\text{curl}\,\vec{\mathbf{F}} = \hat{\mathbf{i}} + x^2\,\hat{\mathbf{j}} + y^2\,\hat{\mathbf{k}}$

9. $\vec{\mathbf{F}}(x, y, z) = 3xyz^2\,\hat{\mathbf{i}} + y^2 \sin z\,\hat{\mathbf{j}} + xe^{2z}\,\hat{\mathbf{k}}$

10. $\vec{\mathbf{F}}(x, y, z) = x^2yz\,\hat{\mathbf{i}} + xy^2z\,\hat{\mathbf{j}} + xyz^2\,\hat{\mathbf{k}}$

Answer

$\text{curl}\,\vec{\mathbf{F}} = (xz^2 - xy^2)\,\hat{\mathbf{i}} + (x^2y - yz^2)\,\hat{\mathbf{j}} + (y^2z - x^2z)\,\hat{\mathbf{k}}$

11. $\vec{\mathbf{F}}(x, y, z) = (x\,\cos y)\,\hat{\mathbf{i}} + xy^2\,\hat{\mathbf{j}}$

12. $\vec{\mathbf{F}}(x, y, z) = (x - y)\,\hat{\mathbf{i}} + (y - z)\,\hat{\mathbf{j}} + (z - x)\,\hat{\mathbf{k}}$

Answer

$\text{curl}\ \vec{\mathbf{F}} = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$

13. $\vec{\mathbf{F}}(x, y, z) = xyz\,\hat{\mathbf{i}} + x^2y^2z^2\,\hat{\mathbf{j}} + y^2z^3\,\hat{\mathbf{k}}$

14. $\vec{\mathbf{F}}(x, y, z) = xy\,\hat{\mathbf{i}} + yz\,\hat{\mathbf{j}} + xz\,\hat{\mathbf{k}}$

Answer

$\text{curl}\ \vec{\mathbf{F}} = -y\,\hat{\mathbf{i}} - z\,\hat{\mathbf{j}} - x\,\hat{\mathbf{k}}$

15. $\vec{\mathbf{F}}(x, y, z) = x^2\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}} + z^2\,\hat{\mathbf{k}}$

16. $\vec{\mathbf{F}}(x, y, z) = ax\,\hat{\mathbf{i}} + by\,\hat{\mathbf{j}} + c\,\hat{\mathbf{k}}$ for constants a, b, c.

Answer

$$\text{curl } \vec{\mathbf{F}} = \vec{\mathbf{0}}$$

For the following exercises, find the divergence of $\vec{\mathbf{F}}$.

17. $\vec{\mathbf{F}}(x,y,z) = x^2z\,\hat{\mathbf{i}} + y^2x\,\hat{\mathbf{j}} + (y+2z)\,\hat{\mathbf{k}}$

18. $\vec{\mathbf{F}}(x,y,z) = 3xyz^2\,\hat{\mathbf{i}} + y^2\sin z\,\hat{\mathbf{j}} + xe^2z\,\hat{\mathbf{k}}$

Answer

$\text{div}\,\vec{\mathbf{F}} = 3yz^2 + 2y\sin z + 2xe^{2z}$

19. $\vec{\mathbf{F}}(x,y) = (\sin x)\,\hat{\mathbf{i}} + (\cos y)\,\hat{\mathbf{j}}$

20. $\vec{\mathbf{F}}(x,y,z) = x^2\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}} + z^2\,\hat{\mathbf{k}}$

Answer

$\text{div}\,\vec{\mathbf{F}} = 2(x+y+z)$

21. $\vec{\mathbf{F}}(x,y,z) = (x-y)\,\hat{\mathbf{i}} + (y-z)\,\hat{\mathbf{j}} + (z-x)\,\hat{\mathbf{k}}$

22. $\vec{\mathbf{F}}(x,y) = \dfrac{x}{\sqrt{x^2+y^2}}\,\hat{\mathbf{i}} + \dfrac{y}{\sqrt{x^2+y^2}}\,\hat{\mathbf{j}}$

Answer

$\text{div}\,\vec{\mathbf{F}} = \dfrac{1}{\sqrt{x^2+y^2}}$

23. $\vec{\mathbf{F}}(x,y) = x\,\hat{\mathbf{i}} - y\,\hat{\mathbf{j}}$

24. $\vec{\mathbf{F}}(x,y,z) = ax\,\hat{\mathbf{i}} + by\,\hat{\mathbf{j}} + c\,\hat{\mathbf{k}}$ for constants $a,\ b,\ c$.

Answer

$\text{div}\,\vec{\mathbf{F}} = a + b$

25. $\vec{\mathbf{F}}(x,y,z) = xyz\,\hat{\mathbf{i}} + x^2y^2z^2\,\hat{\mathbf{j}} + y^2z^3\,\hat{\mathbf{k}}$

26. $\vec{\mathbf{F}}(x,y,z) = xy\,\hat{\mathbf{i}} + yz\,\hat{\mathbf{j}} + xz\,\hat{\mathbf{k}}$

Answer

$\text{div}\,\vec{\mathbf{F}} = x + y + z$

For exercises 27 & 28, determine whether each of the given scalar functions is harmonic.

27. $u(x,y,z) = e^{-x}(\cos y - \sin y)$

28. $w(x,y,z) = (x^2+y^2+z^2)^{-1/2}$

Answer

Harmonic

29. If $\vec{\mathbf{F}}(x,y,z) = 2\,\hat{\mathbf{i}} + 2xj + 3yk$ and $\vec{\mathbf{G}}(x,y,z) = x\,\hat{\mathbf{i}} - y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$, find $\text{curl}\,(\vec{\mathbf{F}} \times \vec{\mathbf{G}})$.

30. If $\vec{\mathbf{F}}(x,y,z) = 2\,\hat{\mathbf{i}} + 2xj + 3yk$ and $\vec{\mathbf{G}}(x,y,z) = x\,\hat{\mathbf{i}} - y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$, find $\text{div}\,(\vec{\mathbf{F}} \times \vec{\mathbf{G}})$.

Answer

$\text{div}\,(\vec{\mathbf{F}} \times \vec{\mathbf{G}}) = 2z + 3x$

31. Find $\text{div}\,\vec{\mathbf{F}}$, given that $\vec{\mathbf{F}} = \vec{\nabla} f$, where $f(x,y,z) = xy^3z^2$.

32. Find the divergence of $\vec{\mathbf{F}}$ for vector field $\vec{\mathbf{F}}(x,y,z) = (y^2+z^2)(x+y)\,\hat{\mathbf{i}} + (z^2+x^2)(y+z)\,\hat{\mathbf{j}} + (x^2+y^2)(z+x)\,\hat{\mathbf{k}}$.

Answer

$\text{div}\,\vec{\mathbf{F}} = 2r^2$

33. Find the divergence of $\vec{\mathbf{F}}$ for vector field $\vec{\mathbf{F}}(x,y,z) = f_1(y,z)\,\hat{\mathbf{i}} + f_2(x,z)\,\hat{\mathbf{j}} + f_3(x,y)\,\hat{\mathbf{k}}$.

For exercises 34 - 36, use $r = |\vec{\mathbf{r}}|$ and $\vec{\mathbf{r}}(x,y,z) = \langle x,y,z\rangle$.

34. Find the $\text{curl}\,\vec{\mathbf{r}}$

Answer

$\text{curl}\,\vec{\mathbf{r}} = \vec{\mathbf{0}}$

35. Find the $\text{curl}\,\dfrac{\vec{\mathbf{r}}}{r}$.

36. Find the $\text{curl}\,\dfrac{\vec{\mathbf{r}}}{r^3}$.

Answer

$\text{curl}\,\dfrac{\vec{\mathbf{r}}}{r^3} = \vec{\mathbf{0}}$

37. Let $\vec{\mathbf{F}}(x,y) = \dfrac{-y\,\hat{\mathbf{i}} + x\,\hat{\mathbf{j}}}{x^2+y^2}$, where $\vec{\mathbf{F}}$ is defined on $\{(x,y) \in \mathbb{R} | (x,y) \neq (0,0)\}$. Find $\text{curl}\,\vec{\mathbf{F}}$.

For the following exercises, use a computer algebra system to find the curl of the given vector fields.

38. [T] $\vec{\mathbf{F}}(x,y,z) = \arctan\left(\dfrac{x}{y}\right)\,\hat{\mathbf{i}} + \ln\sqrt{x^2+y^2}\,\hat{\mathbf{j}} + \hat{\mathbf{k}}$

Answer

$\text{curl}\;\vec{\mathbf{F}} = \dfrac{2x}{x^2+y^2}\,\hat{\mathbf{k}}$

39. [T] $\vec{\mathbf{F}}(x,y,z) = \sin(x-y)\,\hat{\mathbf{i}} + \sin(y-z)\,\hat{\mathbf{j}} + \sin(z-x)\,\hat{\mathbf{k}}$

For the following exercises, find the divergence of $\vec{\mathbf{F}}$ at the given point.

40. $\vec{\mathbf{F}}(x,y,z) = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ at $(2,-1,3)$

Answer

$\text{div}\,\vec{\mathbf{F}} = 0$

41. $\vec{\mathbf{F}}(x,y,z) = xyz\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ at $(1,2,3)$

42. $\vec{\mathbf{F}}(x,y,z) = e^{-xy}\,\hat{\mathbf{i}} + e^{xz}\,\hat{\mathbf{j}} + e^{yz}\,\hat{\mathbf{k}}$ at $(3,2,0)$

Answer

$\text{div}\,\vec{\mathbf{F}} = 2 - 2e^{-6}$

43. $\vec{\mathbf{F}}(x,y,z) = xyz\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ at $(1,2,1)$

44. $\vec{\mathbf{F}}(x,y,z) = e^x \sin y\,\hat{\mathbf{i}} - e^x \cos y\,\hat{\mathbf{j}}$ at $(0,0,3)$

Answer

$\text{div}\,\vec{\mathbf{F}} = 0$

For exercises 45- 49, find the curl of $\vec{\mathbf{F}}$ at the given point.

45. $\vec{\mathbf{F}}(x,y,z) = \hat{\mathbf{i}} + \hat{\mathbf{j}} + \hat{\mathbf{k}}$ at $(2,-1,3)$

46. $\vec{\mathbf{F}}(x,y,z) = xyz\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + x\,\hat{\mathbf{k}}$ at $(1,2,3)$

Answer

$\text{curl } \vec{\mathbf{F}} = \hat{\mathbf{j}} - 3\hat{\mathbf{k}}$

47. $\vec{\mathbf{F}}(x, y, z) = e^{-xy}\,\hat{\mathbf{i}} + e^{xz}\,\hat{\mathbf{j}} + e^{yz}\,\hat{\mathbf{k}}$ at $(3, 2, 0)$

48. $\vec{\mathbf{F}}(x, y, z) = xyz\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ at $(1, 2, 1)$

Answer

$\text{curl } \vec{\mathbf{F}} = 2\hat{\mathbf{j}} - \hat{\mathbf{k}}$

49. $\vec{\mathbf{F}}(x, y, z) = e^{x} \sin y\,\hat{\mathbf{i}} - e^{x} \cos y\,\hat{\mathbf{j}}$ at $(0, 0, 3)$

50. Let $\vec{\mathbf{F}}(x, y, z) = (3x^2y + az)\,\hat{\mathbf{i}} + x^3\,\hat{\mathbf{j}} + (3x + 3z^2)\,\hat{\mathbf{k}}$. For what value of a is $\vec{\mathbf{F}}$ conservative?

Answer

$a = 3$

51. Given vector field $\vec{\mathbf{F}}(x, y) = \dfrac{1}{x^2 + y^2}\langle -y, x\rangle$ on domain $D = \dfrac{\mathbb{R}^2}{\{(0, 0)\}} = \{(x, y) \in \mathbb{R}^2 | (x, y) \neq (0, 0)\}$, is $\vec{\mathbf{F}}$ conservative?

52. Given vector field $\vec{\mathbf{F}}(x, y) = \dfrac{1}{x^2 + y^2}\langle x, y\rangle$ on domain $D = \dfrac{\mathbb{R}^2}{\{(0, 0)\}}$, is $\vec{\mathbf{F}}$ conservative?

Answer

$\vec{\mathbf{F}}$ is conservative.

53. Find the work done by force field $\vec{\mathbf{F}}(x, y) = e^{-y}\,\hat{\mathbf{i}} - xe^{-y}\,\hat{\mathbf{j}}$ in moving an object from $P(0, 1)$ to $Q(2, 0)$. Is the force field conservative?

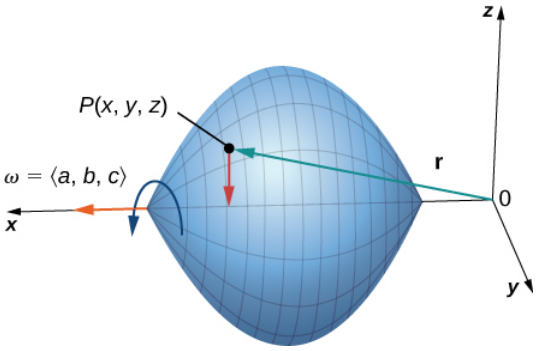
54. Compute divergence $\vec{\mathbf{F}}(x, y, z) = (\sinh x)\,\hat{\mathbf{i}} + (\cosh y)\,\hat{\mathbf{j}} - xyz\,\hat{\mathbf{k}}$.

Answer

$\text{div } \vec{\mathbf{F}} = \cosh x + \sinh y - xy$

55. Compute $\text{curl } \vec{\mathbf{F}} = (\sinh x)\,\hat{\mathbf{i}} + (\cosh y)\,\hat{\mathbf{j}} - xyz\,\hat{\mathbf{k}}$.

For the following exercises, consider a rigid body that is rotating about the x-axis counterclockwise with constant angular velocity $\vec{\omega} = \langle a, b, c\rangle$. If P is a point in the body located at $\vec{\mathbf{r}} = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$, the velocity at P is given by vector field $\vec{\mathbf{F}} = \vec{\omega} \times \vec{\mathbf{r}}$.



56. Express $\vec{\mathbf{F}}$ in terms of $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$, and $\hat{\mathbf{k}}$ vectors.

Answer

$\vec{\mathbf{F}} = (bz - cy)\,\hat{\mathbf{i}} + (cx - az)\,\hat{\mathbf{j}} + (ay - bx)\,\hat{\mathbf{k}}$

57. Find $\text{div } F$.

58. Find $\text{curl } F$

Answer

$$\text{curl } \vec{\mathbf{F}} = 2\vec{\omega}$$

In the following exercises, suppose that $\vec{\nabla} \cdot \vec{\mathbf{F}} = 0$ and $\vec{\nabla} \cdot \vec{\mathbf{G}} = 0$.

59. Does $\vec{\mathbf{F}} + \vec{\mathbf{G}}$ necessarily have zero divergence?

60. Does $\vec{\mathbf{F}} \times \vec{\mathbf{G}}$ necessarily have zero divergence?

Answer

$\vec{\mathbf{F}} \times \vec{\mathbf{G}}$ does not have zero divergence.

In the following exercises, suppose a solid object in $\mathbb{R}^3$ has a temperature distribution given by $T(x, y, z)$. The heat flow vector field in the object is $\vec{\mathbf{F}} = -k\vec{\nabla}T$, where $k > 0$ is a property of the material. The heat flow vector points in the direction opposite to that of the gradient, which is the direction of greatest temperature decrease. The divergence of the heat flow vector is $\vec{\nabla} \cdot \vec{\mathbf{F}} = -k\vec{\nabla} \cdot \vec{\nabla}T = -k\vec{\nabla}^2 T$.

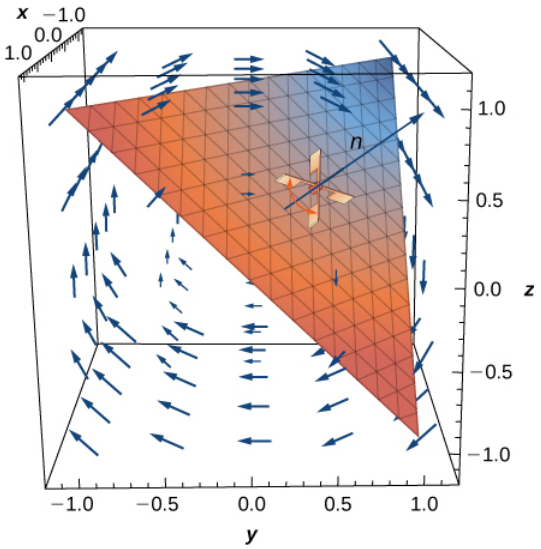
61. Compute the heat flow vector field.

62. Compute the divergence.

Answer

$\vec{\nabla} \cdot \vec{\mathbf{F}} = -200k[1 + 2(x^2 + y^2 + z^2)]e^{-x^2+y^2+z^2}$

63. [T] Consider rotational velocity field $\vec{\mathbf{v}} = \langle 0, 10z, -10y \rangle$. If a paddlewheel is placed in plane $x + y + z = 1$ with its axis normal to this plane, using a computer algebra system, calculate how fast the paddlewheel spins in revolutions per unit time.



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16.6: Surface Integrals

Learning Objectives

- Find the parametric representations of a cylinder, a cone, and a sphere.
- Describe the surface integral of a scalar-valued function over a parametric surface.
- Use a surface integral to calculate the area of a given surface.
- Explain the meaning of an oriented surface, giving an example.
- Describe the surface integral of a vector field.
- Use surface integrals to solve applied problems.

We have seen that a line integral is an integral over a path in a plane or in space. However, if we wish to integrate over a surface (a two-dimensional object) rather than a path (a one-dimensional object) in space, then we need a new kind of integral that can handle integration over objects in higher dimensions. We can extend the concept of a line integral to a surface integral to allow us to perform this integration.

Surface integrals are important for the same reasons that line integrals are important. They have many applications to physics and engineering, and they allow us to develop higher dimensional versions of the Fundamental Theorem of Calculus. In particular, surface integrals allow us to generalize Green's theorem to higher dimensions, and they appear in some important theorems we discuss in later sections.

Parametric Surfaces

A surface integral is similar to a line integral, except the integration is done over a surface rather than a path. In this sense, surface integrals expand on our study of line integrals. Just as with line integrals, there are two kinds of surface integrals: a surface integral of a scalar-valued function and a surface integral of a vector field.

However, before we can integrate over a surface, we need to consider the surface itself. Recall that to calculate a scalar or vector line integral over curve C, we first need to parameterize C. In a similar way, to calculate a surface integral over surface S, we need to parameterize S. That is, we need a working concept of a *parameterized surface* (or a *parametric surface*), in the same way that we already have a concept of a parameterized curve.

A parameterized surface is given by a description of the form

$$\vec{\mathbf{r}}(u,v) = \langle x(u,v),\, y(u,v),\, z(u,v)\rangle.$$

Notice that this parameterization involves two parameters, u and v, because a surface is two-dimensional, and therefore two variables are needed to trace out the surface. The parameters u and v vary over a region called the parameter domain, or parameter space—the set of points in the uv-plane that can be substituted into $\vec{\mathbf{r}}$. Each choice of u and v in the parameter domain gives a point on the surface, just as each choice of a parameter t gives a point on a parameterized curve. The entire surface is created by making all possible choices of u and v over the parameter domain.

Definition: Parameter Domain

Given a parameterization of surface

$$\vec{\mathbf{r}}(u,v) = \langle x(u,v),\, y(u,v),\, z(u,v)\rangle.$$

the *parameter domain* of the parameterization is the set of points in the uv-plane that can be substituted into $\vec{\mathbf{r}}$.

Example 16.6.1: Parameterizing a Cylinder

Describe surface S parameterized by

$$\vec{\mathbf{r}}(u,v) = \langle \cos u,\, \sin u,\, v\rangle,\ -\infty < u < \infty,\ -\infty < v < \infty.$$

Solution

To get an idea of the shape of the surface, we first plot some points. Since the parameter domain is all of $\mathbb{R}^2$, we can choose any value for *u* and *v* and plot the corresponding point. If $u = v = 0$, then $\vec{\mathbf{r}}(0,0) = \langle 1,0,0\rangle$, so point (1, 0, 0) is on S. Similarly, points $\vec{\mathbf{r}}(\pi,2) = (-1,0,2)$ and $\vec{\mathbf{r}}\left(\frac{\pi}{2},4\right) = (0,1,4)$ are on S.

Although plotting points may give us an idea of the shape of the surface, we usually need quite a few points to see the shape. Since it is time-consuming to plot dozens or hundreds of points, we use another strategy. To visualize S, we visualize two families of curves that lie on S. In the first family of curves we hold u constant; in the second family of curves we hold v constant. This allows us to build a "skeleton" of the surface, thereby getting an idea of its shape.

- Suppose that u is a constant K. Then the curve traced out by the parameterization is $\langle \cos K,\ \sin K,\ v\rangle$, which gives a vertical line that goes through point $\langle \cos K, \sin K, v\rangle$ in the xy-plane.
- Suppose that v is a constant K. Then the curve traced out by the parameterization is $\langle \cos u,\ \sin u,\ K\rangle$, which gives a circle in plane $z = K$ with radius 1 and center $(0,0,K)$.

If u is held constant, then we get vertical lines; if v is held constant, then we get circles of radius 1 centered around the vertical line that goes through the origin. Therefore the surface traced out by the parameterization is cylinder $x^2 + y^2 = 1$ (Figure 16.6.1).

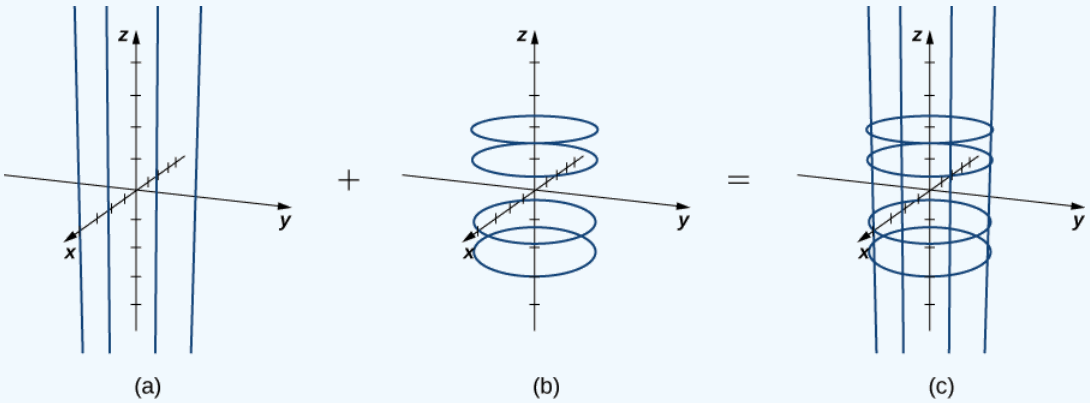


Figure 16.6.1: (a) Lines $\langle \cos K,\ \sin K, v\rangle$ for $K = 0,\ \frac{\pi}{2},\ \pi$, and $\frac{3\pi}{2}$. (b) Circles $\langle \cos u,\ \sin u,\ K\rangle$ for $K = -2, -1, 1,$ and 2 (c) The lines and circles together. As u and v vary, they describe a cylinder.

Notice that if $x = \cos u$ and $y = \sin u$, then $x^2 + y^2 = 1$, so points from *S* do indeed lie on the cylinder. Conversely, each point on the cylinder is contained in some circle $\langle \cos u,\ \sin u,\ k\rangle$ for some k, and therefore each point on the cylinder is contained in the parameterized surface (Figure 16.6.2).

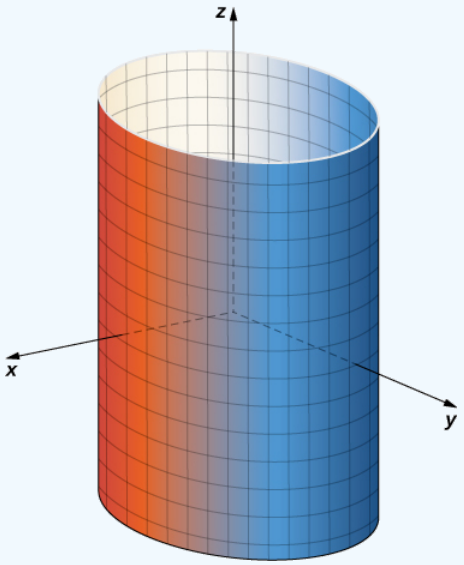


Figure 16.6.2: Cylinder $x^2 + y^2 = r^2$ has parameterization $\vec{\mathbf{r}}(u,v) = \langle r\cos u,\ r\sin u,\ v\rangle$, $0 \le u \le 2\pi$, $-\infty < v < \infty$.

Analysis

Notice that if we change the parameter domain, we could get a different surface. For example, if we restricted the domain to $0 \leq u \leq \pi,\ -\infty < v < 6$, then the surface would be a half-cylinder of height 6.

? Exercise 16.6.1

Describe the surface with parameterization

$$\vec{\mathbf{r}}(u,v) = \langle 2\cos u,\ 2\sin u,\ v\rangle,\ 0 \leq u \leq 2\pi,\ -\infty < v < \infty$$

Hint

Hold u and v constant, and see what kind of curves result.

Answer

Cylinder $x^2 + y^2 = 4$

It follows from Example 16.6.1 that we can parameterize all cylinders of the form $x^2 + y^2 = R^2$. If S is a cylinder given by equation $x^2 + y^2 = R^2$, then a parameterization of S is $\vec{\mathbf{r}}(u,v) = \langle R\cos u,\ R\sin u,\ v\rangle,\ 0 \leq u \leq 2\pi,\ -\infty < v < \infty.$

We can also find different types of surfaces given their parameterization, or we can find a parameterization when we are given a surface.

✓ Example 16.6.2: Describing a Surface

Describe surface S parameterized by $\vec{\mathbf{r}}(u,v) = \langle u\cos v,\ u\sin v,\ u^2\rangle,\ 0 \leq u < \infty,\ 0 \leq v < 2\pi$.

Solution

Notice that if u is held constant, then the resulting curve is a circle of radius u in plane $z = u$. Therefore, as u increases, the radius of the resulting circle increases. If v is held constant, then the resulting curve is a vertical parabola. Therefore, we expect the surface to be an elliptic paraboloid. To confirm this, notice that

$$\begin{aligned} x^2 + y^2 &= (u\cos v)^2 + (u\sin v)^2 \\ &= u^2\cos^2 v + u^2 sin^2 v \\ &= u^2 \\ &= z \end{aligned}$$

Therefore, the surface is the elliptic paraboloid $x^2 + y^2 = z$ (Figure 16.6.3).

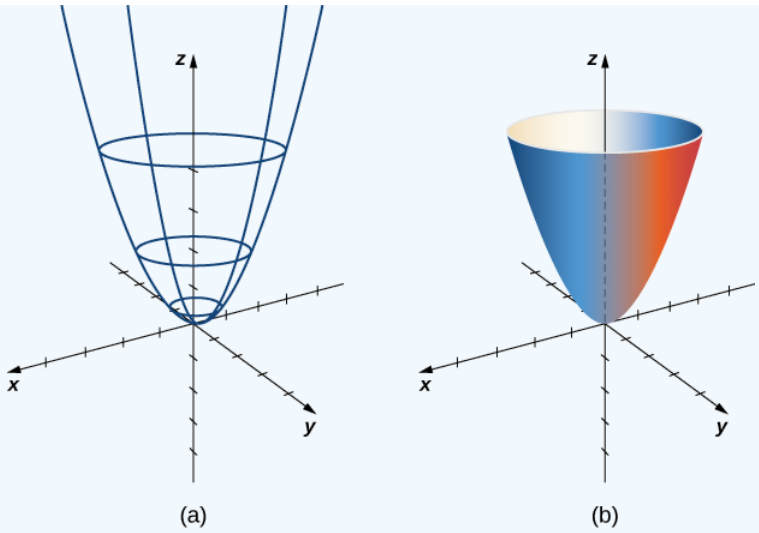


Figure 16.6.3: (a) Circles arise from holding u constant; the vertical parabolas arise from holding v constant. (b) An elliptic paraboloid results from all choices of u and v in the parameter domain.

Exercise 16.6.2

Describe the surface parameterized by $\vec{\mathbf{r}}(u,v) = \langle u\ \cos v,\ u\ \sin v,\ u\rangle,\ -\infty < u < \infty,\ 0 \le v < 2\pi$.

Hint

Hold u constant and see what kind of curves result. Imagine what happens as u increases or decreases.

Answer

Cone $x^2 + y^2 = z^2$

Example 16.6.3: Finding a Parameterization

Give a parameterization of the cone $x^2 + y^2 = z^2$ lying on or above the plane $z = -2$.

Solution

The horizontal cross-section of the cone at height $z = u$ is circle $x^2 + y^2 = u^2$. Therefore, a point on the cone at height u has coordinates $(u\ \cos v,\ u\ \sin v,\ u)$ for angle v. Hence, a parameterization of the cone is $\vec{\mathbf{r}}(u,v) = \langle u\ \cos v,\ u\ \sin v,\ u\rangle$. Since we are not interested in the entire cone, only the portion on or above plane $z = -2$, the parameter domain is given by $-2 < u < \infty,\ 0 \le v < 2\pi$ (Figure 16.6.4).

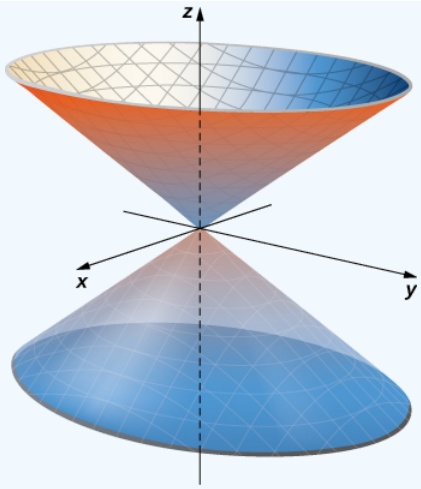


Figure 16.6.4: Cone $x^2 + y^2 = z^2$ has parametrization $\vec{\mathbf{r}}(u, v) = \langle u \cos v,\ u \sin v \rangle,\ -\infty < u < \infty,\ 0 \le v \le 2\pi$.

? Exercise 16.6.3

Give a parameterization for the portion of cone $x^2 + y^2 = z^2$ lying in the first octant.

Hint

Consider the parameter domain for this surface.

Answer

$\vec{\mathbf{r}}(u, v) = \langle u \cos v,\ u \sin v,\ u \rangle,\ 0 < u < \infty,\ 0 \le v < \dfrac{\pi}{2}$

We have discussed parameterizations of various surfaces, but two important types of surfaces need a separate discussion: spheres and graphs of two-variable functions. To parameterize a sphere, it is easiest to use spherical coordinates. The sphere of radius ρ centered at the origin is given by the parameterization

$$\vec{\mathbf{r}}(\phi, \theta) = \langle \rho \cos\theta \sin\phi,\ \rho \sin\theta \sin\phi,\ \rho \cos\phi \rangle,\ 0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi.$$

The idea of this parameterization is that as ϕ sweeps downward from the positive z-axis, a circle of radius $\rho \sin\phi$ is traced out by letting θ run from 0 to 2π. To see this, let ϕ be fixed. Then

$$\begin{aligned} x^2 + y^2 &= (\rho \cos\theta \sin\phi)^2 + (\rho \sin\theta \sin\phi)^2 \\ &= \rho^2 \sin^2\phi(\cos^2\theta + \sin^2\theta) \\ &= \rho^2 \sin^2\phi \\ &= (\rho \sin\phi)^2. \end{aligned}$$

This results in the desired circle (Figure 16.6.5).

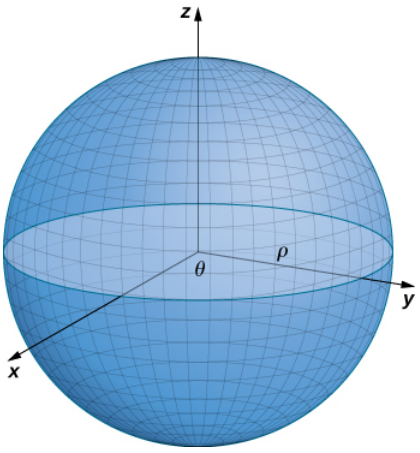


Figure 16.6.5: The sphere of radius ρ has parameterization $\vec{\mathbf{r}}(\phi,\theta) = \langle \rho\cos\theta\sin\phi,\ \rho\sin\theta\sin\phi,\ \rho\cos\phi\rangle,\ 0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi$.

Finally, to parameterize the graph of a two-variable function, we first let $z = f(x,y)$ be a function of two variables. The simplest parameterization of the graph of f is $\vec{\mathbf{r}}(x,y) = \langle x, y, f(x,y)\rangle$, where x and y vary over the domain of f (Figure 16.6.6). For example, the graph of $f(x,y) = x^2y$ can be parameterized by $\vec{\mathbf{r}}(x,y) = \langle x, y, x^2y\rangle$, where the parameters x and y vary over the domain of f. If we only care about a piece of the graph of f - say, the piece of the graph over rectangle $[1,3] \times [2,5]$ - then we can restrict the parameter domain to give this piece of the surface:

$$\vec{\mathbf{r}}(x,y) = \langle x, y, x^2y\rangle,\ 1 \le x \le 3,\ 2 \le y \le 5.$$

Similarly, if S is a surface given by equation $x = g(y,z)$ or equation $y = h(x,z)$, then a parameterization of S is $\vec{\mathbf{r}}(y,z) = \langle g(y,z),\ y, z\rangle$ or $\vec{\mathbf{r}}(x,z) = \langle x, h(x,z), z\rangle$, respectively. For example, the graph of paraboloid $2y = x^2 + z^2$ can be parameterized by $\vec{\mathbf{r}}(x,y) = \left\langle x, \dfrac{x^2+z^2}{2}, z\right\rangle,\ 0 \le x < \infty,\ 0 \le z < \infty$. Notice that we do not need to vary over the entire domain of y because x and z are squared.

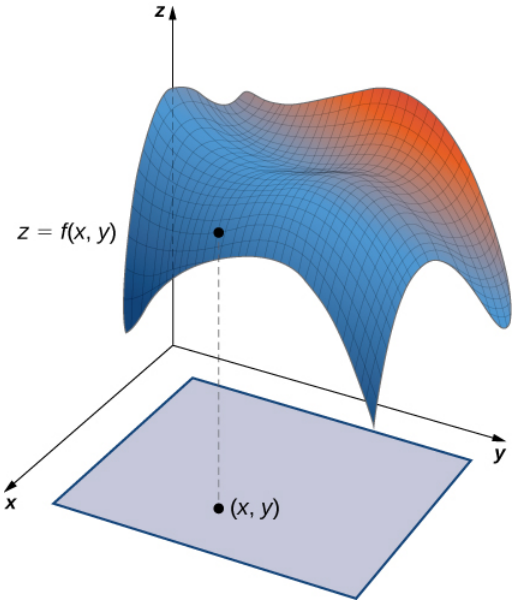


Figure 16.6.6: The simplest parameterization of the graph of a function is $\vec{\mathbf{r}}(x,y) = \langle x, y, f(x,y)\rangle$.

Let's now generalize the notions of smoothness and regularity to a parametric surface. Recall that curve parameterization $\vec{\mathbf{r}}(t),\ a \le t \le b$ is regular (or **smooth**) if $\vec{\mathbf{r}}'(t) \neq \vec{\mathbf{0}}$ for all t in $[a,b]$. For a curve, this condition ensures that the image of $\vec{\mathbf{r}}$ really is a curve, and not just a point. For example, consider curve parameterization $\vec{\mathbf{r}}(t) = \langle 1, 2\rangle,\ 0 \le t \le 5$. The image of this parameterization is simply point $(1,2)$, which is not a curve. Notice also that $\vec{\mathbf{r}}'(t) = \vec{\mathbf{0}}$. The fact that the derivative is the zero vector indicates we are not actually looking at a curve.

Analogously, we would like a notion of regularity (or smoothness) for surfaces so that a surface parameterization really does trace out a surface. To motivate the definition of regularity of a surface parameterization, consider the parameterization

$$\vec{\mathbf{r}}(u,v) = \langle 0,\ \cos v,\ 1\rangle,\ 0 \le u \le 1,\ 0 \le v \le \pi.$$

Although this parameterization appears to be the parameterization of a surface, notice that the image is actually a line (Figure 16.6.7). How could we avoid parameterizations such as this? Parameterizations that do not give an actual surface? Notice that $\vec{\mathbf{r}}_u = \langle 0,0,0\rangle$ and $\vec{\mathbf{r}}_v = \langle 0,-\sin v,0\rangle$, and the corresponding cross product is zero. The analog of the condition $\vec{\mathbf{r}}'(t) = \vec{\mathbf{0}}$ is that $\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v$ is not zero for point (u,v) in the parameter domain, which is a regular parameterization.

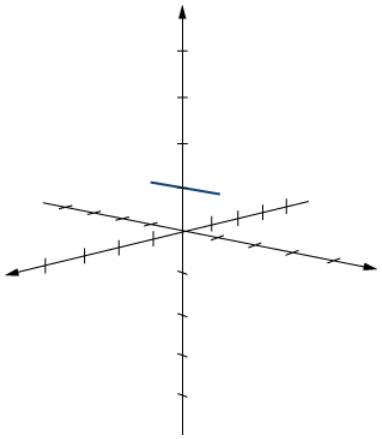


Figure 16.6.7: The image of parameterization $\vec{\mathbf{r}}(u,v) = \langle 0,\cos v,1\rangle,\ 0 \le u \le 1,\ 0 \le v \le \pi$ is a line.

Definition: Regular Parameterization

Parameterization $\vec{\mathbf{r}}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$ is a **regular parameterization** if $\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v$ is not zero for point (u,v) in the parameter domain.

If parameterization $\vec{r}$ is regular, then the image of $\vec{r}$ is a two-dimensional object, as a surface should be. Throughout this chapter, parameterizations $\vec{\mathbf{r}}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$are assumed to be regular.

Recall that curve parameterization $\vec{\mathbf{r}}(t),\ a \le t \le b$ is smooth if $\vec{\mathbf{r}}'(t)$ is continuous and $\vec{\mathbf{r}}'(t) \ne \vec{\mathbf{0}}$ for all t in $[a,b]$. Informally, a curve parameterization is smooth if the resulting curve has no sharp corners. The definition of a smooth surface parameterization is similar. Informally, a surface parameterization is *smooth* if the resulting surface has no sharp corners.

Definition: Smooth Surfaces

A surface parameterization $\vec{\mathbf{r}}(u,v) = \langle x(u,v), y(u,v), z(u,v)\rangle$ is *smooth* if vector $\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v$ is not zero for any choice of u and v in the parameter domain.

A surface may also be *piecewise smooth* if it has smooth faces but also has locations where the directional derivatives do not exist.

Example 16.6.4: Identifying Smooth and Nonsmooth Surfaces

Which of the figures in Figure 16.6.8 is smooth?

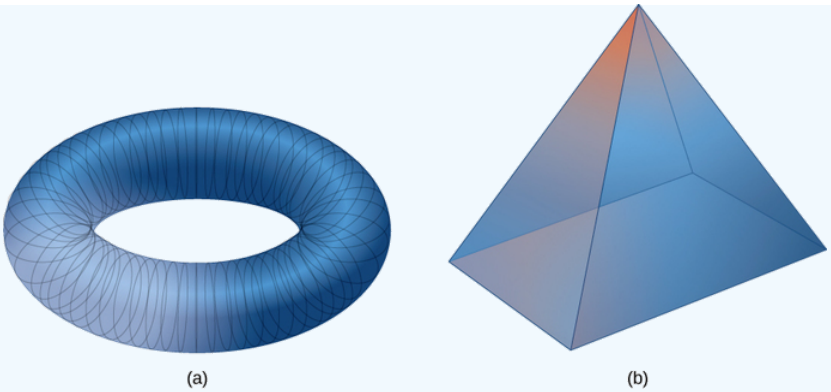


Figure 16.6.8: (a) This surface is smooth. (b) This surface is piecewise smooth.

Solution

The surface in Figure 16.6.8a can be parameterized by

$$\vec{\mathbf{r}}(u,v) = \langle (2+\cos v)\cos u,\ (2+\cos v)\sin u,\ \sin v\rangle,\ 0 \le u < 2\pi,\ 0 \le v < 2\pi$$

(we can use technology to verify). Notice that vectors

$$\vec{\mathbf{r}}_u = \langle -(2+\cos v)\sin u,\ (2+\cos v)\cos u, 0\rangle$$

and

$$\vec{\mathbf{r}}_v = \langle -\sin v\cos u,\ -\sin v\sin u,\ \cos v\rangle$$

exist for any choice of u and v in the parameter domain, and

$$\begin{aligned}\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -(2+\cos v)\sin u & (2+\cos v)\cos u & 0 \\ -\sin v\cos u & -\sin v\sin u & \cos v \end{vmatrix} \\ &= [(2+\cos v)\cos u\cos v]\hat{\mathbf{i}} + [2+\cos v)\sin u\cos v]\hat{\mathbf{j}} \\ &\quad + [(2+\cos v)\sin v\sin^2 u + (2+\cos v)\sin v\cos^2 u]\hat{\mathbf{k}} \\ &= [(2+\cos v)\cos u\cos v]\hat{\mathbf{i}} + [(2+\cos v)\sin u\cos v]\hat{\mathbf{j}} + [(2+\cos v)\sin v]\hat{\mathbf{k}}.\end{aligned}$$

The $\hat{\mathbf{k}}$ component of this vector is zero only if $v = 0$ or $v = \pi$. If $v = 0$ or $v = \pi$, then the only choices for u that make the $\hat{\mathbf{j}}$ component zero are $u = 0$ or $u = \pi$. But, these choices of u do not make the $\hat{\mathbf{i}}$ component zero. Therefore, $\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v$ is not zero for any choice of u and v in the parameter domain, and the parameterization is smooth. Notice that the corresponding surface has no sharp corners.

In the pyramid in Figure 16.6.8b, the sharpness of the corners ensures that directional derivatives do not exist at those locations. Therefore, the pyramid has no smooth parameterization. However, the pyramid consists of four smooth faces, and thus this surface is piecewise smooth.

? Exercise 16.6.4

Is the surface parameterization $\vec{\mathbf{r}}(u,v) = \langle u^{2v}, v+1, \sin u\rangle,\ 0 \le u \le 2,\ 0 \le v \le 3$ smooth?

Hint

Investigate the cross product $\vec{\mathbf{r}}_u \times \vec{\mathbf{r}}_v$.

Answer

Yes

Surface Area of a Parametric Surface

Our goal is to define a surface integral, and as a first step we have examined how to parameterize a surface. The second step is to define the surface area of a parametric surface. The notation needed to develop this definition is used throughout the rest of this chapter.

Let S be a surface with parameterization $\vec{\mathbf{r}}(u,v) = \langle x(u,v),\ y(u,v),\ z(u,v)\rangle$ over some parameter domain D. We assume here and throughout that the surface parameterization $\vec{\mathbf{r}}(u,v) = \langle x(u,v),\ y(u,v),\ z(u,v)\rangle$ is continuously differentiable—meaning, each component function has continuous partial derivatives. Assume for the sake of simplicity that D is a rectangle (although the following material can be extended to handle nonrectangular parameter domains). Divide rectangle D into subrectangles D_{ij} with horizontal width Δu and vertical length Δv. Suppose that *i* ranges from 1 to *m* and *j* ranges from 1 to *n* so that D is subdivided into *mn* rectangles. This division of D into subrectangles gives a corresponding division of surface S into pieces S_{ij}. Choose point P_{ij} in each piece S_{ij}. Point P_{ij} corresponds to point (u_i, v_j) in the parameter domain.

Note that we can form a grid with lines that are parallel to the u-axis and the v-axis in the uv-plane. These grid lines correspond to a set of grid curves on surface S that is parameterized by $\vec{\mathbf{r}}(u,v)$. Without loss of generality, we assume that P_{ij} is located at the corner of two grid curves, as in Figure 16.6.9. If we think of $\vec{\mathbf{r}}$ as a mapping from the uv-plane to $\mathbb{R}^3$, the grid curves are the image of the grid lines under $\vec{\mathbf{r}}$. To be precise, consider the grid lines that go through point (u_i, v_j). One line is given by $x = u_i,\ y = v$; the other is given by $x = u,\ y = v_j$. In the first grid line, the horizontal component is held constant, yielding a vertical line through (u_i, v_j). In the second grid line, the vertical component is held constant, yielding a horizontal line through (u_i, v_j). The corresponding grid curves are $\vec{\mathbf{r}}(u_i, v)$ and (u, v_j) and these curves intersect at point P_{ij}.

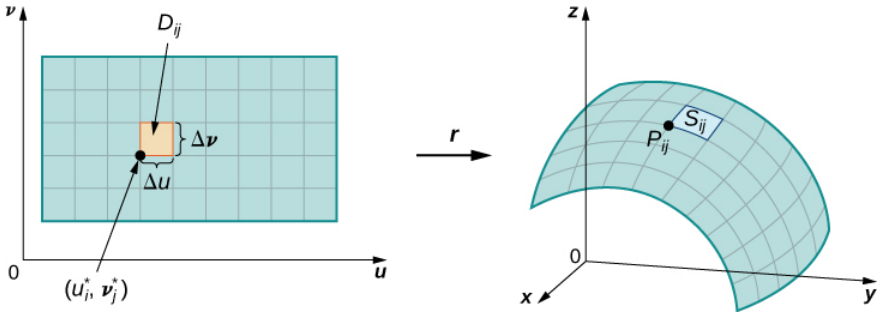


Figure 16.6.9: Grid lines on a parameter domain correspond to grid curves on a surface.

Now consider the vectors that are tangent to these grid curves. For grid curve $\vec{\mathbf{r}}(u_i, v)$, the tangent vector at P_{ij} is

$$\vec{\mathbf{t}}_v(P_{ij}) = \vec{\mathbf{r}}_v(u_i, v_j) = \langle x_v(u_i, v_j),\ y_v(u_i, v_j),\ z_v(u_i, v_j)\rangle.$$

For grid curve $\vec{\mathbf{r}}(u, v_j)$, the tangent vector at P_{ij} is

$$\vec{\mathbf{t}}_u(P_{ij}) = \vec{\mathbf{r}}_u(u_i, v_j) = \langle x_u(u_i, v_j),\ y_u(u_i, v_j),\ z_u(u_i, v_j)\rangle.$$

If vector $\vec{\mathbf{N}} = \vec{\mathbf{t}}_u(P_{ij}) \times \vec{\mathbf{t}}_v(P_{ij})$ exists and is not zero, then the tangent plane at P_{ij} exists (Figure 16.6.10). If piece S_{ij} is small enough, then the tangent plane at point P_{ij} is a good approximation of piece S_{ij}.

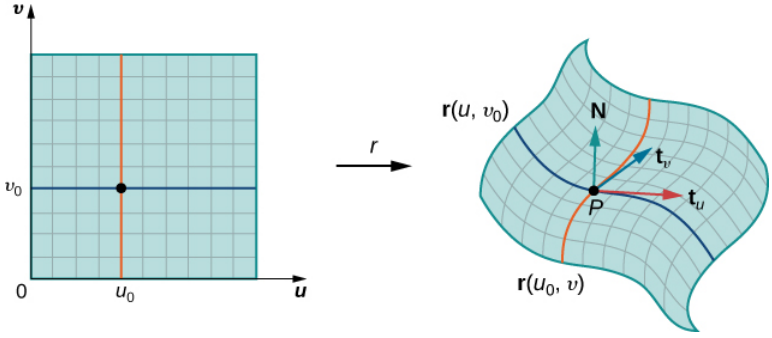


Figure 16.6.10: If the cross product of vectors $\vec{\mathbf{t}}_u$ and $\vec{\mathbf{t}}_v$ exists, then there is a tangent plane.

The tangent plane at P_{ij} contains vectors $\vec{\mathbf{t}}_u(P_{ij})$ and $\vec{\mathbf{t}}_v(P_{ij})$ and therefore the parallelogram spanned by $\vec{\mathbf{t}}_u(P_{ij})$ and $\vec{\mathbf{t}}_v(P_{ij})$ is in the tangent plane. Since the original rectangle in the uv-plane corresponding to S_{ij} has width Δu and length Δv, the parallelogram that we use to approximate S_{ij} is the parallelogram spanned by $\Delta u\vec{\mathbf{t}}_u(P_{ij})$ and $\Delta v\vec{\mathbf{t}}_v(P_{ij})$. In other words, we scale the tangent vectors by the constants Δu and Δv to match the scale of the original division of rectangles in the parameter domain. Therefore, the area of the parallelogram used to approximate the area of S_{ij} is

$$\Delta S_{ij} \approx ||(\Delta u\vec{\mathbf{t}}_u(P_{ij})) \times (\Delta v\vec{\mathbf{t}}_v(P_{ij}))|| = ||\vec{\mathbf{t}}_u(P_{ij}) \times \vec{\mathbf{t}}_v(P_{ij})||\Delta u\,\Delta v.$$

Varying point P_{ij} over all pieces S_{ij} and the previous approximation leads to the following definition of surface area of a parametric surface (Figure 16.6.11).

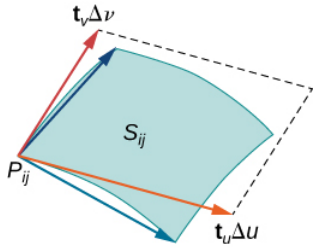


Figure 16.6.11: The parallelogram spanned by $\vec{\mathbf{t}}_u$ *and* $\vec{\mathbf{t}}_v$ *approximates the piece of surface* S_{ij}.

Definition: Smooth Parameterization of Surface

Let $\vec{\mathbf{r}}(u,v) = \langle x(u,v),\, y(u,v),\, z(u,v)\rangle$ with parameter domain D be a **smooth parameterization of surface** S. Furthermore, assume that S is traced out only once as (u,v) varies over D. The surface area of S is

$$\iint_D ||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v||\, dA, \tag{16.6.1}$$

where $\vec{\mathbf{t}}_u = \left\langle \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right\rangle$

and

$$\vec{\mathbf{t}}_v = \left\langle \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right\rangle.$$

Example 16.6.5: Calculating Surface Area

Calculate the lateral surface area (the area of the "side," not including the base) of the right circular cone with height h and radius r.

Solution

Before calculating the surface area of this cone using Equation 16.6.1, we need a parameterization. We assume this cone is in $\mathbb{R}^3$ with its vertex at the origin (Figure 16.6.12). To obtain a parameterization, let α be the angle that is swept out by starting at the positive z-axis and ending at the cone, and let $k = \tan\alpha$. For a height value v with $0 \le v \le h$, the radius of the circle formed by intersecting the cone with plane $z = v$ is kv. Therefore, a parameterization of this cone is

$$\vec{\mathbf{s}}(u,v) = \langle kv\cos u,\, kv\sin u,\, v\rangle,\ 0 \le u < 2\pi,\ 0 \le v \le h.$$

The idea behind this parameterization is that for a fixed v-value, the circle swept out by letting u vary is the circle at height v and radius kv. As v increases, the parameterization sweeps out a "stack" of circles, resulting in the desired cone.

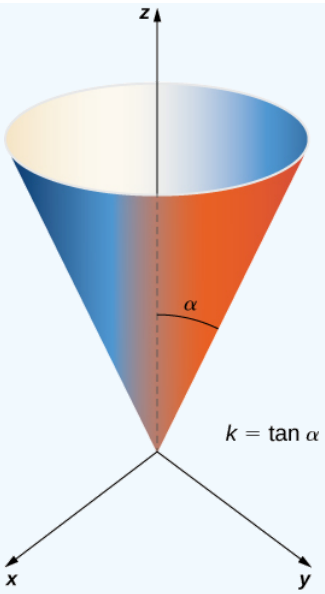


Figure 16.6.12: The right circular cone with radius $r = kh$ and height h has parameterization $\vec{\mathbf{s}}(u, v) = \langle kv\,\cos u,\ kv\,\sin u,\ v\rangle$, $0 \le u < 2\pi$, $0 \le v \le h$.

With a parameterization in hand, we can calculate the surface area of the cone using Equation 16.6.1. The tangent vectors are $\vec{\mathbf{t}}_u = \langle -kv\,\sin u,\ kv\,\cos u,\ 0\rangle$ and $\vec{\mathbf{t}}_v = \langle k\,\cos u,\ k\,\sin u,\ 1\rangle$. Therefore,

$$\begin{aligned}\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -kv\sin u & kv\cos u & 0 \\ k\cos u & k\sin u & 1 \end{vmatrix} \\ &= \langle kv\,\cos u,\ kv\,\sin u,\ -k^2v\,\sin^2 u - k^2v\,\cos^2 u\rangle \\ &= \langle kv\,\cos u,\ kv\,\sin u,\ -k^2v\rangle.\end{aligned}$$

The magnitude of this vector is

$$\begin{aligned}||\langle kv\,\cos u,\ kv\,\sin u,\ -k^2v\rangle|| &= \sqrt{k^2v^2\cos^2 u + k^2v^2\sin^2 u + k^4v^2} \\ &= \sqrt{k^2v^2 + k^4v^2} \\ &= kv\sqrt{1+k^2}.\end{aligned}$$

By Equation 16.6.1, the surface area of the cone is

$$\begin{aligned}\iint_D ||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v||\,dA &= \int_0^h \int_0^{2\pi} kv\sqrt{1+k^2}\,du\,dv \\ &= 2\pi k\sqrt{1+k^2}\int_0^h v\,dv \\ &= 2\pi k\sqrt{1+k^2}\left[\frac{v^2}{2}\right]_0^h \\ &= \pi k h^2\sqrt{1+k^2}.\end{aligned}$$

Since $k = \tan\alpha = r/h$,

$$\begin{aligned}\pi kh^2\sqrt{1+k^2} &= \pi\frac{r}{h}h^2\sqrt{1+\frac{r^2}{h^2}} \\ &= \pi rh\sqrt{1+\frac{r^2}{h^2}} \\ &= \pi r\sqrt{h^2+h^2\left(\frac{r^2}{h^2}\right)} \\ &= \pi r\sqrt{h^2+r^2}.\end{aligned}$$

Therefore, the lateral surface area of the cone is $\pi r\sqrt{h^2+r^2}$.

Analysis

The surface area of a right circular cone with radius r and height h is usually given as $\pi r^2+\pi r\sqrt{h^2+r^2}$. The reason for this is that the circular base is included as part of the cone, and therefore the area of the base πr^2 is added to the lateral surface area $\pi r\sqrt{h^2+r^2}$ that we found.

? Exercise 16.6.5

Find the surface area of the surface with parameterization $\vec{\mathbf{r}}(u,v)=\langle u+v,\, u^2,\, 2v\rangle,\, 0\le u\le 3,\, 0\le v\le 2$.

Hint

Use Equation 16.6.1.

Answer

≈ 43.02

✓ Example 16.6.6: Calculating Surface Area

Show that the surface area of the sphere $x^2+y^2+z^2=r^2$ is $4\pi r^2$.

Solution

The sphere has parameterization

$$r\cos\theta\sin\phi,\, r\sin\theta\sin\phi,\, r\cos\phi\rangle,\, 0\le\theta<2\pi,\, 0\le\phi\le\pi.$$

The tangent vectors are

$$\vec{\mathbf{t}}_\theta=\langle -r\sin\theta\sin\phi,\, r\cos\theta\sin\phi,\, 0\rangle$$

and

$$\vec{\mathbf{t}}_\phi=\langle r\cos\theta\cos\phi,\, r\sin\theta\cos\phi,\, -r\sin\phi\rangle.$$

Therefore,

$$\begin{aligned}\vec{\mathbf{t}}_\phi\times\vec{\mathbf{t}}_\theta &= \langle r^2\cos\theta\sin^2\phi,\, r^2\sin\theta\sin^2\phi,\, r^2\sin^2\theta\sin\phi\cos\phi+r^2\cos^2\theta\sin\phi\cos\phi\rangle \\ &= \langle r^2\cos\theta\sin^2\phi,\, r^2\sin\theta\sin^2\phi,\, r^2\sin\phi\cos\phi\rangle.\end{aligned}$$

Now,

$$\begin{aligned}||\vec{\mathbf{t}}_\phi\times\vec{\mathbf{t}}_\theta|| &= \sqrt{r^4\sin^4\phi\cos^2\theta+r^4\sin^4\phi\sin^2\theta+r^4\sin^2\phi\cos^2\phi} \\ &= \sqrt{r^4\sin^4\phi+r^4\sin^2\phi\cos^2\phi} \\ &= r^2\sqrt{\sin^2\phi} \\ &= r\sin\phi.\end{aligned}$$

Notice that $\sin\phi \geq 0$ on the parameter domain because $0 \leq \phi < \pi$, and this justifies equation $\sqrt{\sin^2\phi} = \sin\phi$. The surface area of the sphere is

$$\int_0^{2\pi}\int_0^{\pi} r^2 \sin\phi \, d\phi \, d\theta = r^2 \int_0^{2\pi} 2 \, d\theta = 4\pi r^2.$$

We have derived the familiar formula for the surface area of a sphere using surface integrals.

Exercise 16.6.6

Show that the surface area of cylinder $x^2 + y^2 = r^2$, $0 \leq z \leq h$ is $2\pi rh$. Notice that this cylinder does not include the top and bottom circles.

Hint

Use the standard parameterization of a cylinder and follow the previous example.

Answer

With the standard parameterization of a cylinder, Equation 16.6.1 shows that the surface area is $2\pi rh$.

In addition to parameterizing surfaces given by equations or standard geometric shapes such as cones and spheres, we can also parameterize surfaces of revolution. Therefore, we can calculate the surface area of a surface of revolution by using the same techniques. Let $y = f(x) \geq 0$ be a positive single-variable function on the domain $a \leq x \leq b$ and let S be the surface obtained by rotating f about the x-axis (Figure 16.6.13). Let θ be the angle of rotation. Then, S can be parameterized with parameters x and θ by

$$\vec{\mathbf{r}}(x,\theta) = \langle x, f(x)\cos\theta, f(x)\sin\theta\rangle, \; a \leq x \leq b, \; 0 \leq x \leq 2\pi.$$

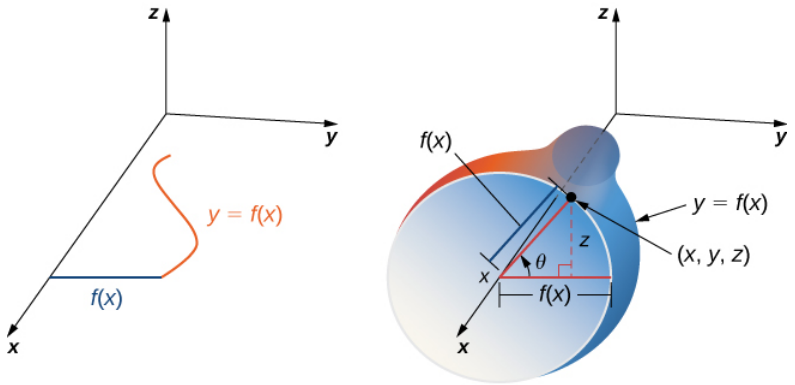


Figure 16.6.13: We can parameterize a surface of revolution by $\vec{\mathbf{r}}(x,\theta) = \langle x, f(x)\cos\theta, f(x)\sin\theta\rangle$, $a \leq x \leq b$, $0 \leq x < 2\pi$.

Example 16.6.7: Calculating Surface Area

Find the area of the surface of revolution obtained by rotating $y = x^2$, $0 \leq x \leq b$ about the x-axis (Figure 16.6.14).

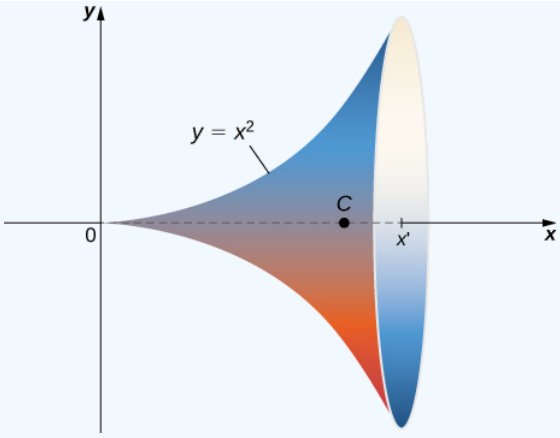


Figure 16.6.14: A surface integral can be used to calculate the surface area of this solid of revolution.

Solution

This surface has parameterization $\vec{\mathbf{r}}(x,\theta) = \langle x,\, x^2\cos\theta,\, x^2\sin\theta\rangle,\ 0 \le x \le b,\ 0 \le x < 2\pi$.

The tangent vectors are $\vec{\mathbf{t}}_x = \langle 1,\, 2x\,\cos\theta,\, 2x\,\sin\theta\rangle$ and $\vec{\mathbf{t}}_\theta = \langle 0,\, -x^2\sin\theta,\, -x^2\cos\theta\rangle$.

Therefore,

$$\begin{aligned}\vec{\mathbf{t}}_x \times \vec{\mathbf{t}}_\theta &= \langle 2x^3\cos^2\theta + 2x^3\sin^2\theta,\, -x^2\cos\theta,\, -x^2\sin\theta\rangle \\ &= \langle 2x^3,\, -x^2\cos\theta,\, -x^2\sin\theta\rangle\end{aligned}$$

and

$$\begin{aligned}\vec{\mathbf{t}}_x \times \vec{\mathbf{t}}_\theta &= \sqrt{4x^6 + x^4\cos^2\theta + x^4\sin^2\theta} \\ &= \sqrt{4x^6 + x^4} \\ &= x^2\sqrt{4x^2+1}\end{aligned}$$

The area of the surface of revolution is

$$\begin{aligned}\int_0^b \int_0^{2\pi} x^2\sqrt{4x^2+1}\,d\theta\,dx &= 2\pi\int_0^b x^2\sqrt{4x^2+1}\,dx \\ &= 2\pi\left[\frac{1}{64}\left(2\sqrt{4x^2+1}(8x^3+x)\,\sinh^{-1}(2x)\right)\right]_0^b \\ &= 2\pi\left[\frac{1}{64}\left(2\sqrt{4b^2+1}(8b^3+b)\,\sinh^{-1}(2b)\right)\right].\end{aligned}$$

? Exercise 16.6.7

Use Equation 16.6.1 to find the area of the surface of revolution obtained by rotating curve $y = \sin x,\ 0 \le x \le \pi$ about the x-axis.

Hint

Use the parameterization of surfaces of revolution given before Example 16.6.7.

Answer

$2\pi(\sqrt{2} + \sinh^{-1}(1))$

Surface Integral of a Scalar-Valued Function

Now that we can parameterize surfaces and we can calculate their surface areas, we are able to define surface integrals. First, let's look at the surface integral of a scalar-valued function. Informally, the surface integral of a scalar-valued function is an analog of a scalar line integral in one higher dimension. The domain of integration of a scalar line integral is a parameterized curve (a one-

dimensional object); the domain of integration of a scalar surface integral is a parameterized surface (a two-dimensional object). Therefore, the definition of a surface integral follows the definition of a line integral quite closely. For scalar line integrals, we chopped the domain curve into tiny pieces, chose a point in each piece, computed the function at that point, and took a limit of the corresponding Riemann sum. For scalar surface integrals, we chop the domain *region* (no longer a curve) into tiny pieces and proceed in the same fashion.

Let S be a piecewise smooth surface with parameterization $\vec{\mathbf{r}}(u,v) = \langle x(u,v),\ y(u,v),\ z(u,v)\rangle$ with parameter domain D and let $f(x,y,z)$ be a function with a domain that contains S. For now, assume the parameter domain D is a rectangle, but we can extend the basic logic of how we proceed to any parameter domain (the choice of a rectangle is simply to make the notation more manageable). Divide rectangle D into subrectangles D_{ij} with horizontal width Δu and vertical length Δv. Suppose that i ranges from 1 to m and j ranges from 1 to n so that D is subdivided into mn rectangles. This division of D into subrectangles gives a corresponding division of S into pieces S_{ij}. Choose point P_{ij} in each piece S_{ij} evaluate P_{ij} at f, and multiply by area S_{ij} to form the Riemann sum

$$\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\,\Delta S_{ij}.$$

To define a surface integral of a scalar-valued function, we let the areas of the pieces of S shrink to zero by taking a limit.

Definition: Surface Integral of a Scalar-Valued Function

The **surface integral of a scalar-valued function** of f over a piecewise smooth surface S is

$$\iint_S f(x,y,z)dA = \lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta S_{ij}.$$

Again, notice the similarities between this definition and the definition of a scalar line integral. In the definition of a line integral we chop a curve into pieces, evaluate a function at a point in each piece, and let the length of the pieces shrink to zero by taking the limit of the corresponding Riemann sum. In the definition of a surface integral, we chop a surface into pieces, evaluate a function at a point in each piece, and let the area of the pieces shrink to zero by taking the limit of the corresponding Riemann sum. Thus, a surface integral is similar to a line integral but in one higher dimension.

The definition of a scalar line integral can be extended to parameter domains that are not rectangles by using the same logic used earlier. The basic idea is to chop the parameter domain into small pieces, choose a sample point in each piece, and so on. The exact shape of each piece in the sample domain becomes irrelevant as the areas of the pieces shrink to zero.

Scalar surface integrals are difficult to compute from the definition, just as scalar line integrals are. To develop a method that makes surface integrals easier to compute, we approximate surface areas ΔS_{ij} with small pieces of a tangent plane, just as we did in the previous subsection. Recall the definition of vectors $\vec{\mathbf{t}}_u$ and $\vec{\mathbf{t}}_v$:

$$\vec{\mathbf{t}}_u = \left\langle \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right\rangle \text{ and } \vec{\mathbf{t}}_v = \left\langle \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right\rangle.$$

From the material we have already studied, we know that

$$\Delta S_{ij} \approx ||\vec{\mathbf{t}}_u(P_{ij}) \times \vec{\mathbf{t}}_v(P_{ij})||\,\Delta u\,\Delta v.$$

Therefore,

$$\iint_S f(x,y,z)\,dS \approx \lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})||\vec{\mathbf{t}}_u(P_{ij}) \times \vec{\mathbf{t}}_v(P_{ij})||\,\Delta u\,\Delta v.$$

This approximation becomes arbitrarily close to $\lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n} f(P_{ij})\Delta S_{ij}$ as we increase the number of pieces S_{ij} by letting m and n go to infinity. Therefore, we have the following equation to calculate **scalar surface integrals**:

$$\iint_S f(x,y,z)\,dS = \iint_D f(\vec{\mathbf{r}}(u,v))||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v||\,dA. \tag{16.6.2}$$

Equation 16.6.2 allows us to calculate a surface integral by transforming it into a double integral. This equation for surface integrals is analogous to the equation for line integrals:

$$\iint_C f(x,y,z)\,ds = \int_a^b f(\vec{\mathbf{r}}(t))||\vec{\mathbf{r}}\,'(t)||\,dt.$$

In this case, vector $\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v$ is perpendicular to the surface, whereas vector $\vec{\mathbf{r}}\,'(t)$ is tangent to the curve.

Example 16.6.8: Calculating a Surface Integral

Calculate surface integral

$$\iint_S 5\,dS,$$

where S is the surface with parameterization $\vec{\mathbf{r}}(u,v) = \langle u,\, u^2,\, v\rangle$ for $0 \le u \le 2$ and $0 \le v \le u$.

Solution

Notice that this parameter domain D is a triangle, and therefore the parameter domain is not rectangular. This is not an issue though, because Equation 16.6.2 does not place any restrictions on the shape of the parameter domain.

To use Equation 16.6.2 to calculate the surface integral, we first find vectors $\vec{\mathbf{t}}_u$ and $\vec{\mathbf{t}}_v$. Note that $\vec{\mathbf{t}}_u = \langle 1, 2u, 0\rangle$ and $\vec{\mathbf{t}}_v = \langle 0, 0, 1\rangle$. Therefore,

$$\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 1 & 2u & 0 \\ 0 & 0 & 1 \end{vmatrix} = \langle 2u,\, -1,\, 0\rangle$$

and

$$||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v|| = \sqrt{1+4u^2}.$$

By Equation 16.6.2,

$$\begin{aligned} \iint_S 5\,dS &= 5\iint_D \sqrt{1+4u^2}\,dA \\ &= 5\int_0^2 \int_0^u \sqrt{1+4u^2}\,dv\,du = 5\int_0^2 u\sqrt{1+4u^2}\,du \\ &= 5\left[\frac{(1+4u^2)^{3/2}}{3}\right]_0^2 \\ &= \frac{5(17^{3/2}-1)}{3} \approx 115.15. \end{aligned}$$

Example 16.6.9: Calculating the Surface Integral of a Cylinder

Calculate surface integral

$$\iint_S (x+y^2)\,dS,$$

where S is cylinder $x^2+y^2=4$, $0 \le z \le 3$ (Figure 16.6.15).

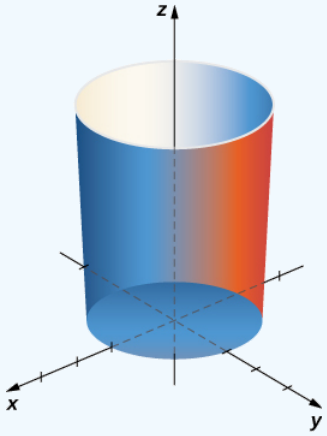


Figure 16.6.15: Integrating function $f(x, y, z) = x + y^2$ over a cylinder.

Solution

To calculate the surface integral, we first need a parameterization of the cylinder. A parameterization is $\vec{\mathbf{r}}(u, v) = \langle \cos u,\ \sin u,\ v \rangle, 0 \le u \le 2\pi,\ 0 \le v \le 3$.

The tangent vectors are $\vec{\mathbf{t}}_u = \langle \sin u,\ \cos u,\ 0 \rangle$ and $\vec{\mathbf{t}}_v = \langle 0, 0, 1 \rangle$. Then,

$$\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -\sin u & \cos u & 0 \\ 0 & 0 & 1 \end{vmatrix} = \langle \cos u,\ \sin u,\ 0 \rangle$$

and $||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v|| = \sqrt{\cos^2 u + \sin^2 u} = 1$. By Equation 16.6.2,

$$\begin{aligned} \iint_S f(x, y, z) dS &= \iint_D f(\vec{\mathbf{r}}(u, v)) ||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v|| \, dA \\ &= \int_0^3 \int_0^{2\pi} (\cos u + \sin^2 u) \, du \, dv \\ &= \int_0^3 \left[\sin u + \frac{u}{2} - \frac{\sin(2u)}{4} \right]_0^{2\pi} dv \\ &= \int_0^3 \pi \, dv = 3\pi. \end{aligned}$$

Exercise 16.6.8

Calculate

$$\iint_S (x^2 - z) \, dS,$$

where S is the surface with parameterization $\vec{\mathbf{r}}(u, v) = \langle v,\ u^2 + v^2,\ 1 \rangle$, $0 \le u \le 2$, $0 \le v \le 3$.

Hint

Use Equation 16.6.2.

Answer

24

Example 16.6.10: Calculating the Surface Integral of a Piece of a Sphere

Calculate surface integral

$$\iint_S f(x, y, z) \, dS,$$

where $f(x, y, z) = z^2$ and S is the surface that consists of the piece of sphere $x^2 + y^2 + z^2 = 4$ that lies on or above plane $z = 1$ and the disk that is enclosed by intersection plane $z = 1$ and the given sphere (Figure 16.6.16).

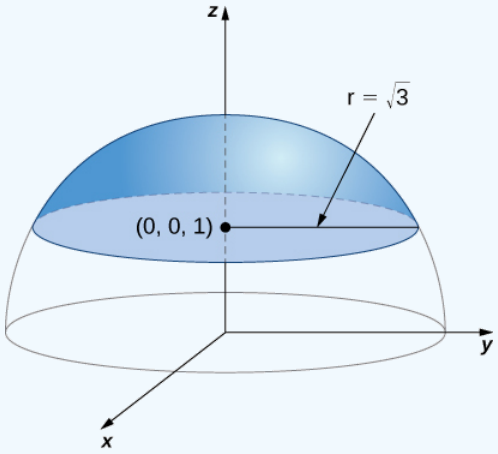


Figure 16.6.16: Calculating a surface integral over surface S.

Solution

Notice that S is not smooth but is piecewise smooth; S can be written as the union of its base S_1 and its spherical top S_2, and both S_1 and S_2 are smooth. Therefore, to calculate

$$\iint_S z^2 dS,$$

we write this integral as

$$\iint_{S_1} z^2\, dS + \iint_{S_2} z^2\, dS$$

and we calculate integrals

$$\iint_{S_1} z^2\, dS$$

and

$$\iint_{S_2} Z^2\, dS.$$

First, we calculate $\iint_{S_1} z^2\, dS$. To calculate this integral we need a parameterization of S_1. This surface is a disk in plane $z = 1$ centered at $(0, 0, 1)$. To parameterize this disk, we need to know its radius. Since the disk is formed where plane $z = 1$ intersects sphere $x^2 + y^2 + z^2 = 4$, we can substitute $z = 1$ into equation $x^2 + y^2 + z^2 = 4$:

$$x^2 + y^2 + 1 = 4 \Rightarrow x^2 + y^2 = 3.$$

Therefore, the radius of the disk is $\sqrt{3}$ and a parameterization of S_1 is $\vec{\mathbf{r}}(u, v) = \langle u\, \cos v,\, u\, \sin v,\, 1\rangle$, $0 \le u \le \sqrt{3}$, $0 \le v \le 2\pi$. The tangent vectors are $\vec{\mathbf{t}}_u = \langle \cos v,\, \sin v,\, 0\rangle$ and $\vec{\mathbf{t}}_v = \langle -u\, \sin v,\, u\, \cos v,\, 0\rangle$, and thus

$$\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \cos v & \sin v & 0 \\ -u \sin v & u \cos v & 0 \end{vmatrix} = \langle 0,\, 0,\, u\, \cos^2 v + u\, \sin^2 v\rangle = \langle 0, 0, u\rangle.$$

The magnitude of this vector is u. Therefore,

$$\begin{aligned}\iint_{S_1} z^2\,dS &= \int_0^{\sqrt{3}}\int_0^{2\pi} f(r(u,v))||t_u \times t_v||\,dv\,du \\ &= \int_0^{\sqrt{3}}\int_0^{2\pi} u\,dv\,du \\ &= 2\pi\int_0^{\sqrt{3}} u\,du \\ &= 2\pi\sqrt{3}.\end{aligned}$$

Now we calculate

$$\iint_{S_2} dS.$$

To calculate this integral, we need a parameterization of S_2. The parameterization of full sphere $x^2+y^2+z^2=4$ is

$$\vec{\mathbf{r}}(\phi,\theta) = \langle 2\cos\theta\sin\phi,\ 2\sin\theta\sin\phi,\ 2\cos\phi\rangle,\ 0\le\theta\le 2\pi, 0\le\phi\le\pi.$$

Since we are only taking the piece of the sphere on or above plane $z=1$, we have to restrict the domain of ϕ. To see how far this angle sweeps, notice that the angle can be located in a right triangle, as shown in Figure 16.6.17 (the $\sqrt{3}$ comes from the fact that the base of S is a disk with radius $\sqrt{3}$). Therefore, the tangent of ϕ is $\sqrt{3}$, which implies that ϕ is $\pi/6$. We now have a parameterization of S_2:

$\vec{\mathbf{r}}(\phi,\theta) = \langle 2\cos\theta\sin\phi,\ 2\sin\theta\sin\phi,\ 2\cos\phi\rangle,\ 0\le\theta\le 2\pi,\ 0\le\phi\le\pi/3.$

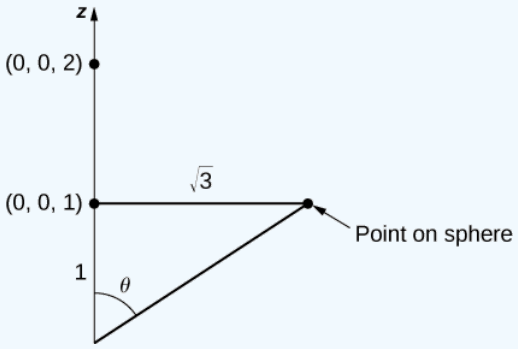


Figure 16.6.17: The maximum value of ϕ has a tangent value of $\sqrt{3}$.

The tangent vectors are $\vec{\mathbf{t}}_\phi = \langle 2\cos\theta\cos\phi,\ 2\sin\theta\cos\phi,\ -2\sin\phi\rangle$ and $\vec{\mathbf{t}}_\theta = \langle -2\sin\theta\sin\phi,\ u\cos\theta\sin\phi,\ 0\rangle$, and thus

$$\begin{aligned}\vec{\mathbf{t}}_\phi \times \vec{\mathbf{t}}_\theta &= \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 2\cos\theta\cos\phi & 2\sin\theta\cos\phi & -2\sin\phi \\ -2\sin\theta\sin\phi & 2\cos\theta\sin\phi & 0 \end{vmatrix} \\ &= \langle 4\cos\theta\sin^2\phi,\ 4\sin\theta\sin^2\phi,\ 4\cos^2\theta\cos\phi\sin\phi + 4\sin^2\theta\cos\phi\sin\phi\rangle \\ &= \langle 4\cos\theta\sin^2\phi,\ 4\sin\theta\sin^2\phi,\ 4\cos\phi\sin\phi\rangle.\end{aligned}$$

The magnitude of this vector is

$$\begin{aligned}\vec{\mathbf{t}}_\phi \times \vec{\mathbf{t}}_\theta &= \sqrt{16\cos^2\theta\sin^4\phi + 16\sin^2\theta\sin^4\phi + 16\cos^2\phi\sin^2\phi} \\ &= 4\sqrt{\sin^4\phi + \cos^2\phi\sin^2\phi}.\end{aligned}$$

Therefore,

$$\begin{aligned}\iint_{S_2} z\,dS &= \int_0^{\pi/6}\int_0^{2\pi} f(\vec{\mathbf{r}}(\phi,\theta))||\vec{\mathbf{t}}_\phi \times \vec{\mathbf{t}}_\theta||\,d\theta\,d\phi \\ &= \int_0^{\pi/6}\int_0^{2\pi} 16\,\cos^2\phi\sqrt{\sin^4\phi+\cos^2\phi\,\sin^2\phi}\,d\theta\,d\phi \\ &= 32\pi\int_0^{\pi/6}\cos^2\phi\sqrt{\sin^4\phi+\cos^2\phi\,\sin^2\phi}\,d\phi \\ &= 32\pi\int_0^{\pi/6}\cos^2\phi\,\sin\phi\sqrt{\sin^2\phi+\cos^2\phi}\,d\phi \\ &= 32\pi\int_0^{\pi/6}\cos^2\phi\,\sin\phi\,d\phi \\ &= 32\pi\left[-\frac{\cos^3\phi}{3}\right]_0^{\pi/6} \\ &= 32\pi\left[\frac{1}{3}-\frac{\sqrt{3}}{8}\right] = \frac{32\pi}{3}-4\sqrt{3}.\end{aligned}$$

Since

$$\iint_S z^2\,dS = \iint_{S_1} z^2\,dS + \iint_{S_2} z^2\,dS,$$

we have

$$\iint_S z^2\,dS = (2\pi-4)\sqrt{3}+\frac{32\pi}{3}.$$

Analysis

In this example we broke a surface integral over a piecewise surface into the addition of surface integrals over smooth subsurfaces. There were only two smooth subsurfaces in this example, but this technique extends to finitely many smooth subsurfaces.

? Exercise 16.6.9

Calculate line integral $\iint_S (x-y)\,dS$, where S is cylinder $x^2+y^2=1$, $0\le z\le 2$, including the circular top and bottom.

Hint

Break the integral into three separate surface integrals.

Answer

0

Scalar surface integrals have several real-world applications. Recall that scalar line integrals can be used to compute the mass of a wire given its density function. In a similar fashion, we can use scalar surface integrals to compute the mass of a sheet given its density function. If a thin sheet of metal has the shape of surface S and the density of the sheet at point (x,y,z) is $\rho(x,y,z)$ then mass m of the sheet is

$$m = \iint_S \rho(x,y,z)\,dS. \tag{16.6.3}$$

✓ Example 16.6.11: Calculating the Mass of a Sheet

A flat sheet of metal has the shape of surface $z=1+x+2y$ that lies above rectangle $0\le x\le 4$ and $0\le y\le 2$. If the density of the sheet is given by $\rho(x,y,z)=x^2yz$, what is the mass of the sheet?

Solution

Let S be the surface that describes the sheet. Then, the mass of the sheet is given by $m = \iint_S x^2 yx \, dS$. To compute this surface integral, we first need a parameterization of S. Since S is given by the function $f(x, y) = 1 + x + 2y$, a parameterization of S is $\vec{\mathbf{r}}(x, y) = \langle x, y, 1 + x + 2y \rangle$, $0 \le x \le 4$, $0 \le y \le 2$.

The tangent vectors are $\vec{\mathbf{t}}_x = \langle 1, 0, 1 \rangle$ and $\vec{\mathbf{t}}_y = \langle 1, 0, 2 \rangle$. Therefore, $\vec{\mathbf{t}}_x + \vec{\mathbf{t}}_y = \langle -1, -2, 1 \rangle$ and $||\vec{\mathbf{t}}_x \times \vec{\mathbf{t}}_y|| = \sqrt{6}$.

By the definition of the line integral (Section 16.2),

$$\begin{aligned} m &= \iint_S x^2 yz \, dS \\ &= \sqrt{6} \int_0^4 \int_0^2 x^2 y(1 + x + 2y) \, dy \, dx \\ &= \sqrt{6} \int_0^4 \frac{22x^2}{3} + 2x^3 \, dx \\ &= \frac{2560\sqrt{6}}{9} \approx 696.74. \end{aligned}$$

? Exercise 16.6.10

A piece of metal has a shape that is modeled by paraboloid $z = x^2 + y^2$, $0 \le z \le 4$, and the density of the metal is given by $\rho(x, y, z) = z + 1$. Find the mass of the piece of metal.

Hint

The mass of a sheet is given by Equation 16.6.3. A useful parameterization of a paraboloid was given in a previous example.

Answer

$38.401\pi \approx 120.640$

Orientation of a Surface

Recall that when we defined a scalar line integral, we did not need to worry about an orientation of the curve of integration. The same was true for scalar surface integrals: we did not need to worry about an "orientation" of the surface of integration.

On the other hand, when we defined vector line integrals, the curve of integration needed an orientation. That is, we needed the notion of an oriented curve to define a vector line integral without ambiguity. Similarly, when we define a surface integral of a vector field, we need the notion of an oriented surface. An oriented surface is given an "upward" or "downward" orientation or, in the case of surfaces such as a sphere or cylinder, an "outward" or "inward" orientation.

Let S be a smooth surface. For any point (x, y, z) on S, we can identify two unit normal vectors $\vec{\mathbf{N}}$ and $-\vec{\mathbf{N}}$. If it is possible to choose a unit normal vector $\vec{\mathbf{N}}$ at every point (x, y, z) on S so that $\vec{\mathbf{N}}$ varies continuously over S, then S is "*orientable*." Such a choice of unit normal vector at each point gives the **orientation of a surface** S. If you think of the normal field as describing water flow, then the side of the surface that water flows toward is the "negative" side and the side of the surface at which the water flows away is the "positive" side. Informally, a choice of orientation gives S an "outer" side and an "inner" side (or an "upward" side and a "downward" side), just as a choice of orientation of a curve gives the curve "forward" and "backward" directions.

Closed surfaces such as spheres are orientable: if we choose the outward normal vector at each point on the surface of the sphere, then the unit normal vectors vary continuously. This is called the *positive orientation of the closed surface* (Figure 16.6.18). We also could choose the inward normal vector at each point to give an "inward" orientation, which is the negative orientation of the surface.

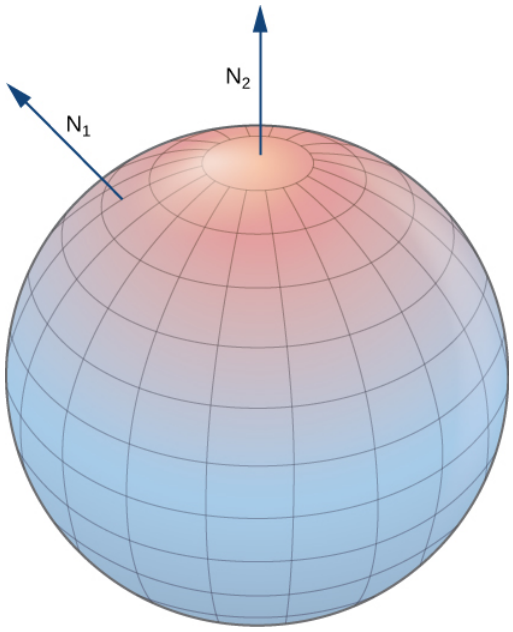


Figure 16.6.18: An oriented sphere with positive orientation.

A portion of the graph of any smooth function $z = f(x, y)$ is also orientable. If we choose the unit normal vector that points "above" the surface at each point, then the unit normal vectors vary continuously over the surface. We could also choose the unit normal vector that points "below" the surface at each point. To get such an orientation, we parameterize the graph of f in the standard way: $\vec{\mathbf{r}}(x, y) = \langle x,\ y,\ f(x, y)\rangle$, where x and y vary over the domain of f. Then, $\vec{\mathbf{t}}_x = \langle 1, 0, f_x\rangle$ and $\vec{\mathbf{t}}_y = \langle 0, 1, f_y\rangle$, and therefore the cross product $\vec{\mathbf{t}}_x \times \vec{\mathbf{t}}_y$ (which is normal to the surface at any point on the surface) is $\langle -f_x,\ -f_y,\ 1\rangle$Since the z-component of this vector is one, the corresponding unit normal vector points "upward," and the upward side of the surface is chosen to be the "positive" side.

Let S be a smooth orientable surface with parameterization $\vec{\mathbf{r}}(u, v)$. For each point $\vec{\mathbf{r}}(a, b)$ on the surface, vectors $\vec{\mathbf{t}}_u$ and $\vec{\mathbf{t}}_v$ lie in the tangent plane at that point. Vector $\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v$ is normal to the tangent plane at $\vec{\mathbf{r}}(a, b)$ and is therefore normal to S at that point. Therefore, the choice of unit normal vector

$$\vec{\mathbf{N}} = \frac{\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v}{||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v||}$$

gives an orientation of surface S.

Example 16.6.12: Choosing an Orientation

Give an orientation of cylinder $x^2 + y^2 = r^2,\ 0 \le z \le h$.

Solution

This surface has parameterization $\vec{\mathbf{r}}(u, v) = \langle r\cos u,\ r\sin u,\ v\rangle,\ 0 \le u < 2\pi,\ 0 \le v \le h$.

The tangent vectors are $\vec{\mathbf{t}}_u = \langle -r\sin u,\ r\cos u,\ 0\rangle$ and $\vec{\mathbf{t}}_v = \langle 0, 0, 1\rangle$. To get an orientation of the surface, we compute the unit normal vector

$$\vec{\mathbf{N}} = \frac{\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v}{||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v||}$$

In this case, $\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \langle r\cos u,\ r\sin u,\ 0\rangle$ and therefore

$$||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v|| = \sqrt{r^2\cos^2 u + r^2\sin^2 u} = r.$$

An orientation of the cylinder is

$$\vec{\mathbf{N}}(u, v) = \frac{\langle r\cos u,\ r\sin u,\ 0\rangle}{r} = \langle \cos u,\ \sin u,\ 0\rangle.$$

Notice that all vectors are parallel to the xy-plane, which should be the case with vectors that are normal to the cylinder. Furthermore, all the vectors point outward, and therefore this is an outward orientation of the cylinder (Figure 16.6.19).



Figure 16.6.19: If all the vectors normal to a cylinder point outward, then this is an outward orientation of the cylinder.

? Exercise 16.6.11

Give the "upward" orientation of the graph of $f(x,y) = xy$.

Hint

Parameterize the surface and use the fact that the surface is the graph of a function.

Answer

$$\vec{\mathbf{N}}(x,y) = \left\langle \frac{-y}{\sqrt{1+x^2+y^2}}, \frac{-x}{\sqrt{1+x^2+y^2}}, \frac{1}{\sqrt{1+x^2+y^2}} \right\rangle$$

Since every curve has a "forward" and "backward" direction (or, in the case of a closed curve, a clockwise and counterclockwise direction), it is possible to give an orientation to any curve. Hence, it is possible to think of every curve as an oriented curve. This is not the case with surfaces, however. Some surfaces cannot be oriented; such surfaces are called *nonorientable*. Essentially, a surface can be oriented if the surface has an "inner" side and an "outer" side, or an "upward" side and a "downward" side. Some surfaces are twisted in such a fashion that there is no well-defined notion of an "inner" or "outer" side.

The classic example of a nonorientable surface is the Möbius strip. To create a Möbius strip, take a rectangular strip of paper, give the piece of paper a half-twist, and the glue the ends together (Figure 16.6.20). Because of the half-twist in the strip, the surface has no "outer" side or "inner" side. If you imagine placing a normal vector at a point on the strip and having the vector travel all the way around the band, then (because of the half-twist) the vector points in the opposite direction when it gets back to its original position. Therefore, the strip really only has one side.

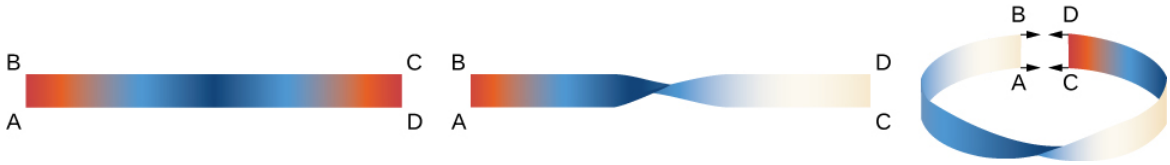


Figure 16.6.20: The construction of a Möbius strip.

Since some surfaces are *nonorientable*, it is not possible to define a vector surface integral on all piecewise smooth surfaces. This is in contrast to vector line integrals, which can be defined on any piecewise smooth curve.

Surface Integral of a Vector Field

With the idea of orientable surfaces in place, we are now ready to define a surface integral of a vector field. The definition is analogous to the definition of the flux of a vector field along a plane curve. Recall that if $\vec{\mathbf{F}}$ is a two-dimensional vector field and C is a plane curve, then the definition of the flux of $\vec{\mathbf{F}}$ along C involved chopping C into small pieces, choosing a point inside each piece, and calculating $\vec{\mathbf{F}} \cdot \vec{\mathbf{N}}$ at the point (where $\vec{\mathbf{N}}$ is the unit normal vector at the point). The definition of a surface integral of a vector field proceeds in the same fashion, except now we chop surface S into small pieces, choose a point in the small (two-dimensional) piece, and calculate $\vec{\mathbf{F}} \cdot \vec{\mathbf{N}}$ at the point.

To place this definition in a real-world setting, let S be an oriented surface with unit normal vector $\vec{\mathbf{N}}$. Let $\vec{\mathbf{v}}$ be a velocity field of a fluid flowing through S, and suppose the fluid has density $\rho(x,y,z)$ Imagine the fluid flows through S, but S is completely

permeable so that it does not impede the fluid flow (Figure 16.6.21). The mass flux of the fluid is the rate of mass flow per unit area. The mass flux is measured in mass per unit time per unit area. How could we calculate the mass flux of the fluid across S?

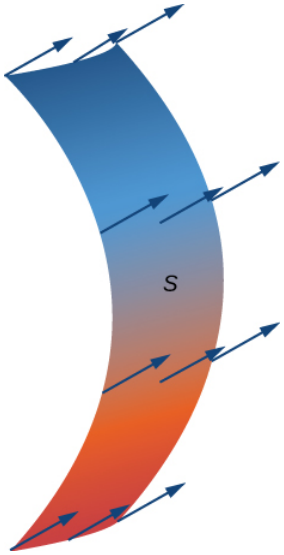


Figure 16.6.21: Fluid flows across a completely permeable surface S.

The rate of flow, measured in mass per unit time per unit area, is $\rho\vec{\mathbf{N}}$. To calculate the mass flux across S, chop S into small pieces S_{ij}. If S_{ij} is small enough, then it can be approximated by a tangent plane at some point P in S_{ij}. Therefore, the unit normal vector at P can be used to approximate $\vec{\mathbf{N}}(x,y,z)$ across the entire piece S_{ij} because the normal vector to a plane does not change as we move across the plane. The component of the vector ρv at P in the direction of $\vec{\mathbf{N}}$ is $\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}}$ at P. Since S_{ij} is small, the dot product $\rho v\cdot N$ changes very little as we vary across S_{ij} and therefore $\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}}$ can be taken as approximately constant across S_{ij}. To approximate the mass of fluid per unit time flowing across S_{ij} (and not just locally at point P), we need to multiply $(\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}})(P)$ by the area of S_{ij}. Therefore, the mass of fluid per unit time flowing across S_{ij} in the direction of $\vec{\mathbf{N}}$ can be approximated by $(\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}})\Delta S_{ij}$ where $\vec{\mathbf{N}}$, ρ and $\vec{\mathbf{v}}$ are all evaluated at P (Figure 16.6.22). This is analogous to the flux of two-dimensional vector field $\vec{\mathbf{F}}$ across plane curve C, in which we approximated flux across a small piece of C with the expression $(\vec{\mathbf{F}}\cdot\vec{\mathbf{N}})\,\Delta s$. To approximate the mass flux across S, form the sum

$$\sum_{i=1}^{m}\sum_{j=1}^{n}(\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}})\Delta S_{ij}.$$

As pieces S_{ij} get smaller, the sum

$$\sum_{i=1}^{m}\sum_{j=1}^{n}(\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}})\Delta S_{ij}$$

gets arbitrarily close to the mass flux. Therefore, the mass flux is

$$\iint_S \rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}}\,dS=\lim_{m,n\to\infty}\sum_{i=1}^{m}\sum_{j=1}^{n}(\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}})\Delta S_{ij}.$$

This is a surface integral of a vector field. Letting the vector field $\rho\vec{\mathbf{v}}$ be an arbitrary vector field $\vec{\mathbf{F}}$ leads to the following definition.

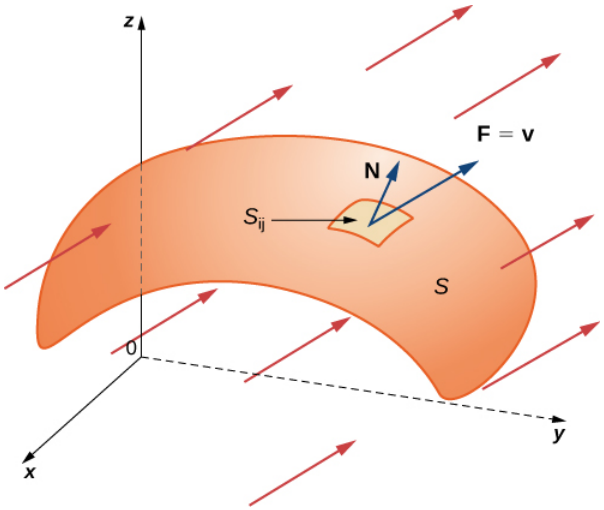


Figure 16.6.22: The mass of fluid per unit time flowing across S_{ij} in the direction of $\vec{\mathbf{N}}$ can be approximated by $(\rho\vec{\mathbf{v}}\cdot\vec{\mathbf{N}})\Delta S_{ij}$.

Definition: Surface Integrals

Let $\vec{\mathbf{F}}$ be a continuous vector field with a domain that contains oriented surface S with unit normal vector $\vec{\mathbf{N}}$. The surface integral of $\vec{\mathbf{F}}$ over S is

$$\iint_S \vec{\mathbf{F}}\cdot d\vec{\mathbf{S}} = \iint_S \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS. \tag{16.6.4}$$

Notice the parallel between this definition and the definition of vector line integral $\int_C \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS$. A surface integral of a vector field is defined in a similar way to a flux line integral across a curve, except the domain of integration is a surface (a two-dimensional object) rather than a curve (a one-dimensional object). Integral $\iint_S \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS$ is called the *flux of* $\vec{\mathbf{F}}$ *across* S, just as integral $\int_C \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS$ is the flux of $\vec{\mathbf{F}}$ across curve C. A surface integral over a vector field is also called a **flux integral**.

Just as with vector line integrals, surface integral $\iint_S \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS$ is easier to compute after surface S has been parameterized. Let $\vec{\mathbf{r}}(u,v)$ be a parameterization of S with parameter domain D. Then, the unit normal vector is given by $\vec{\mathbf{N}} = \dfrac{\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v}{||\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v||}$ and, from Equation 16.6.4, we have

$$\begin{aligned}\int_C \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS &= \iint_S \vec{\mathbf{F}}\cdot\frac{\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v}{||\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v||}\,dS\\ &= \iint_D \left(\vec{\mathbf{F}}(\vec{\mathbf{r}}(u,v))\cdot\frac{\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v}{||\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v||}\right)||\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v||\,dA\\ &= \iint_D (\vec{\mathbf{F}}(\vec{\mathbf{r}}(u,v))\cdot(\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v))\,dA.\end{aligned}$$

Therefore, to compute a surface integral over a vector field we can use the equation

$$\iint_S \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS = \iint_D (\vec{\mathbf{F}}(\vec{\mathbf{r}}(u,v))\cdot(\vec{\mathbf{t}}_u\times\vec{\mathbf{t}}_v))\,dA. \tag{16.6.5}$$

Example 16.6.13: Calculating a Surface Integral

Calculate the surface integral

$$\iint_S \vec{\mathbf{F}}\cdot\vec{\mathbf{N}}\,dS,$$

where $\vec{\mathbf{F}} = \langle -y, x, 0\rangle$ and S is the surface with parameterization

$$\vec{\mathbf{r}}(u,v) = \langle u, v^2 - u, u + v\rangle,\ 0 \le u \le 3,\ 0 \le v \le 4.$$

Solution

The tangent vectors are $\vec{\mathbf{t}}_u = \langle 1, -1, 1\rangle$ and $\vec{\mathbf{t}}_v = \langle 0, 2v, 1\rangle$. Therefore,

$$\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \langle -1 - 2v, -1, 2v\rangle.$$

By Equation 16.6.5,

$$\begin{aligned}
\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} &= \int_0^4 \int_0^3 F(\vec{\mathbf{r}}(u,v)) \cdot (\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v)\, du\, dv \\
&= \int_0^4 \int_0^3 \langle u - v^2,\ u,\ 0\rangle \cdot \langle -1 - 2v,\ -1,\ 2v\rangle\, du\, dv \\
&= \int_0^4 \int_0^3 [(u - v^2)(-1 - 2v) - u]\, du\, dv \\
&= \int_0^4 \int_0^3 (2v^3 + v^2 - 2uv - 2u)\, du\, dv \\
&= \int_0^4 [2v^3 u + v^2 u - v u^2 - u^2]\Big|_0^3\, dv \\
&= \int_0^4 (6v^3 + 3v^2 - 9v - 9)\, dv \\
&= \left[\frac{3v^4}{2} + v^3 - \frac{9v^2}{2} - 9v\right]_0^4 \\
&= 340.
\end{aligned}$$

Therefore, the flux of $\vec{\mathbf{F}}$ across S is 340.

? Exercise 16.6.12

Calculate surface integral

$$\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS,$$

where $\vec{\mathbf{F}} = \langle 0, -z, y\rangle$ and S is the portion of the unit sphere in the first octant with outward orientation.

Hint

Use Equation 16.6.5.

Answer

0

✓ Example 16.6.14: Calculating Mass Flow Rate

Let $\vec{\mathbf{v}}(x,y,z) = \langle 2x,\ 2y,\ z\rangle$ represent a velocity field (with units of meters per second) of a fluid with constant density 80 kg/m^3. Let S be hemisphere $x^2 + y^2 + z^2 = 9$ with $z \le 0$ such that S is oriented outward. Find the mass flow rate of the fluid across S.

Solution

A parameterization of the surface is

$$\vec{\mathbf{r}}(\phi,\theta) = \langle 3\cos\theta\sin\phi,\ 3\sin\theta\sin\phi,\ 3\cos\phi\rangle,\ 0 \le \theta \le 2\pi,\ 0 \le \phi \le \pi/2.$$

As in Example 16.6.7, the tangent vectors are $\vec{\mathbf{t}}_\theta = \langle -3\sin\theta\sin\phi,\ 3\cos\theta\sin\phi,\ 0\rangle$ and $\vec{\mathbf{t}}_\phi = \langle 3\cos\theta\cos\phi,\ 3\sin\theta\cos\phi,\ -3\sin\phi\rangle$, and their cross product is

$$\vec{\mathbf{t}}_\phi \times \vec{\mathbf{t}}_\theta = \langle 9\cos\theta\sin^2\phi,\ 9\sin\theta\sin^2\phi,\ 9\sin\phi\cos\phi\rangle.$$

Notice that each component of the cross product is positive, and therefore this vector gives the outward orientation. Therefore we use the orientation

$\vec{\mathbf{N}} = \langle 9\,\cos\theta\,\sin^2\phi,\ 9\,\sin\theta\,\sin^2\phi,\ 9\,\sin\phi\,\cos\phi\rangle$

for the sphere.

By Equation 16.6.4,

$$\begin{aligned}
\iint_S \rho v\cdot dS &= 80\int_0^{2\pi}\int_0^{\pi/2} v(r(\phi,\theta))\cdot(t_\phi\times t_\theta)\,d\phi\,d\theta\\
&= 80\int_0^{2\pi}\int_0^{\pi/2} \langle 6\,\cos\theta\,\sin\phi,\ 6\,\sin\theta\,\sin\phi,\ 3\,\cos\phi\rangle\\
&\quad\cdot\langle 9\,\cos\theta\,\sin^2\phi,\ 9\,\sin\theta\,\sin^2\phi,\ 9\,\sin\phi\,\cos\phi\rangle\,d\phi\,d\theta\\
&= 80\int_0^{2\pi}\int_0^{\pi/2} 54\,\sin^3\phi + 27\,\cos^2\phi\,\sin\phi\,d\phi\,d\theta\\
&= 80\int_0^{2\pi}\int_0^{\pi/2} 54(1-\cos^2\phi)\,\sin\phi + 27\cos^2\phi\,\sin\phi\,d\phi\,d\theta\\
&= 80\int_0^{2\pi}\int_0^{\pi/2} 54\,\sin\phi - 27\,\cos^2\phi\,\sin\phi\,d\phi\,d\theta\\
&= 80\int_0^{2\pi}\Big[-54\,\cos\phi + 9\,\cos^3\phi\Big]_{\phi=0}^{\phi=2\pi} d\theta\\
&= 80\int_0^{2\pi} 45\,d\theta\\
&= 7200\pi.
\end{aligned}$$

Therefore, the mass flow rate is $7200\pi\,\mathrm{kg/sec/m}^2$.

? Exercise 16.6.13

Let $\vec{\mathbf{v}}(x,y,z) = \langle x^2+y^2,\ z,\ 4y\rangle$ m/sec represent a velocity field of a fluid with constant density 100 kg/m^3. Let S be the half-cylinder $\vec{\mathbf{r}}(u,v) = \langle \cos u,\ \sin u,\ v\rangle$, $0\le u\le\pi$, $0\le v\le 2$ oriented outward. Calculate the mass flux of the fluid across S.

Hint

Use Equation 16.6.4.

Answer

400 kg/sec/m

In Example 16.6.14, we computed the mass flux, which is the rate of mass flow per unit area. If we want to find the flow rate (measured in volume per time) instead, we can use flux integral

$$\iint_S \vec{\mathbf{v}}\cdot\vec{\mathbf{N}}\,dS,$$

which leaves out the density. Since the flow rate of a fluid is measured in volume per unit time, flow rate does not take mass into account. Therefore, we have the following characterization of the flow rate of a fluid with velocity $\vec{\mathbf{v}}$ across a surface S:

$$\text{Flow rate of fluid across S} = \iint_S \vec{\mathbf{v}}\cdot d\vec{\mathbf{S}}.$$

To compute the flow rate of the fluid in Example 16.6.14, we simply remove the density constant, which gives a flow rate of 90π m^3/sec.

Both mass flux and flow rate are important in physics and engineering. Mass flux measures how much mass is flowing across a surface; flow rate measures how much volume of fluid is flowing across a surface.

In addition to modeling fluid flow, surface integrals can be used to model heat flow. Suppose that the temperature at point (x,y,z) in an object is $T(x,y,z)$. Then the heat flow is a vector field proportional to the negative temperature gradient in the object. To be

precise, the heat flow is defined as vector field $F = -k\nabla T$, where the constant k is the *thermal conductivity* of the substance from which the object is made (this constant is determined experimentally). The rate of heat flow across surface S in the object is given by the flux integral

$$\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \iint_S -k\vec{\nabla}T \cdot d\vec{\mathbf{S}}.$$

Example 16.6.15: Calculating Heat Flow

A cast-iron solid cylinder is given by inequalities $x^2 + y^2 \leq 1$, $1 \leq z \leq 4$. The temperature at point (x, y, z) in a region containing the cylinder is $T(x, y, z) = (x^2 + y^2)z$. Given that the thermal conductivity of cast iron is 55, find the heat flow across the boundary of the solid if this boundary is oriented outward.

Solution

Let S denote the boundary of the object. To find the heat flow, we need to calculate flux integral

$$\iint_S -k\vec{\nabla}T \cdot d\vec{\mathbf{S}}.$$

Notice that S is not a smooth surface but is piecewise smooth, since S is the union of three smooth surfaces (the circular top and bottom, and the cylindrical side). Therefore, we calculate three separate integrals, one for each smooth piece of S. Before calculating any integrals, note that the gradient of the temperature is $\vec{\nabla}T = \langle 2xz,\ 2yz,\ x^2 + y^2 \rangle$.

First we consider the circular bottom of the object, which we denote S_1. We can see that S_1 is a circle of radius 1 centered at point $(0, 0, 1)$ sitting in plane $z = 1$. This surface has parameterization $\vec{\mathbf{r}}(u, v) = \langle v\cos u,\ v\sin u,\ 1 \rangle$, $0 \leq u < 2\pi$, $0 \leq v \leq 1$.

Therefore,

$\vec{\mathbf{t}}_u = \langle -v\sin u,\ v\cos u,\ 0 \rangle$ and $\vec{\mathbf{t}}_v = \langle \cos u,\ v\sin u,\ 0 \rangle$, and $\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \langle 0,\ 0,\ -v\sin^2 u - v\cos^2 u \rangle = \langle 0,\ 0,\ -v \rangle$.

Since the surface is oriented outward and S_1 is the bottom of the object, it makes sense that this vector points downward. By Equation 16.6.5 the heat flow across S_1 is

$$\begin{aligned}
\iint_{S_1} -k\vec{\nabla}T \cdot d\vec{\mathbf{S}} &= -55\int_0^{2\pi}\int_0^1 \vec{\nabla}T(u, v) \cdot (\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v)\, dv\, du \\
&= -55\int_0^{2\pi}\int_0^1 \langle 2v\cos u,\ 2v\sin u,\ v^2\cos^2 u + v^2\sin^2 u \rangle \cdot \langle 0, 0, -v \rangle\, dv\, du \\
&= -55\int_0^{2\pi}\int_0^1 \langle 2v\cos u,\ 2v\sin u,\ v^2 \rangle \cdot \langle 0, 0, -v \rangle\, dv\, du \\
&= -55\int_0^{2\pi}\int_0^1 -v^3\, dv\, du \\
&= -55\int_0^{2\pi} -\frac{1}{4}\, du \\
&= \frac{55\pi}{2}.
\end{aligned}$$

Now let's consider the circular top of the object, which we denote S_2. We see that S_2 is a circle of radius 1 centered at point $(0, 0, 4)$, sitting in plane $z = 4$. This surface has parameterization $\vec{\mathbf{r}}(u, v) = \langle v\cos u,\ v\sin u,\ 4 \rangle$, $0 \leq u < 2\pi$, $0 \leq v \leq 1$.

Therefore, $\vec{\mathbf{t}}_u = \langle -v\sin u,\ v\cos u,\ 0 \rangle$ and $\vec{\mathbf{t}}_v = \langle \cos u,\ v\sin u,\ 0 \rangle$, and $\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \langle 0,\ 0,\ -v\sin^2 u - v\cos^2 u \rangle = \langle 0, 0, -v \rangle$.

Since the surface is oriented outward and S_1 is the top of the object, we instead take vector $\vec{\mathbf{t}}_v \times \vec{\mathbf{t}}_u = \langle 0, 0, v \rangle$. By Equation 16.6.5, the heat flow across S_1 is

$$\begin{aligned}\iint_{S_2} -k\vec{\nabla}T \cdot d\vec{\mathbf{S}} &= -55\int_0^{2\pi}\int_0^1 \vec{\nabla}T(u,v)\cdot(\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v)\,dv\,du \\ &= -55\int_0^{2\pi}\int_0^1 \langle 8v\cos u,\ 8v\sin u,\ v^2\cos^2 u + v^2\sin^2 u\rangle \cdot \langle 0,0,-v\rangle\,dv\,du \\ &= -55\int_0^{2\pi}\int_0^1 \langle 8v\cos u,\ 8v\sin u,\ v^2\rangle \cdot \langle 0,0,-v\rangle\ dv\,du \\ &= -55\int_0^{2\pi}\int_0^1 -v^3\,dv\,du = -55\int_0^{2\pi} -\frac{1}{4}\,du = -\frac{55\pi}{2}.\end{aligned}$$

Last, let's consider the cylindrical side of the object. This surface has parameterization $\vec{\mathbf{r}}(u,v) = \langle \cos u,\ \sin u,\ v\rangle$, $0 \le u < 2\pi$, $1 \le v \le 4$. By Example 16.6.9, we know that $\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \langle \cos u,\ \sin u,\ 0\rangle$. By Equation 16.6.5,

$$\begin{aligned}\iint_{S_3} -k\vec{\nabla}T \cdot d\vec{\mathbf{S}} &= -55\int_0^{2\pi}\int_1^4 \vec{\nabla}T(u,v)\cdot(\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v)\,dv\,du \\ &= -55\int_0^{2\pi}\int_1^4 \langle 2v\cos u,\ 2v\sin u,\ \cos^2 u + \sin^2 u\rangle \cdot \langle \cos u,\ \sin u,\ 0\rangle\,dv\,du \\ &= -55\int_0^{2\pi}\int_0^1 \langle 2v\cos^2 u,\ 2v\sin u,\ 1\rangle \cdot \langle \cos u,\ \sin u,\ 0\rangle\,dv\ du \\ &= -55\int_0^{2\pi}\int_0^1 (2v\cos^2 u + 2v\sin^2 u)\,dv\,du \\ &= -55\int_0^{2\pi}\int_0^1 2v\,dv\,du \\ &= -55\int_0^{2\pi} du \\ &= -110\pi.\end{aligned}$$

Therefore, the rate of heat flow across S is

$$\frac{55\pi}{2} - \frac{55\pi}{2} - 110\pi = -110\pi.$$

? Exercise 16.6.14

A cast-iron solid ball is given by inequality $x^2 + y^2 + z^2 \le 1$. The temperature at a point in a region containing the ball is $T(x,y,z) = \frac{1}{3}(x^2+y^2+z^2)$. Find the heat flow across the boundary of the solid if this boundary is oriented outward.

Hint

Follow the steps of Example 16.6.15.

Answer

$$-\frac{440\pi}{3}$$

Key Concepts

- Surfaces can be parameterized, just as curves can be parameterized. In general, surfaces must be parameterized with two parameters.
- Surfaces can sometimes be oriented, just as curves can be oriented. Some surfaces, such as a Möbius strip, cannot be oriented.
- A surface integral is like a line integral in one higher dimension. The domain of integration of a surface integral is a surface in a plane or space, rather than a curve in a plane or space.
  - The integrand of a surface integral can be a scalar function or a vector field. To calculate a surface integral with an integrand that is a function, use Equation 16.6.2. To calculate a surface integral with an integrand that is a vector field, use Equation 16.6.5.

- If S is a surface, then the area of S is

$$\iint_S dS.$$

Key Equations

- **Scalar surface integral**

$$\iint_S f(x,y,z)\,dS = \iint_D f(\vec{\mathbf{r}}(u,v))||\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v||\,dA$$

- **Flux integral**

$$\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\,dS = \iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \iint_D \vec{\mathbf{F}}(\vec{\mathbf{r}}(u,v)) \cdot (\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v)\,dA$$

Glossary

flux integral
another name for a surface integral of a vector field; the preferred term in physics and engineering

grid curves
curves on a surface that are parallel to grid lines in a coordinate plane

heat flow
a vector field proportional to the negative temperature gradient in an object

mass flux
the rate of mass flow of a fluid per unit area, measured in mass per unit time per unit area

orientation of a surface
if a surface has an "inner" side and an "outer" side, then an orientation is a choice of the inner or the outer side; the surface could also have "upward" and "downward" orientations

parameter domain (parameter space)
the region of the uv-plane over which the parameters u and v vary for parameterization $\vec{\mathbf{r}}(u,v) = \langle x(u,v),\, y(u,v),\, z(u,v)\rangle$

parameterized surface (parametric surface)
a surface given by a description of the form $\vec{\mathbf{r}}(u,v) = \langle x(u,v),\, y(u,v),\, z(u,v)\rangle$, where the parameters u and v vary over a parameter domain in the uv-plane

regular parameterization
parameterization $\vec{\mathbf{r}}(u,v) = \langle x(u,v),\, y(u,v),\, z(u,v)\rangle$ such that $r_u \times r_v$ is not zero for point (u,v) in the parameter domain

surface area
the area of surface S given by the surface integral

$$\iint_S dS$$

surface integral
an integral of a function over a surface

surface integral of a scalar-valued function
a surface integral in which the integrand is a scalar function

surface integral of a vector field

a surface integral in which the integrand is a vector field

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16.6E: Exercises for Section 16.6

In exercises 1 - 4, determine whether the statements are *true or false*.

1. If surface S is given by $\{(x, y, z) : 0 \le x \le 1, 0 \le y \le 1, z = 10\}$, then $\iint_S f(x, y, z)\, dS = \int_0^1 \int_0^1 f(x, y, 10)\, dx\, dy.$

Answer

True

2. If surface S is given by $\{(x, y, z) : 0 \le x \le 1, 0 \le y \le 1, z = x\}$, then $\iint_S f(x, y, z)\, dS = \int_0^1 \int_0^1 f(x, y, x)\, dx\, dy.$

3. Surface $\vec{\mathbf{r}} = \langle v \cos u,\ v \sin u,\ v^2 \rangle$, for $0 \le u \le \pi$, $0 \le v \le 2$ is the same surface $\vec{\mathbf{r}} = \langle \sqrt{v} \cos 2u,\ \sqrt{v} \sin 2u,\ v \rangle$, for $0 \le u \le \dfrac{\pi}{2}$, $0 \le v \le 4$.

Answer

True

4. Given the standard parameterization of a sphere, normal vectors $\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v$ are outward normal vectors.

In exercises 5 - 10, find parametric descriptions for the following surfaces.

5. Plane $3x - 2y + z = 2$

Answer

$\vec{\mathbf{r}}(u, v) = \langle u,\ v,\ 2 - 3u + 2v \rangle$ for $-\infty \le u < \infty$ and $-\infty \le v < \infty$.

6. Paraboloid $z = x^2 + y^2$, for $0 \le z \le 9$.

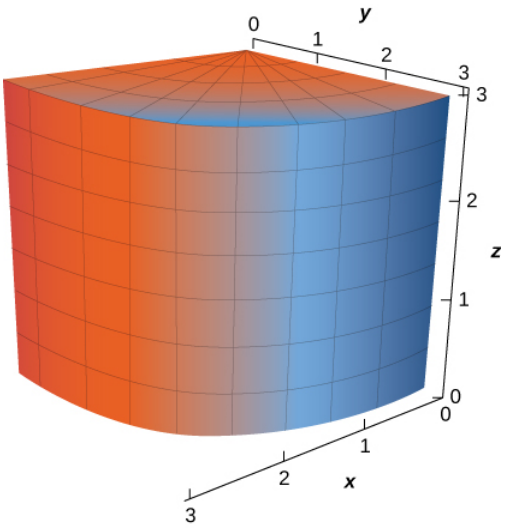
7. Plane $2x - 4y + 3z = 16$

Answer

$\vec{\mathbf{r}}(u, v) = \langle u,\ v,\ \dfrac{1}{3}(16 - 2u + 4v) \rangle$ for $|u| < \infty$ and $|v| < \infty$.

8. The frustum of cone $z^2 = x^2 + y^2$, for $2 \le z \le 8$

9. The portion of cylinder $x^2 + y^2 = 9$ in the first octant, for $0 \le z \le 3$



Answer

$\vec{\mathbf{r}}(u, v) = \langle 3 \cos u,\ 3 \sin u,\ v \rangle$ for $0 \le u \le \dfrac{\pi}{2}$, $0 \le v \le 3$

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10. A cone with base radius r and height h, where r and h are positive constants.

For exercises 11 - 12, use a computer algebra system to approximate the area of the following surfaces using a parametric description of the surface.

11. [T] Half cylinder $\{(r, \theta, z) : r = 4,\ 0 \le \theta \le \pi,\ 0 \le z \le 7\}$

Answer

$A = 87.9646$

12. [T] Plane $z = 10 - z - y$ above square $|x| \le 2,\ |y| \le 2$

In exercises 13 - 15, let S be the hemisphere $x^2 + y^2 + z^2 = 4$, with $z \ge 0$, and evaluate each surface integral, in the counterclockwise direction.

13. $\iint_S z\, dS$

Answer

$\iint_S z\, dS = 8\pi$

14. $\iint_S (x - 2y)\, dS$

15. $\iint_S (x^2 + y^2)\, dS$

Answer

$\iint_S (x^2 + y^2)\, dS = 16\pi$

In exercises 16 - 18, evaluate $\int\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS$ for vector field $\vec{\mathbf{F}}$ where $\vec{\mathbf{N}}$ is an outward normal vector to surface S.

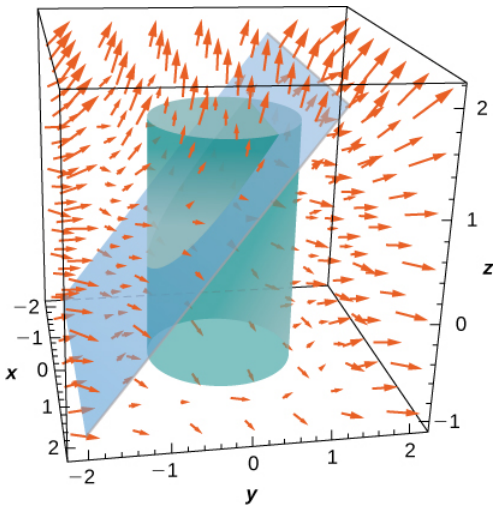
16. $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + 2y\,\hat{\mathbf{j}} = 3z\,\hat{\mathbf{k}}$, and S is that part of plane $15x - 12y + 3z = 6$ that lies above unit square $0 \le x \le 1,\ 0 \le y \le 1$.

17. $\vec{\mathbf{F}}(x, y) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}}$, and S is hemisphere $z = \sqrt{1 - x^2 - y^2}$.

Answer

$\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS = \frac{4\pi}{3}$

18. $\vec{\mathbf{F}}(x, y, z) = x^2\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}} + z^2\,\hat{\mathbf{k}}$, and S is the portion of plane $z = y + 1$ that lies inside cylinder $x^2 + y^2 = 1$.



In exercises 19 - 20, approximate the mass of the homogeneous lamina that has the shape of given surface S. Round to four decimal places.

19. [T] S is surface $z = 4 - x - 2y$, with $z \geq 0,\ x \geq 0,\ y \geq 0;\ \xi = x$.

Answer

$m \approx 13.0639$

20. [T] S is surface $z = x^2 + y^2$, with $z \leq 1;\ \xi = z$.

21. [T] S is surface $x^2 + y^2 + x^2 = 5$, with $z \geq 1;\ \xi = \theta^2$.

Answer

$m \approx 228.5313$

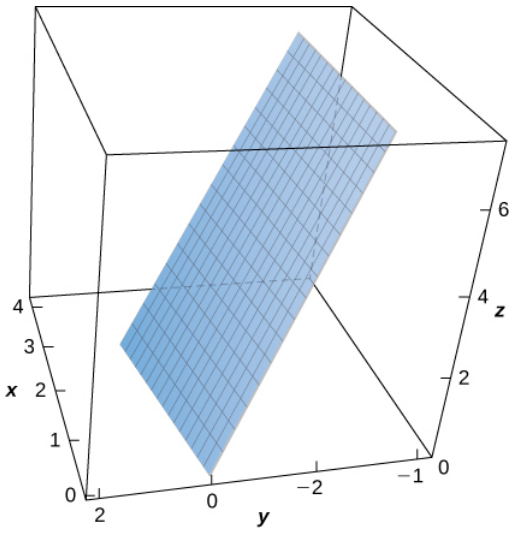
22. Evaluate $\iint_S (y^2 z\,\hat{\mathbf{i}} + y^3\,\hat{\mathbf{j}} + xz\hat{\mathbf{k}}) \cdot dS$, where S is the surface of cube $-1 \leq x \leq 1,\ -1 \leq y \leq 1$, and $0 \leq z \leq 2$ in a counterclockwise direction.

23. Evaluate surface integral $\iint_S g\,dS$, where $g(x, y, z) = xz + 2x^2 - 3xy$ and S is the portion of plane $2x - 3y + z = 6$ that lies over unit square $R : 0 \leq x \leq 1,\ 0 \leq y \leq 1$.

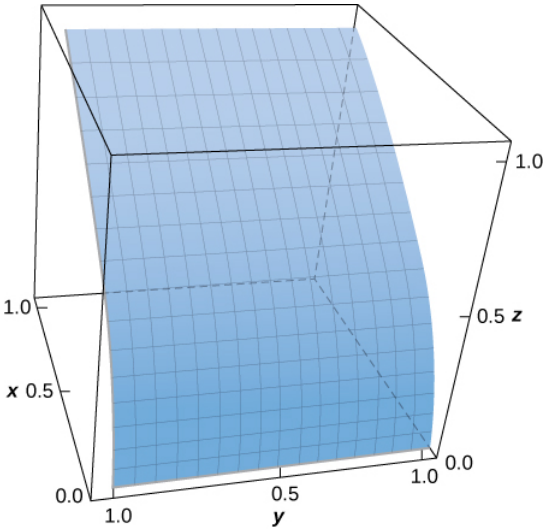
Answer

$\iint_S g\,dS = 3\sqrt{4}$

24. Evaluate $\iint_S (x + y + z)\,dS$, where S is the surface defined parametrically by $\vec{\mathbf{R}}(u, v) = (2u + v)\,\hat{\mathbf{i}} + (u - 2v)\,\hat{\mathbf{j}} + (u + 3v)\,\hat{\mathbf{k}}$ for $0 \leq u \leq 1$, and $0 \leq v \leq 2$.



25. [T] Evaluate $\iint_S (x - y^2 + z)\,dS$, where S is the surface defined parametrically by $\vec{\mathbf{R}}(u, v) = u^2\,\hat{\mathbf{i}} + v\,\hat{\mathbf{j}} + u\,\hat{\mathbf{k}}$ for $0 \leq u \leq 1,\ 0 \leq v \leq 1$.



Answer

$$\iint_S (x^2 + y - z)\, dS \approx 0.9617$$

26. [T] Evaluate where S is the surface defined by $\vec{\mathbf{R}}(u, v) = u\,\hat{\mathbf{i}} - u^2\,\hat{\mathbf{j}} + v\hat{\mathbf{k}}$, $0 \le u \le 2$, $0 \le v \le 1$ for $0 \le u \le 1$, $0 \le v \le 2$.

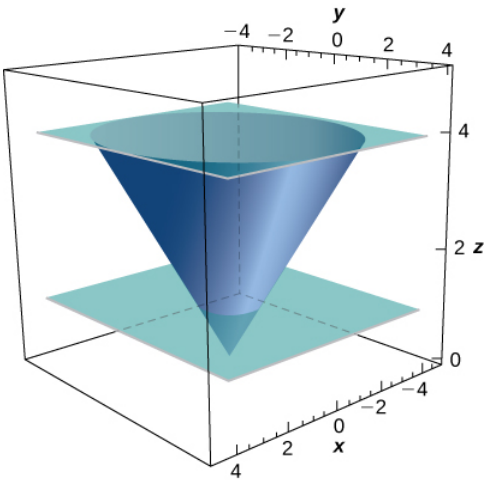
27. Evaluate $\iint_S (x^2 + y^2)\, dS$, where S is the surface bounded above hemisphere $z = \sqrt{1 - x^2 - y^2}$, and below by plane $z = 0$.

Answer

$$\iint_S (x^2 + y^2)\, dS = \frac{4\pi}{3}$$

28. Evaluate $\iint_S (x^2 + y^2 + z^2)\, dS$, where S is the portion of plane that lies inside cylinder $x^2 + y^2 = 1$.

29. [T] Evaluate $\iint_S x^2 z\, dS$, where S is the portion of cone $z^2 = x^2 + y^2$ that lies between planes $z = 1$ and $z = 4$.



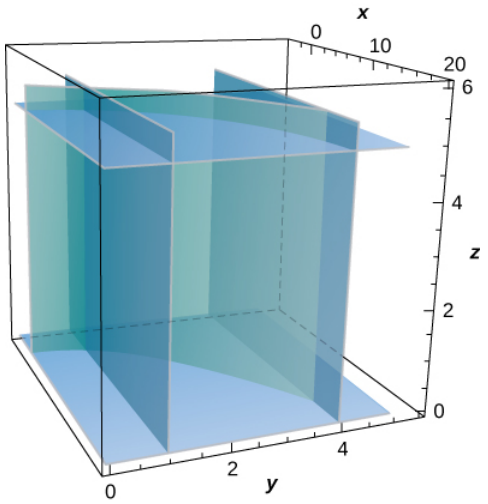
Answer

$$\operatorname{div} \vec{\mathbf{F}} = a + b$$

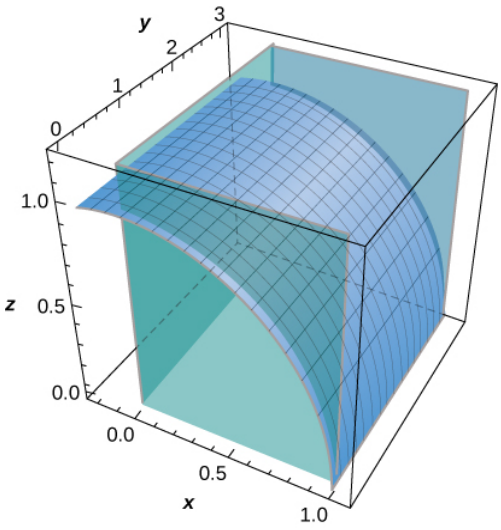
$$\iint_S x^2 z dS = \frac{1023\sqrt{2\pi}}{5}$$

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30. [T] Evaluate $\iint_S \frac{xz}{y}\,dS$, where S is the portion of cylinder $x = y^2$ that lies in the first octant between planes $z = 0,\ z = 5$, and $y = 4$.



31. [T] Evaluate $\iint_S (z + y)\,dS$, where S is the part of the graph of $z = \sqrt{1 - x^2}$ in the first octant between the xy-plane and plane $y = 3$.



Answer

$\iint_S (z + y)\,dS \approx 10.1$

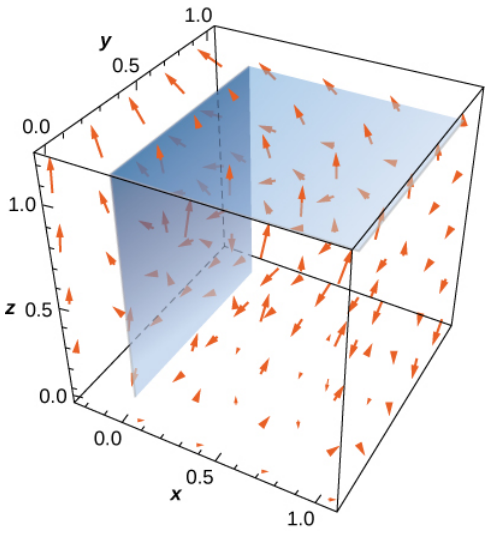
32. Evaluate $\iint_S xyz\,dS$ if S is the part of plane $z = x + y$ that lies over the triangular region in the xy-plane with vertices (0, 0, 0), (1, 0, 0), and (0, 2, 0).

33. Find the mass of a lamina of density $\xi(x, y, z) = z$ in the shape of hemisphere $z = (a^2 - x^2 - y^2)^{1/2}$.

Answer

$m = \pi a^3$

34. Compute $\int\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\,dS$, where $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} - 5y\,\hat{\mathbf{j}} + 4z\,\hat{\mathbf{k}}$ and $\vec{\mathbf{N}}$ is an outward normal vector S, where S is the union of two squares $S_1 : x = 0,\ 0 \le y \le 1,\ 0 \le z \le 1$ and $S_2 : x = 0,\ 0 \le x \le 1,\ 0 \le y \le 1$.



35. Compute $\int\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS$, where $\vec{\mathbf{F}}(x,y,z) = xy\,\hat{\mathbf{i}} + z\,\hat{\mathbf{j}} + (x+y)\,\hat{\mathbf{k}}$ and $\vec{\mathbf{N}}$ is an outward normal vector S, where S is the triangular region cut off from plane $x+y+z=1$ by the positive coordinate axes.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS = \frac{13}{24}$$

36. Compute $\int\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS$, where $\vec{\mathbf{F}}(x,y,z) = 2yz\,\hat{\mathbf{i}} + (\tan^{-1} xz)\,\hat{\mathbf{j}} + e^{xy}\,\hat{\mathbf{k}}$ and $\vec{\mathbf{N}}$ is an outward normal vector S, where S is the surface of sphere $x^2+y^2+z^2=1$.

37. Compute $\int\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS$, where $\vec{\mathbf{F}}(x,y,z) = xyz\,\hat{\mathbf{i}} + xyz\,\hat{\mathbf{j}} + xyz\,\hat{\mathbf{k}}$ and $\vec{\mathbf{N}}$ is an outward normal vector S, where S is the surface of the five faces of the unit cube $0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le z \le 1$ missing $z=0$.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS = \frac{3}{4}$$

For exercises 38 - 39, express the surface integral as an iterated double integral by using a projection on S on the yz-plane.

38. $\iint_S xy^2z^3\, dS$; S is the first-octant portion of plane $2x+3y+4z=12$.

39. $\iint_S (x^2-2y+z)\, dS$; S is the portion of the graph of $4x+y=8$ bounded by the coordinate planes and plane $z=6$.

Answer

$$\int_0^8 \int_0^6 \left(4 - 3y + \frac{1}{16}y^2 + z\right)\left(\frac{1}{4}\sqrt{17}\right)\, dz\, dy$$

For exercises 40 - 41, express the surface integral as an iterated double integral by using a projection on S on the xz-plane.

40. $\iint_S xy^2z^3\, dS$; S is the first-octant portion of plane $2x+3y+4z=12$.

41. $\iint_S (x^2-2y+z)\, dS$; is the portion of the graph of $4x+y=8$ bounded by the coordinate planes and plane $z=6$.

Answer

$$\int_0^2 \int_0^6 \left[x^2 - 2(8-4x) + z\right]\sqrt{17}\, dz\, dx$$

42. Evaluate surface integral $\iint_S yz\, dS$, where S is the first-octant part of plane $x+y+z=\lambda$, where λ is a positive constant.

43. Evaluate surface integral $\iint_S (x^2z + y^2z)\, dS$, where S is hemisphere $x^2 + y^2 + z^2 = a^2,\ z \geq 0$.

Answer

$$\iint_S (x^2z + y^2z)\, dS = \frac{\pi a^5}{2}$$

44. Evaluate surface integral $\iint_S z\, dA$, where S is surface $z = \sqrt{x^2 + y^2},\ 0 \leq z \leq 2$.

45. Evaluate surface integral $\iint_S x^2yz\, dS$, where S is the part of plane $z = 1 + 2x + 3y$ that lies above rectangle $0 \leq x \leq 3$ and $0 \leq y \leq 2$.

Answer

$$\iint_S x^2yz\, dS = 171\sqrt{14}$$

46. Evaluate surface integral $\iint_S yz\, dS$, where S is plane $x + y + z = 1$ that lies in the first octant.

47. Evaluate surface integral $\iint_S yz\, dS$, where S is the part of plane $z = y + 3$ that lies inside cylinder $x^2 + y^2 = 1$.

Answer

$$\iint_S yz\, dS = \frac{\sqrt{2}\pi}{4}$$

For exercises 48 - 50, use geometric reasoning to evaluate the given surface integrals.

48. $\iint_S \sqrt{x^2 + y^2 + z^2}\, dS$, where S is surface $x^2 + y^2 + z^2 = 4,\ z \geq 0$

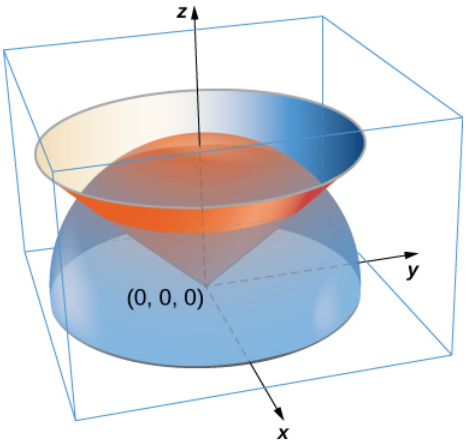
49. $\iint_S (x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}})\cdot dS$, where S is surface $x^2 + y^2 = 4,\ 1 \leq z \leq 3$, oriented with unit normal vectors pointing outward

Answer

$$\iint_S (x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}})\cdot dS = 16\pi$$

50. $\iint_S (z\,\hat{\mathbf{k}})\cdot dS$, where S is disc $x^2 + y^2 \leq 9$ on plane $z = 4$ oriented with unit normal vectors pointing upward

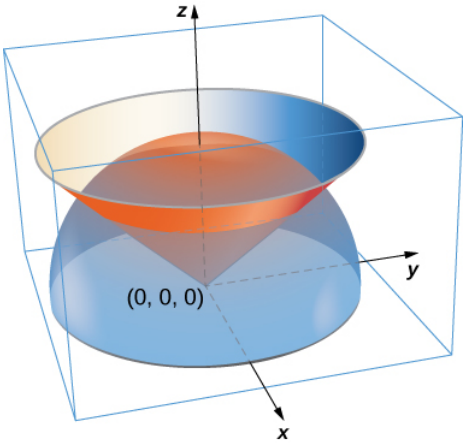
51. A lamina has the shape of a portion of sphere $x^2 + y^2 + z^2 = a^2$ that lies within cone $z = \sqrt{x^2 + y^2}$. Let S be the spherical shell centered at the origin with radius a, and let C be the right circular cone with a vertex at the origin and an axis of symmetry that coincides with the z-axis. Determine the mass of the lamina if $\delta(x, y, z) = x^2y^2z$.



Answer

$$m = \frac{\pi a^7}{192}$$

52. A lamina has the shape of a portion of sphere $x^2 + y^2 + z^2 = a^2$ that lies within cone $z = \sqrt{x^2 + y^2}$. Let S be the spherical shell centered at the origin with radius a, and let C be the right circular cone with a vertex at the origin and an axis of symmetry that coincides with the z-axis. Suppose the vertex angle of the cone is ϕ_0, with $0 \leq \phi_0 < \frac{\pi}{2}$. Determine the mass of that portion of the shape enclosed in the intersection of S and C. Assume $\delta(x, y, z) = x^2 y^2 z$.



53. A paper cup has the shape of an inverted right circular cone of height 6 in. and radius of top 3 in. If the cup is full of water weighing $62.5\, lb/ft^3$, find the magnitude of the total force exerted by the water on the inside surface of the cup.

Answer

$F \approx 4.57\, lb$

For exercises 54 - 55, the heat flow vector field for conducting objects is $\vec{\mathbf{F}} = -k\vec{\nabla}T$, where $T(x, y, z)$ is the temperature in the object and $k > 0$ is a constant that depends on the material. Find the outward flux of $\vec{\mathbf{F}}$ across the following surfaces S for the given temperature distributions and assume $k = 1$.

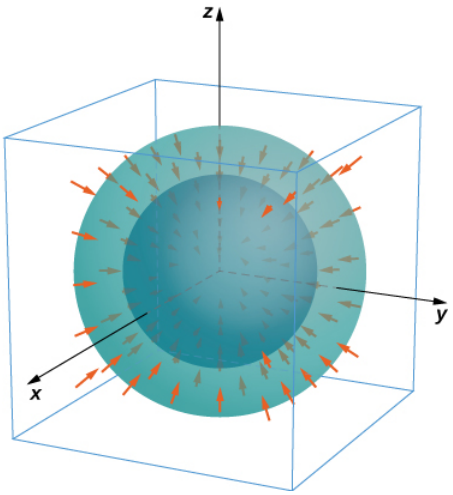
54. $T(x, y, z) = 100e^{-x-y}$; S consists of the faces of cube $|x| \leq 1,\ |y| \leq 1,\ |z| \leq 1$.

55. $T(x, y, z) = -\ln(x^2 + y^2 + z^2)$; S is sphere $x^2 + y^2 + z^2 = a^2$.

Answer

$8\pi a$

For exercises 56 - 57, consider the radial fields $\vec{\mathbf{F}} = \dfrac{\langle x, y, z\rangle}{(x^2 + y^2 + z^2)^{p/2}} = \dfrac{r}{|r|^p}$, where p is a real number. Let S consist of spheres A and B centered at the origin with radii $0 < a < b$. The total outward flux across S consists of the outward flux across the outer sphere B less the flux into S across inner sphere A.



56. Find the total flux across S with $p = 0$.

57. Show that for $p = 3$ the flux across S is independent of a and b.

Answer

The net flux is zero.

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16.7: Stokes' Theorem

Learning Objectives

- Explain the meaning of Stokes' theorem.
- Use Stokes' theorem to evaluate a line integral.
- Use Stokes' theorem to calculate a surface integral.
- Use Stokes' theorem to calculate a curl.

In this section, we study Stokes' theorem, a higher-dimensional generalization of Green's theorem. This theorem, like the Fundamental Theorem for Line Integrals and Green's theorem, is a generalization of the Fundamental Theorem of Calculus to higher dimensions. Stokes' theorem relates a vector surface integral over surface S in space to a line integral around the boundary of S. Therefore, just as the theorems before it, Stokes' theorem can be used to reduce an integral over a geometric object S to an integral over the boundary of S. In addition to allowing us to translate between line integrals and surface integrals, Stokes' theorem connects the concepts of curl and circulation. Furthermore, the theorem has applications in fluid mechanics and electromagnetism. We use Stokes' theorem to derive Faraday's law, an important result involving electric fields.

Stokes' Theorem

Stokes' theorem says we can calculate the flux of $curl\,\vec{\mathbf{F}}$ across surface S by knowing information only about the values of $\vec{\mathbf{F}}$ along the boundary of S. Conversely, we can calculate the line integral of vector field $\vec{\mathbf{F}}$ along the boundary of surface S by translating to a double integral of the curl of $\vec{\mathbf{F}}$ over S.

Let S be an oriented smooth surface with unit normal vector $\vec{\mathbf{N}}$. Furthermore, suppose the boundary of S is a simple closed curve C. The orientation of S induces the positive orientation of C if, as you walk in the positive direction around C with your head pointing in the direction of $\vec{\mathbf{N}}$, the surface is always on your left. With this definition in place, we can state *Stokes' theorem*.

Theorem 16.7.1: Stokes' Theorem

Let S be a piecewise smooth oriented surface with a boundary that is a simple closed curve C with positive orientation (Figure 16.7.1). If $\vec{\mathbf{F}}$ is a vector field with component functions that have continuous partial derivatives on an open region containing S, then

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_S curl\,\vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}. \tag{16.7.1}$$

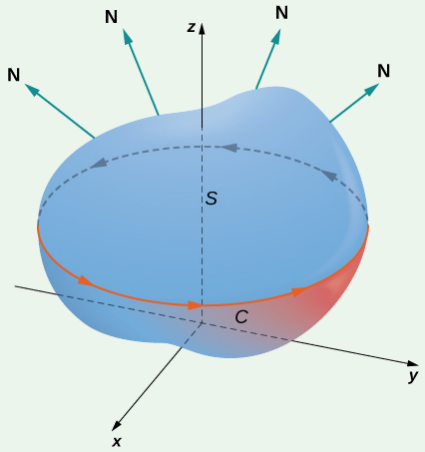


Figure 16.7.1: Stokes' theorem relates the flux integral over the surface to a line integral around the boundary of the surface. Note that the orientation of the curve is positive.

Suppose surface S is a flat region in the xy-plane with upward orientation. Then the unit normal vector is $\vec{\mathbf{k}}$ and surface integral

$$\iint_S curl\,\vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

is actually the double integral

$$\iint_S curl\,\vec{\mathbf{F}} \cdot \vec{\mathbf{k}}\, dA.$$

In this special case, Stokes' theorem gives

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_S curl\,\vec{\mathbf{F}} \cdot \vec{\mathbf{k}}\, dA.$$

However, this is the flux form of Green's theorem, which shows us that Green's theorem is a special case of Stokes' theorem. Green's theorem can only handle surfaces in a plane, but Stokes' theorem can handle surfaces in a plane or in space.

The complete proof of Stokes' theorem is beyond the scope of this text. We look at an intuitive explanation for the truth of the theorem and then see proof of the theorem in the special case that surface S is a portion of a graph of a function, and S, the boundary of S, and $\vec{\mathbf{F}}$ are all fairly tame.

Proof

First, we look at an informal proof of the theorem. This proof is not rigorous, but it is meant to give a general feeling for why the theorem is true. Let S be a surface and let D be a small piece of the surface so that D does not share any points with the boundary of S. We choose D to be small enough so that it can be approximated by an oriented square E. Let D inherit its orientation from S, and give E the same orientation. This square has four sides; denote them E_l, E_r, E_u, and E_d for the left, right, up, and down sides, respectively. On the square, we can use the flux form of Green's theorem:

$$\int_{E_l+E_d+E_r+E_u} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_E curl\,\vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, d\vec{\mathbf{S}} = \iint_E curl\,\vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}.$$

To approximate the flux over the entire surface, we add the values of the flux on the small squares approximating small pieces of the surface (Figure 16.7.2).

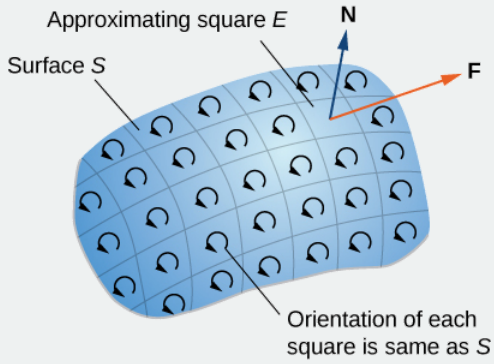


Figure 16.7.2: Chop the surface into small pieces. The pieces should be small enough that they can be approximated by a square.

By Green's theorem, the flux across each approximating square is a line integral over its boundary. Let F be an approximating square with an orientation inherited from S and with a right side E_l (so F is to the left of E). Let F_r denote the right side of F; then, $E_l = -F_r$. In other words, the right side of F is the same curve as the left side of E, just oriented in the opposite direction. Therefore,

$$\int_{E_l} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = -\int_{F_r} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

As we add up all the fluxes over all the squares approximating surface S, line integrals

$$\int_{E_l} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

and

$$\int_{F_r} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

cancel each other out. The same goes for the line integrals over the other three sides of E. These three line integrals cancel out with the line integral of the lower side of the square above E, the line integral over the left side of the square to the right of E, and the line integral over the upper side of the square below E (Figure 16.7.3). After all this cancelation occurs over all the approximating squares, the only line integrals that survive are the line integrals over sides approximating the boundary of S. Therefore, the sum of all the fluxes (which, by Green's theorem, is the sum of all the line integrals around the boundaries of approximating squares) can be approximated by a line integral over the boundary of S. In the limit, as the areas of the approximating squares go to zero, this approximation gets arbitrarily close to the flux.

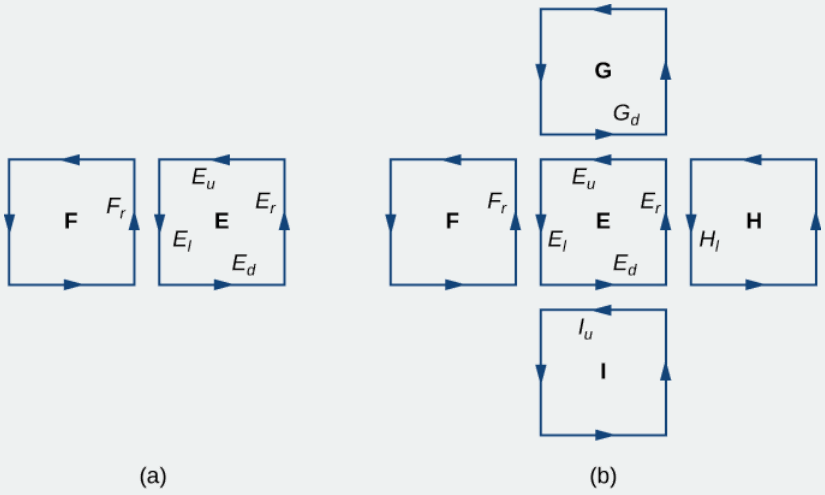


Figure 16.7.3: (a) The line integral along E_l cancels out the line integral along F_r because $E_l = -F_r$. (b) The line integral along any of the sides of E cancels out with the line integral along a side of an adjacent approximating square.

Let's now look at a rigorous proof of the theorem in the special case that S is the graph of function $z = f(x, y)$, where x and y vary over a bounded, simply connected region D of finite area (Figure 16.7.4). Furthermore, assume that f has continuous second-order partial derivatives. Let C denote the boundary of S and let C' denote the boundary of D. Then, D is the "shadow" of S in the plane and C' is the "shadow" of C. Suppose that S is oriented upward. The counterclockwise orientation of C is positive, as is the counterclockwise orientation of C'. Let $\vec{\mathbf{F}}(x, y, z) = \langle P, Q, R \rangle$ be a vector field with component functions that have continuous partial derivatives.

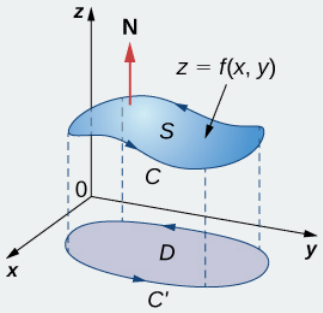


Figure 16.7.4: D is the "shadow," or projection, of S in the plane and C' is the projection of C.

We take the standard parameterization of $S: x = x,\ y = y,\ z = g(x, y)$. The tangent vectors are $\vec{\mathbf{t}}_x = \langle 1, 0, g_x \rangle$ and $\vec{\mathbf{t}}_y = \langle 0, 1, g_y \rangle$, and therefore $\vec{\mathbf{t}}_x \times \vec{\mathbf{t}}_y = \langle -g_x, -g_y, 1 \rangle$.

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \iint_D [-(R_y - Q_z)z_x - (P_z - R_x)z_y + (Q_x - P_y)]\, dA,$$

where the partial derivatives are all evaluated at $(x, y, g(x, y))$, making the integrand depend on x and y only. Suppose $\langle x(t),\ y(t) \rangle,\ a \le t \le b$ is a parameterization of C'. Then, a parameterization of C is $\langle x(t),\ y(t),\ g(x(t),\ y(t)) \rangle,\ a \le t \le b$. Armed with

these parameterizations, the Chain rule, and Green's theorem, and keeping in mind that P, Q and R are all functions of x and y, we can evaluate line integral

$$\begin{aligned}\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_a^b (Px'(t) + Qy'(t) + Rz'(t))\, dt \\ &= \int_a^b \left[Px'(t) + Qy'(t) + R\left(\frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}\right)\right] dt \\ &= \int_a^b \left[\left(P + R\frac{\partial z}{\partial x}\right)x'(t) + \left(Q + R\frac{\partial z}{\partial y}\right)y'(t)\right] dt \\ &= \int_{C'} \left(P + R\frac{\partial z}{\partial x}\right) dx + \left(Q + R\frac{\partial z}{\partial y}\right) dy \\ &= \iint_D \left[\frac{\partial}{\partial x}\left(Q + R\frac{\partial z}{\partial y}\right) - \frac{\partial}{\partial y}\left(P + R\frac{\partial z}{\partial x}\right)\right] dA \\ &= \iint_D \left(\frac{\partial Q}{\partial x} + \frac{\partial Q}{\partial z}\frac{\partial z}{\partial x} + \frac{\partial R}{\partial x}\frac{\partial z}{\partial y} + \frac{\partial R}{\partial z}\frac{\partial z}{\partial x}\frac{\partial z}{\partial y} + R\frac{\partial^2 z}{\partial x \partial y}\right) \\ &\quad - \left(\frac{\partial P}{\partial y} + \frac{\partial P}{\partial z}\frac{\partial z}{\partial y} + \frac{\partial R}{\partial z}\frac{\partial z}{\partial y}\frac{\partial z}{\partial x} + R\frac{\partial^2 z}{\partial y \partial x}\right)\end{aligned}$$

By Clairaut's theorem,

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x}$$

Therefore, four of the terms disappear from this double integral, and we are left with

$$\iint_D [-(R_y - Q_z)Z_x - (P_z - R_x)z_y + (Q_x - P_y)]\, dA,$$

which equals

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}.$$

□

We have shown that Stokes' theorem is true in the case of a function with a domain that is a simply connected region of finite area. We can quickly confirm this theorem for another important case: when vector field $\vec{\mathbf{F}}$ is a conservative field. If $\vec{\mathbf{F}}$ is conservative, the curl of $\vec{\mathbf{F}}$ is zero, so

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = 0.$$

Since the boundary of S is a closed curve, the integral

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

is also zero.

Example 16.7.1: Verifying Stokes' Theorem for a Specific Case

Verify that Stokes' theorem is true for vector field $\vec{\mathbf{F}}(x, y) = \langle -z, x, 0\rangle$ and surface S, where S is the hemisphere, oriented outward, with parameterization $\vec{\mathbf{r}}(\phi, \theta) = \langle \sin\phi\, \cos\theta,\, \sin\phi\, \sin\theta,\, \cos\phi\rangle$, $0 \le \theta \le \pi$, $0 \le \phi \le \pi$ as shown in Figure 16.7.5.

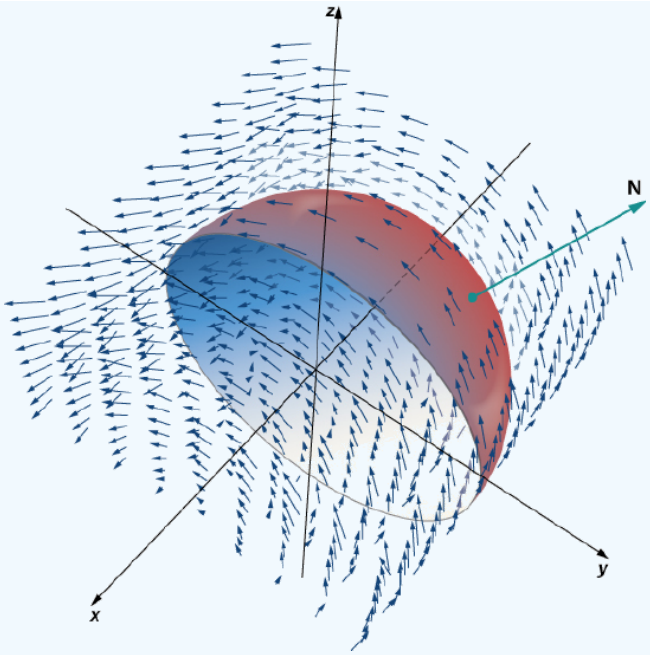


Figure 16.7.5: Verifying Stokes' theorem for a hemisphere in a vector field.

Solution

Let C be the boundary of S. Note that C is a circle of radius 1, centered at the origin, sitting in plane $y = 0$. This circle has parameterization $\langle \cos t,\ 0,\ \sin t \rangle$, $0 \le t \le 2\pi$. the equation for scalar surface integrals

$$\begin{aligned} \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \int_0^{2\pi} \langle -\sin t,\ \cos t,\ 0 \rangle \cdot \langle -\sin t,\ 0,\ \cos t \rangle \, dt \\ &= \int_0^{2\pi} \sin^2 t \, dt \\ &= \pi. \end{aligned}$$

By the equation for vector line integrals,

$$\begin{aligned} \iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} &= \iint_D curl\, \vec{\mathbf{F}}(\vec{\mathbf{r}}(\phi,\theta)) \cdot (\vec{\mathbf{t}}_\phi \times \vec{\mathbf{t}}_\theta) \, dA \\ &= \iint_D \langle 0, -1, 1 \rangle \cdot \langle \cos\theta \sin^2\phi,\ \sin\theta \sin^2\phi,\ \sin\phi \cos\phi \rangle \, dA \\ &= \int_0^{\pi} \int_0^{\pi} (\sin\phi \cos\phi - \sin\theta \sin^2\phi) \, d\phi d\theta \\ &= \frac{\pi}{2} \int_0^{\pi} \sin\theta \, d\theta \\ &= \pi. \end{aligned}$$

Therefore, we have verified Stokes' theorem for this example.

Exercise 16.7.1

Verify that Stokes' theorem is true for vector field $\vec{\mathbf{F}}(x,y,z) = \langle y, x, -z \rangle$ and surface S, where S is the upwardly oriented portion of the graph of $f(x,y) = x^2 y$ over a triangle in the xy-plane with vertices $(0,0)$, $(2,0)$, and $(0,2)$.

Hint

Calculate the double integral and line integral separately.

Answer

Both integrals give $-\frac{136}{45}$:

Applying Stokes' Theorem

Stokes' theorem translates between the flux integral of surface S to a line integral around the boundary of S. Therefore, the theorem allows us to compute surface integrals or line integrals that would ordinarily be quite difficult by translating the line integral into a surface integral or vice versa. We now study some examples of each kind of translation.

Example 16.7.2: Calculating a Surface Integral

Calculate surface integral

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}},$$

where S is the surface, oriented outward, in Figure 16.7.6 and $\vec{\mathbf{F}} = \langle z,\ 2xy,\ x+y \rangle$.

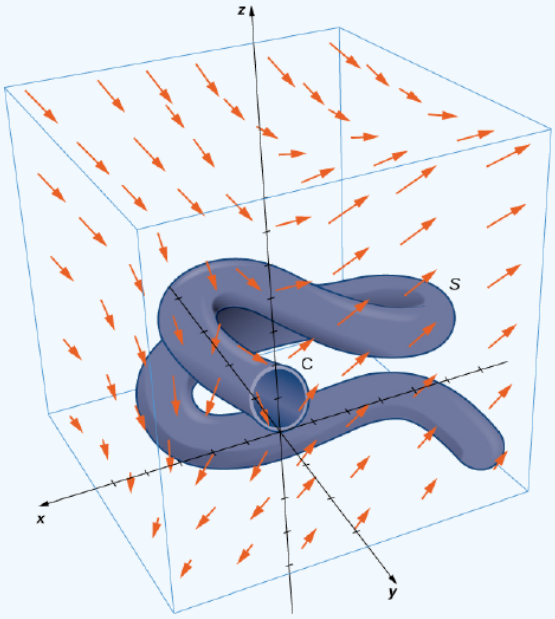


Figure 16.7.6: A complicated surface in a vector field.

Solution

Note that to calculate

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

without using Stokes' theorem, we would need the equation for scalar surface integrals. Use of this equation requires a parameterization of S. Surface S is complicated enough that it would be extremely difficult to find a parameterization. Therefore, the methods we have learned in previous sections are not useful for this problem. Instead, we use Stokes' theorem, noting that the boundary C of the surface is merely a single circle with radius 1.

The curl of $\vec{\mathbf{F}}$ is $\langle 1, 1, 2y \rangle$. By Stokes' theorem,

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}},$$

where C has parameterization $\langle \cos t,\ \sin t,\ 1 \rangle, 0 \le t \le 2\pi$. By the equation for vector line integrals,

$$\begin{aligned}\iint_S curl\, F \cdot d\vec{\mathbf{S}} &= \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} \\ &= \int_0^2 \langle 1,\, \sin t\, \cos t,\, \cos t + \sin t \rangle \cdot \langle -\sin t,\, \cos t,\, 0 \rangle\, dt \\ &= \int_0^{2\pi} (-\sin t + 2\, \sin t\, \cos^2 t)\, dt \\ &= \left[\cos t - \frac{2\, \cos^3 t}{3}\right]_0^{2\pi} \\ &= \cos(2\pi) - \frac{2\, \cos^3(2\pi)}{3} - \left(\cos(0) - \frac{2\, \cos^3(0)}{3}\right) \\ &= 0.\end{aligned}$$

An amazing consequence of Stokes' theorem is that if S' is any other smooth surface with boundary C and the same orientation as S, then

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 0$$

because Stokes' theorem says the surface integral depends on the line integral around the boundary only.

In Example 16.7.2, we calculated a surface integral simply by using information about the boundary of the surface. In general, let S_1 and S_2 be smooth surfaces with the same boundary C and the same orientation. By Stokes' theorem,

$$\iint_{S_1} curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_{S_2} curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}. \tag{16.7.2}$$

Therefore, if

$$\iint_{S_1} curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

is difficult to calculate but

$$\iint_{S_2} curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

is easy to calculate, Stokes' theorem allows us to calculate the easier surface integral. In Example 16.7.2, we could have calculated

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

by calculating

$$\iint_{S'} curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}},$$

where $\vec{\mathbf{S}}'$ is the disk enclosed by boundary curve C (a much more simple surface with which to work).

Equation 16.7.2 shows that flux integrals of curl vector fields are surface independent in the same way that line integrals of gradient fields are path independent. Recall that if $\vec{\mathbf{F}}$ is a two-dimensional conservative vector field defined on a simply connected domain, f is a *potential function* for $\vec{\mathbf{F}}$, and C is a curve in the domain of $\vec{\mathbf{F}}$, then

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

depends only on the endpoints of C. Therefore if C' is any other curve with the same starting point and endpoint as C (that is, C' has the same orientation as C), then

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \int_{C'} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$$

In other words, the value of the integral depends on the boundary of the path only; it does not really depend on the path itself.

Analogously, suppose that S and S' are surfaces with the same boundary and same orientation, and suppose that $\vec{\mathbf{G}}$ is a three-dimensional vector field that can be written as the curl of another vector field $\vec{\mathbf{F}}$ (so that $\vec{\mathbf{F}}$ is like a "potential field" of $\vec{\mathbf{G}}$). By Equation 16.7.2,

$$\iint_S \vec{\mathbf{G}} \cdot d\vec{\mathbf{S}} = \iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_{S'} curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \iint_{S'} \vec{\mathbf{G}} \cdot d\vec{\mathbf{S}}.$$

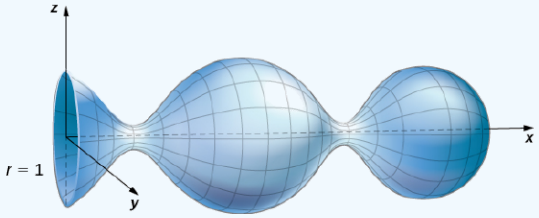
Therefore, the flux integral of $\vec{\mathbf{G}}$ does not depend on the surface, only on the boundary of the surface. Flux integrals of vector fields that can be written as the curl of a vector field are surface independent in the same way that line integrals of vector fields that can be written as the gradient of a scalar function are path independent.

? Exercise 16.7.2

Use Stokes' theorem to calculate surface integral

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}},$$

where $\vec{\mathbf{F}}(x, y, z) = \langle z, x, y \rangle$ and S is the surface as shown in the following figure. The boundary curve, C is oriented clockwise when looking along the positive y-axis.



Hint

Parameterize the boundary of S and translate to a line integral.

Answer

$-\pi$

✓ Example 16.7.3: Calculating a Line Integral

Calculate the line integral

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}},$$

where $\vec{\mathbf{F}} = \langle xy,\ x^2 + y^2 + z^2,\ yz \rangle$ and C is the boundary of the parallelogram with vertices $(0, 0, 1),\ (0, 1, 0),\ (2, 0, -1)$, and $(2, 1, -2)$.

Solution

To calculate the line integral directly, we need to parameterize each side of the parallelogram separately, calculate four separate line integrals, and add the result. This is not overly complicated, but it is time-consuming.

By contrast, let's calculate the line integral using Stokes' theorem. Let S denote the surface of the parallelogram. Note that S is the portion of the graph of $z = 1 - x - y$ for (x, y) varying over the rectangular region with vertices $(0, 0),\ (0, 1),\ (2, 0)$, and $(2, 1)$ in the xy-plane. Therefore, a parameterization of S is $\langle x, y,\ 1 - x - y \rangle,\ 0 \le x \le 2,\ 0 \le y \le 1$. The curl of $\vec{\mathbf{F}}$ is $\langle -z,\ 0,\ x \rangle$,and Stokes' theorem and the equation for scalar surface integrals

$$\begin{aligned}\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} &= \iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} \\ &= \int_0^2 \int_0^1 curl\, \vec{\mathbf{F}}(x,y) \cdot (\vec{\mathbf{t}}_x \times \vec{\mathbf{t}}_y)\, dy\, dx \\ &= \int_0^2 \int_0^1 \langle -(1-x-y),\, 0,\, x\rangle \cdot (\langle 1,\, 0, -1\rangle \times \langle 0,\, 1,\, -1\rangle)\, dy\, dx \\ &= \int_0^2 \int_0^1 \langle x+y-1,\, 0,\, x\rangle \cdot \langle 1,1,1\rangle\, dy\, dx \\ &= \int_0^2 \int_0^1 2x+y-1\, dy\, dx \\ &= 3.\end{aligned}$$

? Exercise 16.7.3

Use Stokes' theorem to calculate line integral

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}},$$

where $\vec{\mathbf{F}} = \langle z, x, y\rangle$ and C is the boundary of a triangle with vertices $(0,0,1)$, $(3,0,-2)$, and $(0,1,2)$.

Hint

This triangle lies in plane $z = 1 - x + y$.

Answer

$\frac{3}{2}$

Interpretation of Curl

In addition to translating between line integrals and flux integrals, Stokes' theorem can be used to justify the physical interpretation of curl that we have learned. Here we investigate the relationship between curl and circulation, and we use Stokes' theorem to state Faraday's law—an important law in electricity and magnetism that relates the curl of an electric field to the rate of change of a magnetic field.

Recall that if C is a closed curve and $\vec{\mathbf{F}}$ is a vector field defined on C, then the circulation of $\vec{\mathbf{F}}$ around C is line integral

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

If $\vec{\mathbf{F}}$ represents the velocity field of a fluid in space, then the circulation measures the tendency of the fluid to move in the direction of C.

Let $\vec{\mathbf{F}}$ be a continuous vector field and let D_τ be a small disk of radius r with center P_0 (Figure 16.7.7). If D_τ is small enough, then $(curl\, \vec{\mathbf{F}})(P) \approx (curl\, \vec{\mathbf{F}})(P_0)$ for all points P in D_τ because the curl is continuous. Let C_τ be the boundary circle of D_τ: By Stokes' theorem,

$$\int_{C_\tau} \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_{D_\tau} curl\, \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, d\vec{\mathbf{S}} \approx \iint_{D_\tau} (curl\, \vec{\mathbf{F}})(P_0) \cdot \vec{\mathbf{N}}(P_0)\, d\vec{\mathbf{S}}.$$

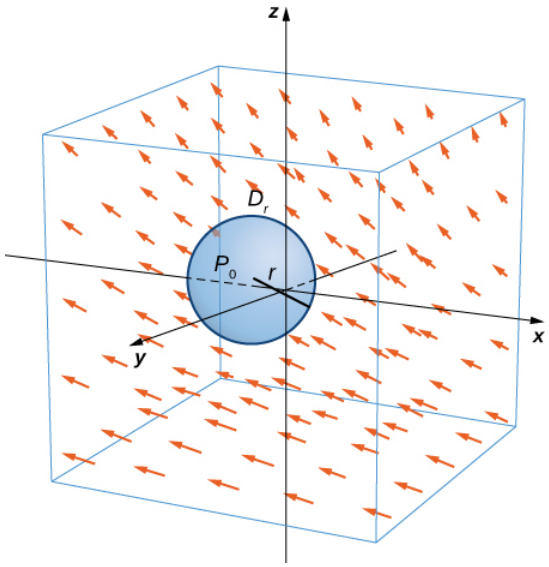


Figure 16.7.7: Disk D_τ is a small disk in a continuous vector field.

The quantity $(curl\,\vec{\mathbf{F}})(P_0)\cdot\vec{\mathbf{N}}(P_0)$ is constant, and therefore

$$\iint_{D_\tau}(curl\,\vec{\mathbf{F}})(P_0)\cdot\vec{\mathbf{N}}(P_0)\,d\vec{\mathbf{S}}=\pi r^2[(curl\,\vec{\mathbf{F}})(P_0)\cdot\vec{\mathbf{N}}(P_0)].$$

Thus

$$\int_{C_\tau}\vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}\approx\pi r^2[(curl\,\vec{\mathbf{F}})(P_0)\cdot\vec{\mathbf{N}}(P_0)],$$

and the approximation gets arbitrarily close as the radius shrinks to zero. Therefore Stokes' theorem implies that

$$(curl\,\vec{\mathbf{F}})(P_0)\cdot\vec{\mathbf{N}}(P_0)=\lim_{r\to0^+}\frac{1}{\pi r^2}\int_{C_\tau}\vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}.$$

This equation relates the curl of a vector field to the circulation. Since the area of the disk is πr^2, this equation says we can view the curl (in the limit) as the circulation per unit area. Recall that if $\vec{\mathbf{F}}$ is the velocity field of a fluid, then circulation

$$\oint_{C_\tau}\vec{\mathbf{F}}\cdot d\vec{\mathbf{r}}=\oint_{C_\tau}\vec{\mathbf{F}}\cdot\vec{\mathbf{T}}\,ds$$

is a measure of the tendency of the fluid to move around C_τ: The reason for this is that $\vec{\mathbf{F}}\cdot\vec{\mathbf{T}}$ is a component of $\vec{\mathbf{F}}$ in the direction of $\vec{\mathbf{T}}$, and the closer the direction of $\vec{\mathbf{F}}$ is to $\vec{\mathbf{T}}$, the larger the value of $\vec{\mathbf{F}}\cdot\vec{\mathbf{T}}$ (remember that if $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ are vectors and $\vec{\mathbf{b}}$ is fixed, then the dot product $\vec{\mathbf{a}}\cdot\vec{\mathbf{b}}$ is maximal when $\vec{\mathbf{a}}$ points in the same direction as $\vec{\mathbf{b}}$). Therefore, if $\vec{\mathbf{F}}$ is the velocity field of a fluid, then $curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}}$ is a measure of how the fluid rotates about axis $\vec{\mathbf{N}}$. The effect of the curl is largest about the axis that points in the direction of $\vec{\mathbf{N}}$, because in this case $curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}}$ is as large as possible.

To see this effect in a more concrete fashion, imagine placing a tiny paddlewheel at point P_0 (Figure 16.7.8). The paddlewheel achieves its maximum speed when the axis of the wheel points in the direction of curl $\vec{\mathbf{F}}$. This justifies the interpretation of the curl we have learned: curl is a measure of the rotation in the vector field about the axis that points in the direction of the normal vector $\vec{\mathbf{N}}$, and Stokes' theorem justifies this interpretation.

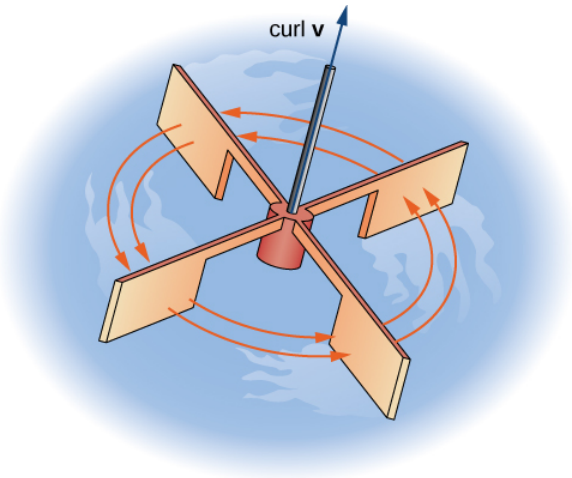


Figure 16.7.8: To visualize curl at a point, imagine placing a tiny paddlewheel at that point in the vector field.

Now that we have learned about Stokes' theorem, we can discuss applications in the area of electromagnetism. In particular, we examine how we can use Stokes' theorem to translate between two equivalent forms of Faraday's law. Before stating the two forms of Faraday's law, we need some background terminology.

Let C be a closed curve that models a thin wire. In the context of electric fields, the wire may be moving over time, so we write $C(t)$ to represent the wire. At a given time t, curve $C(t)$ may be different from original curve C because of the movement of the wire, but we assume that $C(t)$ is a closed curve for all times t. Let $D(t)$ be a surface with $C(t)$ as its boundary, and orient $C(t)$ so that $D(t)$ has positive orientation. Suppose that $C(t)$is in a magnetic field $\vec{\mathbf{B}}(t)$ that can also change over time. In other words, $\vec{\mathbf{B}}$ has the form

$$\vec{\mathbf{B}}(x,y,z) = \langle P(x,y,z),\, Q(x,y,z),\, R(x,y,z)\rangle,$$

where P, Q, and R can all vary continuously over time. We can produce current along the wire by changing field $\vec{\mathbf{B}}(t)$ (this is a consequence of Ampere's law). Flux $\phi(t) = \iint_{D(t)} \vec{\mathbf{B}}(t)\cdot d\vec{\mathbf{S}}$ creates electric field $\vec{\mathbf{E}}(t)$ that does work. The integral form of Faraday's law states that

$$Work = \int_{C(t)} \vec{\mathbf{E}}(t)\cdot d\vec{\mathbf{r}} = -\frac{\partial \phi}{\partial t}.$$

In other words, the work done by $\vec{\mathbf{E}}$ is the line integral around the boundary, which is also equal to the rate of change of the flux with respect to time. The differential form of Faraday's law states that

$$curl\, \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}.$$

Using Stokes' theorem, we can show that the differential form of Faraday's law is a consequence of the integral form. By Stokes' theorem, we can convert the line integral in the integral form into surface integral

$$-\frac{\partial \phi}{\partial t} = \int_{C(t)} \vec{\mathbf{E}}(t)\cdot d\vec{\mathbf{r}} = \iint_{D(t)} curl\, \vec{\mathbf{E}}(t)\cdot d\vec{\mathbf{S}}.$$

Since

$$\phi(t) = \iint_{D(t)} B(t)\cdot d\vec{\mathbf{S}},$$

then as long as the integration of the surface does not vary with time we also have

$$-\frac{\partial \phi}{\partial t} = \iint_{D(t)} -\frac{\partial \vec{\mathbf{B}}}{\partial t}\cdot d\vec{\mathbf{S}}.$$

Therefore,

$$\iint_{D(t)} -\frac{\partial \vec{\mathbf{B}}}{\partial t} \cdot d\vec{\mathbf{S}} = \iint_{D(t)} curl\, \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}}.$$

To derive the differential form of Faraday's law, we would like to conclude that $curl\, \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}$: In general, the equation

$$\iint_{D(t)} -\frac{\partial \vec{\mathbf{B}}}{\partial t} \cdot d\vec{\mathbf{S}} = \iint_{D(t)} curl\, \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}}$$

is not enough to conclude that $curl\, \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}$: The integral symbols do not simply "cancel out," leaving equality of the integrands. To see why the integral symbol does not just cancel out in general, consider the two single-variable integrals $\int_0^1 x\, dx$ and $\int_0^1 f(x)\, dx$, where

$$f(x) = \begin{cases} 1, & \text{if } 0 \le x \le 1/2 \\ 0, & \text{if } 1/2 \le x \le 1. \end{cases}$$

Both of these integrals equal $\frac{1}{2}$, so $\int_0^1 x\, dx = \int_0^1 f(x)\, dx$.

However, $x \neq f(x)$. Analogously, with our equation

$$\iint_{D(t)} -\frac{\partial \vec{\mathbf{B}}}{\partial t} \cdot d\vec{\mathbf{S}} = \iint_{D(t)} curl\, \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}},$$

we cannot simply conclude that $curl\, \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}$ just because their integrals are equal. However, in our context, equation

$$\iint_{D(t)} -\frac{\partial \vec{\mathbf{B}}}{\partial t} \cdot d\vec{\mathbf{S}} = \iint_{D(t)} curl\, \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}}$$

is true for *any* region, however small (this is in contrast to the single-variable integrals just discussed). If $\vec{\mathbf{F}}$ and $\vec{\mathbf{G}}$ are three-dimensional vector fields such that

$$\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \iint_S \vec{\mathbf{G}} \cdot d\vec{\mathbf{S}}$$

for any surface S, then it is possible to show that $\vec{\mathbf{F}} = \vec{\mathbf{G}}$ by shrinking the area of S to zero by taking a limit (the smaller the area of S, the closer the value of $\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$ to the value of $\vec{\mathbf{F}}$ at a point inside S). Therefore, we can let area $D(t)$ shrink to zero by taking a limit and obtain the differential form of Faraday's law:

$$curl\, \vec{\mathbf{E}} = -\frac{\partial \vec{\mathbf{B}}}{\partial t}.$$

In the context of electric fields, the curl of the electric field can be interpreted as the negative of the rate of change of the corresponding magnetic field with respect to time.

Example 16.7.4: Using Faraday's Law

Calculate the curl of electric field $\vec{\mathbf{E}}$ if the corresponding magnetic field is constant field $\vec{\mathbf{B}}(t) = \langle 1, -4, 2 \rangle$.

Solution

Since the magnetic field does not change with respect to time, $-\frac{\partial \vec{\mathbf{B}}}{\partial t} = \vec{0}$. By Faraday's law, the curl of the electric field is therefore also zero.

Analysis

A consequence of Faraday's law is that the curl of the electric field corresponding to a constant magnetic field is always zero.

Exercise 16.7.4

Calculate the curl of electric field $\vec{\mathbf{E}}$ if the corresponding magnetic field is $\vec{\mathbf{B}}(t) = \langle tx,\, ty,\, -2tz \rangle,\, 0 \leq t < \infty$.

Hint

- Use the differential form of Faraday's law.
- Notice that the curl of the electric field does not change over time, although the magnetic field does change over time.

Answer

$curl\, \vec{\mathbf{E}} = \langle x,\, y,\, -2z \rangle$

Key Concepts

- Stokes' theorem relates a flux integral over a surface to a line integral around the boundary of the surface. Stokes' theorem is a higher dimensional version of Green's theorem, and therefore is another version of the Fundamental Theorem of Calculus in higher dimensions.
- Stokes' theorem can be used to transform a difficult surface integral into an easier line integral, or a difficult line integral into an easier surface integral.
- Through Stokes' theorem, line integrals can be evaluated using the simplest surface with boundary C.
- Faraday's law relates the curl of an electric field to the rate of change of the corresponding magnetic field. Stokes' theorem can be used to derive Faraday's law.

Key Equations

- **Stokes' theorem**

$$\int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = \iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

Glossary

Stokes' theorem

relates the flux integral over a surface S to a line integral around the boundary C of the surface S

surface independent

flux integrals of curl vector fields are surface independent if their evaluation does not depend on the surface but only on the boundary of the surface

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16.7E: Exercises for Section 16.7

In exercises 1 - 6, without using Stokes' theorem, calculate directly both the flux of $curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}}$ over the given surface and the circulation integral around its boundary, assuming all are oriented clockwise.

1. $\vec{\mathbf{F}}(x,y,z)=y^2\,\hat{\mathbf{i}}+z^2\,\hat{\mathbf{j}}+x^2\,\hat{\mathbf{k}}$; S is the first-octant portion of plane $x+y+z=1$.

2. $\vec{\mathbf{F}}(x,y,z)=z\,\hat{\mathbf{i}}+x\,\hat{\mathbf{j}}+y\,\hat{\mathbf{k}}$; S is hemisphere $z=(a^2-x^2-y^2)^{1/2}$.

Answer

$$\iint_S (curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}})\,dS=\pi a^2$$

3. $\vec{\mathbf{F}}(x,y,z)=y^2\,\hat{\mathbf{i}}+2x\,\hat{\mathbf{j}}+5\,\hat{\mathbf{k}}$; S is hemisphere $z=(4-x^2-y^2)^{1/2}$.

4. $\vec{\mathbf{F}}(x,y,z)=z\,\hat{\mathbf{i}}+2x\,\hat{\mathbf{j}}+3y\,\hat{\mathbf{k}}$; S is upper hemisphere $z=\sqrt{9-x^2-y^2}$.

Answer

$$\iint_S (curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}})\,dS=18\pi$$

5. $\vec{\mathbf{F}}(x,y,z)=(x+2z)\,\hat{\mathbf{i}}+(y-x)\,\hat{\mathbf{j}}+(z-y)\,\hat{\mathbf{k}}$; S is a triangular region with vertices $(3,0,0)$, $(0,3/2,0)$, and $(0,0,3)$.

6. $\vec{\mathbf{F}}(x,y,z)=2y\,\hat{\mathbf{i}}+6z\,\hat{\mathbf{j}}+3x\,\hat{\mathbf{k}}$; S is a portion of paraboloid $z=4-x^2-y^2$ and is above the xy-plane.

Answer

$$\iint_S (curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}})\,dS=-8\pi$$

In exercises 7 - 9, use Stokes' theorem to evaluate $\iint_S (curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}})\,dS$ for the vector fields and surface.

7. $\vec{\mathbf{F}}(x,y,z)=xy\,\hat{\mathbf{i}}-z\,\hat{\mathbf{j}}$ and S is the surface of the cube $0\le x\le 1$, $0\le y\le 1$, $0\le z\le 1$, except for the face where $z=0$ and using the outward unit normal vector.

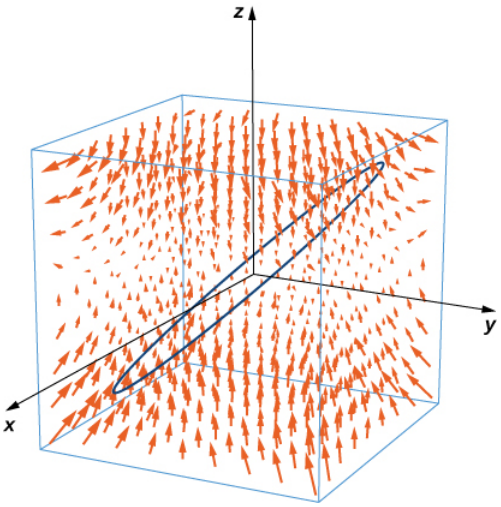
8. $\vec{\mathbf{F}}(x,y,z)=xy\,\hat{\mathbf{i}}+x^2\,\hat{\mathbf{j}}+z^2\,\hat{\mathbf{k}}$; and C is the intersection of paraboloid $z=x^2+y^2$ and plane $z=y$, and using the outward normal vector.

Answer

$$\iint_S (curl\,\vec{\mathbf{F}}\cdot\vec{\mathbf{N}})\,dS=0$$

9. $\vec{\mathbf{F}}(x,y,z)=4y\,\hat{\mathbf{i}}+z\,\hat{\mathbf{j}}+2y\,\hat{\mathbf{k}}$; and C is the intersection of sphere $x^2+y^2+z^2=4$ with plane $z=0$, and using the outward normal vector.

10. Use Stokes' theorem to evaluate $\int_C \left[2xy^2z\,dx+2x^2yz\,dy+(x^2y^2-2z)\,dz\right]$, where C is the curve given by $x=\cos t,\ y=\sin t,\ 0\le t\le 2\pi$, traversed in the direction of increasing t.



Answer

$$\int_C \vec{\mathbf{F}} \cdot dS = 0$$

11. [T] Use a computer algebraic system (CAS) and Stokes' theorem to approximate line integral $\int_C (y\,dx + z\,dy + x\,dz)$, where C is the intersection of plane $x + y = 2$ and surface $x^2 + y^2 + z^2 = 2(x + y)$, traversed counterclockwise viewed from the origin.

12. [T] Use a CAS and Stokes' theorem to approximate line integral $\int_C (3y\,dx + 2z\,dy - 5x\,dz)$, where C is the intersection of the xy-plane and hemisphere $z = \sqrt{1 - x^2 - y^2}$, traversed counterclockwise viewed from the top—that is, from the positive z-axis toward the xy-plane.

Answer

$$\int_C \vec{\mathbf{F}} \cdot dS = -9.4248$$

13. [T] Use a CAS and Stokes' theorem to approximate line integral $\int_C [(1 + y)\,z\,dx + (1 + z)x\,dy + (1 + x)y\,dz]$, where C is a triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ oriented counterclockwise.

14. Use Stokes' theorem to evaluate $\iint_S curl\ \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = e^{xy} cos\ z\ \hat{\mathbf{i}} + x^2 z\ \hat{\mathbf{j}} + xy\ \hat{\mathbf{k}}$, and S is half of sphere $x = \sqrt{1 - y^2 - z^2}$, oriented out toward the positive x-axis.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = 0$$

15. [T] Use a CAS and Stokes' theorem to evaluate $\iint_S (curl\ \vec{\mathbf{F}} \cdot \vec{\mathbf{N}})\,dS$, where $\vec{\mathbf{F}}(x, y, z) = x^2 y\,\hat{\mathbf{i}} + xy^2\,\hat{\mathbf{j}} + z^3\,\hat{\mathbf{k}}$ and C is the curve of the intersection of plane $3x + 2y + z = 6$ and cylinder $x^2 + y^2 = 4$, oriented clockwise when viewed from above.

16. [T] Use a CAS and Stokes' theorem to evaluate $\iint_S curl\ \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = \left(\sin(y + z) - yx^2 - \frac{y^3}{3}\right)\hat{\mathbf{i}} + x\ \cos(y + z)\,\hat{\mathbf{j}} + \cos(2y)\,\hat{\mathbf{k}}$ and S consists of the top and the four sides but not the bottom of the cube with vertices $(\pm 1, \pm 1, \pm 1)$, oriented outward.

Answer

$$\iint_S curl\ \vec{\mathbf{F}} \cdot dS = 2.6667$$

17. [T] Use a CAS and Stokes' theorem to evaluate $\iint_S curl\ \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = z^2\,\hat{\mathbf{i}} + 3xy\,\hat{\mathbf{j}} + x^3 y^3\,\hat{\mathbf{k}}$ and S is the top part of $z = 5 - x^2 - y^2$ above plane $z = 1$ and S is oriented upward.

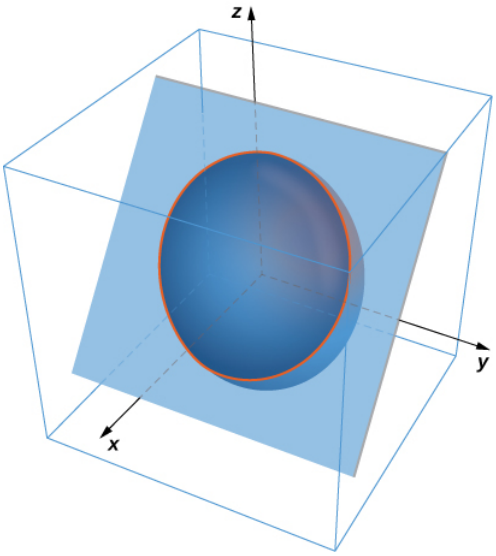
18. Use Stokes' theorem to evaluate $\iint_S (curl\,\vec{\mathbf{F}} \cdot \vec{\mathbf{N}})dS$, where $\vec{\mathbf{F}}(x,y,z) = z^2\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}} + x\,\hat{\mathbf{k}}$ and S is a triangle with vertices $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$ with counterclockwise orientation.

Answer

$$\iint_S (curl\,\vec{\mathbf{F}} \cdot \vec{\mathbf{N}})dS = -\frac{1}{6}$$

19. Use Stokes' theorem to evaluate line integral $\int_C (z\,dx + x\,dy + y\,dz)$, where C is a triangle with vertices $(3,0,0)$, $(0,0,2)$, and $(0,6,0)$ traversed in the given order.

20. Use Stokes' theorem to evaluate $\int_C \left(\frac{1}{2}y^2\,dx + z\,dy + x\,dz\right)$, where C is the curve of intersection of plane $x + z = 1$ and ellipsoid $x^2 + 2y^2 + z^2 = 1$, oriented clockwise from the origin.



Answer

$$\int_C \left(\frac{1}{2}y^2\,dx + z\,dy + x\,dz\right) = -\frac{\pi}{4}$$

21. Use Stokes' theorem to evaluate $\iint_S (curl\,\vec{\mathbf{F}} \cdot \vec{\mathbf{N}})dS$, where $\vec{\mathbf{F}}(x,y,z) = x\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}} + ze^{xy}\,\hat{\mathbf{k}}$ and S is the part of surface $z = 1 - x^2 - 2y^2$ with $z \geq 0$, oriented counterclockwise.

22. Use Stokes' theorem for vector field $\vec{\mathbf{F}}(x,y,z) = z\,\hat{\mathbf{i}} + 3x\,\hat{\mathbf{j}} + 2z\hat{\mathbf{k}}$ where S is surface $z = 1 - x^2 - 2y^2,\ z \geq 0$, C is boundary circle $x^2 + y^2 = 1$, and S is oriented in the positive z-direction.

Answer

$$\iint_S (curl\,\vec{\mathbf{F}} \cdot \vec{\mathbf{N}})dS = -3\pi$$

23. Use Stokes' theorem for vector field $\vec{\mathbf{F}}(x,y,z) = -\frac{3}{2}y^2\,\hat{\mathbf{i}} - 2xy\,\hat{\mathbf{j}} + yz\hat{\mathbf{k}}$, where S is that part of the surface of plane $x + y + z = 1$ contained within triangle C with vertices $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$, traversed counterclockwise as viewed from above.

24. A certain closed path C in plane $2x + 2y + z = 1$ is known to project onto unit circle $x^2 + y^2 = 1$ in the xy-plane. Let C be a constant and let $\vec{\mathbf{R}}(x,y,z) = x\,\hat{\mathbf{i}} + y\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$. Use Stokes' theorem to evaluate $\int_C (c\,\hat{\mathbf{k}} \times \vec{\mathbf{R}}) \cdot dS$.

Answer

$$\int_C (c\,\hat{\mathbf{k}} \times \vec{\mathbf{R}}) \cdot dS = 2\pi c$$

25. Use Stokes' theorem and let C be the boundary of surface $z = x^2 + y^2$ with $0 \le x \le 2$ and $0 \le y \le 1$ oriented with upward facing normal. Define $\vec{\mathbf{F}}(x, y, z) = \left(\sin(x^3) + xz\right)\hat{\mathbf{i}} + (x - yz)\hat{\mathbf{j}} + \cos(z^4)\hat{\mathbf{k}}$ and evaluate $\int_C \vec{\mathbf{F}} \cdot dS$.

26. Let S be hemisphere $x^2 + y^2 + z^2 = 4$ with $z \ge 0$, oriented upward. Let $\vec{\mathbf{F}}(x, y, z) = x^2 e^{yz}\hat{\mathbf{i}} + y^2 e^{xz}\hat{\mathbf{j}} + z^2 e^{xy}\hat{\mathbf{k}}$ be a vector field. Use Stokes' theorem to evaluate $\iint_S curl\, \vec{\mathbf{F}} \cdot dS$.

Answer

$\iint_S curl\, \vec{\mathbf{F}} \cdot dS = 0$

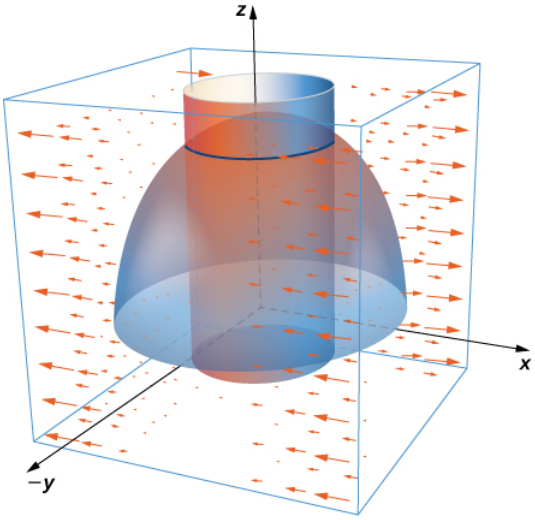
27. Let $\vec{\mathbf{F}}(x, y, z) = xy\,\hat{\mathbf{i}} + (e^{z^2} + y)\,\hat{\mathbf{j}} + (x + y)\,\hat{\mathbf{k}}$ and let S be the graph of function $y = \frac{x^2}{9} + \frac{z^2}{9} - 1$ with $z \le 0$ oriented so that the normal vector S has a positive y component. Use Stokes' theorem to compute integral $\iint_S curl\, \vec{\mathbf{F}} \cdot dS$.

28. Use Stokes' theorem to evaluate $\oint \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = y\,\hat{\mathbf{i}} + z\,\hat{\mathbf{j}} + x\,\hat{\mathbf{k}}$ and C is a triangle with vertices $(0, 0, 0)$, $(2, 0, 0)$ and $0, -2, 2)$ oriented counterclockwise when viewed from above.

Answer

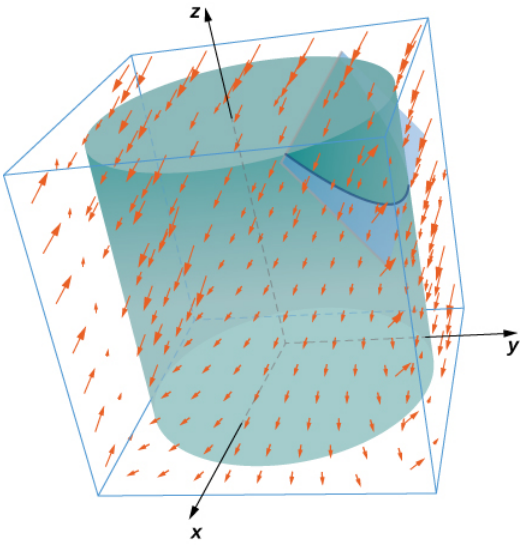
$\oint \vec{\mathbf{F}} \cdot dS = -4$

29. Use the surface integral in Stokes' theorem to calculate the circulation of field $\vec{\mathbf{F}}$, $\vec{\mathbf{F}}(x, y, z) = x^2 y^3\,\hat{\mathbf{i}} + \hat{\mathbf{j}} + z\hat{\mathbf{k}}$ around C, which is the intersection of cylinder $x^2 + y^2 = 4$ and hemisphere $x^2 + y^2 + z^2 = 16$, $z \ge 0$, oriented counterclockwise when viewed from above.



30. Use Stokes' theorem to compute $\iint_S curl\, \vec{\mathbf{F}} \cdot dS$. where $\vec{\mathbf{F}}(x, y, z) = \hat{\mathbf{i}} + xy^2\,\hat{\mathbf{j}} + xy^2\,\hat{\mathbf{k}}$ and S is a part of plane $y + z = 2$ inside cylinder $x^2 + y^2 = 1$ and oriented counterclockwise.

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Answer

$$\iint_S curl\, \vec{\mathbf{F}} \cdot dS = 0$$

31. Use Stokes' theorem to evaluate $\iint_S curl\, \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = -y^2\, \hat{\mathbf{i}} + x\, \hat{\mathbf{j}} + z^2\, \hat{\mathbf{k}}$ and S is the part of plane $x + y + z = 1$ in the positive octant and oriented counterclockwise $x \geq 0,\ y \geq 0,\ z \geq 0$.

32. Let $\vec{\mathbf{F}}(x, y, z) = xy\, \hat{\mathbf{i}} + 2z\, \hat{\mathbf{j}} - 2y\, \hat{\mathbf{k}}$ and let C be the intersection of plane $x + z = 5$ and cylinder $x^2 + y^2 = 9$, which is oriented counterclockwise when viewed from the top. Compute the line integral of $\vec{\mathbf{F}}$ over C using Stokes' theorem.

Answer

$$\iint_S curl\, \vec{\mathbf{F}} \cdot dS = -36\pi$$

33. [T] Use a CAS and let $\vec{\mathbf{F}}(x, y, z) = xy^2\, \hat{\mathbf{i}} + (yz - x)\, \hat{\mathbf{j}} + e^{yxz}\, \hat{\mathbf{k}}$. Use Stokes' theorem to compute the surface integral of curl $\vec{\mathbf{F}}$ over surface S with inward orientation consisting of cube $[0, 1] \times [0, 1] \times [0, 1]$ with the right side missing.

34. Let S be ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1$ oriented counterclockwise and let $\vec{\mathbf{F}}$ be a vector field with component functions that have continuous partial derivatives.

Answer

$$\iint_S curl\, \vec{\mathbf{F}} \cdot \vec{\mathbf{N}} = 0$$

35. Let S be the part of paraboloid $z = 9 - x^2 - y^2$ with $z \geq 0$. Verify Stokes' theorem for vector field $\vec{\mathbf{F}}(x, y, z) = 3z\, \hat{\mathbf{i}} + 4x\, \hat{\mathbf{j}} + 2y\, \hat{\mathbf{k}}$.

36. [T] Use a CAS and Stokes' theorem to evaluate $\oint \vec{\mathbf{F}} \cdot dS$, if $\vec{\mathbf{F}}(x, y, z) = (3z - \sin x)\, \hat{\mathbf{i}} + (x^2 + e^y)\, \hat{\mathbf{j}} + (y^3 - \cos z)\, \hat{\mathbf{k}}$, where C is the curve given by $x = \cos t,\ y = \sin t,\ z = 1;\ 0 \leq t \leq 2\pi$.

Answer

$$\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 0$$

37. [T] Use a CAS and Stokes' theorem to evaluate $\vec{\mathbf{F}}(x, y, z) = 2y\, \hat{\mathbf{i}} + e^z\, \hat{\mathbf{j}} - \arctan x\, \hat{\mathbf{k}}$ with S as a portion of paraboloid $z = 4 - x^2 - y^2$ cut off by the xy-plane oriented counterclockwise.

38. [T] Use a CAS to evaluate $\iint_S curl(F) \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = 2z\, \hat{\mathbf{i}} + 3x\, \hat{\mathbf{j}} + 5y\, \hat{\mathbf{k}}$ and S is the surface parametrically by $\vec{\mathbf{r}}(r, \theta) = r \cos\theta\, \hat{\mathbf{i}} + r \sin\theta\, \hat{\mathbf{j}} + (4 - r^2)\, \hat{\mathbf{k}}\ (0 \leq \theta \leq 2\pi,\ 0 \leq r \leq 3)$.

Answer

$$\iint_S curl(F)\cdot dS = 84.8230$$

39. Let S be paraboloid $z = a(1 - x^2 - y^2)$, for $z \geq 0$, where $a > 0$ is a real number. Let $\vec{\mathbf{F}}(x, y, z) = \langle x - y,\ y + z,\ z - x\rangle$. For what value(s) of a (if any) does $\iint_S (\vec{\nabla} \times \vec{\mathbf{F}}) \cdot \vec{\mathbf{n}}\, dS$ have its maximum value?

For application exercises 40 - 41, the goal is to evaluate $A = \iint_S (\vec{\nabla} \times \vec{\mathbf{F}}) \cdot \vec{\mathbf{n}}\, dS$, where $\vec{\mathbf{F}} = \langle xz,\ -xz,\ xy\rangle$ and S is the upper half of ellipsoid $x^2 + y^2 + 8z^2 = 1$, where $z \geq 0$.

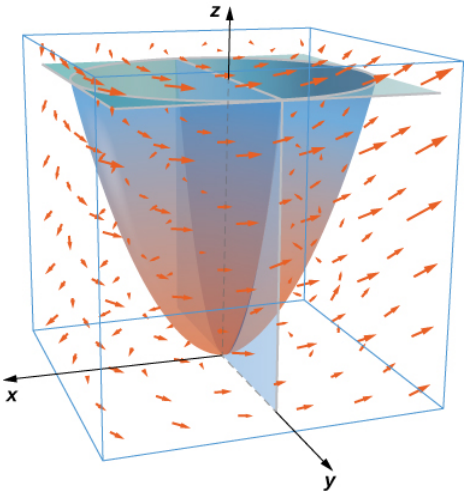
40. Evaluate a surface integral over a more convenient surface to find the value of A.

Answer

$$A = \iint_S (\vec{\nabla} \times \vec{\mathbf{F}}) \cdot \vec{\mathbf{n}}\, dS = 0$$

41. Evaluate A using a line integral.

42. Take paraboloid $z = x^2 + y^2$, for $0 \leq z \leq 4$, and slice it with plane $y = 0$. Let S be the surface that remains for $y \geq 0$, including the planar surface in the xz-plane. Let C be the semicircle and line segment that bounded the cap of S in plane $z = 4$ with counterclockwise orientation. Let $\vec{\mathbf{F}} = \langle 2z + y,\ 2x + z,\ 2y + x\rangle$. Evaluate $\iint_S (\vec{\nabla} \times \vec{\mathbf{F}}) \cdot \vec{\mathbf{n}}\, dS$.



Answer

$$\iint_S (\vec{\nabla} \times \vec{\mathbf{F}}) \cdot \vec{\mathbf{n}}\, dS = 2\pi$$

For exercises 43 - 45, let S be the disk enclosed by curve $C : \vec{\mathbf{r}}(t) = \langle \cos\varphi\ \cos t,\ \sin t,\ \sin\varphi\ \cos t\rangle$, for $0 \leq t \leq 2\pi$, where $0 \leq \varphi \leq \frac{\pi}{2}$ is a fixed angle.

43. What is the length of C in terms of φ?

44. What is the circulation of C of vector field $\vec{\mathbf{F}} = \langle -y,\ -z,\ x\rangle$ as a function of φ?

Answer

$$C = \pi(\cos\varphi - \sin\varphi)$$

45. For what value of φ is the circulation a maximum?

46. Circle C in plane $x + y + z = 8$ has radius 4 and center $(2, 3, 3)$. Evaluate $\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}$ for $\vec{\mathbf{F}} = \langle 0,\ -z,\ 2y\rangle$, where C has a counterclockwise orientation when viewed from above.

Answer

$$\oint_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}} = 48\pi$$

47. Velocity field $v = \langle 0,\ 1 - x^2,\ 0 \rangle$, for $|x| \leq 1$ and $|z| \leq 1$, represents a horizontal flow in the y-direction. Compute the curl of $\vec{\mathbf{v}}$ in a clockwise rotation.

48. Evaluate integral $\iint_S (\vec{\nabla} \times \vec{\mathbf{F}}) \cdot \vec{\mathbf{n}}\, dS$, where $\vec{\mathbf{F}} = -xz\hat{\mathbf{i}} + yz\hat{\mathbf{j}} + xye^z\hat{\mathbf{k}}$ and S is the cap of paraboloid $z = 5 - x^2 - y^2$ above plane $z = 3$, and $\vec{\mathbf{n}}$ points in the positive z-direction on S.

Answer

$$\iint_S (\vec{\nabla} \times \vec{\mathbf{F}}) \cdot \vec{\mathbf{n}} = 0$$

In exercises 49 - 50, use Stokes' theorem to find the circulation of the following vector fields around any smooth, simple closed curve C.

49. $\vec{\mathbf{F}} = \vec{\nabla}(x\ \sin ye^z)$

50. $\vec{\mathbf{F}} = \langle y^2z^3,\ z2xyz^3, 3xy^2z^2 \rangle$

Answer

0

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16.8: The Divergence Theorem

Learning Objectives

- Explain the meaning of the divergence theorem.
- Use the divergence theorem to calculate the flux of a vector field.
- Apply the divergence theorem to an electrostatic field.

We have examined several versions of the Fundamental Theorem of Calculus in higher dimensions that relate the integral around an oriented boundary of a domain to a "derivative" of that entity on the oriented domain. In this section, we state the divergence theorem, which is the final theorem of this type that we will study. The divergence theorem has many uses in physics; in particular, the divergence theorem is used in the field of partial differential equations to derive equations modeling heat flow and conservation of mass. We use the theorem to calculate flux integrals and apply it to electrostatic fields.

Overview of Theorems

Before examining the divergence theorem, it is helpful to begin with an overview of the versions of the Fundamental Theorem of Calculus we have discussed:

1. The Fundamental Theorem of Calculus:

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

This theorem relates the integral of derivative f' over line segment $[a, b]$ along the x-axis to a difference of f evaluated on the boundary.

2. The Fundamental Theorem for Line Integrals:

$$\int_C \vec{\nabla} f \cdot d\vec{\mathbf{r}} = f(P_1) - f(P_0),$$

where P_0 is the initial point of C and P_1 is the terminal point of C. The Fundamental Theorem for Line Integrals allows path C to be a path in a plane or in space, not just a line segment on the x-axis. If we think of the gradient as a derivative, then this theorem relates an integral of derivative $\vec{\nabla} f$ over path C to a difference of f evaluated on the boundary of C.

3. Green's theorem, circulation form:

$$\iint_D (Q_x - P_y)\,dA = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

Since $Q_x - P_y = \text{curl } \vec{\mathbf{F}} \cdot \hat{\mathbf{k}}$ and curl is a derivative of sorts, Green's theorem relates the integral of derivative curl $\vec{\mathbf{F}}$ over planar region D to an integral of $\vec{\mathbf{F}}$ over the boundary of D.

4. Green's theorem, flux form:

$$\iint_D (P_x + Q_y)\,dA = \int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\,dS.$$

Since $P_x + Q_y = \text{div } \vec{\mathbf{F}}$ and divergence is a derivative of sorts, the flux form of Green's theorem relates the integral of derivative div $\vec{\mathbf{F}}$ over planar region D to an integral of $\vec{\mathbf{F}}$ over the boundary of D.

5. Stokes' theorem:

$$\iint_S curl\, \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \int_C \vec{\mathbf{F}} \cdot d\vec{\mathbf{r}}.$$

If we think of the curl as a derivative of sorts, then Stokes' theorem relates the integral of derivative curl $\vec{\mathbf{F}}$ over surface S (not necessarily planar) to an integral of $\vec{\mathbf{F}}$ over the boundary of S.

Stating the Divergence Theorem

The divergence theorem follows the general pattern of these other theorems. If we think of divergence as a derivative of sorts, then the divergence theorem relates a triple integral of derivative div $\vec{\mathbf{F}}$ over a solid to a flux integral of $\vec{\mathbf{F}}$ over the boundary of the solid. More specifically, the divergence theorem relates a flux integral of vector field $\vec{\mathbf{F}}$ over a closed surface S to a triple integral of the divergence of $\vec{\mathbf{F}}$ over the solid enclosed by S.

The Divergence Theorem

Let S be a piecewise, smooth closed surface that encloses solid E in space. Assume that S is oriented outward, and let $\vec{\mathbf{F}}$ be a vector field with continuous partial derivatives on an open region containing E (Figure 16.8.1). Then

$$\iiint_E \operatorname{div} \vec{\mathbf{F}}\, dV = \iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}. \tag{16.8.1}$$

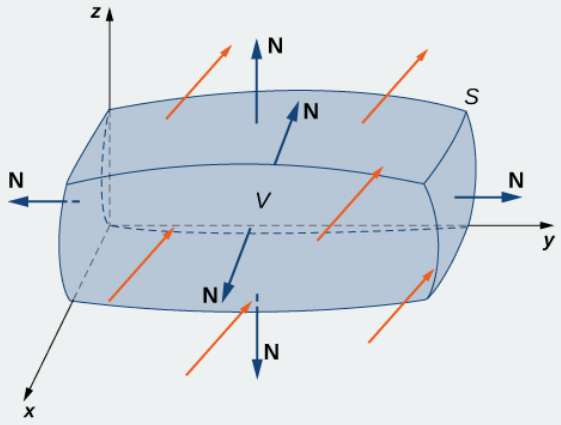


Figure 16.8.1: The divergence theorem relates a flux integral across a closed surface S to a triple integral over solid E enclosed by the surface.

Recall that the flux form of Green's theorem states that

$$\iint_D \operatorname{div} \vec{\mathbf{F}}\, dA = \int_C \vec{\mathbf{F}} \cdot \vec{\mathbf{N}}\, dS.$$

Therefore, the divergence theorem is a version of Green's theorem in one higher dimension.

The proof of the divergence theorem is beyond the scope of this text. However, we look at an informal proof that gives a general feel for why the theorem is true, but does not prove the theorem with full rigor. This explanation follows the informal explanation given for why Stokes' theorem is true.

Proof

Let B be a small box with sides parallel to the coordinate planes inside E (Figure 16.8.2a). Let the center of B have coordinates (x, y, z) and suppose the edge lengths are Δx, Δy, and Δz. (Figure 16.8.1b). The normal vector out of the top of the box is $\hat{\mathbf{k}}$ and the normal vector out of the bottom of the box is $-\hat{\mathbf{k}}$. The dot product of $\vec{\mathbf{F}} = \langle P, Q, R \rangle$ with $\hat{\mathbf{k}}$ is R and the dot product with $-\hat{\mathbf{k}}$ is $-R$. The area of the top of the box (and the bottom of the box) ΔS is $\Delta x \Delta y$.

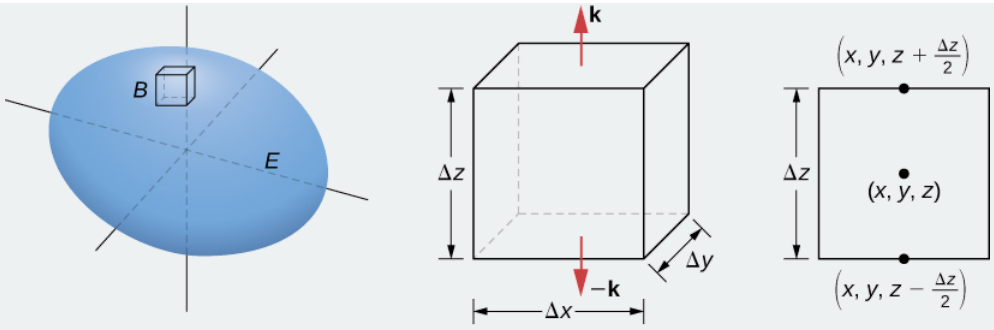


Figure 16.8.2: (a) A small box B inside surface E has sides parallel to the coordinate planes. (b) Box B has side lengths Δx, Δy, and Δz (c) If we look at the side view of B, we see that, since (x, y, z) is the center of the box, to get to the top of the box we must travel a vertical distance of $\frac{\Delta z}{2}$ up from (x, y, z). Similarly, to get to the bottom of the box we must travel a distance $\frac{\Delta z}{2}$ down from (x, y, z).

The flux out of the top of the box can be approximated by $R\left(x,\, y,\, z+\frac{\Delta z}{2}\right)\, \Delta x\, \Delta y$ (Figure 16.8.2c) and the flux out of the bottom of the box is $-R\left(x,\, y,\, z-\frac{\Delta z}{2}\right)\, \Delta x\, \Delta y$. If we denote the difference between these values as ΔR, then the net flux in the vertical direction can be approximated by $\Delta R\, \Delta x\, \Delta y$. However,

$$\Delta R\, \Delta x\, \Delta y = \left(\frac{\Delta R}{\Delta z}\right)\, \Delta x\, \Delta y \Delta z \approx \left(\frac{\partial R}{\partial z}\right)\, \Delta V.$$

Therefore, the net flux in the vertical direction can be approximated by $\left(\frac{\partial R}{\partial z}\right) \Delta V$. Similarly, the net flux in the x-direction can be approximated by $\left(\frac{\partial P}{\partial x}\right)\, \Delta V$ and the net flux in the y-direction can be approximated by $\left(\frac{\partial Q}{\partial y}\right)\, \Delta V$. Adding the fluxes in all three directions gives an approximation of the total flux out of the box:

$$\text{Total flux } \approx \left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}+\frac{\partial R}{\partial z}\right) \Delta V = \text{div } \vec{\mathbf{F}}\, \Delta V.$$

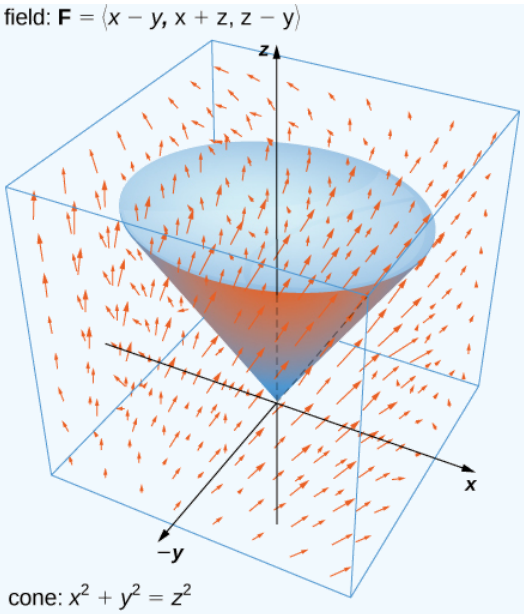
This approximation becomes arbitrarily close to the value of the total flux as the volume of the box shrinks to zero.

The sum of $\text{div } \vec{\mathbf{F}}\, \Delta V$ over all the small boxes approximating E is approximately $\iiint_E \text{div } \vec{\mathbf{F}}\, dV$. On the other hand, the sum of $\text{div } \vec{\mathbf{F}}\, \Delta V$ over all the small boxes approximating E is the sum of the fluxes over all these boxes. Just as in the informal proof of Stokes' theorem, adding these fluxes over all the boxes results in the cancelation of a lot of the terms. If an approximating box shares a face with another approximating box, then the flux over one face is the negative of the flux over the shared face of the adjacent box. These two integrals cancel out. When adding up all the fluxes, the only flux integrals that survive are the integrals over the faces approximating the boundary of E. As the volumes of the approximating boxes shrink to zero, this approximation becomes arbitrarily close to the flux over S.

□

Example 16.8.1: Verifying the Divergence Theorem

Verify the divergence theorem for vector field $\vec{\mathbf{F}} = \langle x-y,\, x+z,\, z-y\rangle$ and surface S that consists of cone $x^2+y^2=z^2$, $0 \le z \le 1$, and the circular top of the cone (see the following figure). Assume this surface is positively oriented.



Solution

Let E be the solid cone enclosed by S. To verify the theorem for this example, we show that

$$\iiint_E \text{div } \vec{\mathbf{F}}\, dV = \iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

by calculating each integral separately.

To compute the triple integral, note that $\text{div } \vec{\mathbf{F}} = P_x + Q_y + R_z = 2$, and therefore the triple integral is

$$\begin{aligned}\iiint_E \text{div } \vec{\mathbf{F}}\, dV &= 2 \iiint_E dV \\ &= 2\,(volume\ of\ E).\end{aligned}$$

The volume of a right circular cone is given by $\pi r^2 \frac{h}{3}$. In this case, $h = r = 1$. Therefore,

$$\iiint_E \text{div } \vec{\mathbf{F}}\, dV = 2\,(volume\ of\ E) = \frac{2\pi}{3}.$$

To compute the flux integral, first note that S is piecewise smooth; S can be written as a union of smooth surfaces. Therefore, we break the flux integral into two pieces: one flux integral across the circular top of the cone and one flux integral across the remaining portion of the cone. Call the circular top S_1 and the portion under the top S_2. We start by calculating the flux across the circular top of the cone. Notice that S_1 has parameterization

$$\vec{\mathbf{r}}(u, v) = \langle u\, \cos v,\, u\, \sin v,\, 1\rangle,\ 0 \le u \le 1,\ 0 \le v \le 2\pi.$$

Then, the tangent vectors are $\vec{\mathbf{t}}_u = \langle \cos v,\, \sin v,\, 0\rangle$ and $\vec{\mathbf{t}}_v = \langle -u\, \sin v,\, u\, \cos v, 0\rangle$. Therefore, the flux across S_1 is

$$\begin{aligned}\iint_{S_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} &= \int_0^1 \int_0^{2\pi} \vec{\mathbf{F}}(\vec{\mathbf{r}}(u, v)) \cdot (\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v)\, dA \\ &= \int_0^1 \int_0^{2\pi} \langle u\, \cos v - u\, \sin v,\, u\, \cos v + 1,\, 1 - u\, \sin v\rangle \cdot \langle 0, 0, u\rangle\, dv\, du \\ &= \int_0^1 \int_0^{2\pi} u - u^2 \sin v\, dv du \\ &= \pi.\end{aligned}$$

We now calculate the flux over S_2. A parameterization of this surface is

$$\vec{\mathbf{r}}(u, v) = \langle u\, \cos v,\, u\, \sin v,\, u\rangle,\ 0 \le u \le 1,\ 0 \le v \le 2\pi.$$

The tangent vectors are $\vec{\mathbf{t}}_u = \langle \cos v,\ \sin v,\ 1\rangle$ and $\vec{\mathbf{t}}_v = \langle -u\ \sin v,\ u\ \cos v, 0\rangle$, so the cross product is

$$\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v = \langle -u\ \cos v,\ -u\ \sin v,\ u\rangle.$$

Notice that the negative signs on the x and y components induce the negative (or inward) orientation of the cone. Since the surface is positively oriented, we use vector $\vec{\mathbf{t}}_v \times \vec{\mathbf{t}}_u = \langle u\ \cos v,\ u\ \sin v,\ -u\rangle$ in the flux integral. The flux across S_2 is then

$$\begin{aligned}\iint_{S_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} &= \int_0^1 \int_0^{2\pi} \vec{\mathbf{F}}(\vec{\mathbf{r}}(u,v)) \cdot (\vec{\mathbf{t}}_u \times \vec{\mathbf{t}}_v)\, dA \\ &= \int_0^1 \int_0^{2\pi} \langle u\ \cos v - u\ \sin v,\ u\ \cos v + u,\ u - u \sin v\rangle \cdot \langle u\ \cos v,\ u\ \sin v,\ -u\rangle\, dv\, du \\ &= \int_0^1 \int_0^{2\pi} u^2 \cos^2 v + 2u^2 \sin v - u^2\, dv\, du \\ &= -\frac{\pi}{3}\end{aligned}$$

The total flux across S is

$$\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \iint_{S_1} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} + \iint_{S_2} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \frac{2\pi}{3} = \iiint_E \operatorname{div} \vec{\mathbf{F}}\, dV,$$

and we have verified the divergence theorem for this example.

Exercise 16.8.1

Verify the divergence theorem for vector field $\vec{\mathbf{F}}(x,y,z) = \langle x + y + z,\ y,\ 2x - y\rangle$ and surface S given by the cylinder $x^2 + y^2 = 1$, $0 \le z \le 3$ plus the circular top and bottom of the cylinder. Assume that S is positively oriented.

Hint

Calculate both the flux integral and the triple integral with the divergence theorem and verify they are equal.

Answer

Both integrals equal 6π.

Recall that the divergence of continuous field $\vec{\mathbf{F}}$ at point P is a measure of the "outflowing-ness" of the field at P. If $\vec{\mathbf{F}}$ represents the velocity field of a fluid, then the divergence can be thought of as the rate per unit volume of the fluid flowing out less the rate per unit volume flowing in. The divergence theorem confirms this interpretation. To see this, let P be a point and let B_τ be a ball of small radius r centered at P (Figure 16.8.3). Let S_τ be the boundary sphere of B_τ. Since the radius is small and $\vec{\mathbf{F}}$ is continuous, $\operatorname{div} \vec{\mathbf{F}}(Q) \approx \operatorname{div} \vec{\mathbf{F}}(P)$ for all other points Q in the ball. Therefore, the flux across S_τ can be approximated using the divergence theorem:

$$\iint_{S_\tau} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} = \iiint_{B_\tau} \operatorname{div} \vec{\mathbf{F}}\, dV \approx \iiint_{B_\tau} \operatorname{div} \vec{\mathbf{F}}(P)\, dV.$$

Since $divF(P)$ is a constant,

$$\iiint_{B_\tau} \operatorname{div} \vec{\mathbf{F}}(P)\, dV = \operatorname{div} \vec{\mathbf{F}}(P)\, V(B_\tau).$$

Therefore, flux

$$\iint_{S_\tau} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

can be approximated by $\vec{\mathbf{F}}(P)\, V(B_\tau)$. This approximation gets better as the radius shrinks to zero, and therefore

$$\operatorname{div} \vec{\mathbf{F}}(P) = \lim_{\tau \to 0} \frac{1}{V(B_\tau)} \iint_{S_\tau} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}.$$

This equation says that the divergence at P is the net rate of outward flux of the fluid per unit volume.

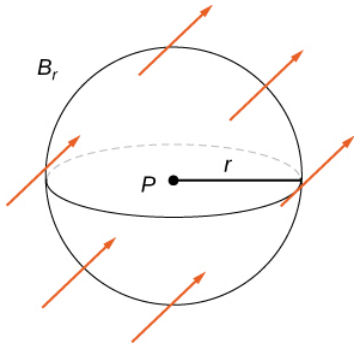


Figure 16.8.3: Ball B_r of small radius r centered at P.

Using the Divergence Theorem

The divergence theorem translates between the flux integral of closed surface S and a triple integral over the solid enclosed by S. Therefore, the theorem allows us to compute flux integrals or triple integrals that would ordinarily be difficult to compute by translating the flux integral into a triple integral and vice versa.

Example 16.8.2: Applying the Divergence Theorem

Calculate the surface integral

$$\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}},$$

where S is cylinder $x^2 + y^2 = 1$, $0 \le z \le 2$, including the circular top and bottom, and $\vec{\mathbf{F}} = \left\langle \frac{x^3}{3} + yz,\ \frac{y^3}{3} - \sin(xz),\ z - x - y \right\rangle$.

Solution

We could calculate this integral without the divergence theorem, but the calculation is not straightforward because we would have to break the flux integral into three separate integrals: one for the top of the cylinder, one for the bottom, and one for the side. Furthermore, each integral would require parameterizing the corresponding surface, calculating tangent vectors and their cross product..

By contrast, the divergence theorem allows us to calculate the single triple integral

$$\iiint_E \operatorname{div} \vec{\mathbf{F}}\, dV,$$

where E is the solid enclosed by the cylinder. Using the divergence theorem (Equation 16.8.1) and converting to cylindrical coordinates, we have

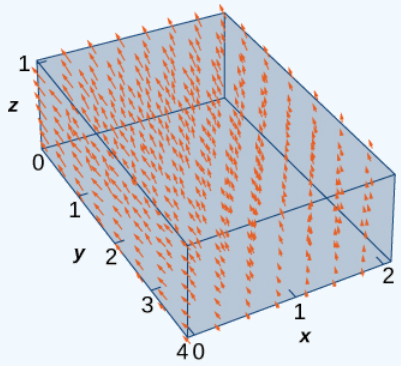
$$\begin{aligned}
\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} &= \iiint_E \operatorname{div} \vec{\mathbf{F}}\, dV, \\
&= \iiint_E (x^2 + y^2 + 1)\, dV \\
&= \int_0^{2\pi} \int_0^1 \int_0^2 (r^2 + 1)\, r\, dz\, dr\, d\theta \\
&= \frac{3}{2} \int_0^{2\pi} d\theta \\
&= 3\pi.
\end{aligned}$$

Exercise 16.8.2

Use the divergence theorem to calculate flux integral

$$\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}},$$

where S is the boundary of the box given by $0 \le x \le 2,\ 0 \le y \le 4,\ 0 \le z \le 1$ and $\vec{\mathbf{F}} = \langle x^2 + yz,\ y - z,\ 2x + 2y + 2z \rangle$ (see the following figure).



Hint

Calculate the corresponding triple integral.

Answer

40

Example 16.8.3: Applying the Divergence Theorem

Let $\vec{\mathbf{v}} = \langle -\frac{y}{z},\ \frac{x}{z},\ 0 \rangle$ be the velocity field of a fluid. Let C be the solid cube given by $1 \le x \le 4,\ 2 \le y \le 5,\ 1 \le z \le 4$, and let S be the boundary of this cube (see the following figure). Find the flow rate of the fluid across S.

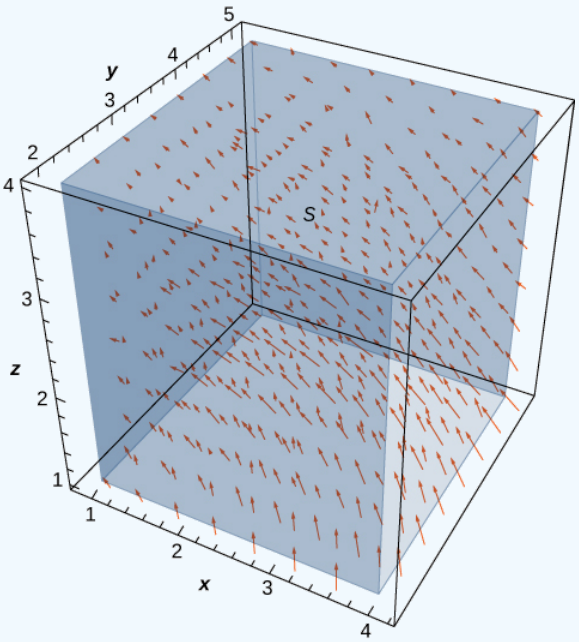


Figure 16.8.4: Vector field $\vec{\mathbf{v}} = \langle -\frac{y}{z},\ \frac{x}{z},\ 0 \rangle$.

Solution

The flow rate of the fluid across S is $\iint_S \vec{\mathbf{v}} \cdot d\vec{\mathbf{S}}$. Before calculating this flux integral, let's discuss what the value of the integral should be. Based on Figure 16.8.4, we see that if we place this cube in the fluid (as long as the cube doesn't encompass the origin), then the rate of fluid entering the cube is the same as the rate of fluid exiting the cube. The field is rotational in nature and, for a given circle parallel to the xy-plane that has a center on the z-axis, the vectors along that circle are all the same magnitude. That is how we can see that the flow rate is the same entering and exiting the cube. The flow into the cube cancels with the flow out of the cube, and therefore the flow rate of the fluid across the cube should be zero.

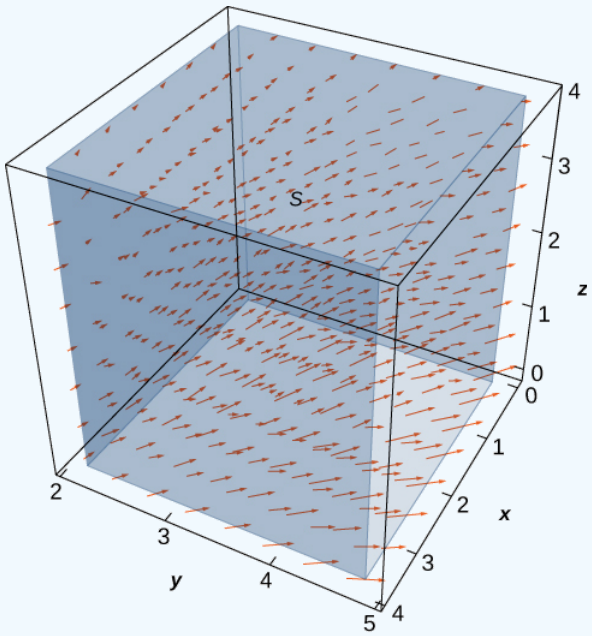
To verify this intuition, we need to calculate the flux integral. Calculating the flux integral directly requires breaking the flux integral into six separate flux integrals, one for each face of the cube. We also need to find tangent vectors, compute their cross product. However, using the divergence theorem makes this calculation go much more quickly:

$$\begin{aligned} \iint_S \vec{\mathbf{v}} \cdot d\vec{\mathbf{S}} &= \iiint_C \operatorname{div} \vec{\mathbf{v}}\, dV \\ &= \iiint_C 0\, dV = 0. \end{aligned}$$

Therefore the flux is zero, as expected.

? Exercise 16.8.3

Let $\vec{\mathbf{v}} = \langle \frac{x}{z}, \frac{y}{z}, 0 \rangle$ be the velocity field of a fluid. Let C be the solid cube given by $1 \le x \le 4$, $2 \le y \le 5$, $1 \le z \le 4$, and let S be the boundary of this cube (see the following figure). Find the flow rate of the fluid across S.



Hint

Use the divergence theorem and calculate a triple integral

Answer

$9 \ln(16)$

Example illustrates a remarkable consequence of the divergence theorem. Let S be a piecewise, smooth closed surface and let $\vec{\mathbf{F}}$ be a vector field defined on an open region containing the surface enclosed by S. If $\vec{\mathbf{F}}$ has the form $F = \langle f(y,z),\ g(x,z),\ h(x,y) \rangle$, then the divergence of $\vec{\mathbf{F}}$ is zero. By the divergence theorem, the flux of $\vec{\mathbf{F}}$ across S is also zero. This makes certain flux integrals incredibly easy to calculate. For example, suppose we wanted to calculate the flux integral $\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$ where S is a cube and

$$\vec{\mathbf{F}} = \langle \sin(y)\, e^{yz},\ x^2 z^2,\ \cos(xy)\, e^{\sin x} \rangle.$$

Calculating the flux integral directly would be difficult, if not impossible, using techniques we studied previously. At the very least, we would have to break the flux integral into six integrals, one for each face of the cube. But, because the divergence of this field is zero, the divergence theorem immediately shows that the flux integral is zero.

We can now use the divergence theorem to justify the physical interpretation of divergence that we discussed earlier. Recall that if $\vec{\mathbf{F}}$ is a continuous three-dimensional vector field and P is a point in the domain of $\vec{\mathbf{F}}$, then the divergence of $\vec{\mathbf{F}}$ at P is a measure of the "outflowing-ness" of $\vec{\mathbf{F}}$ at P. If $\vec{\mathbf{F}}$ represents the velocity field of a fluid, then the divergence of $\vec{\mathbf{F}}$ at P is a measure of the net flow rate out of point P (the flow of fluid out of P less the flow of fluid in to P). To see how the divergence theorem justifies this interpretation, let B_τ be a ball of very small radius r with center P, and assume that B_τ is in the domain of $\vec{\mathbf{F}}$. Furthermore, assume that B_τ has a positive, outward orientation. Since the radius of B_τ is small and $\vec{\mathbf{F}}$ is continuous, the divergence of $\vec{\mathbf{F}}$ is approximately constant on B_τ. That is, ifv P' is any point in B_τ, then $\text{div } \vec{\mathbf{F}}(P) \approx \text{div } \vec{\mathbf{F}}(P')$. Let S_τ denote the boundary sphere of B_τ. We can approximate the flux across S_τ using the divergence theorem as follows:

$$\begin{aligned} \iint_{S_\tau} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}} &= \iiint_{B_\tau} \text{div } \vec{\mathbf{F}}\, dV \\ &\approx \iiint_{B_\tau} \text{div } \vec{\mathbf{F}}(P)\, dV \\ &= \text{div } \vec{\mathbf{F}}(P)\, V(B_\tau). \end{aligned}$$

As we shrink the radius r to zero via a limit, the quantity $\text{div } \vec{\mathbf{F}}(P)\, V(B_\tau)$ gets arbitrarily close to the flux. Therefore,

$$\text{div } \vec{\mathbf{F}}(P) = \lim_{\tau \to 0} \frac{1}{V(B_\tau)} \iint_{S_\tau} \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

and we can consider the divergence at P as measuring the net rate of outward flux per unit volume at P. Since "outflowing-ness" is an informal term for the net rate of outward flux per unit volume, we have justified the physical interpretation of divergence we discussed earlier, and we have used the divergence theorem to give this justification.

Application to Electrostatic Fields

The divergence theorem has many applications in physics and engineering. It allows us to write many physical laws in both an integral form and a differential form (in much the same way that Stokes' theorem allowed us to translate between an integral and differential form of Faraday's law). Areas of study such as fluid dynamics, electromagnetism, and quantum mechanics have equations that describe the conservation of mass, momentum, or energy, and the divergence theorem allows us to give these equations in both integral and differential forms.

One of the most common applications of the divergence theorem is to **electrostatic fields**. An important result in this subject is **Gauss' law**. This law states that if S is a closed surface in electrostatic field $\vec{\mathbf{E}}$, then the flux of $\vec{\mathbf{E}}$ across S is the total charge enclosed by S (divided by an electric constant). We now use the divergence theorem to justify the special case of this law in which the electrostatic field is generated by a stationary point charge at the origin.

If (x, y, z) is a point in space, then the distance from the point to the origin is $r = \sqrt{x^2 + y^2 + z^2}$. Let $\vec{\mathbf{F}}_\tau$ denote radial vector field $\vec{\mathbf{F}}_\tau = \frac{1}{\tau^2} \left\langle \frac{x}{\tau}, \frac{y}{\tau}, \frac{z}{\tau} \right\rangle$.The vector at a given position in space points in the direction of unit radial vector $\left\langle \frac{x}{\tau}, \frac{y}{\tau}, \frac{z}{\tau} \right\rangle$ and is scaled by the quantity $1/\tau^2$. Therefore, the magnitude of a vector at a given point is inversely proportional to the square of the vector's distance from the origin. Suppose we have a stationary charge of q Coulombs at the origin, existing in a vacuum. The charge generates electrostatic field $\vec{\mathbf{E}}$ given by

$$\vec{\mathbf{E}} = \frac{q}{4\pi\epsilon_0} \vec{\mathbf{F}}_\tau,$$

where the approximation $\epsilon_0 = 8.854 \times 10^{-12}$ farad (F)/m is an electric constant. (The constant ϵ_0 is a measure of the resistance encountered when forming an electric field in a vacuum.) Notice that $\vec{\mathbf{E}}$ is a radial vector field similar to the gravitational field described in [link]. The difference is that this field points outward whereas the gravitational field points inward. Because

$$\vec{\mathbf{E}} = \frac{q}{4\pi\epsilon_0}\vec{\mathbf{F}}_\tau = \frac{q}{4\pi\epsilon_0}\left(\frac{1}{\tau^2}\left\langle\frac{x}{\tau}, \frac{y}{\tau}, \frac{z}{\tau}\right\rangle\right),$$

we say that electrostatic fields obey an inverse-square law. That is, the electrostatic force at a given point is inversely proportional to the square of the distance from the source of the charge (which in this case is at the origin). Given this vector field, we show that the flux across closed surface S is zero if the charge is outside of S, and that the flux is $q/epsilon_0$ if the charge is inside of S. In other words, the flux across S is the charge inside the surface divided by constant ϵ_0. This is a special case of Gauss' law, and here we use the divergence theorem to justify this special case.

To show that the flux across S is the charge inside the surface divided by constant ϵ_0, we need two intermediate steps. First we show that the divergence of $\vec{\mathbf{F}}_\tau$ is zero and then we show that the flux of $\vec{\mathbf{F}}_\tau$ across any smooth surface S is either zero or 4π. We can then justify this special case of Gauss' law.

Example 16.8.4: The Divergence of F_τ is Zero

Verify that the divergence of $\vec{\mathbf{F}}_\tau$ is zero where $\vec{\mathbf{F}}_\tau$ is defined (away from the origin).

Solution

Since $\tau = \sqrt{x^2+y^2+z^2}$, the quotient rule gives us

$$\begin{aligned}\frac{\partial}{\partial x}\left(\frac{x}{\tau^3}\right) &= \frac{\partial}{\partial x}\left(\frac{x}{(x^2+y^2+z^2)^{3/2}}\right)\\ &= \frac{(x^2+y^2+z^2)^{3/2} - x\left[\frac{3}{2}(x^2+y^2+z^2)^{1/2}2x\right]}{(x^2+y^2+z^2)^3}\\ &= \frac{\tau^3 - 3x^2\tau}{\tau^6} = \frac{\tau^2-3x^2}{\tau^5}.\end{aligned}$$

Similarly,

$$\frac{\partial}{\partial y}\left(\frac{y}{\tau^3}\right) = \frac{\tau^2-3y^2}{\tau^5} \; and \; \frac{\partial}{\partial z}\left(\frac{z}{\tau^3}\right) = \frac{\tau^2-3z^2}{\tau^5}.$$

Therefore,

$$\begin{aligned}\text{div } \vec{\mathbf{F}}_\tau &= \frac{\tau^2-3x^2}{\tau^5} + \frac{\tau^2-3y^2}{\tau^5} + \frac{\tau^2-3z^2}{\tau^5}\\ &= \frac{3\tau^2 - 3(x^2+y^2+z^2)}{\tau^5}\\ &= \frac{3\tau^2-3\tau^2}{\tau^5} = 0.\end{aligned}$$

Notice that since the divergence of $\vec{\mathbf{F}}_\tau$ is zero and $\vec{\mathbf{E}}$ is $\vec{\mathbf{F}}_\tau$ scaled by a constant, the divergence of electrostatic field $\vec{\mathbf{E}}$ is also zero (except at the origin).

Flux across a Smooth Surface

Let S be a connected, piecewise smooth closed surface and let $\vec{\mathbf{F}}_\tau = \frac{1}{\tau^2}\left\langle\frac{x}{\tau}, \frac{y}{\tau}, \frac{z}{\tau}\right\rangle$. Then,

$$\iint_S \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}} = \begin{cases}0, & \text{if } S \text{ does not encompass the origin}\\ 4\pi, & \text{if } S \text{ encompasses the origin.}\end{cases}$$

In other words, this theorem says that the flux of $\vec{\mathbf{F}}_\tau$ across any piecewise smooth closed surface S depends only on whether the origin is inside of S.

Proof

The logic of this proof follows the logic of [link], only we use the divergence theorem rather than Green's theorem.

First, suppose that S does not encompass the origin. In this case, the solid enclosed by S is in the domain of $\vec{\mathbf{F}}_\tau$, and since the divergence of $\vec{\mathbf{F}}_\tau$ is zero, we can immediately apply the divergence theorem and find that

$$\iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

is zero.

Now suppose that S does encompass the origin. We cannot just use the divergence theorem to calculate the flux, because the field is not defined at the origin. Let S_a be a sphere of radius a inside of S centered at the origin. The outward normal vector field on the sphere, in spherical coordinates, is

$$\vec{\mathbf{t}}_\phi \times \vec{\mathbf{t}}_\theta = \langle a^2 \cos\theta \sin^2\phi,\ a^2 \sin\theta \sin^2\phi,\ a^2 \sin\phi \cos\phi \rangle$$

(see [link]). Therefore, on the surface of the sphere, the dot product $\vec{\mathbf{F}}_\tau \cdot \vec{\mathbf{N}}$ (in spherical coordinates) is

$$\begin{aligned} \vec{\mathbf{F}}_\tau \cdot \vec{\mathbf{N}} &= \left\langle \frac{\sin\phi\cos\theta}{a^2}, \frac{\sin\phi\sin\theta}{a^2}, \frac{\cos\phi}{a^2} \right\rangle \cdot \langle a^2\cos\theta\sin^2\phi, a^2\sin\theta\sin^2\phi,\ a^2\sin\phi\cos\phi\rangle \\ &= \sin\phi(\langle \sin\phi\cos\theta,\ \sin\phi\sin\theta,\ \cos\phi\rangle \cdot \langle \sin\phi\cos\theta, \sin\phi\sin\theta,\ \cos\phi\rangle) \\ &= \sin\phi. \end{aligned}$$

The flux of $\vec{\mathbf{F}}_\tau$ across S_a is

$$\iint_{S_a} \vec{\mathbf{F}}_\tau \cdot \vec{\mathbf{N}} dS = \int_0^{2\pi}\int_0^{\pi} \sin\phi\, d\phi\, d\theta = 4\pi.$$

Now, remember that we are interested in the flux across S, not necessarily the flux across S_a. To calculate the flux across S, let E be the solid between surfaces S_a and S. Then, the boundary of E consists of S_a and S. Denote this boundary by $S - S_a$ to indicate that S is oriented outward but now S_a is oriented inward. We would like to apply the divergence theorem to solid E. Notice that the divergence theorem, as stated, can't handle a solid such as E because E has a hole. However, the divergence theorem can be extended to handle solids with holes, just as Green's theorem can be extended to handle regions with holes. This allows us to use the divergence theorem in the following way. By the divergence theorem,

$$\begin{aligned} \iint_{S-S_a} \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}} &= \iint_S \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}} - \iint_{S_a} \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}} \\ &= \iiint_E \operatorname{div} \vec{\mathbf{F}}_\tau\, dV \\ &= \iiint_E 0\, dV = 0. \end{aligned}$$

Therefore,

$$\iint_S \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}} = \iint_{S_a} \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}} = 4\pi,$$

and we have our desired result.

□

Now we return to calculating the flux across a smooth surface in the context of electrostatic field $\vec{\mathbf{E}} = \frac{q}{4\pi\epsilon_0}\vec{\mathbf{F}}_\tau$ of a point charge at the origin. Let S be a piecewise smooth closed surface that encompasses the origin. Then

$$\iint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}} = \iint_S \frac{q}{4\pi\epsilon_0} \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}}$$
$$= \frac{q}{4\pi\epsilon_0} \iint_S \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}}$$
$$= \frac{q}{\epsilon_0}.$$

If S does not encompass the origin, then

$$\iint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}} = \frac{q}{4\pi\epsilon_0} \iint_S \vec{\mathbf{F}}_\tau \cdot d\vec{\mathbf{S}} = 0.$$

Therefore, we have justified the claim that we set out to justify: the flux across closed surface S is zero if the charge is outside of S, and the flux is q/ϵ_0 if the charge is inside of S.

This analysis works only if there is a single point charge at the origin. In this case, Gauss' law says that the flux of $\vec{\mathbf{E}}$ across S is the total charge enclosed by S. Gauss' law can be extended to handle multiple charged solids in space, not just a single point charge at the origin. The logic is similar to the previous analysis, but beyond the scope of this text. In full generality, Gauss' law states that if S is a piecewise smooth closed surface and Q is the total amount of charge inside of S, then the flux of $\vec{\mathbf{E}}$ across S is Q/ϵ_0.

Example 16.8.5: Using Gauss' law

Suppose we have four stationary point charges in space, all with a charge of 0.002 Coulombs (C). The charges are located at $(0,0,1)$, $(1,1,4)$, $(-1,0,0)$, and $(-2,-2,2)$. Let $\vec{\mathbf{E}}$ denote the electrostatic field generated by these point charges. If S is the sphere of radius 2 oriented outward and centered at the origin, then find

$$\iint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}}.$$

Solution

According to Gauss' law, the flux of $\vec{\mathbf{E}}$ across S is the total charge inside of S divided by the electric constant. Since S has radius 2, notice that only two of the charges are inside of S: the charge at $(0,1,1)$ and the charge at $(-1,0,0)$. Therefore, the total charge encompassed by S is 0.004 and, by Gauss' law,

$$\iint_S \vec{\mathbf{E}} \cdot d\vec{\mathbf{S}} = \frac{0.004}{8.854 \times 10^{-12}} \approx 4.418 \times 10^9\, V - m.$$

Exercise 16.8.4

Work the previous example for surface S that is a sphere of radius 4 centered at the origin, oriented outward.

Hint

Use Gauss' law.

Answer

$\approx 6.777 \times 10^9$

Key Concepts

- The divergence theorem relates a surface integral across closed surface S to a triple integral over the solid enclosed by S. The divergence theorem is a higher dimensional version of the flux form of Green's theorem, and is therefore a higher dimensional version of the Fundamental Theorem of Calculus.
- The divergence theorem can be used to transform a difficult flux integral into an easier triple integral and vice versa.
- The divergence theorem can be used to derive Gauss' law, a fundamental law in electrostatics.

Key Equations

- **Divergence theorem**

$$\iiint_E \text{div } \vec{\mathbf{F}}\, dV = \iint_S \vec{\mathbf{F}} \cdot d\vec{\mathbf{S}}$$

Glossary

divergence theorem
a theorem used to transform a difficult flux integral into an easier triple integral and vice versa

Gauss' law
if S is a piecewise, smooth closed surface in a vacuum and Q is the total stationary charge inside of S, then the flux of electrostatic field $\vec{\mathbf{E}}$ across S is Q/ϵ_0

inverse-square law
the electrostatic force at a given point is inversely proportional to the square of the distance from the source of the charge

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16.8E: Exercises for Section 16.8

For exercises 1 - 9, use a computer algebraic system (CAS) and the divergence theorem to evaluate surface integral $\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, ds$ for the given choice of $\vec{\mathbf{F}}$ and the boundary surface S. For each closed surface, assume $\vec{\mathbf{N}}$ is the outward unit normal vector.

1. [T] $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$; S is the surface of cube $0 \le x \le 1$, $0 \le y \le 1$, $0 < z \le 1$.

2. [T] $\vec{\mathbf{F}}(x, y, z) = (\cos yz)\,\hat{\mathbf{i}} + e^{xz}\,\hat{\mathbf{j}} + 3z^2\,\hat{\mathbf{k}}$; S is the surface of hemisphere $z = \sqrt{4 - x^2 - y^2}$ together with disk $x^2 + y^2 \le 4$ in the xy-plane.

Answer

$$\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, ds = 75.3982$$

3. [T] $\vec{\mathbf{F}}(x, y, z) = (x^2 + y^2 - x^2)\,\hat{\mathbf{i}} + x^2 y\,\hat{\mathbf{j}} + 3z\,\hat{\mathbf{k}}$; S is the surface of the five faces of unit cube $0 \le x \le 1$, $0 \le y \le 1$, $0 < z \le 1$.

4. [T] $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$; S is the surface of paraboloid $z = x^2 + y^2$ for $0 \le z \le 9$.

Answer

$$\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, ds = 127.2345$$

5. [T] $\vec{\mathbf{F}}(x, y, z) = x^2\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}} + z^2\,\hat{\mathbf{k}}$; S is the surface of sphere $x^2 + y^2 + z^2 = 4$.

6. [T] $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + (z^2 - 1)\,\hat{\mathbf{k}}$; S is the surface of the solid bounded by cylinder $x^2 + y^2 = 4$ and planes $z = 0$ and $z = 1$.

Answer

$$\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, ds = 37.699$$

7. [T] $\vec{\mathbf{F}}(x, y, z) = x^2\,\hat{\mathbf{i}} + y^2\,\hat{\mathbf{j}} + z^2\,\hat{\mathbf{k}}$; S is the surface bounded above by sphere $\rho = 2$ and below by cone $\varphi = \frac{\pi}{4}$ in spherical coordinates. (Think of S as the surface of an "ice cream cone.")

8. [T] $\vec{\mathbf{F}}(x, y, z) = x^3\,\hat{\mathbf{i}} + y^3\,\hat{\mathbf{j}} + 3a^2 z\,\hat{\mathbf{k}}$ $(constant\ a > 0)$; S is the surface bounded by cylinder $x^2 + y^2 = a^2$ and planes $z = 0$ and $z = 1$.

Answer

$$\int_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, ds = \frac{9\pi a^4}{2}$$

9. [T] Surface integral $\iint_S \vec{\mathbf{F}} \cdot dS$, where S is the solid bounded by paraboloid $z = x^2 + y^2$ and plane $z = 4$, and $\vec{\mathbf{F}}(x, y, z) = (x + y^2 z^2)\,\hat{\mathbf{i}} + (y + z^2 x^2)\,\hat{\mathbf{j}} + (z + x^2 y^2)\,\hat{\mathbf{k}}$

10. Use the divergence theorem to calculate surface integral $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = (e^{y^2}\,\hat{\mathbf{i}} + (y + \sin(z^2))\,\hat{\mathbf{j}} + (z - 1)\,\hat{\mathbf{k}}$ and S is upper hemisphere $x^2 + y^2 + z^2 = 1$, $z \ge 0$, oriented upward.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = \frac{\pi}{3}$$

11. Use the divergence theorem to calculate surface integral $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = x^4\,\hat{\mathbf{i}} - x^3 z^2\,\hat{\mathbf{j}} + 4xy^2 z\,\hat{\mathbf{k}}$ and S is the surface bounded by cylinder $x^2 + y^2 = 1$ and planes $z = x + 2$ and $z = 0$.

12. Use the divergence theorem to calculate surface integral $\iint_S \vec{\mathbf{F}} \cdot dS$, when $\vec{\mathbf{F}}(x, y, z) = x^2 z^3\,\hat{\mathbf{i}} + 2xyz^3\,\hat{\mathbf{j}} + xz^4\,\hat{\mathbf{k}}$ and S is the surface of the box with vertices $(\pm 1,\ \pm 2,\ \pm 3)$.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = 0$$

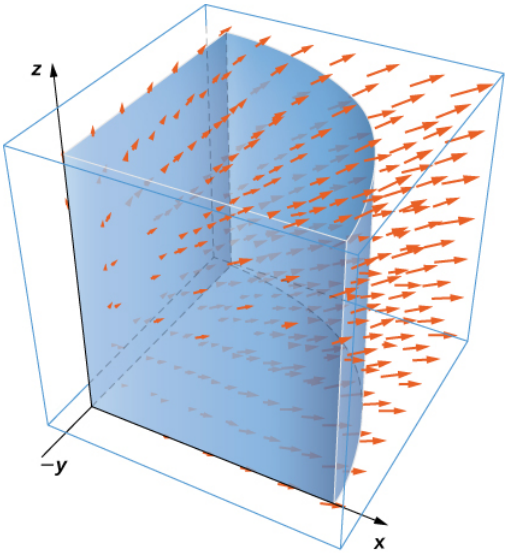
13. Use the divergence theorem to calculate surface integral $\iint_S \vec{\mathbf{F}} \cdot dS$, when $\vec{\mathbf{F}}(x, y, z) = z \tan^{-1}(y^2)\,\hat{\mathbf{i}} + z^3 \ln(x^2 + 1)\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ and S is a part of paraboloid $x^2 + y^2 + z = 2$ that lies above plane $z = 1$ and is oriented upward.

14. [T] Use a CAS and the divergence theorem to calculate flux $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = (x^3 + y^3)\,\hat{\mathbf{i}} + (y^3 + z^3)\,\hat{\mathbf{j}} + (z^3 + x^3)\,\hat{\mathbf{k}}$ and S is a sphere with center $(0, 0)$ and radius 2.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = 241.2743$$

15. Use the divergence theorem to compute the value of flux integral $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = (y^3 + 3x)\,\hat{\mathbf{i}} + (xz + y)\,\hat{\mathbf{j}} + (z + x^4 \cos(x^2 y))\,\hat{\mathbf{k}}$ and S is the area of the region bounded by $x^2 + y^2 = 1,\ x \geq 0,\ y \geq 0$, and $0 \leq z \leq 1$.



16. Use the divergence theorem to compute flux integral $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = y\,\hat{\mathbf{j}} - z\,\hat{\mathbf{k}}$ and S consists of the union of paraboloid $y = x^2 + z^2,\ 0 \leq y \leq 1$, and disk $x^2 + z^2 \leq 1,\ y = 1$, oriented outward. What is the flux through just the paraboloid?

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = -\pi$$

17. Use the divergence theorem to compute flux integral $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z^4\,\hat{\mathbf{k}}$ and S is a part of cone $z = \sqrt{x^2 + y^2}$ beneath top plane $z = 1$ oriented downward.

18. Use the divergence theorem to calculate surface integral $\iint_S \vec{\mathbf{F}} \cdot dS$ for $\vec{\mathbf{F}}(x, y, z) = x^4\,\hat{\mathbf{i}} - x^3 z^2\,\hat{\mathbf{j}} + 4xy^2 z\,\hat{\mathbf{k}}$, where S is the surface bounded by cylinder $x^2 + y^2 = 1$ and planes $z = x + 2$ and $z = 0$.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = \frac{2\pi}{3}$$

19. Consider $\vec{\mathbf{F}}(x, y, z) = x^2\,\hat{\mathbf{i}} + xy\,\hat{\mathbf{j}} + (z + 1)\,\hat{\mathbf{k}}$. Let E be the solid enclosed by paraboloid $z = 4 - x^2 - y^2$ and plane $z = 0$ with normal vectors pointing outside E. Compute flux $\vec{\mathbf{F}}$ across the boundary of E using the divergence theorem.

In exercises 20 - 23, use a CAS along with the divergence theorem to compute the net outward flux for the fields across the given surfaces S.

20. [T] $\vec{\mathbf{F}} = \langle x, -2y, 3z\rangle$; S is sphere $\{(x, y, z) : x^2 + y^2 + z^2 = 6\}$.

Answer

$15\sqrt{6}\pi$

21. [T] $\vec{\mathbf{F}} = \langle x, 2y, z\rangle$; S is the boundary of the tetrahedron in the first octant formed by plane $x + y + z = 1$.

22. [T] $\vec{\mathbf{F}} = \langle y - 2x, x^3 - y, y^2 - z\rangle$; S is sphere $\{(x, y, z) : x^2 + y^2 + z^2 = 4\}$.

Answer

$-\frac{128}{3}\pi$

23. [T] $\vec{\mathbf{F}} = \langle x, y, z\rangle$; S is the surface of paraboloid $z = 4 - x^2 - y^2$, for $z \geq 0$, plus its base in the xy-plane.

For exercises 24 - 26, use a CAS and the divergence theorem to compute the net outward flux for the vector fields across the boundary of the given regions D.

24. [T] $\vec{\mathbf{F}} = \langle z - x, x - y, 2y - z\rangle$; D is the region between spheres of radius 2 and 4 centered at the origin.

Answer

-703.7168

25. [T] $\vec{\mathbf{F}} = \dfrac{\vec{\mathbf{r}}}{\|\vec{\mathbf{r}}\|} = \dfrac{\langle x, y, z\rangle}{\sqrt{x^2 + y^2 + z^2}}$; D is the region between spheres of radius 1 and 2 centered at the origin.

26. [T] $\vec{\mathbf{F}} = \langle x^2, -y^2, z^2\rangle$; D is the region in the first octant between planes $z = 4 - x - y$ and $z = 2 - x - y$.

Answer

20

27. Let $\vec{\mathbf{F}}(x, y, z) = 2x\,\hat{\mathbf{i}} - 3xy\,\hat{\mathbf{j}} + xz^2\,\hat{\mathbf{k}}$. Use the divergence theorem to calculate $\iint_S \vec{\mathbf{F}} \cdot dS$, where S is the surface of the cube with corners at $(0, 0, 0)$, $(1, 0, 0)$, $(0, 1, 0)$, $(1, 1, 0)$, $(0, 0, 1)$, $(1, 0, 1)$, $(0, 1, 1)$, and $(1, 1, 1)$, oriented outward.

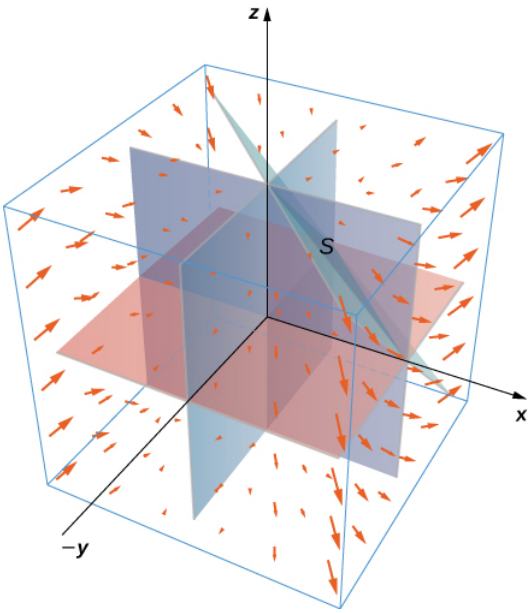
28. Use the divergence theorem to find the outward flux of field $\vec{\mathbf{F}}(x, y, z) = (x^3 - 3y)\,\hat{\mathbf{i}} + (2yz + 1)\,\hat{\mathbf{j}} + xyz\,\hat{\mathbf{k}}$ through the cube bounded by planes $x = \pm 1$, $y = \pm 1$, and $z = \pm 1$.

Answer

$\iint_S \vec{\mathbf{F}} \cdot dS = 8$

29. Let $\vec{\mathbf{F}}(x, y, z) = 2x\,\hat{\mathbf{i}} - 3y\,\hat{\mathbf{j}} + 5z\,\hat{\mathbf{k}}$ and let S be hemisphere $z = \sqrt{9 - x^2 - y^2}$ together with disk $x^2 + y^2 \leq 9$ in the xy-plane. Use the divergence theorem.

30. Evaluate $\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, dS$, where $\vec{\mathbf{F}}(x, y, z) = x^2\,\hat{\mathbf{i}} + xy\,\hat{\mathbf{j}} + x^3y^3\,\hat{\mathbf{k}}$ and S is the surface consisting of all faces except the tetrahedron bounded by plane $x + y + z = 1$ and the coordinate planes, with outward unit normal vector $\vec{\mathbf{N}}$.



Answer

$$\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, dS = \frac{1}{8}$$

31. Find the net outward flux of field $\vec{\mathbf{F}} = \langle bz - cy,\ cx - az,\ ay - bx \rangle$ across any smooth closed surface in R^3 where a, b, and c are constants.

32. Use the divergence theorem to evaluate $\iint_S ||\vec{\mathbf{R}}||\vec{\mathbf{R}} \cdot \vec{\mathbf{n}}\, ds$, where $\vec{\mathbf{R}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ and S is sphere $x^2 + y^2 + z^2 = a^2$, with constant $a > 0$.

Answer

$$\iint_S ||\vec{\mathbf{R}}||\vec{\mathbf{R}} \cdot \vec{\mathbf{n}}\, ds = 4\pi a^4$$

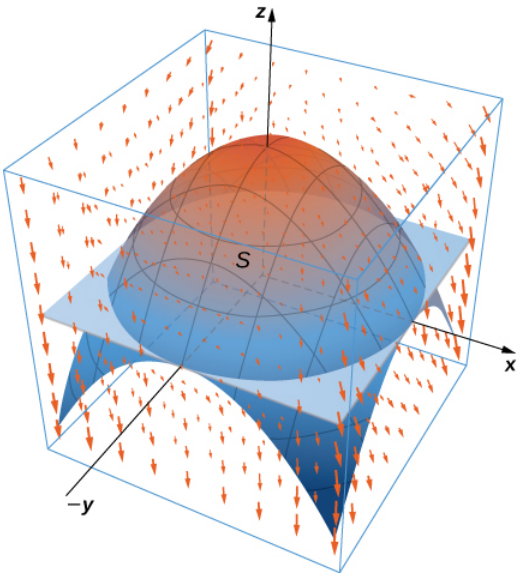
33. Use the divergence theorem to evaluate $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = y^2 z\,\hat{\mathbf{i}} + y^3\,\hat{\mathbf{j}} + xz\,\hat{\mathbf{k}}$ and S is the boundary of the cube defined by $-1 \le x \le 1$, $-1 \le y \le 1$, and $0 \le z \le 2$.

34. Let R be the region defined by $x^2 + y^2 + z^2 \le 1$. Use the divergence theorem to find $\iiint_R z^2\, dV$.

Answer

$$\iiint_R z^2 dV = \frac{4\pi}{15}$$

35. Let E be the solid bounded by the xy-plane and paraboloid $z = 4 - x^2 - y^2$ so that S is the surface of the paraboloid piece together with the disk in the xy-plane that forms its bottom. If $\vec{\mathbf{F}}(x, y, z) = (xz\sin(yz) + x^3)\,\hat{\mathbf{i}} + \cos(yz)\,\hat{\mathbf{j}} + (3zy^2 - e^{x^2+y^2})\,\hat{\mathbf{k}}$, find $\iint_S \vec{\mathbf{F}} \cdot dS$ using the divergence theorem.



36. Let E be the solid unit cube with diagonally opposite corners at the origin and $(1, 1, 1)$, and faces parallel to the coordinate planes. Let S be the surface of E, oriented with the outward-pointing normal. Use a CAS to find $\iint_S \vec{\mathbf{F}} \cdot dS$ using the divergence theorem if $\vec{\mathbf{F}}(x, y, z) = 2xy\,\hat{\mathbf{i}} + 3ye^z\,\hat{\mathbf{j}} + x \sin z\,\hat{\mathbf{k}}$.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = 6.5759$$

37. Use the divergence theorem to calculate the flux of $\vec{\mathbf{F}}(x, y, z) = x^3\,\hat{\mathbf{i}} + y^3\,\hat{\mathbf{j}} + z^3\,\hat{\mathbf{k}}$ through sphere $x^2 + y^2 + z^2 = 1$.

38. Find $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x, y, z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ and S is the outwardly oriented surface obtained by removing cube $[1, 2] \times [1, 2] \times [1, 2]$ from cube $[0, 2] \times [0, 2] \times [0, 2]$.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = 21$$

39. Consider radial vector field $\vec{\mathbf{F}} = \dfrac{\vec{\mathbf{r}}}{\|\vec{\mathbf{r}}\|} = \dfrac{\langle x, y, z \rangle}{(x^2 + y^2 + z^2)^{1/2}}$. Compute the surface integral, where S is the surface of a sphere of radius *a* centered at the origin.

40. Compute the flux of water through parabolic cylinder $S : y = x^2$, from $0 \le x \le 2$, $0 \le z \le 3$, if the velocity vector is $\vec{\mathbf{F}}(x, y, z) = 3z^2\,\hat{\mathbf{i}} + 6\,\hat{\mathbf{j}} + 6xz\,\hat{\mathbf{k}}$.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = 72$$

41. [T] Use a CAS to find the flux of vector field $\vec{\mathbf{F}}(x, y, z) = z\,\hat{\mathbf{i}} + z\,\hat{\mathbf{j}} + \sqrt{x^2 + y^2}\,\hat{\mathbf{k}}$ across the portion of hyperboloid $x^2 + y^2 = z^2 + 1$ between planes $z = 0$ and $z = \dfrac{\sqrt{3}}{3}$, oriented so the unit normal vector points away from the z-axis.

42. Use a CAS to find the flux of vector field $\vec{\mathbf{F}}(x, y, z) = (e^y + x)\,\hat{\mathbf{i}} + (3\cos(xz) - y)\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ through surface S, where S is given by $z^2 = 4x^2 + 4y^2$ from $0 \le z \le 4$, oriented so the unit normal vector points downward.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = -33.5103$$

43. [T] Use a CAS to compute $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x,y,z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + 2z\,\hat{\mathbf{k}}$ and S is a part of sphere $x^2 + y^2 + z^2 = 2$ with $0 \le z \le 1$.

44. Evaluate $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x,y,z) = bxy^2\,\hat{\mathbf{i}} + bx^2y\,\hat{\mathbf{j}} + (x^2+y^2)z^2\,\hat{\mathbf{k}}$ and S is a closed surface bounding the region and consisting of solid cylinder $x^2 + y^2 \le a^2$ and $0 \le z \le b$.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = \pi a^4 b^2$$

45. [T] Use a CAS to calculate the flux of $\vec{\mathbf{F}}(x,y,z) = (x^3 + y\,\sin z)\,\hat{\mathbf{i}} + (y^3 + z\,\sin x)\,\hat{\mathbf{j}} + 3z\,\hat{\mathbf{k}}$ across surface S, where S is the boundary of the solid bounded by hemispheres $z = \sqrt{4 - x^2 - y^2}$ and $z = \sqrt{1 - x^2 - y^2}$, and plane $z = 0$.

46. Use the divergence theorem to evaluate $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x,y,z) = xy\,\hat{\mathbf{i}} - \frac{1}{2}y^2\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ and S is the surface consisting of three pieces: $z = 4 - 3x^2 - 3y^2,\ 1 \le z \le 4$ on the top; $x^2 + y^2 = 1,\ 0 \le z \le 1$ on the sides; and $z = 0$ on the bottom.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = \frac{5}{2}\pi$$

47. [T] Use a CAS and the divergence theorem to evaluate $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x,y,z) = (2x + y\,\cos z)\,\hat{\mathbf{i}} + (x^2 - y)\,\hat{\mathbf{j}} + y^2 z\,\hat{\mathbf{k}}$ and S is sphere $x^2 + y^2 + z^2 = 4$ orientated outward.

48. Use the divergence theorem to evaluate $\iint_S \vec{\mathbf{F}} \cdot dS$, where $\vec{\mathbf{F}}(x,y,z) = x\,\hat{\mathbf{i}} + y\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$ and S is the boundary of the solid enclosed by paraboloid $y = x^2 + z^2 - 2$, cylinder $x^2 + z^2 = 1$, and plane $x + y = 2$, and S is oriented outward.

Answer

$$\iint_S \vec{\mathbf{F}} \cdot dS = \frac{21\pi}{2}$$

For the following exercises, **Fourier's law of heat transfer** states that the heat flow vector $\vec{\mathbf{F}}$ at a point is proportional to the negative gradient of the temperature; that is, $\vec{\mathbf{F}} = -k\vec{\nabla}T$, which means that heat energy flows hot regions to cold regions. The constant $k > 0$ is called the *conductivity*, which has metric units of joules per meter per second-kelvin or watts per meter-kelvin. A temperature function for region D s given. Use the divergence theorem to find net outward heat flux $\iint_S \vec{\mathbf{F}} \cdot \vec{\mathbf{n}}\, dS = -k \iint_S \vec{\nabla}T \cdot N\, dS$ across the boundary S of D, where $k = 1$.

49. $T(x,y,z) = 100 + x + 2y + z$;

$D = \{(x,y,z) : 0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le z \le 1\}$

50. $T(x,y,z) = 100 + e^{-z}$;

$D = \{(x,y,z) : 0 \le x \le 1,\ 0 \le y \le 1,\ 0 \le z \le 1\}$

Answer

$-(1 - e^{-1})$

51. $T(x,y,z) = 100e^{-x^2-y^2-z^2}$; D is the sphere of radius a centered at the origin.

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16.R: Chapter 16 Review Exercises

***True or False?* Justify your answer with a proof or a counterexample.**

1. The vector field $\vec{\mathbf{F}}(x,y) = x^2y\,\hat{\mathbf{i}} + y^2x\,\hat{\mathbf{j}}$ is conservative.

Answer

False

2. For vector field $\vec{\mathbf{F}}(x,y) = P(x,y)\,\hat{\mathbf{i}} + Q(x,y)\,\hat{\mathbf{j}}$, if $P_y(x,y) = Q_z(x,y)$ in open region D, then $\int_{\partial D} P\,dx + Q\,dy = 0$.

3. The divergence of a vector field is a vector field.

Answer

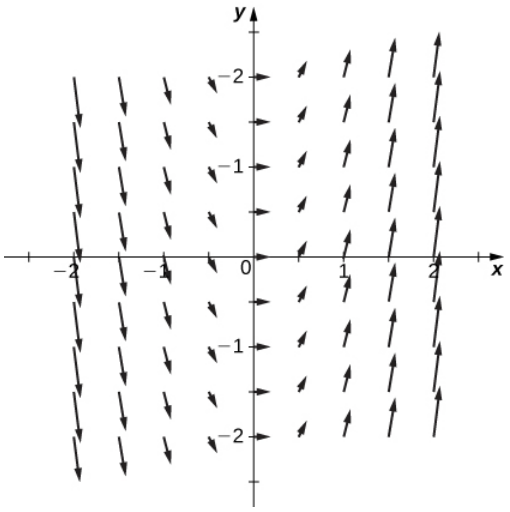
False

4. If $curl\,\vec{\mathbf{F}} = \vec{0}$, then $\vec{\mathbf{F}}$ is a conservative vector field.

Draw the following vector fields.

5. $\vec{\mathbf{F}}(x,y) = \frac{1}{2}\,\hat{\mathbf{i}} + 2x\,\hat{\mathbf{j}}$

Answer



6. $\vec{\mathbf{F}}(x,y) = \sqrt{\dfrac{y\,\hat{\mathbf{i}} + 3x\,\hat{\mathbf{j}}}{x^2+y^2}}$

Are the following the vector fields conservative? If so, find the potential function $\vec{\mathbf{F}}$ such that $\vec{\mathbf{F}} = \vec{\nabla} f$.

7. $\vec{\mathbf{F}}(x,y) = y\,\hat{\mathbf{i}} + (x - 2e^y)\,\hat{\mathbf{j}}$

Answer

Conservative, $f(x,y) = xy - 2e^y$

8. $\vec{\mathbf{F}}(x,y) = (6xy)\,\hat{\mathbf{i}} + (3x^2 - ye^y)\,\hat{\mathbf{j}}$

9. $\vec{\mathbf{F}}(x,y) = (2xy + z^2)\,\hat{\mathbf{i}} + (x^2 + 2yz)\,\hat{\mathbf{j}} + (2xz + y^2)\,\hat{\mathbf{k}}$

Answer

Conservative, $f(x,y,z) = x^2y + y^2z + z^2x$

10. $\vec{\mathbf{F}}(x,y,z) = (e^x y)\,\hat{\mathbf{i}} + (e^x + z)\,\hat{\mathbf{j}} + (e^x + y^2)\,\hat{\mathbf{k}}$

Evaluate the following integrals.

11. $\displaystyle\int_C x^2\,dy + (2x - 3xy)\,dx$, along $C : y = \frac{1}{2}x$ from $(0,0)$ to $(4,2)$

Answer

$-\frac{16}{3}$

12. $\displaystyle\int_C y\,dx + xy^2\,dy$, where $C : x = \sqrt{t},\ y = t - 1,\ 0 \le t \le 1$

13. $\displaystyle\iint_S xy^2\,dS$, where S is the surface $z = x^2 - y,\ 0 \le x \le 1,\ 0 \le y \le 4$

Answer

$\frac{32\sqrt{2}}{9}(3\sqrt{3} - 1)$

Find the divergence and curl for the following vector fields.

14. $\vec{\mathbf{F}}(x,y,z) = 3xyz\,\hat{\mathbf{i}} + xye^x\,\hat{\mathbf{j}} - 3xy\,\hat{\mathbf{k}}$

15. $\vec{\mathbf{F}}(x,y,z) = e^x\,\hat{\mathbf{i}} + e^{xy}\,\hat{\mathbf{j}} - e^{xyz}\,\hat{\mathbf{k}}$

Answer

Divergence: $e^x + x\,e^{xy} + xy\,e^{xyz}$
Curl: $xze^{xyz}\,\hat{\mathbf{i}} - yze^{xyz}\,\hat{\mathbf{j}} + ye^{xy}\,\hat{\mathbf{k}}$

Use Green's theorem to evaluate the following integrals.

16. $\displaystyle\int_C 3xy\,dx + 2xy^2\,dy$, where C is a square with vertices $(0,0)$, $(0,2)$, $(2,2)$ and $(2,0)$.

17. $\displaystyle\oint_C 3y\,dx + (x + e^y)\,dy$, where C is a circle centered at the origin with radius 3.

Answer

-2π

Use Stokes' theorem to evaluate $\iint_S curl\,\vec{\mathbf{F}}\cdot dS$.

18. $\vec{\mathbf{F}}(x,y,z) = y\,\hat{\mathbf{i}} - x\,\hat{\mathbf{j}} + z\,\hat{\mathbf{k}}$, where S is the upper half of the unit sphere

19. $\vec{\mathbf{F}}(x,y,z) = y\,\hat{\mathbf{i}} + xyz\,\hat{\mathbf{j}} - 2zx\,\hat{\mathbf{k}}$, where S is the upward-facing paraboloid $z = x^2 + y^2$ lying in cylinder $x^2 + y^2 = 1$

Answer

$-\pi$

Use the divergence theorem to evaluate $\iint_S \vec{\mathbf{F}}\cdot dS$.

20. $\vec{\mathbf{F}}(x,y,z) = (x^3 y)\,\hat{\mathbf{i}} + (3y - e^x)\,\hat{\mathbf{j}} + (z + x)\,\hat{\mathbf{k}}$, over cube S defined by $-1 \le x \le 1,\ 0 \le y \le 2,\ 0 \le z \le 2$

21. $\vec{\mathbf{F}}(x,y,z) = (2xy)\,\hat{\mathbf{i}} + (-y^2)\,\hat{\mathbf{j}} + (2z^3)\,\hat{\mathbf{k}}$, where S is bounded by paraboloid $z = x^2 + y^2$ and plane $z = 2$

Answer

$31\pi/2$

22. Find the amount of work performed by a 50-kg woman ascending a helical staircase with radius 2 m and height 100 m. The woman completes five revolutions during the climb.

23. Find the total mass of a thin wire in the shape of a semicircle with radius $\sqrt{2}$, and a density function of $\rho(x,y) = y + x^2$.

Answer

$\sqrt{2}(2+\pi)$

24. Find the total mass of a thin sheet in the shape of a hemisphere with radius 2 for $z \geq 0$ with a density function $\rho(x,y,z) = x + y + z$.

25. Use the divergence theorem to compute the value of the flux integral over the unit sphere with $\vec{\mathbf{F}}(x,y,z) = 3z\hat{\mathbf{i}} + 2y\hat{\mathbf{j}} + 2x\hat{\mathbf{k}}$.

Answer

$2\pi/3$

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CHAPTER OVERVIEW

17: Second-Order Differential Equations

We have already studied the basics of differential equations, including separable first-order equations. In this chapter, we go a little further and look at second-order equations, which are equations containing second derivatives of the dependent variable. The solution methods we examine are different from those discussed earlier, and the solutions tend to involve trigonometric functions as well as exponential functions. Here we concentrate primarily on second-order equations with constant coefficients.

17.0: Prelude to Second-Order Differential Equations

17.1: Second-Order Linear Equations

17.1E: Exercises for Section 17.1

17.2: Nonhomogeneous Linear Equations

17.2E: Exercises for Section 17.2

17.3: Applications of Second-Order Differential Equations

17.3E: Exercises for Section 17.3

17.4: Series Solutions of Differential Equations

17.4E: Exercises for Section 17.4

17.R: Chapter 17 Review Exercises

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17.0: Prelude to Second-Order Differential Equations

We have already studied the basics of differential equations, including separable first-order equations. In this chapter, we go a little further and look at second-order equations, which are equations containing second derivatives of the dependent variable. The solution methods we examine are different from those discussed earlier, and the solutions tend to involve trigonometric functions as well as exponential functions. Here we concentrate primarily on second-order equations with constant coefficients.



Figure 17.0.1: A motorcycle suspension system is an example of a damped spring-mass system. The spring absorbs bumps and keeps the tire in contact with the road. The shock absorber damps the motion so the motorcycle does not continue to bounce after going over each bump. (credit: nSeika, Flickr)

Such equations have many practical applications. The operation of certain electrical circuits, known as resistor–inductor–capacitor (*RLC*) circuits, can be described by second-order differential equations with constant coefficients. These circuits are found in all kinds of modern electronic devices—from computers to smartphones to televisions. Such circuits can be used to select a range of frequencies from the entire radio wave spectrum, and are they commonly used for tuning AM/FM radios. We look at these circuits more closely in Applications.

Spring-mass systems, such as motorcycle shock absorbers, are a second common application of second-order differential equations. For motocross riders, the suspension systems on their motorcycles are very important. The off-road courses on which they ride often include jumps, and losing control of the motorcycle when landing could cost them the race. The movement of the shock absorber depends on the amount of damping in the system. In this chapter, we model forced and unforced spring-mass systems with varying amounts of damping.

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17.1: Second-Order Linear Equations

Learning Objectives

- Recognize homogeneous and nonhomogeneous linear differential equations.
- Determine the characteristic equation of a homogeneous linear equation.
- Use the roots of the characteristic equation to find the solution to a homogeneous linear equation.
- Solve initial-value and boundary-value problems involving linear differential equations.

When working with differential equations, usually the goal is to find a solution. In other words, we want to find a function (or functions) that satisfies the differential equation. The technique we use to find these solutions varies, depending on the form of the differential equation with which we are working. Second-order differential equations have several important characteristics that can help us determine which solution method to use. In this section, we examine some of these characteristics and the associated terminology.

Homogeneous Linear Equations

Consider the second-order differential equation

$$xy'' + 2x^2y' + 5x^3y = 0.$$

Notice that y and its derivatives appear in a relatively simple form. They are multiplied by functions of x, but are not raised to any powers themselves, nor are they multiplied together. As discussed in previously, first-order equations with similar characteristics are said to be linear. The same is true of second-order equations. Also note that all the terms in this differential equation involve either y or one of its derivatives. There are no terms involving only functions of x. Equations like this, in which every term contains y or one of its derivatives, are called **homogeneous**.

Not all differential equations are homogeneous. Consider the differential equation

$$xy'' + 2x^2y' + 5x^3y = x^2.$$

The x^2 term on the right side of the equal sign does not contain y or any of its derivatives. Therefore, this differential equation is *nonhomogeneous*.

Definition: Homogeneous and Nonhomogeneous Linear Equations

A second-order differential equation is linear if it can be written in the form

$$a_2(x)y'' + a)1(x)y' + a_0(x)y = r(x), \tag{17.1.1}$$

where $a_2(x), a_1(x), a_0(x)$, and $r(x)$ are real-valued functions and $a_2(x)$ is not identically zero. If $r(x) \equiv 0$—in other words, if $r(x) = 0$ for every value of x—the equation is said to be a *homogeneous linear equation*. If $r(x) \neq 0$ for some value of x, the equation is said to be a *nonhomogeneous linear equation*.

In linear differential equations, y and its derivatives can be raised only to the first power and they may not be multiplied by one another. Terms involving y^2 or $\sqrt{y'}$ make the equation nonlinear. Functions of y and its derivatives, such as $\sin y$ or $e^{y'}$, are similarly prohibited in linear differential equations.

Note that equations may not always be given in standard form (the form shown in the definition). It can be helpful to rewrite them in that form to decide whether they are linear, or whether a linear equation is homogeneous.

Example 17.1.1: Classifying Second-Order Equations

Classify each of the following equations as linear or nonlinear. If the equation is linear, determine further whether it is homogeneous or nonhomogeneous.

a. $y'' + 3x^4y' + x^2y^2 = x^3$

b. $(\sin x)y'' + (\cos x)y' + 3y = 0$
c. $4t^2x'' + 3txx' + 4x = 0$
d. $5y'' + y = 4x^5$
e. $(\cos x)y'' - \sin y' + (\sin x)y - \cos x = 0$
f. $8ty'' - 6t^2y' + 4ty - 3t^2 = 0$
g. $\sin(x^2)y'' - (\cos x)y' + x^2y = y' - 3$
h. $y'' + 5xy' - 3y = \cos y$

Solution

a. This equation is nonlinear because of the y^2 term.
b. This equation is linear. There is no term involving a power or function of y, and the coefficients are all functions of x.The equation is already written in standard form, and $r(x)$ is identically zero, so the equation is homogeneous.
c. This equation is nonlinear. Note that, in this case, x is the dependent variable and $t$is the independent variable. The second term involves the product of x and x', so the equation is nonlinear.
d. This equation is linear. Since $r(x) = 4x^5$, the equation is nonhomogeneous.
e. This equation is nonlinear, because of the $\sin y'$ term.
f. This equation is linear. Rewriting it in standard form gives

$$8t^2y'' - 6t^2y' + 4ty = 3t^2.$$

With the equation in standard form, we can see that $r(t) = 3t^2$,so the equation is nonhomogeneous.
g. This equation looks like it's linear, but we should rewrite it in standard form to be sure. We get

$$\sin(x^2)y'' - (\cos x + 1)y' + x^2y = -3.$$

This equation is, indeed, linear. With $r(x) = -3$, it is nonhomogeneous.
h. This equation is nonlinear because of the $\cos y$ term.

? Exercise 17.1.1

Classify each of the following equations as linear or nonlinear. If the equation is linear, determine further whether it is homogeneous or nonhomogeneous.

a. $(y'')2 - y' + 8x^3y = 0$
b. $(\sin t)y'' + \cos t - 3ty' = 0$

Hint

Write the equation in standard form (Equation 17.1.1) if necessary. Check for powers or functions of y and its derivatives.

Answer a

Nonlinear Linear

Answer b

nonhomogeneous

Later in this section, we will see some techniques for solving specific types of differential equations. Before we get to that, however, let's get a feel for how solutions to linear differential equations behave. In many cases, solving differential equations depends on making educated guesses about what the solution might look like. Knowing how various types of solutions behave will be helpful.

✓ Example 17.1.2: Verifying a Solution

Consider the linear, homogeneous differential equation

$$x^2y'' - xy' - 3y = 0.$$

Looking at this equation, notice that the coefficient functions are polynomials, with higher powers of x associated with higher-order derivatives of y. Show that $y = x^3$ is a solution to this differential equation.

Solution

Let $y = x^3$. Then $y' = 3x^2$ and $y'' = 6x$. Substituting into the differential equation, we see that

$$\begin{aligned} x^2y'' - xy' - 3y &= x^2(6x) - x(3x^2) - 3(x^3) \\ &= 6x^3 - 3x^3 - 3x^3 \\ &= 0. \end{aligned}$$

Exercise 17.1.2

Show that $y = 2x^2$ is a solution to the differential equation

$$\frac{1}{2}x^2y'' - xy' + y = 0. \tag{17.1.2}$$

Hint

Calculate the derivatives and substitute them into the differential equation.

Answer

This requires calculating y' and y''.

$$y' = \frac{dy}{dx} = 4x$$

and

$$y'' = \frac{dy'}{dx} = 4$$

Inserting these derivatives along with $y = 2x^2$ into Equation 17.1.2.

$$\begin{aligned} \frac{1}{2}x^2y'' - xy' + y &\stackrel{?}{=} 0 \\ \frac{1}{2}x^2(4) - x(4x) + 2x^2 &\stackrel{?}{=} 0 \\ 2x^2 - 4x^2 + 2x^2 &\stackrel{\checkmark}{=} 0 \end{aligned}$$

Yes, this is a solution to the differential equation in Equation 17.1.2.

Although simply finding any solution to a differential equation is important, mathematicians and engineers often want to go beyond finding *one* solution to a differential equation to finding *all* solutions to a differential equation. In other words, we want to find a general solution**.** Just as with first-order differential equations, a general solution (or family of solutions) gives the entire set of solutions to a differential equation. An important difference between first-order and second-order equations is that, with second-order equations, we typically need to find two different solutions to the equation to find the general solution. If we find two solutions, then any linear combination of these solutions is also a solution. We state this fact as the following theorem.

Theorem: SUPERPOSITION PRINCIPLE

If $y_1(x)$ and $y_2(x)$ are solutions to a linear homogeneous differential equation, then the function

$$y(x) = c_1y_1(x) + c_2y_2(x), \tag{17.1.3}$$

where c_1 and c_2 are constants, is also a solution.

The proof of this superposition principle theorem is left as an exercise.

Example 17.1.3: Verifying the Superposition Principle

Consider the differential equation

$$y'' - 4y' - 5y = 0.$$

Given that e^{-x} and e^{5x} are solutions to this differential equation, show that $4e^{-x} + e^{5x}$ is a solution.

Solution

Although this can be done through a simple application of the Superposition principle (Equation 17.1.3), but we can also confirm it is a solution via an approach like in Example 17.1.2. We have

$$\begin{aligned} y(x) &= 4e^{-x} + e^{5x} \\ y'(x) &= -4e^{-x} + 5e^{5x} \\ y''(x) &= 4e^{-x} + 25e^{5x}. \end{aligned}$$

Then

$$\begin{aligned} y'' - 4y' - 5y &\stackrel{?}{=} (4e^{-x} + 25e^{5x}) - 4(-4e^{-x} + 5e^{5x}) - 5(4e^{-x} + e^{5x}) \\ &\stackrel{?}{=} 4e^{-x} + 25e^{5x} + 16e^{-x} - 20e^{5x} - 20e^{-x} - 5e^{5x} \\ &\stackrel{\checkmark}{=} 0. \end{aligned}$$

Thus, $y(x) = 4e^{-x} + e^{5x}$ is a solution.

Exercise 17.1.3

Consider the differential equation

$$y'' + 5y' + 6y = 0.$$

Given that e^{-2x} and e^{-3x} are solutions to this differential equation, show that $3e^{-2x} + 6e^{-3x}$ is a solution.

Hint

Differentiate the function and substitute into the differential equation.

Answer

Although this can be a simple application of the Superposition principle (Equation 17.1.3), we can also set through it like in Example 17.1.2. We have

$$\begin{aligned} y(x) &= 3e^{-2x} + 6e^{-3x} \\ y'(x) &= -6e^{-2x} - 18e^{-3x} \\ y''(x) &= 12e^{-2x} + 54e^{3x}. \end{aligned}$$

Then

$$\begin{aligned} y'' + 5y' + 6y &= (12e^{-2x} + 54e^{3x}) + 5(-6e^{-2x} - 18e^{-3x}) + 6(3e^{-2x} + 6e^{3x}) \\ &\stackrel{?}{=} \cancel{12e^{-2x}} + \cancel{54e^{3x}} - \cancel{30e^{-2x}} - \cancel{90e^{3x}} + \cancel{18e^{-2x}} + \cancel{36e^{3x}} \\ &\stackrel{\checkmark}{=} 0. \end{aligned}$$

Thus, $3e^{-2x} + 6e^{-3x}$ is a solution to the differential equation

Unfortunately, to find the general solution to a second-order differential equation, it is not enough to find any two solutions and then combine them. Consider the differential equation

$$x'' + 7x' + 12x = 0.$$

Both e^{-3t} and $2e^{-3t}$ are solutions (you can check this). However,

$$x(t) = c_1 e^{-3t} + c_2(2e^{-3t})$$

is *not* the general solution. This expression does not account for all solutions to the differential equation. In particular, it fails to account for the function e^{-4t}, which is also a solution to the differential equation. It turns out that to find the general solution to a second-order differential equation, we must find two *linearly independent solutions*. We define that terminology here.

Definition: Linearly Dependent functions

A set of functions $f_1(x),\ f_2(x), \ldots, f_n(x)$ is said to be *linearly dependent* if there are constants $c_1,\ c_2, \ldots c_n,$, not all zero, such that

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0$$

for all x over the interval of interest. A set of functions that is not linearly dependent is said to be *linearly independent*.

In this chapter, we usually test sets of only two functions for linear independence, which allows us to simplify this definition. From a practical perspective, we see that two functions are linearly dependent if either one of them is identically zero or if they are constant multiples of each other.

First we show that if the functions meet the conditions given previously, then they are linearly dependent. If one of the functions is identically zero—say, $f_2(x) \equiv 0$—then choose $c_1 = 0$ and $c_2 = 1$, and the condition for linear dependence is satisfied. If, on the other hand, neither $f_1(x)$ nor $f_2(x)$ is identically zero, but $f_1(x) = Cf_2(x)$ for some constant C, then choose $c_1 = C$ and $c_2 = -1$, and again, the condition is satisfied.

Next, we show that if two functions are linearly dependent, then either one is identically zero or they are constant multiples of one another. Assume $f_1(x)$ and $f_2(x)$ are linearly independent. Then, there are constants, c_1 and c_2, not both zero, such that

$$c_1 f_1(x) + c_2 f_2(x) = 0$$

for all x over the interval of interest. Then,

$$c_1 f_1(x) = -c_2 f_2(x).$$

Now, since we stated that c_1 and c_2 can't both be zero, assume $c_2 \neq 0$. Then, there are two cases: either $c_1 = 0$ or $c_1 \neq 0$. If $c_1 = 0$, then

$$\begin{aligned} 0 &= -c_2 f_2(x) \\ 0 &= f_2(x), \end{aligned}$$

so one of the functions is identically zero. Now suppose $c_1 \neq 0$. Then,

$$f_1(x) = \left(-\frac{c_2}{c_1}\right) f_2(x)$$

and we see that the functions are constant multiples of one another.

Theorem: Linear Dependence of Two Functions

Two functions, $f_1(x)$ and $f_2(x)$, are said to be linearly dependent if either one of them is identically zero or if $f_1(x) = Cf_2(x)$ for some constant C and for all x over the interval of interest. Functions that are not linearly dependent are said to be *linearly independent*.

Example 17.1.4: Testing for Linear Dependence

Determine whether the following pairs of functions are linearly dependent or linearly independent.

a. $f_1(x) = x^2$ and $f_2(x) = 5x^2$
b. $f_1(x) = \sin x$ and $f_2(x) = \cos x$
c. $f_1(x) = e^{3x}$ and $f_2(x) = e^{-3x}$

d. $f_1(x) = 3x$ and $f_2(x) = 3x + 1$

Solution

a. $f_2(x) = 5f_1(x)$, so the functions are linearly dependent.
b. There is no constant C such that $f_1(x) = Cf_2(x)$, so the functions are linearly independent.
c. There is no constant C such that $f_1(x) = Cf_2(x)$, so the functions are linearly independent. Don't get confused by the fact that the exponents are constant multiples of each other. With two exponential functions, unless the exponents are equal, the functions are linearly independent.
d. There is no constant C such that $f_1(x) = Cf_2(x)$, so the functions are linearly independent.

Exercise 17.1.4

Determine whether the following pairs of functions are linearly dependent or linearly independent: $f_1(x) = e^x$ and $f_2(x) = 3e^{3x}$.

Hint

Are the functions constant multiples of one another?

Answer

Linearly independent

If we are able to find two linearly independent solutions to a second-order differential equation, then we can combine them to find the general solution. This result is formally stated in the following theorem.

Theorem: General Solution to a Homogeneous Equation

If $y_1(x)$ and $y_2(x)$ are linearly independent solutions to a second-order, linear, homogeneous differential equation, then the general solution is given by

$$y(x) = c_1y_1(x) + c_2y_2(x),$$

where c_1 and c_2 are constants.

When we say a family of functions is the general solution to a differential equation, we mean that

1. every expression of that form is a solution and
2. every solution to the differential equation can be written in that form, which makes this theorem extremely powerful.

If we can find two linearly independent solutions to a second order differential equation, we have, effectively, found *all* solutions to the second order differential equation—quite a remarkable statement. The proof of this theorem is beyond the scope of this text.

Example 17.1.5: Writing the General Solution

If $y_1(t) = e^{3t}$ and $y_2(t) = e^{-3t}$ are solutions to $y'' - 9y = 0$, what is the general solution?

Solution

Note that y_1 and y_2 are not constant multiples of one another, so they are linearly independent. Then, the general solution to the differential equation is

$y(t) = c_1e^{3t} + c_2e^{-3t}$.

Exercise 17.1.5

If $y_1(x) = e^{3x}$ and $y_2(x) = xe^{3x}$ are solutions to $y'' - 6y' + 9y = 0$, what is the general solution?

Hint

Check for linear independence first.

Answer

$y(x) = c_1 e^{3x} + c_2 x e^{3x}$

Second-Order Equations with Constant Coefficients

Now that we have a better feel for linear differential equations, we are going to concentrate on solving second-order equations of the form

$$ay'' + by' + cy = 0, \tag{17.2}$$

where $a, b,$ and c are constants.

Since all the coefficients are constants, the solutions are probably going to be functions with derivatives that are constant multiples of themselves. We need all the terms to cancel out, and if taking a derivative introduces a term that is not a constant multiple of the original function, it is difficult to see how that term cancels out. Exponential functions have derivatives that are constant multiples of the original function, so let's see what happens when we try a solution of the form $y(x) = e^{\lambda x}$, where λ (the lowercase Greek letter lambda) is some constant.

If $y(x) = e^{\lambda x}$, then $y'(x) = \lambda e^{\lambda x}$ and $y'' = \lambda^2 e^{\lambda x}$. Substituting these expressions into Equation 17.1.1, we get

$$\begin{aligned} ay'' + by' + cy &= a(\lambda^2 e^{\lambda x}) + b(\lambda e^{\lambda x}) + ce^{\lambda x} \\ &= e^{\lambda x}(a\lambda^2 + b\lambda + c). \end{aligned}$$

Since $e^{\lambda x}$ is never zero, this expression can be equal to zero for all x only if

$$a\lambda^2 + b\lambda + c = 0.$$

We call this the *characteristic equation* of the differential equation.

Definition: characteristic equation

The *characteristic equation* of the second order differential equation $ay'' + by' + cy = 0$ is

$$a\lambda^2 + b\lambda + c = 0.$$

The characteristic equation is very important in finding solutions to differential equations of this form. We can solve the characteristic equation either by factoring or by using the quadratic formula

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

This gives three cases. The characteristic equation has

1. distinct real roots;
2. a single, repeated real root; or
3. complex conjugate roots.

We consider each of these cases separately.

Case 1: Distinct Real Roots

If the characteristic equation has distinct real roots λ_1 and λ_2, then $e^{\lambda_1 x}$ and $e^{\lambda_2 x}$ are linearly independent solutions to Example 17.1.1, and the general solution is given by

$$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x},$$

where c_1 and c_2 are constants.

For example, the differential equation $y'' + 9y' + 14y = 0$ has the associated characteristic equation $\lambda^2 + 9\lambda + 14 = 0$. This factors into $(\lambda + 2)(\lambda + 7) = 0$, which has roots $\lambda_1 = -2$ and $\lambda_2 = -7$. Therefore, the general solution to this differential equation is

$$y(x) = c_1 e^{-2x} + c_2 e^{-7x}.$$

Case 2: Single Repeated Real Root

Things are a little more complicated if the characteristic equation has a repeated real root, λ. In this case, we know $e^{\lambda x}$ is a solution to Equation 17.1.1, but it is only one solution and we need two linearly independent solutions to determine the general solution. We might be tempted to try a function of the form $ke^{\lambda x}$, where k is some constant, but it would not be linearly independent of $e^{\lambda x}$. Therefore, let's try $xe^{\lambda x}$ as the second solution. First, note that by the quadratic formula,

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

But, λ is a repeated root, so the discriminate $(b^2 - 4ac)$ is zero and $\lambda = \frac{-b}{2a}$. Thus, if $y = xe^{\lambda x}$, we have

$$\begin{aligned} y' &= e^{\lambda x} + \lambda x e^{\lambda x} \\ y'' &= 2\lambda e^{\lambda x} + \lambda^2 x e^{\lambda x}. \end{aligned}$$

Substituting both expressions into Equation 17.1.1, we see that

$$\begin{aligned} ay'' + by' + cy &= a(2\lambda e^{\lambda x} + \lambda^2 x e^{\lambda x}) + b(e^{\lambda x} + \lambda x e^{\lambda x}) + cxe^{\lambda x} \\ &= xe^{\lambda x}(a\lambda^2 + b\lambda + c) + e^{\lambda x}(2a\lambda + b) \\ &= xe^{\lambda x}(0) + e^{\lambda x}(2a(-b2a) + b) \\ &= 0 + e^{\lambda x}(0) \\ &\overset{\checkmark}{=} 0. \end{aligned}$$

This shows that $xe^{\lambda x}$ is a solution to Equation 17.1.1. Since $e^{\lambda x}$ and $xe^{\lambda x}$ are linearly independent, when the characteristic equation has a repeated root λ, the general solution to Equation 17.1.1 is given by

$$y(x) = c_1 e^{\lambda x} + c_2 x e^{\lambda x},$$

where c_1 and c_2 are constants.

For example, the differential equation $y'' + 12y' + 36y = 0$ has the associated characteristic equation

$$\lambda^2 + 12\lambda + 36 = 0.$$

This factors into $(\lambda + 6)^2 = 0$, which has a repeated root $\lambda = -6$. Therefore, the general solution to this differential equation is

$$y(x) = c_1 e^{-6x} + c_2 x e^{-6x}.$$

Case 3: Complex Conjugate Roots

The third case we must consider is when $b^2 - 4ac < 0$. In this case, when we apply the quadratic formula, we are taking the square root of a negative number. We must use the imaginary number $i = \sqrt{-1}$ to find the roots, which take the form $\lambda_1 = \alpha + \beta i$ and $\lambda_2 = \alpha - \beta i$. The **complex number** $\alpha + \beta i$ is called the *conjugate* of $\alpha - \beta i$. Thus, we see that when the discriminate $b^2 - 4ac$ is negative, the roots of our characteristic equation are always **complex conjugates**.

This creates a little bit of a problem for us. If we follow the same process we used for distinct real roots—using the roots of the characteristic equation as the coefficients in the exponents of exponential functions—we get the functions $e^{(\alpha+\beta i)x}$ and $e^{(\alpha-\beta i)x}$ as our solutions. However, there are problems with this approach. First, these functions take on complex (imaginary) values, and a complete discussion of such functions is beyond the scope of this text. Second, even if we were comfortable with complex-value functions, in this course we do not address the idea of a derivative for such functions. So, if possible, we'd like to find two linearly independent *real-value* solutions to the differential equation. For purposes of this development, we are going to manipulate and differentiate the functions $e^{(\alpha+\beta i)x}$ and $e^{(\alpha-\beta i)x}$ as if they were real-value functions. For these particular functions, this approach is valid mathematically, but be aware that there are other instances when complex-value functions do not follow the same rules as real-value functions. Those of you interested in a more in-depth discussion of complex-value functions should consult a complex analysis text.

Based on the roots $\alpha \pm \beta i$ of the characteristic equation, the functions $e^{(\alpha+\beta i)x}$ and $e^{(\alpha-\beta i)x}$ are linearly independent solutions to the differential equation and the general solution is given by

$$y(x) = c_1 e^{(\alpha+\beta i)x} + c_2 e^{(\alpha-\beta i)x}.$$

Using some smart choices for c_1 and c_2, and a little bit of algebraic manipulation, we can find two linearly independent, real-value solutions to Equation 17.1.1 and express our general solution in those terms.

We encountered exponential functions with complex exponents earlier. One of the key tools we used to express these exponential functions in terms of sines and cosines was Euler's formula, which tells us that

$$\underbrace{e^{i\theta} = \cos\theta + i\sin\theta}_{\text{Euler's formula}} \tag{17.1.4}$$

for all real numbers θ.

Going back to the general solution, we have

$$\begin{aligned} y(x) &= c_1 e^{(\alpha+\beta i)x} + c_2 e^{(\alpha-\beta i)x} \\ &= c_1 e^{\alpha x} e^{\beta i x} + c_2 e^{\alpha x} e^{-\beta i x} \\ &= e^{\alpha x}(c_1 e^{\beta i x} + c_2 e^{-\beta i x}). \end{aligned}$$

Applying Euler's formula (Equation 17.1.4) together with the identities $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$, we get

$$\begin{aligned} y(x) &= e^{\alpha x}[c_1(\cos\beta x + i\sin\beta x) + c_2(\cos(-\beta x) + i\sin(-\beta x))] \\ &= e^{\alpha x}[(c_1 + c_2)\cos\beta x + (c_1 - c_2)i\sin\beta x]. \end{aligned} \tag{17.1.5}$$

Now, if we choose $c_1 = c_2 = \frac{1}{2}$, the second term is zero and we get

$$y(x) = e^{\alpha x}\cos\beta x$$

as a real-value solution to Equation 17.1.1. Similarly, if we choose $c_1 = -\frac{i}{2}$ and $c_2 = \frac{i}{2}$, the first term of Equation 17.1.5 is zero and we get

$$y(x) = e^{\alpha x}\sin\beta x$$

as a second, linearly independent, real-value solution to Equation 17.1.1.

Based on this, we see that if the characteristic equation has complex conjugate roots $\alpha \pm \beta i$, then the general solution to Equation 17.1.1 is given by

$$\begin{aligned} y(x) &= c_1 e^{\alpha x}\cos\beta x + c_2 e^{\alpha x}\sin\beta x \\ &= e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x), \end{aligned}$$

where c_1 and c_2 are constants.

For example, the differential equation $y'' - 2y' + 5y = 0$ has the associated characteristic equation $\lambda^2 - 2\lambda + 5 = 0$. By the quadratic formula, the roots of the characteristic equation are $1 \pm 2i$. Therefore, the general solution to this differential equation is

$$y(x) = e^x(c_1\cos 2x + c_2\sin 2x).$$

Summary of Results

We can solve second-order, linear, homogeneous differential equations with constant coefficients by finding the roots of the associated characteristic equation. The form of the general solution varies, depending on whether the characteristic equation has distinct, real roots; a single, repeated real root; or complex conjugate roots. The three cases are summarized in Table 17.1.1.

Table 17.1.1: Summary of Characteristic Equation Cases

Characteristic Equation Roots	General Solution to the Differential Equation
Distinct real roots, λ_1 and λ_2	$y(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$
A repeated real root, λ	$y(x) = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$
Complex conjugate roots $\alpha \pm \beta i$	$y(x) = e^{\alpha x}(c_1\cos\beta x + c_2\sin\beta x)$

PROBLEM-SOLVING STRATEGY: USING THE CHARACTERISTIC EQUATION TO SOLVE SECOND-ORDER DIFFERENTIAL EQUATIONS WITH CONSTANT COEFFICIENTS

1. Write the differential equation in the form $a'' + by' + cy = 0$.
2. Find the corresponding characteristic equation $a\lambda^2 + b\lambda + c = 0$.
3. Either factor the characteristic equation or use the quadratic formula to find the roots.
4. Determine the form of the general solution based on whether the characteristic equation has distinct, real roots; a single, repeated real root; or complex conjugate roots.

Example 17.1.6: Solving Second-Order Equations with Constant Coefficients

Find the general solution to the following differential equations. Give your answers as functions of x.

a. $y'' + 3y' - 4y = 0$
b. $y'' + 6y' + 13y = 0$
c. $y'' + 2y' + y = 0$
d. $y'' - 5y' = 0$
e. $y'' - 16y = 0$
f. $y'' + 16y = 0$

Solution

Note that all these equations are already given in standard form (step 1).

1. The characteristic equation is $\lambda^2 + 3\lambda - 4 = 0$ (step 2). This factors into $(\lambda + 4)(\lambda - 1) = 0$, so the roots of the characteristic equation are $\lambda_1 = -4$ and $\lambda_2 = 1$ (step 3). Then the general solution to the differential equation is

$$y(x) = c_1 e^{-4x} + c_2 e^{x}. \qquad \text{(step 1)}$$

2. The characteristic equation is $\lambda^2 + 6\lambda + 13 = 0$ (step 2). Applying the quadratic formula, we see this equation has complex conjugate roots $-3 \pm 2i$ (step 3). Then the general solution to the differential equation is

$$y(t) = e^{-3t}(c_1 \cos 2t + c_2 \sin 2t). \qquad \text{(step 2)}$$

3. The characteristic equation is $\lambda^2 + 2\lambda + 1 = 0$ (step 2). This factors into $(\lambda + 1)2 = 0$, so the characteristic equation has a repeated real root $\lambda = -1$ (step 3). Then the general solution to the differential equation is

$$y(t) = c_1 e^{-t} + c_2 t e^{-t}. \qquad \text{(step 3)}$$

4. The characteristic equation is $\lambda^2 - 5\lambda$ (step 2). This factors into $\lambda(\lambda - 5) = 0$, so the roots of the characteristic equation are $\lambda_1 = 0$ and $\lambda_2 = 5$ (step 3). Note that $e^{0x} = e^0 = 1$, so our first solution is just a constant. Then the general solution to the differential equation is

$$y(x) = c_1 + c_2 e^{5x}. \qquad \text{(step 4)}$$

5. The characteristic equation is $\lambda^2 - 16 = 0$ (step 2). This factors into $(\lambda + 4)(\lambda - 4) = 0$, so the roots of the characteristic equation are $\lambda_1 = 4$ and $\lambda_2 = -4$ (step 3). Then the general solution to the differential equation is

$$y(x) = c_1 e^{4x} + c_2 e^{-4x}. \qquad \text{(step 5)}$$

6. The characteristic equation is $\lambda^2 + 16 = 0$ (step 2). This has complex conjugate roots $\pm 4i$ (step 3). Note that $e^{0x} = e^0 = 1$, so the exponential term in our solution is just a constant. Then the general solution to the differential equation is

$$y(t) = c_1 \cos 4t + c_2 \sin 4t. \qquad \text{(step 6)}$$

Exercise 17.1.6

Find the general solution to the following differential equations:

a. $y'' - 2y' + 10y = 0$

b. $y'' + 14y' + 49y = 0$

Hint

Find the roots of the characteristic equation.

Answer a

$y(x) = e^x(c_1 \cos 3x + c_2 \sin 3x)$

Answer b

$y(x) = c_1 e^{-7x} + c_2 x e^{-7x}$

Initial-Value Problems and Boundary-Value Problems

So far, we have been finding general solutions to differential equations. However, differential equations are often used to describe physical systems, and the person studying that physical system usually knows something about the state of that system at one or more points in time. For example, if a constant-coefficient differential equation is representing how far a motorcycle shock absorber is compressed, we might know that the rider is sitting still on his motorcycle at the start of a race, time $t = t_0$. This means the system is at equilibrium, so $y(t_0) = 0$, and the compression of the shock absorber is not changing, so $y'(t_0) = 0$. With these two initial conditions and the general solution to the differential equation, we can find the *specific* solution to the differential equation that satisfies both initial conditions. This process is known as *solving an initial-value problem*. (Recall that we discussed initial-value problems in Introduction to Differential Equations.) Note that second-order equations have two arbitrary constants in the general solution, and therefore we require two initial conditions to find the solution to the initial-value problem.

Sometimes we know the condition of the system at two different times. For example, we might know $y(t_0) = y_0$ and $y(t_1) = y_1$. These conditions are called boundary conditions, and finding the solution to the differential equation that satisfies the boundary conditions is called solving a boundary-value problem.

Mathematicians, scientists, and engineers are interested in understanding the conditions under which an initial-value problem or a boundary-value problem has a unique solution. Although a complete treatment of this topic is beyond the scope of this text, it is useful to know that, within the context of constant-coefficient, second-order equations, initial-value problems are guaranteed to have a unique solution as long as two initial conditions are provided. Boundary-value problems, however, are not as well behaved. Even when two boundary conditions are known, we may encounter boundary-value problems with unique solutions, many solutions, or no solution at all.

Example 17.1.7: Solving an Initial-Value Problem

Solve the following initial-value problem: $y'' + 3y' - 4y = 0,\ y(0) = 1,\ y'(0) = -9.$

Solution

We already solved this differential equation in Example 17.6a. and found the general solution to be

$$y(x) = c_1 e^{-4x} + c_2 e^x.$$

Then

$$y'(x) = -4c_1 e^{-4x} + c_2 e^x.$$

When $x = 0$, we have $y(0) = c_1 + c_2$ and $y'(0) = -4c_1 + c_2$. Applying the initial conditions, we have

$$c_1 + c_2 = 1$$
$$-4c_1 + c_2 = -9.$$

Then $c_1 = 1 - c_2$. Substituting this expression into the second equation, we see that

$$-4(1-c_2)+c_2=-9$$
$$-4+4c_2+c_2=-9$$
$$5c_2=-5$$
$$c_2=-1.$$

So, $c_1=2$ and the solution to the initial-value problem is

$$y(x)=2e^{-4x}-e^{x}.$$

? Exercise 17.1.7

Solve the initial-value problem $y''-3y'-10y=0, \quad y(0)=0,\ y'(0)=7.$

Hint

Use the initial conditions to determine values for c_1 and c_2.

Answer

$$y(x)=-e^{-2x}+e^{5x}$$

✓ Example 17.1.8: Solving an Initial-Value Problem and Graphing the Solution

Solve the following initial-value problem and graph the solution:

$$y''+6y'+13y=0, \quad y(0)=0,\ y'(0)=2$$

Solution

We already solved this differential equation in Example 17.1.6*b*. and found the general solution to be

$$y(x)=e^{-3x}(c_1\cos 2x+c_2\sin 2x).$$

Then

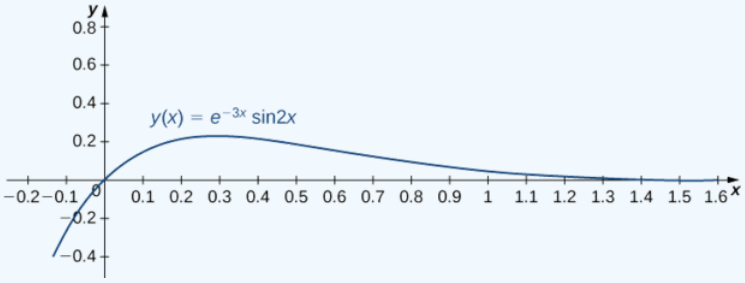
$$y'(x)=e^{-3x}(-2c_1\sin 2x+2c_2\cos 2x)-3e^{-3x}(c_1\cos 2x+c_2\sin 2x).$$

When $x=0$, we have $y(0)=c_1$ and $y'(0)=2c_2-3c_1$. Applying the initial conditions, we obtain

$$c_1=0$$
$$-3c_1+2c_2=2.$$

Therefore, $c_1=0$, $c_2=1$, and the solution to the initial value problem is shown in the following graph.

$$y=e^{-3x}\sin 2x.$$



? Exercise 17.1.8

Solve the following initial-value problem and graph the solution: $y''-2y'+10y=0, \quad y(0)=2,\ y'(0)=-1$

Hint

Use the initial conditions to determine values for c_1 and c_2.

Answer

$$y(x) = e^x(2\cos 3x - \sin 3x)$$

Example 17.1.9: Initial-Value Problem Representing a Spring-Mass System

The following initial-value problem models the position of an object with mass attached to a spring. Spring-mass systems are examined in detail in Applications. The solution to the differential equation gives the position of the mass with respect to a neutral (equilibrium) position (in meters) at any given time. (Note that for spring-mass systems of this type, it is customary to define the downward direction as positive.)

$$y'' + 2y' + y = 0, \quad y(0) = 1,\ y'(0) = 0$$

Solve the initial-value problem and graph the solution. What is the position of the mass at time $t = 2$ sec? How fast is the mass moving at time $t = 1$ sec? In what direction?

Solution

In Example Example 17.1.6c. we found the general solution to this differential equation to be

$$y(t) = c_1e^{-t} + c_2te^{-t}.$$

Then

$$y'(t) = -c_1e^{-t} + c_2(-te^{-t} + e^{-t}).$$

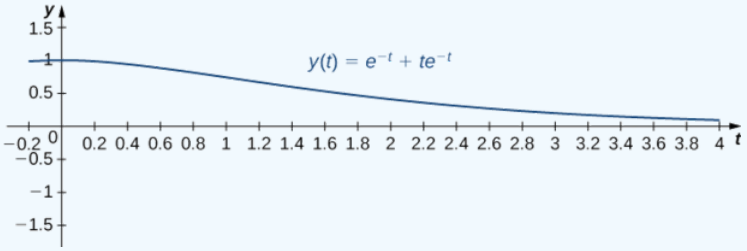
When $t = 0$, we have $y(0) = c_1$ and $y'(0) = c_1 + c_2$. Applying the initial conditions, we obtain

$$\begin{aligned} c_1 &= 1 \\ -c_1 + c_2 &= 0. \end{aligned}$$

Thus, $c_1 = 1, c_2 = 1$, and the solution to the initial value problem is

$$y(t) = e^{-t} + te^{-t}.$$

This solution is represented in the following graph. At time $t = 2$, the mass is at position $y(2) = e^{-2} + 2e^{-2} = 3e^{-2} \approx 0.406$ m below equilibrium.



To calculate the velocity at time $t = 1$, we need to find the derivative. We have $y(t) = e^{-t} + te^{-t}$, so

$$y'(t) = -e^{-t} + e^{-t} - te^{-t} = -te^{-t}.$$

Then $y'(1) = -e^{-1} \approx -0.3679$. At time $t = 1$, the mass is moving upward at 0.3679 m/sec.

Exercise 17.1.9

Suppose the following initial-value problem models the position (in feet) of a mass in a spring-mass system at any given time. Solve the initial-value problem and graph the solution. What is the position of the mass at time $t = 0.3$ sec? How fast is it moving at time $t = 0.1$ sec? In what direction?

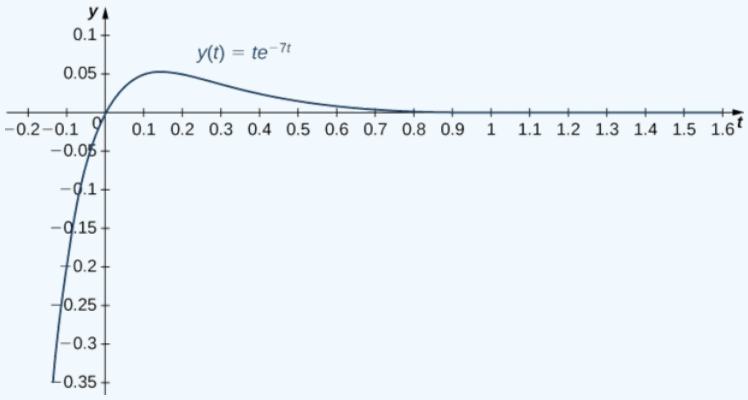
$$y'' + 14y' + 49y = 0, \quad y(0) = 0,\ y'(0) = 1$$

Hint

Use the initial conditions to determine values for c_1 and c_2.

Answer

$$y(t) = te^{-7t}$$



At time $t = 0.3,\ y(0.3) = 0.3e^{(-7*0.3)} = 0.3e^{-2.1} \approx 0.0367$. The mass is 0.0367ft below equilibrium. At time $t = 0.1,\ y'(0.1) = 0.3e^{-0.7} \approx 0.1490$. The mass is moving downward at a speed of 0.1490 ft/sec.

Example 17.1.10: Solving a Boundary-Value Problem

In Example 17.6f. we solved the differential equation $y'' + 16y = 0$ and found the general solution to be $y(t) = c_1 \cos 4t + c_2 \sin 4t$. If possible, solve the boundary-value problem if the boundary conditions are the following:

a. $y(0) = 0, y(\frac{\pi}{4}) = 0$
b. $y(0) = 1, y(0) = 1, y(\frac{\pi}{8}) = 0$
c. $y(\frac{\pi}{8}) = 0, y(\frac{3\pi}{8}) = 2$

Solution

We have

$$y(x) = c_1 \cos 4t + c_2 \sin 4t.$$

1. Applying the first boundary condition given here, we get $y(0) = c_1 = 0$. So the solution is of the form $y(t) = c_2 \sin 4t$. When we apply the second boundary condition, though, we get $y(\frac{\pi}{4}) = c_2 \sin(4(\frac{\pi}{4})) = c_2 \sin \pi = 0$ for all values of c_2. The boundary conditions are not sufficient to determine a value for c_2, so this boundary-value problem has infinitely many solutions. Thus, $y(t) = c_2 \sin 4t$ is a solution for any value of c_2.
2. Applying the first boundary condition given here, we get $y(0) = c_1 = 1$. Applying the second boundary condition gives $y(\frac{\pi}{8}) = c_2 = 0$, so $c_2 = 0$. In this case, we have a unique solution: $y(t) = \cos 4t$.
3. Applying the first boundary condition given here, we get $y(\frac{\pi}{8}) = c_2 = 0$. However, applying the second boundary condition gives $y(\frac{3\pi}{8}) = -c_2 = 2$, so $c_2 = -2$. We cannot have $c_2 = 0 = -2$, so this boundary value problem has no solution.

Key Concepts

- Second-order differential equations can be classified as linear or nonlinear, homogeneous or nonhomogeneous.
- To find a general solution for a homogeneous second-order differential equation, we must find two linearly independent solutions. If $y_1(x)$ and $y_2(x)$ are linearly independent solutions to a second-order, linear, homogeneous differential equation, then the general solution is given by

$$y(x) = c_1 y_1(x) + c_2 y_2(x).$$

- To solve homogeneous second-order differential equations with constant coefficients, find the roots of the characteristic equation. The form of the general solution varies depending on whether the characteristic equation has distinct, real roots; a

single, repeated real root; or complex conjugate roots.

- Initial conditions or boundary conditions can then be used to find the specific solution to a differential equation that satisfies those conditions, except when there is no solution or infinitely many solutions.

Key Equations

- **Linear second-order differential equation**

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = r(x)$$

- **Second-order equation with constant coefficients**

$$ay'' + by' + cy = 0$$

Glossary

boundary conditions
the conditions that give the state of a system at different times, such as the position of a spring-mass system at two different times

boundary-value problem
a differential equation with associated boundary conditions

characteristic equation
the equation $a\lambda^2 + b\lambda + c = 0$ for the differential equation $ay''+by'+cy = 0$

homogeneous linear equation
a second-order differential equation that can be written in the form $a_2(x)y''+a_1(x)y'+a_0(x)y = r(x)$, but $r(x) = 0$ for every value of x

nonhomogeneous linear equation
a second-order differential equation that can be written in the form $a_2(x)y''+a_1(x)y'+a_0(x)y = r(x)$, but $r(x) \neq 0$ for some value of x

linearly dependent
a set of functions $f_1(x),\ f_2(x),\ \ldots,\ f_n(x)$ for whichthere are constants $c_1,\ c_2,\ \ldots,\ c_n$, not all zero, such that $c_1f_1(x) + c_2f_2(x) + \cdots + c_nf_n(x) = 0$ for all x in the interval of interest

linearly independent
a set of functions $f_1(x),\ f_2(x),\ \ldots,\ f_n(x)$ for which there are no constants $c_1,\ c_2,\ \ldots,\ c_n$, such that $c_1f_1(x) + c_2f_2(x) + \cdots + c_nf_n(x) = 0$ for all x in the interval of interest

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17.1E: Exercises for Section 17.1

In exercises 1 - 6, classify each of the following equations as linear or nonlinear. If the equation is linear, determine whether it is homogeneous or nonhomogeneous.

1. $x^3y'' + (x-1)y' - 8y = 0$

Answer

linear, homogenous

2. $(1+y^2)y'' + xy' - 3y = \cos x$

3. $xy'' + e^y y' = x$

Answer

nonlinear

4. $y'' + \dfrac{4}{x}y' - 8xy = 5x^2 + 1$

5. $y'' + (\sin x)y' - xy = 4y$

Answer

linear, homogenous

6. $y'' + \left(\dfrac{x+3}{y}\right)y' = 0$

In exercises 7 - 10, verify that the given function is a solution to the differential equation. Use a graphing utility to graph the particular solutions for several values of c_1 and c_2. What do the solutions have in common?

7. **[T]** $y'' + 2y' - 3y = 0; \quad y(x) = c_1e^x + c_2e^{-3x}$

8. **[T]** $x^2y'' - 2y - 3x^2 + 1 = 0; \quad y(x) = c_1x^2 + c_2x^{-1} + x^2\ln(x) + \frac{1}{2}$

9. **[T]** $y'' + 14y + 49y = 0; \quad y(x) = c_1e^{-7x} + c_2xe^{-7x}$

10. **[T]** $6y'' - 49y' + 8y = 0; \quad y(x) = c_1e^{x/6} + c_2e^{8x}$

In exercises 11 - 30, find the general solution to the linear differential equation.

11. $y'' - 3y' - 10y = 0$

Answer

$y = c_1e^{5x} + c_2e^{-2x}$

12. $y'' - 7y' + 12y = 0$

13. $y'' + 4y' + 4y = 0$

Answer

$y = c_1e^{-2x} + c_2xe^{-2x}$

14. $4y'' - 12y' + 9y = 0$

15. $2y'' - 3y' - 5y = 0$

Answer

$y = c_1e^{5x/2} + c_2e^{-x}$

16. $3y'' - 14y' + 8y = 0$

17. $y'' + y' + y = 0$

Answer

$y = e^{-x/2}\left(c_1 \cos \frac{\sqrt{3}x}{2} + c_2 \sin \frac{\sqrt{3}x}{2}\right)$

18. $5y'' + 2y' + 4y = 0$

19. $y'' - 121y = 0$

Answer

$y = c_1 e^{-11x} + c_2 e^{11x}$

20. $8y'' + 14y' - 15y = 0$

21. $y'' + 81y = 0$

Answer

$y = c_1 \cos 9x + c_2 \sin 9x$

22. $y'' - y' + 11y = 0$

23. $2y'' = 0$

Answer

$y = c_1 + c_2 x$

24. $y'' - 6y' + 9y = 0$

25. $3y'' - 2y' - 7y = 0$

Answer

$y = c_1 e^{\left((1+\sqrt{22})/3\right)x} + c_2 e^{\left((1-\sqrt{22})/3\right)x}$

26. $4y'' - 10y' = 0$

27. $36\frac{d^2y}{dx^2} + 12\frac{dy}{dx} + y = 0$

Answer

$y = c_1 e^{-x/6} + c_2 x e^{-x/6}$

28. $25\frac{d^2y}{dx^2} - 80\frac{dy}{dx} + 64y = 0$

29. $\frac{d^2y}{dx^2} - 9\frac{dy}{dx} = 0$

Answer

$y = c_1 + c_2 e^{9x}$

30. $4\frac{d^2y}{dx^2} + 8y = 0$

In exercises 31 - 38, solve the initial-value problem.

31. $y'' + 5y' + 6y = 0, \quad y(0) = 0,\ y'(0) = -2$

Answer

$y = -2e^{-2x} + 2e^{-3x}$

32. $y'' + 2y' - 8y = 0, \quad y(0) = 5,\ y'(0) = 4$

33. $y'' + 4y = 0, \quad y(0) = 3,\ y'(0) = 10$

Answer

$y = 3\cos(2x) + 5\sin(2x)$

34. $y'' - 18y' + 81y = 0, \quad y(0) = 1,\ y'(0) = 5$

35. $y'' - y' - 30y = 0, \quad y(0) = 1,\ y'(0) = -16$

Answer

$y = -e^{6x} + 2e^{-5x}$

36. $4y'' + 4y' - 8y = 0, \quad y(0) = 2,\ y'(0) = 1$

37. $25y'' + 10y' + y = 0, \quad y(0) = 2,\ y'(0) = 1$

Answer

$y = 2e^{-x/5} + \frac{7}{5}xe^{-x/5}$

38. $y'' + y = 0, \quad y(\pi) = 1,\ y'(\pi) = -5$

In exercises 39 - 46, solve the boundary-value problem, if possible.

39. $y'' + y' - 42y = 0, \quad y(0) = 0,\ y(1) = 2$

Answer

$y = \left(\frac{2}{e^6 - e^{-7}}\right)e^{6x} - \left(\frac{2}{e^6 - e^{-7}}\right)e^{-7x}$

40. $9y'' + y = 0, \quad y(3\pi^2) = 6,\ y(0) = -8$

41. $y'' + 10y' + 34y = 0, \quad y(0) = 6,\ y(\pi) = 2$

Answer

No solutions exist.

42. $y'' + 7y' - 60y = 0, \quad y(0) = 4,\ y(2) = 0$

43. $y'' - 4y' + 4y = 0, \quad y(0) = 2,\ y(1) = -1$

Answer

$y = 2e^{2x} - \left(\frac{2e^2 + 1}{e^2}\right)xe^{2x}$

44. $y'' - 5y' = 0, \quad y(0) = 3,\ y(-1) = 2$

45. $y'' + 9y = 0, \quad y(0) = 4,\ y(\pi^3) = -4$

Answer

$y = 4\cos 3x + c_2 \sin 3x$, infinitely many solutions

46. $4y'' + 25y = 0, \quad y(0) = 2,\ y(2\pi) = -2$

47. Find a differential equation with a general solution that is $y = c_1e^{x/5} + c_2e^{-4x}$.

Answer

$5y'' + 19y' - 4y = 0$

48. Find a differential equation with a general solution that is $y = c_1e^{x} + c_2e^{-4x/3}$.

For each differential equation in exercises 49 - 51:

a. **Solve the initial value problem.**

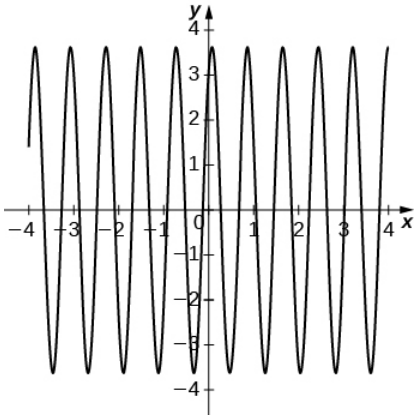
b. **[T] Use a graphing utility to graph the particular solution.**

49. $y'' + 64y = 0; \quad y(0) = 3,\ y'(0) = 16$

Answer

a. $y = 3\cos 8x + 2\sin 8x$

b.



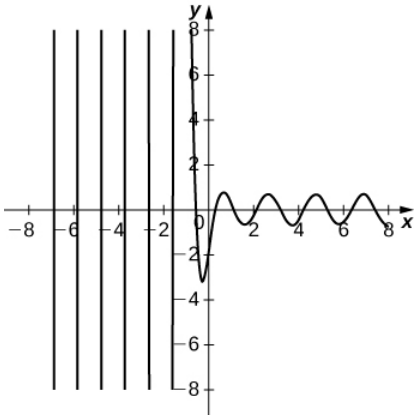
50. $y'' - 2y' + 10y = 0; \quad y(0) = 1,\ y'(0) = 13$

51. $y'' + 5y' + 15y = 0; \quad y(0) = -2,\ y'(0) = 7$

Answer

a. $y = e^{-5/2}\left[-2\cos\left(\frac{\sqrt{35}}{2}x\right) + \frac{4\sqrt{35}}{35}\sin\left(\frac{\sqrt{35}}{2}x\right)\right]$

b.



52. (Principle of superposition) Prove that if $y_1(x)$ and $y_2(x)$ are solutions to a linear homogeneous differential equation, $y'' + p(x)y' + q(x)y = 0$, then the function $y(x) = c_1y_1(x) + c_2y_2(x)$, where c_1 and c_2 are constants, is also a solution.

53. Prove that if a, b and c are positive constants, then all solutions to the second-order linear differential equation $ay'' + by' + cy = 0$ approach zero as $x \to \infty$. (*Hint:* Consider three cases: two distinct roots, repeated real roots, and complex conjugate roots.)

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17.2: Nonhomogeneous Linear Equations

Learning Objectives

- Write the general solution to a nonhomogeneous differential equation.
- Solve a nonhomogeneous differential equation by the method of undetermined coefficients.
- Solve a nonhomogeneous differential equation by the method of variation of parameters.

In this section, we examine how to solve nonhomogeneous differential equations. The terminology and methods are different from those we used for homogeneous equations, so let's start by defining some new terms.

General Solution to a Nonhomogeneous Linear Equation

Consider the nonhomogeneous linear differential equation

$$a_2(x)y''+a_1(x)y'+a_0(x)y = r(x).$$

The associated homogeneous equation

$$a_2(x)y''+a_1(x)y'+a_0(x)y = 0$$

is called the *complementary equation*. We will see that solving the complementary equation is an important step in solving a nonhomogeneous differential equation.

Definition: Particular Solution

A solution $y_p(x)$ of a differential equation that contains no arbitrary constants is called a *particular solution* to the equation.

GENERAL Solution TO A NONHOMOGENEOUS EQUATION

Let $y_p(x)$ be any particular solution to the nonhomogeneous linear differential equation

$$a_2(x)y''+a_1(x)y'+a_0(x)y = r(x).$$

Also, let $c_1y_1(x) + c_2y_2(x)$ denote the general solution to the complementary equation. Then, the *general solution* to the nonhomogeneous equation is given by

$$y(x) = c_1y_1(x) + c_2y_2(x) + y_p(x).$$

Proof

To prove $y(x)$ is the general solution, we must first show that it solves the differential equation and, second, that any solution to the differential equation can be written in that form. Substituting $y(x)$ into the differential equation, we have

$$\begin{aligned} a_2(x)y''+a_1(x)y'+a_0(x)y &= a_2(x)(c_1y_1 + c_2y_2 + y_p)''+a_1(x)(c_1y_1 + c_2y_2 + y_p)' \\ &\quad +a_0(x)(c_1y_1 + c_2y_2 + y_p) \\ &= [a_2(x)(c_1y_1 + c_2y_2)''+a_1(x)(c_1y_1 + c_2y_2)'+a_0(x)(c_1y_1 + c_2y_2)] \\ &\quad +a_2(x)y_p''+a_1(x)y_p'+a_0(x)y_p \\ &= 0 + r(x) \\ &= r(x). \end{aligned}$$

So $y(x)$ is a solution.

Now, let $z(x)$ be any solution to $a_2(x)y'' + a_1(x)y'+a_0(x)y = r(x)$. Then

$$\begin{aligned} a_2(x)(z-y_p)''+a_1(x)(z-y_p)'+a_0(x)(z-y_p) &= (a_2(x)z''+a_1(x)z'+a_0(x)z) \\ &\quad -(a_2(x)y_p''+a_1(x)y_p'+a_0(x)y_p) \\ &= r(x)-r(x) \\ &= 0, \end{aligned}$$

so $z(x)-y_p(x)$ is a solution to the complementary equation. But, $c_1y_1(x)+c_2y_2(x)$ is the general solution to the complementary equation, so there are constants c_1 and c_2 such that

$$z(x)-y_p(x)=c_1y_1(x)+c_2y_2(x).$$

Hence, we see that

$$z(x)=c_1y_1(x)+c_2y_2(x)+y_p(x).$$

Example 17.2.1: Verifying the General Solution

Given that $y_p(x)=x$ is a particular solution to the differential equation $y''+y=x$, write the general solution and check by verifying that the solution satisfies the equation.

Solution

The complementary equation is $y''+y=0$, which has the general solution $c_1\cos x+c_2\sin x$. So, the general solution to the nonhomogeneous equation is

$$y(x)=c_1\cos x+c_2\sin x+x.$$

To verify that this is a solution, substitute it into the differential equation. We have

$$y'(x)=-c_1\sin x+c_2\cos x+1$$

and

$$y''(x)=-c_1\cos x-c_2\sin x.$$

Then

$$\begin{aligned} y''(x)+y(x) &= -c_1\cos x-c_2\sin x+c_1\cos x+c_2\sin x+x \\ &= x. \end{aligned}$$

So, $y(x)$ is a solution to $y''+y=x$.

Exercise 17.2.1

Given that $y_p(x)=-2$ is a particular solution to $y''-3y'-4y=8$, write the general solution and verify that the general solution satisfies the equation.

Hint

Find the general solution to the complementary equation.

Answer

$y(x)=c_1e^{-x}+c_2e^{4x}-2$

In the preceding section, we learned how to solve homogeneous equations with constant coefficients. Therefore, for nonhomogeneous equations of the form $ay''+by'+cy=r(x)$, we already know how to solve the complementary equation, and the problem boils down to finding a particular solution for the nonhomogeneous equation. We now examine two techniques for this: the method of undetermined coefficients and the method of variation of parameters.

Undetermined Coefficients

The method of *undetermined coefficients* involves making educated guesses about the form of the particular solution based on the form of $r(x)$. When we take derivatives of polynomials, exponential functions, sines, and cosines, we get polynomials, exponential functions, sines, and cosines. So when $r(x)$ has one of these forms, it is possible that the solution to the nonhomogeneous differential equation might take that same form. Let's look at some examples to see how this works.

Example 17.2.2: Undetermined Coefficients When $r(x)$ Is a Polynomial

Find the general solution to $y''+4y'+3y = 3x$.

Solution

The complementary equation is $y''+4y'+3y = 0$, with general solution $c_1e^{-x} + c_2e^{-3x}$. Since $r(x) = 3x$, the particular solution might have the form $y_p(x) = Ax + B$. If this is the case, then we have $y_p'(x) = A$ and $y_p''(x) = 0$. For y_p to be a solution to the differential equation, we must find values for A and B such that

$$\begin{aligned} y''+4y'+3y &= 3x \\ 0 + 4(A) + 3(Ax + B) &= 3x \\ 3Ax + (4A + 3B) &= 3x. \end{aligned}$$

Setting coefficients of like terms equal, we have

$$\begin{aligned} 3A &= 3 \\ 4A + 3B &= 0. \end{aligned}$$

Then, $A = 1$ and $B = -\frac{4}{3}$, so $y_p(x) = x - \frac{4}{3}$ and the general solution is

$$y(x) = c_1e^{-x} + c_2e^{-3x} + x - \frac{4}{3}.$$

In Example 17.2.2, notice that even though $r(x)$ did not include a constant term, it was necessary for us to include the constant term in our guess. If we had assumed a solution of the form $y_p = Ax$ (with no constant term), we would not have been able to find a solution. (Verify this!) If the function $r(x)$ is a polynomial, our guess for the particular solution should be a polynomial of the same degree, and it must include all lower-order terms, regardless of whether they are present in $r(x)$.

Example 17.2.3: Undetermined Coefficients When $r(x)$ Is an Exponential

Find the general solution to $y''-y'-2y = 2e^{3x}$.

Solution

The complementary equation is $y''-y'-2y = 0$, with the general solution $c_1e^{-x} + c_2e^{2x}$. Since $r(x) = 2e^{3x}$, the particular solution might have the form $y_p(x) = Ae^{3x}$. Then, we have $yp'(x) = 3Ae^{3x}$ and $y_p''(x) = 9Ae^{3x}$. For y_p to be a solution to the differential equation, we must find a value for A such that

$$\begin{aligned} y''-y'-2y &= 2e^{3x} \\ 9Ae^{3x} - 3Ae^{3x} - 2Ae^{3x} &= 2e^{3x} \\ 4Ae^{3x} &= 2e^{3x}. \end{aligned}$$

So, $4A = 2$ and $A = 1/2$. Then, $y_p(x) = (\frac{1}{2})e^{3x}$, and the general solution is

$$y(x) = c_1e^{-x} + c_2e^{2x} + \frac{1}{2}e^{3x}.$$

Exercise 17.2.2

Find the general solution to $y''-4y'+4y = 7\sin t - \cos t$.

Hint

Use $y_p(t) = A\sin t + B\cos t$ as a guess for the particular solution.

Answer

$y(t) = c_1e^{2t} + c_2te^{2t} + \sin t + \cos t$

In the previous checkpoint, $r(x)$ included both sine and cosine terms. However, even if $r(x)$ included a sine term only or a cosine term only, both terms must be present in the guess. The method of undetermined coefficients also works with products of polynomials, exponentials, sines, and cosines. Some of the key forms of $r(x)$ and the associated guesses for $y_p(x)$ are summarized in Table 17.2.1.

Table 17.2.1: Key Forms for the Method of Undetermined Coefficients

$r(x)$	**Initial guess for $y_p(x)$**
k (a constant)	A (a constant)
$ax + b$	$Ax + B$ (*Note*: The guess must include both terms even if $b = 0$.)
$ax^2 + bx + c$	$Ax^2 + Bx + C$ (*Note*: The guess must include all three terms even if b or c are zero.)
Higher-order polynomials	Polynomial of the same order as $r(x)$
$ae^{\lambda x}$	$Ae^{\lambda x}$
$a\cos\beta x + b\sin\beta x$	$A\cos\beta x + B\sin\beta x$ (*Note*: The guess must include both terms even if either $a = 0$ or $b = 0$.)
$ae^{\alpha x}\cos\beta x + be^{\alpha x}\sin\beta x$	$Ae^{\alpha x}\cos\beta x + Be^{\alpha x}\sin\beta x$
$(ax^2 + bx + c)e^{\lambda x}$	$(Ax^2 + Bx + C)e^{\lambda x}$
$(a_2x^2 + a_1x + a0)\cos\beta x$ $+ (b_2x^2 + b_1x + b_0)\sin\beta x$	$(A_2x^2 + A_1x + A_0)\cos\beta x$ $+ (B_2x^2 + B_1x + B_0)\sin\beta x$
$(a_2x^2 + a_1x + a_0)e^{\alpha x}\cos\beta x$ $+ (b_2x^2 + b_1x + b_0)e^{\alpha x}\sin\beta x$	$(A_2x^2 + A_1x + A_0)e^{\alpha x}\cos\beta x$ $+ (B_2x^2 + B_1x + B_0)e^{\alpha x}\sin\beta x$

Keep in mind that there is a key pitfall to this method. Consider the differential equation $y''+5y'+6y = 3e^{-2x}$. Based on the form of $r(x)$, we guess a particular solution of the form $y_p(x) = Ae^{-2x}$. But when we substitute this expression into the differential equation to find a value for A,we run into a problem. We have

$$y_p{}'(x) = -2Ae^{-2x}$$

and

$$y_p'' = 4Ae^{-2x},$$

so we want

$$\begin{aligned} y''+5y'+6y &= 3e^{-2x} \\ 4Ae^{-2x} + 5(-2Ae^{-2x}) + 6Ae^{-2x} &= 3e^{-2x} \\ 4Ae^{-2x} - 10Ae^{-2x} + 6Ae^{-2x} &= 3e^{-2x} \\ 0 &= 3e^{-2x}, \end{aligned}$$

which is not possible.

Looking closely, we see that, in this case, the general solution to the complementary equation is $c_1e^{-2x} + c_2e^{-3x}$. The exponential function in $r(x)$ is actually a solution to the complementary equation, so, as we just saw, all the terms on the left side of the equation cancel out. We can still use the method of undetermined coefficients in this case, but we have to alter our guess by multiplying it by x. Using the new guess, $y_p(x) = Axe^{-2x}$, we have

$$y_p'(x) = A(e^{-2x} - 2xe^{-2x}$$

and

$$y_p''(x) = -4Ae^{-2x} + 4Axe^{-2x}.$$

Substitution gives

$$\begin{aligned} y''+5y'+6y &= 3e^{-2x} \\ (-4Ae^{-2x} + 4Axe^{-2x}) + 5(Ae^{-2x} - 2Axe^{-2x}) + 6Axe^{-2x} &= 3e^{-2x} \\ -4Ae^{-2x} + 4Axe^{-2x} + 5Ae^{-2x} - 10Axe^{-2x} + 6Axe^{-2x} &= 3e^{-2x} \\ Ae^{-2x} &= 3e^{-2x}. \end{aligned}$$

So, $A = 3$ and $y_p(x) = 3xe^{-2x}$. This gives us the following general solution

$$y(x) = c_1e^{-2x} + c_2e^{-3x} + 3xe^{-2x}.$$

Note that if xe^{-2x} were also a solution to the complementary equation, we would have to multiply by x again, and we would try $y_p(x) = Ax^2e^{-2x}$.

PROBLEM-SOLVING STRATEGY: METHOD OF UNDETERMINED COEFFICIENTS

1. Solve the complementary equation and write down the general solution.
2. Based on the form of $r(x)$, make an initial guess for $y_p(x)$.
3. Check whether any term in the guess for$y_p(x)$ is a solution to the complementary equation. If so, multiply the guess by x. Repeat this step until there are no terms in $y_p(x)$ that solve the complementary equation.
4. Substitute $y_p(x)$ into the differential equation and equate like terms to find values for the unknown coefficients in $y_p(x)$.
5. Add the general solution to the complementary equation and the particular solution you just found to obtain the general solution to the nonhomogeneous equation.

Example 17.2.4: Solving Nonhomogeneous Equations

Find the general solutions to the following differential equations.

a. $y'' - 9y = -6\cos 3x$
b. $x''+2x'+x = 4e^{-t}$
c. $y''-2y'+5y = 10x^2 - 3x - 3$
d. $y'' - 3y' = -12t$

Solution

a. The complementary equation is $y''-9y = 0$, which has the general solution $c_1e^{3x} + c_2e^{-3x}$(step 1). Based on the form of $r(x) = -6\cos 3x$, our initial guess for the particular solution is $y_p(x) = A\cos 3x + B\sin 3x$ (step 2). None of the terms in $y_p(x)$ solve the complementary equation, so this is a valid guess (step 3).
Now we want to find values for A and B, so substitute y_p into the differential equation. We have

$$y_p'(x) = -3A\sin 3x + 3B\cos 3x \text{ and } y_p''(x) = -9A\cos 3x - 9B\sin 3x,$$

so we want to find values of A and B such that

$$\begin{aligned} y''-9y &= -6\cos 3x \\ -9A\cos 3x - 9B\sin 3x - 9(A\cos 3x + B\sin 3x) &= -6\cos 3x \\ -18A\cos 3x - 18B\sin 3x &= -6\cos 3x. \end{aligned}$$

Therefore,

$$\begin{aligned} -18A &= -6 \\ -18B &= 0. \end{aligned}$$

This gives $A = \frac{1}{3}$ and $B = 0$, so $y_p(x) = (\frac{1}{3})\cos 3x$ (step 4).
Putting everything together, we have the general solution

$$y(x) = c_1 e^{3x} + c_2 e^{-3x} + \frac{1}{3}\cos 3x.$$

b. The complementary equation is $x'' + 2x' + x = 0$, which has the general solution $c_1 e^{-t} + c_2 te^{-t}$ (step 1). Based on the form $r(t) = 4e^{-t}$, our initial guess for the particular solution is $x_p(t) = Ae^{-t}$ (step 2). However, we see that this guess solves the complementary equation, so we must multiply by t, which gives a new guess: $x_p(t) = Ate^{-t}$ (step 3). Checking this new guess, we see that it, too, solves the complementary equation, so we must multiply by *t* again, which gives $x_p(t) = At^2e^{-t}$ (step 3 again). Now, checking this guess, we see that $x_p(t)$ does not solve the complementary equation, so this is a valid guess (step 3 yet again).
We now want to find a value for A, so we substitute x_p into the differential equation. We have

$$x_p(t) = At^2e^{-t}, \text{ so}$$
$$x_p'(t) = 2Ate^{-t} - At^2e^{-t}$$

and

$$x_p''(t) = 2Ae^{-t} - 2Ate^{-t} - (2Ate^{-t} - At^2e^{-t}) = 2Ae^{-t} - 4Ate^{-t} + At^2e^{-t}.$$

Substituting into the differential equation, we want to find a value of A so that

$$\begin{aligned} x''+2x'+x &= 4e^{-t} \\ 2Ae^{-t} - 4Ate^{-t} + At^2e^{-t} + 2(2Ate^{-t} - At^2e^{-t}) + At^2e^{-t} &= 4e^{-t} \\ 2Ae^{-t} &= 4e^{-t}. \end{aligned}$$

This gives $A = 2$, so $x_p(t) = 2t^2e^{-t}$ (step 4). Putting everything together, we have the general solution

$$x(t) = c_1 e^{-t} + c_2 te^{-t} + 2t^2e^{-t}.$$

c. The complementary equation is $y''-2y'+5y = 0$, which has the general solution $c_1 e^x \cos 2x + c_2 e^x \sin 2x$ (step 1). Based on the form $r(x) = 10x^2 - 3x - 3$, our initial guess for the particular solution is $y_p(x) = Ax^2 + Bx + C$ (step 2). None of the terms in $y_p(x)$ solve the complementary equation, so this is a valid guess (step 3). We now want to find values for A, B, and C, so we substitute y_p into the differential equation. We have $y_p'(x) = 2Ax + B$ and $y_p''(x) = 2A$, so we want to find values of A, B, and C such that

$$\begin{aligned} y''-2y'+5y &= 10x^2 - 3x - 3 \\ 2A - 2(2Ax + B) + 5(Ax^2 + Bx + C) &= 10x^2 - 3x - 3 \\ 5Ax^2 + (5B - 4A)x + (5C - 2B + 2A) &= 10x^2 - 3x - 3. \end{aligned}$$

Therefore,

$$\begin{aligned} 5A &= 10 \\ 5B - 4A &= -3 \\ 5C - 2B + 2A &= -3. \end{aligned}$$

This gives $A = 2$, $B = 1$, and $C = -1$, so $y_p(x) = 2x^2 + x - 1$ (step 4). Putting everything together, we have the general solution

$$y(x) = c_1 e^x \cos 2x + c_2 e^x \sin 2x + 2x^2 + x - 1.$$

d. The complementary equation is $y''-3y' = 0$, which has the general solution $c_1 e^{3t} + c_2$ (step 1). Based on the form r(t)=−12t,r(t)=−12t, our initial guess for the particular solution is $y_p(t) = At + B$ (step 2). However, we see that the constant term in this guess solves the complementary equation, so we must multiply by t, which gives a new guess: $y_p(t) = At^2 + Bt$ (step 3). Checking this new guess, we see that none of the terms in $y_p(t)$ solve the complementary equation, so this is a

valid guess (step 3 again). We now want to find values for A and B, so we substitute y_p into the differential equation. We have $y_p'(t) = 2At + B$ and $y_p''(t) = 2A$, so we want to find values of AA and BB such that

$$\begin{aligned} y''-3y' &= -12t \\ 2A - 3(2At + B) &= -12t \\ -6At + (2A - 3B) &= -12t. \end{aligned}$$

Therefore,

$$\begin{aligned} -6A &= -12 \\ 2A - 3B &= 0. \end{aligned}$$

This gives $A = 2$ and $B = 4/3$, so $y_p(t) = 2t^2 + (4/3)t$ (step 4). Putting everything together, we have the general solution

$$y(t) = c_1e^{3t} + c_2 + 2t^2 + \frac{4}{3}t.$$

? Exercise 17.2.3

Find the general solution to the following differential equations.

a. $y''-5y'+4y = 3e^x$
b. $y''+y'-6y = 52\cos 2t$

Hint

Use the problem-solving strategy.

Answer a

$y(x) = c_1e^{4x} + c_2e^x - xe^x$

Answer b

$y(t) = c_1e^{-3t} + c_2e^{2t} - 5\cos 2t + \sin 2t$

Variation of Parameters

Sometimes, $r(x)$ is not a combination of polynomials, exponentials, or sines and cosines. When this is the case, the method of undetermined coefficients does not work, and we have to use another approach to find a particular solution to the differential equation. We use an approach called the *method of variation of parameters*.

To simplify our calculations a little, we are going to divide the differential equation through by a, so we have a leading coefficient of 1. Then the differential equation has the form

$$y''+py'+qy = r(x),$$

where p and q are constants.

If the general solution to the complementary equation is given by $c_1y_1(x) + c_2y_2(x)$, we are going to look for a particular solution of the form

$$y_p(x) = u(x)y_1(x) + v(x)y_2(x).$$

In this case, we use the two linearly independent solutions to the complementary equation to form our particular solution. However, we are assuming the coefficients are functions of x, rather than constants. We want to find functions $u(x)$ and $v(x)$ such that $y_p(x)$ satisfies the differential equation. We have

$$\begin{aligned} y_p &= uy_1 + vy_2 \\ y_p' &= u'y_1 + uy_1'+v'y_2 + vy_2' \\ y_p'' &= (u'y_1 + v'y_2)'+u'y_1'+uy_1''+v'y_2'+vy_2''. \end{aligned}$$

Substituting into the differential equation, we obtain

$$\begin{aligned} y_p''+py_p'+qy_p &= [(u'y_1+v'y_2)'+u'y_1'+uy_1''+v'y_2'+vy_2'']\\ &\quad +p[u'y_1+uy_1'+v'y_2+vy_2']+q[uy_1+vy_2]\\ &= u[y_1''+py_1'+qy_1]+v[y_2''+py_2'+qy_2]\\ &\quad +(u'y_1+v'y_2)'+p(u'y_1+v'y_2)+(u'y_1'+v'y_2'). \end{aligned}$$

Note that y_1 and y_2 are solutions to the complementary equation, so the first two terms are zero. Thus, we have

$$(u'y_1+v'y_2)'+p(u'y_1+v'y_2)+(u'y_1'+v'y_2')=r(x).$$

If we simplify this equation by imposing the additional condition $u'y_1+v'y_2=0$, the first two terms are zero, and this reduces to $u'y_1'+v'y_2'=r(x)$. So, with this additional condition, we have a system of two equations in two unknowns:

$$\begin{aligned} u'y_1+v'y_2&=0\\ u'y_1'+v'y_2'&=r(x). \end{aligned}$$

Solving this system gives us u' and v', which we can integrate to find u and v.

Then, $y_p(x)=u(x)y_1(x)+v(x)y_2(x)$ is a particular solution to the differential equation. Solving this system of equations is sometimes challenging, so let's take this opportunity to review Cramer's rule, which allows us to solve the system of equations using determinants.

RULE: CRAMER'S RULE

The system of equations

$$\begin{aligned} a_1z_1+b_1z_2&=r_1\\ a_2z_1+b_2z_2&=r_2 \end{aligned}$$

has a unique solution if and only if the determinant of the coefficients is not zero. In this case, the solution is given by

$$z_1=\frac{\begin{vmatrix} r_1 & b_1\\ r_2 & b_2\end{vmatrix}}{\begin{vmatrix} a_1 & b_1\\ a_2 & b_2\end{vmatrix}} \quad \text{and} \quad z_2=\frac{\begin{vmatrix} a_1 & r_1\\ a_2 & r_2\end{vmatrix}}{\begin{vmatrix} a_1 & b_1\\ a_2 & b_2\end{vmatrix}}. \tag{17.2.1}$$

Example 17.2.5: Using Cramer's Rule

Use Cramer's rule to solve the following system of equations.

$$\begin{aligned} x^2z_1+2xz_2&=0\\ z_1-3x^2z_2&=2x \end{aligned}$$

Solution

We have

$$\begin{aligned} a_1(x)&=x^2\\ a_2(x)&=1\\ b_1(x)&=2x\\ b_2(x)&=-3x^2\\ r_1(x)&=0\\ r_2(x)&=2x. \end{aligned}$$

Then,

$$\begin{vmatrix} a_1 & b_1\\ a_2 & b_2\end{vmatrix}=\begin{vmatrix} x^2 & 2x\\ 1 & -3x^2\end{vmatrix}=-3x^4-2x$$

and

$$\begin{vmatrix} r_1b_1 \\ r_2b_2 \end{vmatrix} = \begin{vmatrix} 02x \\ 2x-3x^2 \end{vmatrix} = 0-4x^2 = -4x^2.$$

Thus,

$$z1 = \frac{\begin{vmatrix} r_1b_1 \\ r_2b_2 \end{vmatrix}}{\begin{vmatrix} a_1b_1 \\ a_2b_2 \end{vmatrix}} = \frac{-4x^2}{-3x^4-2x} = \frac{4x}{3x^3+2}.$$

In addition,

$$\begin{vmatrix} a_1r_1 \\ a_2r_2 \end{vmatrix} = \begin{vmatrix} x^20 \\ 12x \end{vmatrix} = 2x^3 - 0 = 2x^3.$$

Thus,

$$z2 = \frac{\begin{vmatrix} a_1r_1 \\ a_2r_2 \end{vmatrix}}{\begin{vmatrix} a_1b_1 \\ a_2b_2 \end{vmatrix}} = \frac{2x^3}{-3x^4-2x} = \frac{-2x^2}{3x^3+2}.$$

Exercise 17.2.4

Use Cramer's rule to solve the following system of equations.

$$2xz_1 - 3z_2 = 0$$
$$x^2z_1 + 4xz_2 = x+1$$

Hint

Use the process from the previous example.

Answer

$z_1 = \frac{3x+3}{11x^2}, z_2 = \frac{2x+2}{11x}$

PROBLEM-SOLVING STRATEGY: METHOD OF VARIATION OF PARAMETERS

1. Solve the complementary equation and write down the general solution
$$c_1y_1(x) + c_2y_2(x).$$
2. Use Cramer's rule or another suitable technique to find functions $u'(x)$ and $v'(x)$ satisfying
$$u'y_1 + v'y_2 = 0$$
$$u'y_1' + v'y_2' = r(x).$$
3. Integrate u' and v' to find $u(x)$ and $v(x)$. Then, $y_p(x) = u(x)y_1(x) + v(x)y_2(x)$ is a particular solution to the equation.
4. Add the general solution to the complementary equation and the particular solution found in step 3 to obtain the general solution to the nonhomogeneous equation.

Example 17.2.6: Using the Method of Variation of Parameters

Find the general solution to the following differential equations.

a. $y''-2y'+y = \frac{e^t}{t^2}$
b. $y''+y = 3\sin^2 x$

Solution

a. The complementary equation is $y''-2y'+y=0$ with associated general solution $c_1e^t+c_2te^t$. Therefore, $y_1(t)=e^t$ and $y_2(t)=te^t$. Calculating the derivatives, we get $y_1'(t)=e^t$ and $y_2'(t)=e^t+te^t$ (step 1). Then, we want to find functions $u'(t)$ and $v'(t)$ so that

$$\begin{aligned} u'e^t+v'te^t&=0\\ u'e^t+v'(e^t+te^t)&=\frac{e^t}{t^2}. \end{aligned}$$

Applying Cramer's rule (Equation 17.2.1), we have

$$u'=\frac{\begin{vmatrix} 0 & te^t\\ \frac{e^t}{t^2} & e^t+te^t\end{vmatrix}}{\begin{vmatrix} te^t & te^t\\ e^t & e^t+te^t\end{vmatrix}}=\frac{0-te^t(\frac{e^t}{t^2})}{e^t(e^t+te^t)-e^tte^t}=\frac{-\frac{e^{2t}}{t}}{e^{2t}}=-\frac{1}{t}$$

and

$$v'=\frac{\begin{vmatrix} e^t & 0\\ e^t & \frac{e^t}{t^2}\end{vmatrix}}{\begin{vmatrix} te^t & te^t\\ e^t & e^t+te^t\end{vmatrix}}=\frac{e^t(\frac{e^t}{t^2})}{e^{2t}}=\frac{1}{t^2}\quad\text{(step 2)}.$$

Integrating, we get

$$\begin{aligned} u&=-\int\frac{1}{t}dt=-\ln|t|\\ v&=\int\frac{1}{t^2}dt=-\frac{1}{t}\qquad\text{(step 3)} \end{aligned}$$

Then we have

$$\begin{aligned} y_p&=-e^t\ln|t|-\frac{1}{t}te^t\\ &=-e^t\ln|t|-e^t.\qquad\text{(step 4)} \end{aligned}$$

The e^t term is a solution to the complementary equation, so we don't need to carry that term into our general solution explicitly. The general solution is

$$y(t)=c_1e^t+c_2te^t-e^t\ln|t|\qquad\text{(step 5)}$$

b. The complementary equation is $y''+y=0$ with associated general solution $c_1\cos x+c_2\sin x$. So, $y_1(x)=\cos x$ and $y_2(x)=\sin x$ (step 1). Then, we want to find functions $u'(x)$ and $v'(x)$ such that

$$\begin{aligned} u'\cos x+v'\sin x&=0\\ -u'\sin x+v'\cos x&=3\sin_2 x \end{aligned}.$$

Applying Cramer's rule, we have

$$u'=\frac{\begin{vmatrix} 0 & \sin x\\ 3\sin^2x\cos x & \cos x\sin x\end{vmatrix}}{\begin{vmatrix} \cos x\sin x & \sin x\\ -\sin x\cos x & \cos x\end{vmatrix}}=\frac{0-3\sin^3x}{\cos^2x+\sin^2x}=-3\sin^3x$$

and

$$v' = \frac{\begin{vmatrix} \cos x & 0 \\ -\sin x & 3\sin^2 x \end{vmatrix}}{\begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}} = \frac{3\sin^2 x \cos x}{1} = 3\sin^2 x \cos x \text{(step 2)}.$$

Integrating first to find u, we get

$$u = \int -3\sin^3 x dx = -3\left[-\frac{1}{3}\sin^2 x \cos x + \frac{2}{3}\int \sin x dx\right] = \sin^2 x \cos x + 2\cos x.$$

Now, we integrate to find v. Using substitution (with $w = \sin x$), we get

$$v = \int 3\sin^2 x \cos x dx = \int 3w^2 dw = w^3 = sin^3 x.$$

Then,

$$\begin{aligned} y_p &= (\sin^2 x \cos x + 2\cos x)\cos x + (\sin^3 x)\sin x \\ &= \sin_2 x \cos_2 x + 2\cos_2 x + \sin_4 x \\ &= 2\cos_2 x + \sin_2 x(\cos^2 x + \sin^2 x) \qquad \text{(step 4)}. \\ &= 2\cos_2 x + \sin_2 x \\ &= \cos_2 x + 1 \end{aligned}$$

The general solution is

$$y(x) = c_1 \cos x + c_2 \sin x + 1 + \cos^2 x \text{(step 5)}.$$

Exercise 17.2.5

Find the general solution to the following differential equations.

a. $y'' + y = \sec x$

b. $x'' - 2x' + x = \dfrac{e^t}{t}$

Hint

Follow the problem-solving strategy.

Answer a

$y(x) = c_1 \cos x + c_2 \sin x + \cos x \ln|\cos x| + x \sin x$

Answer b

$x(t) = c_1 e^t + c_2 t e^t + t e^t \ln|t|$

Key Concepts

- To solve a nonhomogeneous linear second-order differential equation, first find the general solution to the complementary equation, then find a particular solution to the nonhomogeneous equation.
- Let $y_p(x)$ be any particular solution to the nonhomogeneous linear differential equation

  $$a_2(x)y'' + a_1(x)y' + a_0(x)y = r(x),$$

  and let $c_1 y_1(x) + c_2 y_2(x)$ denote the general solution to the complementary equation. Then, the general solution to the nonhomogeneous equation is given by

  $$y(x) = c_1 y_1(x) + c_2 y_2(x) + y_p(x).$$

- When $r(x)$ is a combination of polynomials, exponential functions, sines, and cosines, use the method of undetermined coefficients to find the particular solution. To use this method, assume a solution in the same form as $r(x)$, multiplying by x as

necessary until the assumed solution is linearly independent of the general solution to the complementary equation. Then, substitute the assumed solution into the differential equation to find values for the coefficients.

- When $r(x)$ is *not* a combination of polynomials, exponential functions, or sines and cosines, use the method of variation of parameters to find the particular solution. This method involves using Cramer's rule or another suitable technique to find functions and $v'(x)$ satisfying

$$u'y_1 + v'y_2 = 0$$
$$u'y_1{}' + v'y_2{}' = r(x).$$

Then, $y_p(x) = u(x)y_1(x) + v(x)y_2(x)$ is a particular solution to the differential equation.

Key Equations

- **Complementary equation**
  $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$
- **General solution to a nonhomogeneous linear differential equation**
  $y(x) = c_1y_1(x) + c_2y_2(x) + y_p(x)$

Glossary

complementary equation

for the nonhomogeneous linear differential equation $a + 2(x)y'' + a_1(x)y' + a_0(x)y = r(x)$, the associated homogeneous equation, called the *complementary equation*, is $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$

method of undetermined coefficients

a method that involves making a guess about the form of the particular solution, then solving for the coefficients in the guess

method of variation of parameters

a method that involves looking for particular solutions in the form $y_p(x) = u(x)y_1(x) + v(x)y_2(x)$, where y_1 and y_2 are linearly independent solutions to the complementary equations, and then solving a system of equations to find $u(x)$ and $v(x)$

particular solution

a solution $y_p(x)$ of a differential equation that contains no arbitrary constants

---

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17.2E: Exercises for Section 17.2

Solve the following equations using the method of undetermined coefficients.

1. $2y'' - 5y' - 12y = 6$

2. $3y'' + y' - 4y = 8$

Answer

$y = c_1 e^{-4x/3} + c_2 e^{x} - 2$

3. $y'' - 6y' + 5y = e^{-x}$

4. $y'' + 16y = e^{-2x}$

Answer

$y = c_1 \cos 4x + c_2 \sin 4x + \frac{1}{20} e^{-2x}$

5. $y'' - 4y = x^2 + 1$

6. $y'' - 4y' + 4y = 8x^2 + 4x$

Answer

$y = c_1 e^{2x} + c_2 x e^{2x} + 2x^2 + 5x$

7. $y'' - 2y' - 3y = \sin 2x$

8. $y'' + 2y' + y = \sin x + \cos x$

Answer

$y = c_1 e^{-x} + c_2 x e^{-x} + \frac{1}{2} \sin x - \frac{1}{2} \cos x$

9. $y'' + 9y = e^{x} \cos x$

10. $y'' + y = 3 \sin 2x + x \cos 2x$

Answer

$y = c_1 \cos x + c_2 \sin x - \frac{1}{3} x \cos 2x - \frac{5}{9} \sin 2x$

11. $y'' + 3y' - 28y = 10e^{4x}$

12. $y'' + 10y' + 25y = xe^{-5x} + 4$

Answer

$y = c_1 e^{-5x} + c_2 x e^{-5x} + \frac{1}{6} x^3 e^{-5x} + \frac{4}{25}$

In exercises 13 - 18,

a. **Write the form for the particular solution $y_p(x)$ for the method of undetermined coefficients.**
b. **[T]** Use a computer algebra system to find a particular solution to the given equation.

13. $y'' - y' - y = x + e^{-x}$

14. $y'' - 3y = x^2 - 4x + 11$

Answer

a. $y_p(x) = Ax^2 + Bx + C$

b. $y_p(x) = -\frac{1}{3} x^2 + \frac{4}{3} x - \frac{35}{9}$

15. $y'' - y' - 4y = e^x \cos 3x$

16. $2y'' - y' + y = (x^2 - 5x)e^{-x}$

Answer

a. $y_p(x) = (Ax^2 + Bx + C)e^{-x}$

b. $y_p(x) = (\frac{1}{4}x^2 - \frac{5}{8}x - \frac{33}{32})e^{-x}$

17. $4y'' + 5y' - 2y = e^{2x} + x \sin x$

18. $y'' - y' - 2y = x^2 e^x \sin x$

Answer

a. $y_p(x) = (Ax^2 + Bx + C)e^x \cos x + (Dx^2 + Ex + F)e^x \sin x$

b. $y_p(x) = (-\frac{1}{10}x^2 - \frac{11}{25}x - \frac{27}{250})e^x \cos x + (-\frac{3}{10}x^2 + \frac{2}{25}x + \frac{39}{250})e^x \sin x$

Solve the differential equation using either the method of undetermined coefficients or the variation of parameters.

19. $y'' + 3y' - 4y = 2e^x$

20. $y'' + 2y' = e^{3x}$

Answer

$y = c_1 + c_2 e^{-2x} + \frac{1}{15}e^{3x}$

21. $y'' + 6y' + 9y = e^{-x}$

22. $y'' + 2y' - 8y = 6e^{2x}$

Answer

$y = c_1 e^{2x} + c_2 e^{-4x} + xe^{2x}$

Solve the differential equation using the method of variation of parameters.

23. $4y'' + y = 2 \sin x$

24. $y'' - 9y = 8x$

Answer

$y = c_1 e^{3x} + c_2 e^{-3x} - \frac{8x}{9}$

25. $y'' + y = \sec x, \quad 0 < x < \pi/2$

26. $y'' + 4y = 3 \csc 2x, \quad 0 < x < \pi/2$

Answer

$y = c_1 \cos 2x + c_2 \sin 2x - \frac{3}{2}x \cos 2x + \frac{3}{4}\sin 2x \ln(\sin 2x)$

Find the unique solution satisfying the differential equation and the initial conditions given, where $y_p(x)$ is the particular solution.

27. $y'' - 2y' + y = 12e^x, \quad y_p(x) = 6x^2 e^x, \ y(0) = 6, \ y'(0) = 0$

28. $y'' - 7y' = 4xe^{7x}, \quad y_p(x) = \frac{2}{7}x^2 e^{7x} - \frac{4}{49}xe^{7x}, \ y(0) = -1, \ y'(0) = 0$

Answer

$y = -\frac{347}{343} + \frac{4}{343}e^{7x} + \frac{2}{7}x^2 e^{7x} - \frac{4}{49}xe^{7x}$

29. $y'' + y = \cos x - 4 \sin x, \quad y_p(x) = 2x \cos x + \frac{1}{2}x \sin x, \ y(0) = 8, \ y'(0) = -4$

30. $y''-5y'=e^{5x}+8e^{-5x}, \quad y_p(x)=\frac{1}{5}xe^{5x}+\frac{4}{25}e^{-5x},\ y(0)=-2,\ y'(0)=0$

Answer

$y=-\frac{57}{25}+\frac{3}{25}e^{5x}+\frac{1}{5}xe^{5x}+\frac{4}{25}e^{-5x}$

In problems 31 - 32, two linearly independent solutions—y_1 and y_2—are given that satisfy the corresponding homogeneous equation. Use the method of variation of parameters to find a particular solution to the given nonhomogeneous equation. Assume $x>0$ in each exercise.

31. $x^2y''+2xy'-2y=3x, \quad y_1(x)=x,\ y2(x)=x^{-2}$

32. $x^2y''-2y=10x^2-1, \quad y_1(x)=x^2,\ y_2(x)=x^{-1}$

Answer

$y_p=\frac{1}{2}+\frac{10}{3}x^2\ln x$

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17.3: Applications of Second-Order Differential Equations

Learning Objectives

- Solve a second-order differential equation representing simple harmonic motion.
- Solve a second-order differential equation representing damped simple harmonic motion.
- Solve a second-order differential equation representing forced simple harmonic motion.
- Solve a second-order differential equation representing charge and current in an RLC series circuit.

We saw in the chapter introduction that second-order linear differential equations are used to model many situations in physics and engineering. In this section, we look at how this works for systems of an object with mass attached to a vertical spring and an electric circuit containing a resistor, an inductor, and a capacitor connected in series. Models such as these can be used to approximate other more complicated situations; for example, bonds between atoms or molecules are often modeled as springs that vibrate, as described by these same differential equations.

Simple Harmonic Motion

Consider a mass suspended from a spring attached to a rigid support. (This is commonly called a **spring-mass system**.) Gravity is pulling the mass downward and the restoring force of the spring is pulling the mass upward. As shown in Figure 17.3.1, when these two forces are equal, the mass is said to be at the equilibrium position. If the mass is displaced from equilibrium, it oscillates up and down. This behavior can be modeled by a second-order constant-coefficient differential equation.

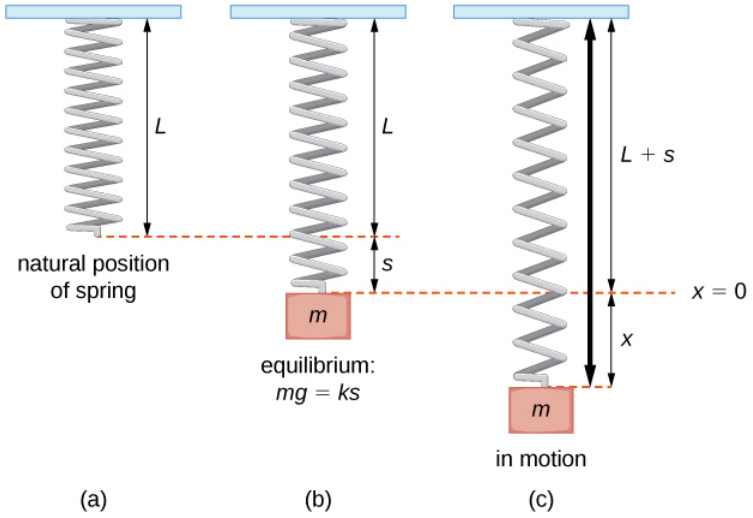


Figure 17.3.1: A spring in its natural position (a), at equilibrium with a mass *m* attached (b), and in oscillatory motion (c).

Let $x(t)$ denote the displacement of the mass from equilibrium. Note that for spring-mass systems of this type, it is customary to adopt the convention that down is positive. Thus, a positive displacement indicates the mass is *below* the equilibrium point, whereas a negative displacement indicates the mass is *above* equilibrium. Displacement is usually given in feet in the English system or meters in the metric system.

Consider the forces acting on the mass. The force of gravity is given by mg.mg. In the English system, mass is in slugs and the acceleration resulting from gravity is in feet per second squared. The acceleration resulting from gravity is constant, so in the English system, $g = 32\ ft/sec^2$. Recall that 1 slug-foot/sec^2 is a pound, so the expression *mg* can be expressed in pounds. Metric system units are kilograms for mass and m/sec^2 for gravitational acceleration. In the metric system, we have $g = 9.8$ m/sec^2.

According to **Hooke's law**, the restoring force of the spring is proportional to the displacement and acts in the opposite direction from the displacement, so the restoring force is given by $-k(s + x)$. The spring constant is given in pounds per foot in the English system and in newtons per meter in the metric system.

Now, by Newton's second law, the sum of the forces on the system (gravity plus the restoring force) is equal to mass times acceleration, so we have

$$\begin{aligned} mx'' &= -k(s+x) + mg \\ &= -ks - kx + mg. \end{aligned}$$

However, by the way we have defined our equilibrium position, $mg = ks$, the differential equation becomes

$$mx'' + kx = 0.$$

It is convenient to rearrange this equation and introduce a new variable, called the **angular frequency**, ω. Letting $\omega = \sqrt{k/m}$, we can write the equation as

$$x'' + \omega^2 x = 0.$$

This differential equation has the general solution

$$x(t) = c_1 \cos \omega t + c_2 \sin \omega t, \tag{17.3.1}$$

which gives the position of the mass at any point in time. The motion of the mass is called simple **harmonic motion**. The period of this motion (the time it takes to complete one oscillation) is $T = \dfrac{2\pi}{\omega}$ and the frequency is $f = \dfrac{1}{T} = \dfrac{\omega}{2\pi}$ (Figure 17.3.2).

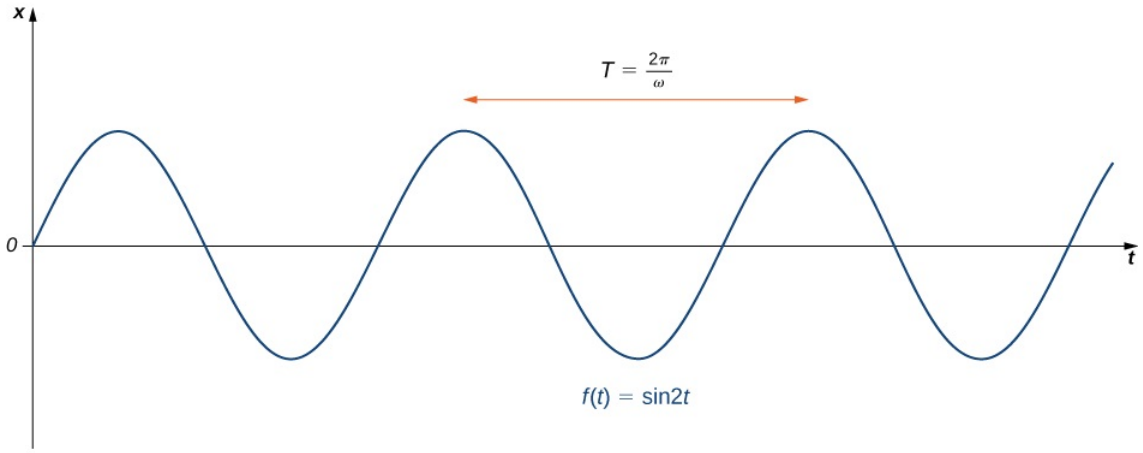


Figure 17.3.2: A graph of vertical displacement versus time for simple harmonic motion.

Example 17.3.1: Simple Harmonic Motion

Assume an object weighing 2 lb stretches a spring 6 in. Find the equation of motion if the spring is released from the equilibrium position with an upward velocity of 16 ft/sec. What is the period of the motion?

Solution

We first need to find the spring constant. We have

$$\begin{aligned} mg &= ks \\ 2 &= k\left(\frac{1}{2}\right) \\ k &= 4. \end{aligned}$$

We also know that weight W equals the product of mass m and the acceleration due to gravity g. In English units, the acceleration due to gravity is 32 ft/sec^2.

$$\begin{aligned} W &= mg \\ 2 &= m(32) \\ m &= \frac{1}{16} \end{aligned}$$

Thus, the differential equation representing this system is

$$\frac{1}{16}x''+4x=0.$$

Multiplying through by 16, we get $x''+64x=0$, which can also be written in the form $x''+(8^2)x=0$. This equation has the general solution

$$x(t)=c_1\cos(8t)+c_2\sin(8t).$$

The mass was released from the equilibrium position, so $x(0)=0$, and it had an initial upward velocity of 16 ft/sec, so $x'(0)=-16$. Applying these initial conditions to solve for c_1 and c_2. gives

$$x(t)=-2\sin 8t.$$

The period of this motion is $\frac{2\pi}{8}=\frac{\pi}{4}$ sec.

? Exercise 17.3.1

A 200-g mass stretches a spring 5 cm. Find the equation of motion of the mass if it is released from rest from a position 10 cm below the equilibrium position. What is the frequency of this motion?

Hint

First find the spring constant.

Answer

$x(t)=0.1\cos(14t)$ (in meters); frequency is $\frac{14}{2\pi}$ Hz.

Writing the general solution in the form $x(t)=c_1\cos(\omega t)+c_2\sin(\omega t)$ (Equation 17.3.1) has some advantages. It is easy to see the link between the differential equation and the solution, and the period and frequency of motion are evident. This form of the function tells us very little about the amplitude of the motion, however. In some situations, we may prefer to write the solution in the form

$$x(t)=A\sin(\omega t+\phi).$$

Although the link to the differential equation is not as explicit in this case, the period and frequency of motion are still evident. Furthermore, the amplitude of the motion, A, is obvious in this form of the function. The constant ϕ is called a **phase shift** and has the effect of shifting the graph of the function to the left or right.

To convert the solution to this form, we want to find the values of A and ϕ such that

$$c_1\cos(\omega t)+c_2\sin(\omega t)=A\sin(\omega t+\phi).$$

We first apply the trigonometric identity

$$\sin(\alpha+\beta)=\sin\alpha\cos\beta+\cos\alpha\sin\beta$$

to get

$$\begin{aligned}c_1\cos(\omega t)+c_2\sin(\omega t)&=A(\sin(\omega t)\cos\phi+\cos(\omega t)\sin\phi)\\&=A\sin\phi(\cos(\omega t))+A\cos\phi(\sin(\omega t)).\end{aligned}$$

Thus,

$$c_1=A\sin\phi \text{ and } c_2=A\cos\phi.$$

If we square both of these equations and add them together, we get

$$\begin{aligned}c_1^2+c_2^2&=A^2\sin_2\phi+A^2\cos_2\phi\\&=A^2(\sin^2\phi+\cos^2\phi)\\&=A^2.\end{aligned}$$

Thus,

$$A = \sqrt{c_1^2 + c_2^2}.$$

Now, to find ϕ, go back to the equations for c_1 and c_2, but this time, divide the first equation by the second equation to get

$$\begin{aligned} \frac{c_1}{c_2} &= \frac{A\sin\phi}{A\cos\phi} \\ &= \tan\phi. \end{aligned}$$

Then,

$$\tan\phi = \frac{c_1}{c_2}.$$

We summarize this finding in the following theorem.

SOLUTION TO THE EQUATION FOR SIMPLE HARMONIC MOTION

The function $x(t) = c_1\cos(\omega t) + c_2\sin(\omega t)$ can be written in the form $x(t) = A\sin(\omega t + \phi)$, where $A = \sqrt{c_1^2 + c_2^2}$ and $\tan\phi = \frac{c_1}{c_2}$.

Note that when using the formula $\tan\phi = \frac{c_1}{c_2}$ to find ϕ, we must take care to ensure ϕ is in the right quadrant (Figure 17.3.3).

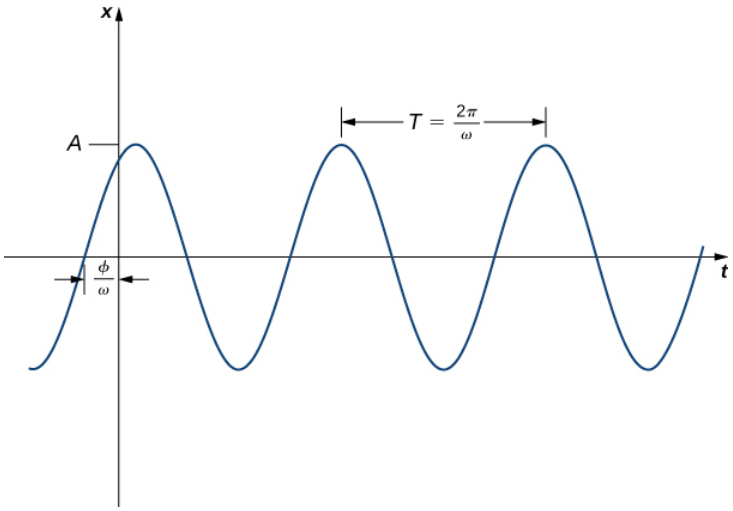


Figure 17.3.3: A graph of vertical displacement versus time for simple harmonic motion with a phase change.

Example 17.3.2: Expressing the Solution with a Phase Shift

Express the following functions in the form $A\sin(\omega t + \phi)$. What is the frequency of motion? The amplitude?

a. $x(t) = 2\cos(3t) + \sin(3t)$
b. $x(t) = 3\cos(2t) - 2\sin(2t)$

Solution

We have

$$A = \sqrt{c_1^2 + c_2^2} = \sqrt{2^2 + 1^2} = \sqrt{5}$$

and

$$\tan\phi = \frac{c_1}{c_2} = \frac{2}{1} = 2.$$

Note that both c_1 and c_2 are positive, so ϕ is in the first quadrant. Thus,

$$\phi \approx 1.107 \text{ rad},$$

so we have

$$x(t) = 2\cos(3t) + \sin(3t) = 5\sin(3t + 1.107).$$

The frequency is $\frac{\omega}{2\pi} = \frac{3}{2\pi} \approx 0.477$. The amplitude is $\sqrt{5}$.

1. We have

$$A = \sqrt{c_1^2 + c_2^2} = \sqrt{3^2 + 2^2} = \sqrt{13}$$

and

$$\tan\phi = \frac{c_1}{c_2} = \frac{3}{-2} = -\frac{3}{2}.$$

Note that c_1 is positive but c_2 is negative, so ϕ is in the fourth quadrant. Thus,

$$\phi \approx -0.983 \text{ rad},$$

so we have

$$\begin{aligned} x(t) &= 3\cos(2t) - 2\sin(2t) \\ &= \sqrt{13}\sin(2t - 0.983). \end{aligned}$$

The frequency is $\frac{\omega}{2\pi} = \frac{2}{2\pi} \approx 0.318$. The amplitude is $\sqrt{13}$.

? Exercise 17.3.2

Express the function $x(t) = \cos(4t) + 4\sin(4t)$ in the form $A\sin(\omega t + \phi)$. What is the frequency of motion? The amplitude?

Hint

Use the process from the Example 17.3.2.

Answer

$x(t) = \sqrt{17}\sin(4t + 0.245)$, frequency $= \frac{4}{2\pi} \approx 0.637$, $A = \sqrt{17}$

Damped Vibrations

With the model just described, the motion of the mass continues indefinitely. Clearly, this doesn't happen in the real world. In the real world, there is almost always some friction in the system, which causes the oscillations to die off slowly—an effect called *damping*. So now let's look at how to incorporate that damping force into our differential equation.

Physical spring-mass systems almost always have some damping as a result of friction, air resistance, or a physical damper, called a *dashpot* (a pneumatic cylinder; Figure 17.3.4).



Figure 17.3.4: A dashpot is a pneumatic cylinder that dampens the motion of an oscillating system.

Because damping is primarily a friction force, we assume it is proportional to the velocity of the mass and acts in the opposite direction. So the damping force is given by $-bx'$ for some constant $b > 0$. Again applying Newton's second law, the differential equation becomes

$$mx''+bx'+kx = 0.$$

Then the associated characteristic equation is

$$m\lambda^2 + b\lambda + k = 0.$$

Applying the quadratic formula, we have

$$\lambda = \frac{-b \pm \sqrt{b^2 - 4mk}}{2m}.$$

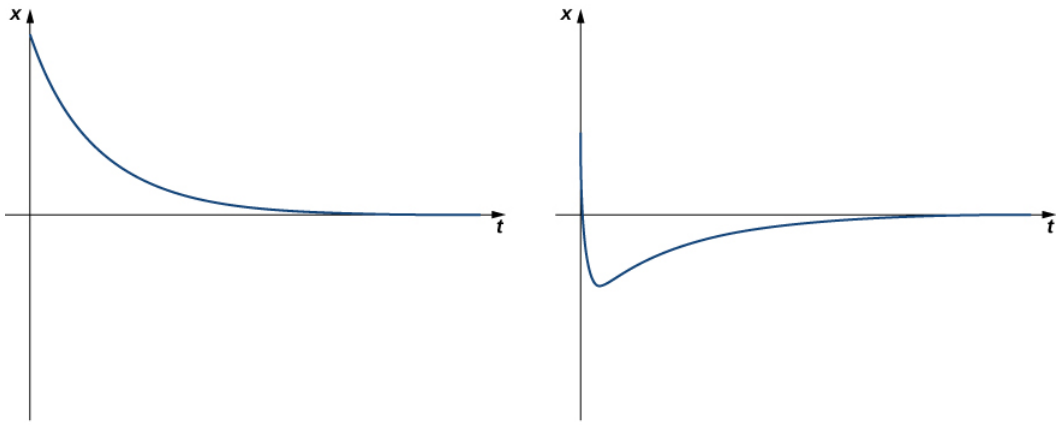
Just as in Second-Order Linear Equations we consider three cases, based on whether the characteristic equation has distinct real roots, a repeated real root, or complex conjugate roots.

Case 1: Overdamped Vibrations

When $b^2 > 4mk$, we say the system is ***overdamped***. The general solution has the form

$$x(t) = c_1 e^{\lambda_1 t} + c_2 e^{\lambda_2 t},$$

where both λ_1 and λ_2 are less than zero. Because the exponents are negative, the displacement decays to zero over time, usually quite quickly. Overdamped systems do not oscillate (no more than one change of direction), but simply move back toward the equilibrium position. Figure 17.3.5 shows what typical critically damped behavior looks like.



(a) (b)

Figure 17.3.5: Behavior of an overdamped spring-mass system, with no change in direction (a) and only one change in direction (b).

Example 17.3.3: Overdamped Spring-Mass System

A 16-lb mass is attached to a 10-ft spring. When the mass comes to rest in the equilibrium position, the spring measures 15 ft 4 in. The system is immersed in a medium that imparts a damping force equal to 5252 times the instantaneous velocity of the mass. Find the equation of motion if the mass is pushed upward from the equilibrium position with an initial upward velocity of 5 ft/sec. What is the position of the mass after 10 sec? Its velocity?

Solution

The mass stretches the spring 5 ft 4 in., or $\frac{16}{3}$ ft. Thus, $16 = \left(\frac{16}{3}\right)k$, so $k = 3$. We also have $m = \frac{16}{32} = \frac{1}{2}$, so the differential equation is

$$\frac{5}{2}x' + 3x = 0.$$

Multiplying through by 2 gives $x'' + 5x' + 6x = 0$, which has the general solution

$$x(t) = c_1 e^{-2t} + c_2 e^{-3t}.$$

Applying the initial conditions, $x(0) = 0$ and $x'(0) = -5$, we get

$$x(t) = -5e^{-2t} + 5e^{-3t}.$$

After 10 sec the mass is at position

$$x(10) = -5e^{-20} + 5e^{-30} \approx -1.0305 \times 10^{-8} \approx 0,$$

so it is, effectively, at the equilibrium position. We have $x'(t) = 10e^{-2t} - 15e^{-3t}$, so after 10 sec the mass is moving at a velocity of

$$x'(10) = 10e^{-20} - 15e^{-30} \approx 2.061 \times 10^{-8} \approx 0.$$

After only 10 sec, the mass is barely moving.

Exercise 17.3.3

A 2-kg mass is attached to a spring with spring constant 24 N/m. The system is then immersed in a medium imparting a damping force equal to 16 times the instantaneous velocity of the mass. Find the equation of motion if it is released from rest at a point 40 cm below equilibrium.

Hint

Follow the process from the previous example.

Answer

$x(t) = 0.6e^{-2t} - 0.2e^{-6t}$

Case 2: Critically Damped Vibrations

When $b^2 = 4mk$, we say the system is *critically damped*. The general solution has the form

$$x(t) = c_1 e^{\lambda_1 t} + c_2 t e^{\lambda_1 t},$$

where λ_1 is less than zero. The motion of a critically damped system is very similar to that of an overdamped system. It does not oscillate. However, with a critically damped system, if the damping is reduced even a little, oscillatory behavior results. From a practical perspective, physical systems are almost always either overdamped or underdamped (case 3, which we consider next). It is impossible to fine-tune the characteristics of a physical system so that b^2 and $4mk$ are exactly equal. Figure 17.3.6 shows what typical critically damped behavior looks like.

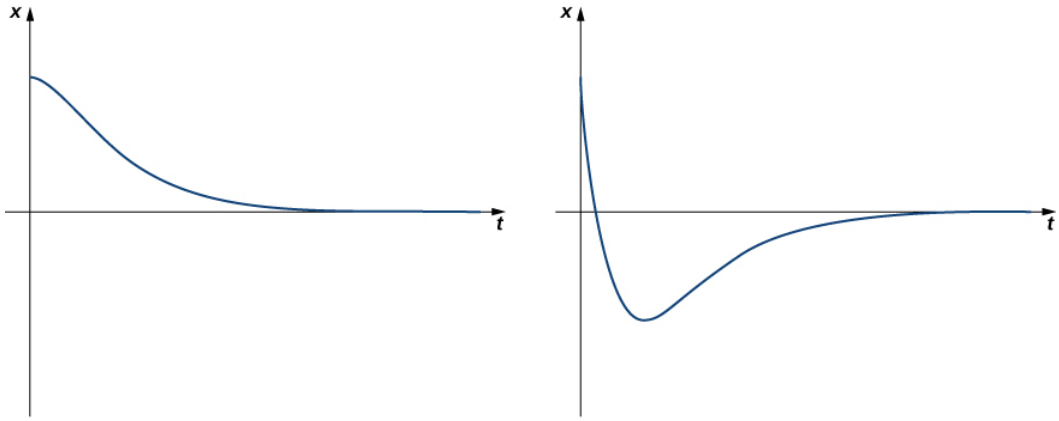


Figure 17.3.6: Behavior of a critically damped spring-mass system. The system graphed in part (a) has more damping than the system graphed in part (b).

Example 17.3.4: Critically Damped Spring-Mass System

A 1-kg mass stretches a spring 20 cm. The system is attached to a dashpot that imparts a damping force equal to 14 times the instantaneous velocity of the mass. Find the equation of motion if the mass is released from equilibrium with an upward velocity of 3 m/sec.

Solution

We have $mg = 1(9.8) = 0.2k$, so $k = 49$. Then, the differential equation is

$$x'' + 14x' + 49x = 0,$$

which has general solution

$$x(t) = c_1 e^{-7t} + c_2 t e^{-7t}.$$

Applying the initial conditions $x(0) = 0$ and $x'(0) = -3$ gives

$$x(t) = -3te^{-7t}.$$

Exercise 17.3.4

A 1-lb weight stretches a spring 6 in., and the system is attached to a dashpot that imparts a damping force equal to half the instantaneous velocity of the mass. Find the equation of motion if the mass is released from rest at a point 6 in. below equilibrium.

Hint

First find the spring constant.

Answer

$x(t) = \frac{1}{2}e^{-8t} + 4te^{-8t}$

Case 3: Undamped Vibrations

When $b^2 < 4mk$, we say the system is *underdamped*. The general solution has the form

$$x(t) = e^{\alpha t}(c_1 \cos(\beta t) + c_2 \sin(\beta t)),$$

where α is less than zero. Underdamped systems do oscillate because of the sine and cosine terms in the solution. However, the exponential term dominates eventually, so the amplitude of the oscillations decreases over time. Figure 17.3.7 shows what typical underdamped behavior looks like.

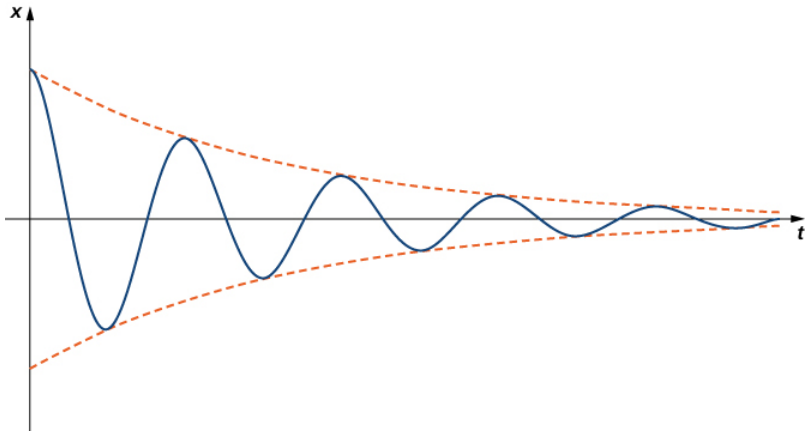


Figure 17.3.7: Behavior of an underdamped spring-mass system.

Note that for all damped systems, $\lim_{t\to\infty} x(t) = 0$. The system always approaches the equilibrium position over time.

Example 17.3.5: Underdamped Spring-Mass System

A 16-lb weight stretches a spring 3.2 ft. Assume the damping force on the system is equal to the instantaneous velocity of the mass. Find the equation of motion if the mass is released from rest at a point 9 in. below equilibrium.

Solution

We have $k = \frac{16}{3.2} = 5$ and $m = \frac{16}{32} = \frac{1}{2}$, so the differential equation is

$$\frac{1}{2}x'' + x' + 5x = 0, \text{ or } x'' + 2x' + 10x = 0.$$

This equation has the general solution

$$x(t) = e^{-t}(c_1 \cos(3t) + c_2 \sin(3t)).$$

Applying the initial conditions, $x(0) = \frac{3}{4}$ and $x'(0) = 0$, we get

$$x(t) = e^{-t}\left(\frac{3}{4}\cos(3t) + \frac{1}{4}\sin(3t)\right).$$

Exercise 17.3.5

A 1-kg mass stretches a spring 49 cm. The system is immersed in a medium that imparts a damping force equal to four times the instantaneous velocity of the mass. Find the equation of motion if the mass is released from rest at a point 24 cm above equilibrium.

Hint

First find the spring constant.

Answer

$x(t) = -0.24e^{-2t}\cos(4t) - 0.12e^{-2t}\sin(4t)$

Example 17.3.6: Chapter Opener: Modeling a Motorcycle Suspension System

For motocross riders, the suspension systems on their motorcycles are very important. The off-road courses on which they ride often include jumps, and losing control of the motorcycle when they land could cost them the race.



Figure 17.3.8: (credit: modification of work by nSeika, Flickr)

This suspension system can be modeled as a damped spring-mass system. We define our frame of reference with respect to the frame of the motorcycle. Assume the end of the shock absorber attached to the motorcycle frame is fixed. Then, the “mass” in our spring-mass system is the motorcycle wheel. We measure the position of the wheel with respect to the motorcycle frame. This may seem counterintuitive, since, in many cases, it is actually the motorcycle frame that moves, but this frame of reference preserves the development of the differential equation that was done earlier. As with earlier development, we define the downward direction to be positive.

When the motorcycle is lifted by its frame, the wheel hangs freely and the spring is uncompressed. This is the spring’s natural position. When the motorcycle is placed on the ground and the rider mounts the motorcycle, the spring compresses and the system is in the equilibrium position (Figure 17.3.9).

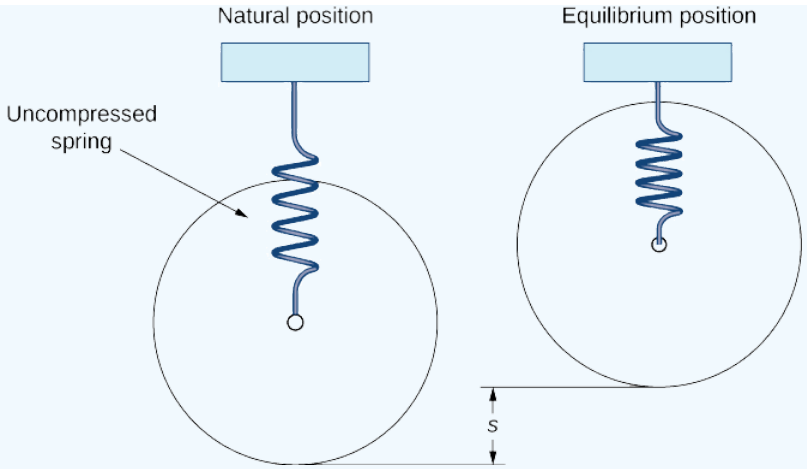


Figure 17.3.9: We can use a spring-mass system to model a motorcycle suspension.

This system can be modeled using the same differential equation we used before:

$$mx''+bx'+kx = 0.$$

A motocross motorcycle weighs 204 lb, and we assume a rider weight of 180 lb. When the rider mounts the motorcycle, the suspension compresses 4 in., then comes to rest at equilibrium. The suspension system provides damping equal to 240 times the instantaneous vertical velocity of the motorcycle (and rider).

1. Set up the differential equation that models the behavior of the motorcycle suspension system.
2. We are interested in what happens when the motorcycle lands after taking a jump. Let time

$$t = 0$$

denote the time when the motorcycle first contacts the ground. If the motorcycle hits the ground with a velocity of 10 ft/sec downward, find the equation of motion of the motorcycle after the jump.
3. Graph the equation of motion over the first second after the motorcycle hits the ground.

Solution

1. We have defined equilibrium to be the point where $mg = ks$, so we have

$$\begin{aligned} mg &= ks \\ 384 &= k\left(\frac{1}{3}\right) \\ k &= 1152. \end{aligned}$$

We also have

$$\begin{aligned} W &= mg \\ 384 &= m(32) \\ m &= 12. \end{aligned}$$

Therefore, the differential equation that models the behavior of the motorcycle suspension is

$$12x''+240x'+1152x = 0.$$

Dividing through by 12, we get

$$x'' + 20x'+96x = 0.$$

2. The differential equation found in part a. has the general solution

$$x(t) = c_1e^{-8t} + c_2e^{-12t}.$$

Now, to determine our initial conditions, we consider the position and velocity of the motorcycle wheel when the wheel first contacts the ground. Since the motorcycle was in the air prior to contacting the ground, the wheel was hanging freely